

# Unit Functional Design Specification - Refinery 1

## Wilmar Industrial Complex PT MNA Serang, Indonesia

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# 1 EQUIPMENT MODULE CLASS DESIGN – EM\_RF\_STARTUP1 (STARTUP)

## 1.1 Functional Description

EM\_RF\_STARTUP1 is a class based equipment module that communicates with four other sequences to complete production process after checking parameters and fatty acid circuit operation. Refer FDS for the functional design requirements including faceplate and detail display. Initially Operator shall be provided only with Hold and Aux On. Production, Drain and Aux Off command shall be enabled during the execution of Aux On State. All devices used in EM shall be acquired in Aux On Command and It shall be released in Aux Off Command. Hold Request shall be enabled for all acquired devices at the end of Hold command. Commands which require operator intervention shall be configured as Operator Prompts wherever possible.

## 1.2 Instances

The following Table contains the module instances of the equipment module class.

| Device Name | Module Class     | Instance Name  |
|-------------|------------------|----------------|
| EM          | EM_RF_STARTUP1   | RF1EQ_STARTUP1 |
| DEV1        | _MVS11_IL_130_SO | RF1_PU1734CA   |
| DEV2        | _MRSS11_130_SE   | RF1_PU1732B    |
| DEV3        | _MRSS11_130_SE   | RF1_PU1741X1   |
| DEV4        | _MRSS11_130_SE   | RF1_PU1741X2   |
| DEV5        | _V12_IP_130_SE   | RF1_V1714ICB   |
| DEV6        | _CMT_130_SE      | RF1_FT1734CA   |
| DEV7        | EM_IC_VAC_PROD1  | RF1EQ_819IC1   |
| DEV8        | EM_IC_VAC_PROD1  | RF1EQ_819IC2   |
| DEV9        | EM_IC_VAC_PROD2  | RF1EQ_878_32IC |
| DEV10       | EM_DEOD_FA_CKT3  | RF1EQ_814_23   |
| DEV11       | EM_DEOD_FA_CKT1  | RF1EQ_802      |

**Table 1: Instances**

## 1.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command

### 1.3.1 Command 00 – Hold

| Step   | Description                               | Action | Action Description                                      |
|--------|-------------------------------------------|--------|---------------------------------------------------------|
| H0010  | Common Hold Start                         | A0010  | Store Fail Message                                      |
|        |                                           | A0020  | Set Hold Request                                        |
| T1000  | Actions Complete or Time Out & Hold=True  |        | Go to step H1000                                        |
| T2000  | Actions Complete or Time Out & Hold=False |        | Go to step H2000                                        |
|        |                                           |        |                                                         |
| H1000  | Hold Command 1                            | A0010  | Stopping and Disabling Conditional Logic for DEV1       |
| T9900A | Actions Complete or Time Out              |        | Go to step H9900A                                       |
|        |                                           |        |                                                         |
| H2000  | General Reset                             | A0010  | Stopping and Disabling Conditional Logic for DEV1       |
|        |                                           | A0020  | Stopping and Disabling Conditional Logic for DEV2       |
|        |                                           | A0030  | Stopping and Disabling Conditional Logic for DEV3       |
|        |                                           | A0040  | Stopping and Disabling Conditional Logic for DEV4       |
|        |                                           | A0050  | Closing and Disabling Conditional Logic for DEV5        |
|        |                                           | A0070  | Set DEV6 Mode to Remote Output                          |
|        |                                           | A0080  | Set DEV6 output to Closed and Disable Conditional Logic |
| T9900B | Actions Complete or Time Out              |        | Go to step H9900B                                       |
|        |                                           |        |                                                         |
| H9900A | Command Hold End                          | A0010  | Completed                                               |
| END    | Command Complete                          |        | True                                                    |
|        |                                           |        |                                                         |
| H9900B | Release Devices                           | A0010  | Release Devices Message                                 |
|        |                                           | A0020  | Release Private Devices                                 |

| Step | Description      | Action | Action Description      |
|------|------------------|--------|-------------------------|
|      |                  | A0030  | Release Private Devices |
|      |                  | A0040  | Release Private Devices |
|      |                  | A0050  | Release Private Devices |
|      |                  | A0060  | Release Private Devices |
|      |                  | A0070  | Reset Parameters        |
| END1 | Command Complete |        | True                    |

**Table 2: Hold Sequence**

## 1.3.2 Command 01 – Aux On

| Step   | Description          | Action | Action Description                                   |
|--------|----------------------|--------|------------------------------------------------------|
| S0000  | Start                | A0010  | Set Message: Starting Aux On Command                 |
| T0010  | First Pass = False   |        | If First Pass = False then go to step S0010          |
| T0020A | First Pass = True    |        | If First Pass = True then go to step S0020           |
|        |                      |        |                                                      |
| S0010  | Initializations      | A0010  | First Pass Logic                                     |
|        |                      | A0020  | First Pass = True                                    |
|        |                      | A0030  | Set Task Pointer                                     |
|        |                      | A0040  | Acquire EM                                           |
| T0020B | Acquire Devices      |        | If no pending actions in S0010 then go to step S0020 |
|        |                      |        |                                                      |
| S0020  | Acquire Devices      | A0010  | Acquire Devices Message                              |
|        |                      | A0020  | Acquire Private Devices                              |
|        |                      | A0030  | Acquire Private Devices                              |
|        |                      | A0040  | Acquire Private Devices                              |
|        |                      | A0050  | Acquire Private Devices                              |
|        |                      | A0060  | Acquire Private Devices                              |
|        |                      | A0070  | Acquire Private Devices                              |
|        |                      | A0080  | Acquire Private Devices                              |
|        |                      | A0090  | Acquire Private Devices                              |
|        |                      | A0100  | Acquire Private Devices                              |
|        |                      | A0110  | Reset Hold Request                                   |
| T0030  | Standby              |        | If Task Pointer = 1 then go to step S0030            |
| T0100B | Start Aux            |        | If Task Pointer = 2 then go to step S0100            |
| T0200B | Check Parameters     |        | If Task Pointer = 3 then go to step S0200            |
| T0300B | Start Fatty Acid Ckt |        | If Task Pointer = 4 then go to step S0300            |
| T0400B | Section Ready        |        | If Task Pointer = 5 then go to step S0400            |
|        |                      |        |                                                      |



| Step   | Description          | Action | Action Description                                                         |
|--------|----------------------|--------|----------------------------------------------------------------------------|
| S0030  | Standby              | A0010  | Stopping and Disabling Conditional Logic for DEV1                          |
|        |                      | A0020  | Stopping and Disabling Conditional Logic for DEV2                          |
|        |                      | A0030  | Stopping and Disabling Conditional Logic for DEV3                          |
|        |                      | A0040  | Stopping and Disabling Conditional Logic for DEV4                          |
|        |                      | A0050  | Closing and Disabling Conditional Logic for DEV5                           |
|        |                      | A0060  | Set DEV6 Mode to Remote Output                                             |
|        |                      | A0070  | Set DEV4 output to Closed and Disable Conditional Logic                    |
| T0100A | Start Aux            | S0030  | True                                                                       |
| S0100  | Start Aux            | A0010  | Enabling Conditional Logic for DEV1                                        |
|        |                      | A0020  | Enabling Conditional Logic for DEV2                                        |
|        |                      | A0030  | Enabling Conditional Logic for DEV3 and DEV4                               |
|        |                      | A0040  | Enabling Conditional Logic for DEV5                                        |
|        |                      | A0050  | Set DEV6 Mode to Remote Cascade                                            |
|        |                      | A0060  | Setting DEV6 setpoint (FT_734CA Setpoint) and Enabling Conditional Logic   |
|        |                      | A0070  | Start DEV7 and DEV8                                                        |
|        |                      | A0080  | Start DEV9                                                                 |
|        |                      | A0090  | Waiting for Auxilary Circuit to run                                        |
|        |                      | A0100  | 1 min Delay (TP001)                                                        |
| T0200A | Check Parameters     |        | If 1 min delay time (TP001) is over then go to step S0200                  |
| S0200  | Check Parameters     | A0010  | Enabling Conditional Logic for DEV1                                        |
|        |                      | A0020  | Enabling Conditional Logic for DEV2                                        |
|        |                      | A0030  | Enabling Conditional Logic for DEV3 and DEV4                               |
|        |                      | A0040  | Enabling Conditional Logic for DEV5                                        |
|        |                      | A0050  | Set DEV6 Mode to Cascade                                                   |
|        |                      | A0060  | Setting DEV6 setpoint (FT_734CA Setpoint) and Enabling Conditional Logic   |
|        |                      | A0070  | Set Message: Waiting for Vacuum Setpoint                                   |
| T0300A | Start Fatty Acd. Ckt |        | If Vacuum PT_722AB is below Vacuum set point (TP002) then go to step S0300 |
| S0300  | Start Fatty Acd. Ckt | A0010  | Enabling Conditional Logic for DEV1                                        |
|        |                      | A0020  | Enabling Conditional Logic for DEV2                                        |
|        |                      | A0030  | Enabling Conditional Logic for DEV3 and DEV4                               |

| Step   | Description      | Action | Action Description                                                       |
|--------|------------------|--------|--------------------------------------------------------------------------|
|        |                  | A0040  | Enabling Conditional Logic for DEV5                                      |
|        |                  | A0050  | Set DEV6 Mode to Cascade                                                 |
|        |                  | A0060  | Setting DEV6 setpoint (FT_734CA Setpoint) and Enabling Conditional Logic |
|        |                  | A0070  | Acquire DEV10                                                            |
|        |                  | A0080  | Set Message: Starting DEV10                                              |
|        |                  | A0090  | Set Message: Checking Fatty acid tank is in step Production              |
| T0400A | Section Ready    |        | If Fatty acid tank DEV10 is in step <Production> then go to step S0400   |
| S0400  | Section Ready    | A0010  | Enabling Conditional Logic for DEV1                                      |
|        |                  | A0020  | Enabling Conditional Logic for DEV2                                      |
|        |                  | A0030  | Enabling Conditional Logic for DEV3 and DEV4                             |
|        |                  | A0040  | Enabling Conditional Logic for DEV5                                      |
|        |                  | A0050  | Set DEV6 Mode to Cascade                                                 |
|        |                  | A0060  | Setting DEV6 setpoint (FT_734CA Setpoint) and Enabling Conditional Logic |
| T9900  | Complete         |        | If no pending actions in S0400 then go to step S9900                     |
| S9900  | Complete         | A0010  | Set Message: Aux On Command Complete                                     |
| END    | Command Complete |        | True                                                                     |

Table 3: Aux On Sequence

## 1.3.3 Command 02 – Production

| Step   | Description | Action | Action Description                                                       |
|--------|-------------|--------|--------------------------------------------------------------------------|
| S0000  | Start       | A0010  | Set Message: Starting Production Command                                 |
| T0100  | Standby     |        | If Task Pointer = 5 then go to step S0100                                |
| T0400B | Start Aux   |        | If Task Pointer = 6 then go to step S0200                                |
| S0100  | Startup     | A0010  | Enabling Conditional Logic for DEV1                                      |
|        |             | A0020  | Enabling Conditional Logic for DEV2                                      |
|        |             | A0030  | Enabling Conditional Logic for DEV3 and DEV4                             |
|        |             | A0040  | Enabling Conditional Logic for DEV5                                      |
|        |             | A0050  | Set DEV6 Mode to Cascade                                                 |
|        |             | A0060  | Setting DEV6 setpoint (FT_734CA Setpoint) and Enabling Conditional Logic |
|        |             | A0070  | Acquire DEV11                                                            |
|        |             | A0080  | Starting DEV11                                                           |

| Step  | Description      | Action | Action Description                                                       |
|-------|------------------|--------|--------------------------------------------------------------------------|
| T0200 | Production       |        | If DEOD_FA_CKT2 is in step <Production> then go to step S0200            |
|       |                  |        |                                                                          |
| S0200 | Production       | A0010  | Enabling Conditional Logic for DEV1                                      |
|       |                  | A0020  | Enabling Conditional Logic for DEV2                                      |
|       |                  | A0030  | Enabling Conditional Logic for DEV3 and DEV4                             |
|       |                  | A0040  | Enabling Conditional Logic for DEV5                                      |
|       |                  | A0050  | Set DEV6 Mode to Cascade                                                 |
|       |                  | A0060  | Setting DEV6 setpoint (FT_734CA Setpoint) and Enabling Conditional Logic |
| T0200 | Complete         |        | If DEOD_FA_CKT2 is in step <Production> then go to step S0200            |
|       |                  |        |                                                                          |
| S9900 | Complete         | A0010  | Set Message: Production Command Complete                                 |
| END   | Command Complete |        | True                                                                     |

**Table 4: Production Sequence**

## 1.3.4 Command 03 – Drain

| Step  | Description      | Action | Action Description                                                       |
|-------|------------------|--------|--------------------------------------------------------------------------|
| S0000 | Start            | A0010  | Set Message: Starting Drain Command                                      |
| T0100 | Drain            |        | If Task Pointer = 6 then go to step S0100                                |
|       |                  |        |                                                                          |
| S0100 | Drain            | A0010  | Enabling Conditional Logic for DEV1                                      |
|       |                  | A0020  | Enabling Conditional Logic for DEV2                                      |
|       |                  | A0030  | Enabling Conditional Logic for DEV3 and DEV4                             |
|       |                  | A0040  | Enabling Conditional Logic for DEV5                                      |
|       |                  | A0050  | Set DEV6 Mode to Cascade                                                 |
|       |                  | A0060  | Setting DEV6 setpoint (FT_734CA Setpoint) and Enabling Conditional Logic |
| T9900 | Complete         |        | If no pending actions in S0100 then go to step S9900                     |
|       |                  |        |                                                                          |
| S9900 | Complete         | A0010  | Set Message: Drain Command Complete                                      |
| END   | Command Complete |        | True                                                                     |

**Table 5: Drain Sequence**

## 1.3.5 Command 04 – Aux Off

| Step  | Description       | Action | Action Description                                                             |
|-------|-------------------|--------|--------------------------------------------------------------------------------|
| S0000 | Start             | A0010  | Set Message: Starting Aux Off Command                                          |
| T0100 | Aux Off           |        | If DEV11 is in <Standby> or DEOD_FA_CKT2 is in <Standby> then go to step S0100 |
|       |                   |        |                                                                                |
| S0100 | Aux Off           | A0010  | Stopping and Disabling Conditional Logic for DEV1                              |
|       |                   | A0020  | Stopping and Disabling Conditional Logic for DEV2                              |
|       |                   | A0030  | Stopping and Disabling Conditional Logic for DEV3                              |
|       |                   | A0040  | Stopping and Disabling Conditional Logic for DEV4                              |
|       |                   | A0050  | Closing and Disabling Conditional Logic for DEV5                               |
|       |                   | A0060  | Reset AUX (OP003), RUN (OP004), DRAIN (OP005) Command                          |
|       |                   | A0070  | Set DEV6 Mode to Remote Output                                                 |
|       |                   | A0080  | Set DEV6 output to Closed and Disable Conditional Logic                        |
| T0200 | Disable HM and SM |        | If no pending actions in S0100 then go to step S0200                           |
|       |                   |        |                                                                                |
| S0200 | Disable HM        | A0010  | Disable HM 01 (Module Ownership Lost)                                          |
| T0300 | Release Devices   |        | If no pending actions in S0200 then go to step S0300                           |
|       |                   |        |                                                                                |
| S0300 | Release Devices   | A0010  | Release Devices Message                                                        |
|       |                   | A0020  | Release Private Devices                                                        |
|       |                   | A0030  | Release Private Devices                                                        |
|       |                   | A0040  | Release Private Devices                                                        |
|       |                   | A0050  | Release Private Devices                                                        |
|       |                   | A0060  | Release Private Devices                                                        |
|       |                   | A0070  | Release Private Devices                                                        |
| T9900 | Complete          |        | If no pending actions in S0300 then go to step S9900                           |
|       |                   |        |                                                                                |
| S9900 | Complete          | A0010  | Set Message: Aux Off Command Complete                                          |
| END   | Command Complete  |        | True                                                                           |

Table 6: Aux Off Sequence

## 1.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

| Parameter            | Description                | Type           | Units | Default Value/Reference        |
|----------------------|----------------------------|----------------|-------|--------------------------------|
| Operating Parameters |                            |                |       |                                |
| OP001                | Vaccum Equipment Selection | Named Set      | NA    | PU1741X = 0<br>(EJECTOR = 1)   |
| OP002                | Vaccum Pump Selection      | Named Set      | NA    | PU1741X1 = 0<br>(PU1741X2 = 1) |
| OP003                | AUX                        | Named Set      | NA    | OFF                            |
| OP004                | DRAIN                      | Named Set      | NA    | OFF                            |
| OP005                | HOLD                       | Named Set      | NA    | OFF                            |
| Tuning Parameters    |                            |                |       |                                |
| TP001                | Delay Timer                | Floating Point | sec   | 60                             |
| TP002                | Vacuum Setpoint            | Floating Point | mbar  | 0.003                          |
| TP003                | FT_734CA RSP               | Floating Point | sec   | 10                             |

**Table 7: Operating and Tuning Parameters**

## 1.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

| Parameter Name | Parameter Type     | Value/Refrence             |
|----------------|--------------------|----------------------------|
| MY_TAG         |                    | RF1EQ_STARTUP1             |
| OPR_CMD_MASK   |                    | 3                          |
| LT_734CA       | External Refernece | RF1_LT1734CA/AI1/LO_LO_ACT |
| LT_734NA       | External Refernece | RF1_LT1734NA/AI1/LO_LO_ACT |
| LSL_732A       | External Refernece | RF1_LSL1732A/DI1/DISC_ACT  |
| PT_722AB       | External Refernece | RF1_PT1722AB/PV            |
| 819IC1         | Boolean            | False                      |
| 819IC2         | Boolean            | False                      |
| DY_REF_1       | Dynamic Reference  |                            |
| DEOD_FA_CKT1   | String             | RF1EQ_802                  |
| DEOD_FA_CKT2   | String             | RF1EQ_822QE                |
| DEOD_FA_CKT3   | String             | RF1EQ_814_23               |
| DEOD3_STEP     | Boolean            | True                       |

**Table 8: Configurable Parameters**

## 1.6 Named Set

Following Named set shall be used for configuration of EM.

| Name           | Description                                 |
|----------------|---------------------------------------------|
| NS_RF_STARTUP1 | Refinery Area: Deodorising & Ice Condensing |

**Table 9: Named Set Details**

Details of states for NS\_RF\_STARTUP1 named set are as follows,

| Named Set Value | Named Set Name |
|-----------------|----------------|
| 00              | Hold           |
| 01              | Aux On         |
| 02              | Production     |
| 03              | Drain          |
| 04              | Aux Off        |

**Table 10: Name Set State Details**

## 2 EQUIPMENT MODULE CLASS DESIGN – EM\_DEOD\_FA\_CKT1 (1-802)

### 2.1 Functional Description

EM\_DEOD\_FA\_CKT1 is a class based equipment module that performs filling operation to produce the material and afterwards drains the material. EM\_DEOD\_FA\_CKT1 is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. Initially Operator shall be provided only with Hold and Production command. All devices used in EM shall be acquired and released in Production Command. Hold Request shall be enabled for all acquired devices at the end of Hold command. Commands which require operator intervention shall be configured as Operator Prompts wherever possible.

### 2.2 Instances

The following Table contains the module instances of the equipment module class.

| Device Name | Module Class    | Instance Name |
|-------------|-----------------|---------------|
| EM          | EM_DEOD_FA_CKT1 | RF1EQ_802     |
| DEV1        | EM_DEOD_FA_CKT2 | RF1EQ_822QE   |
| DEV2        | _V12_IP_130_SE  | RF1_V1702F    |
| DEV3        | _V12_IP_130_SE  | RF1_V1721XS   |

**Table 11: Instances**

### 2.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

#### 2.3.1 Command 00 – Hold

| Step  | Description                               | Action | Action Description |
|-------|-------------------------------------------|--------|--------------------|
| H0010 | Common Hold Start                         | A0010  | Store Fail Message |
|       |                                           | A0020  | Reset Hold Request |
| T1000 | Actions Complete or Time Out & Hold=True  |        | Go to step H1000   |
| T2000 | Actions Complete or Time Out & Hold=False |        | Go to step H2000   |
|       |                                           |        |                    |

| Step   | Description                  | Action | Action Description      |
|--------|------------------------------|--------|-------------------------|
| H1000  | Hold Command 1               | A0010  | Set Message             |
| T9900A | Actions Complete or Time Out |        | Go to step H9900A       |
|        |                              |        |                         |
| H2000  | General Reset                | A0010  | Closing DEV2            |
|        |                              | A0020  | Closing DEV3            |
| T9900B | Actions Complete or Time Out |        | Go to step H9900B       |
|        |                              |        |                         |
| H9900A | Command Hold End             | A0010  | Completed               |
| END    | Command Complete             |        | True                    |
|        |                              |        |                         |
| H9900B | Release Devices              | A0010  | Release Devices Message |
|        |                              | A0020  | Release Private Device  |
|        |                              | A0030  | Release Private Device  |
|        |                              | A0040  | Reset Parameters        |
| END1   | Command Complete             |        | True                    |

**Table 12: Hold Sequence**

## 2.3.2

## Command 01 – Production

| Step   | Description        | Action | Action Description                                   |
|--------|--------------------|--------|------------------------------------------------------|
| S0000  | Start              | A0010  | Set Message: Starting Production Command             |
| T0010  | First Pass = False |        | If First Pass = False then go to step S0010          |
| T0020A | First Pass = True  |        | If First Pass = True then go to step S0020           |
|        |                    |        |                                                      |
| S0010  | Initializations    | A0010  | First Pass Logic                                     |
|        |                    | A0020  | First Pass = True                                    |
|        |                    | A0030  | Set Task pointer                                     |
| T0020B | Actions Complete   |        | If no pending actions in S0010 then go to step S0020 |
|        |                    |        |                                                      |
| S0020  | Acquire Devices    | A0010  | Acquire Devices Message                              |
|        |                    | A0020  | Acquire Private Device                               |
|        |                    | A0030  | Acquire Private Device                               |
|        |                    | A0040  | Reset Hold Request                                   |
| T0100  | Start              |        | If Task Pointer = 1 then go to step S0100            |

| Step   | Description | Action | Action Description                                                                                      |
|--------|-------------|--------|---------------------------------------------------------------------------------------------------------|
| T0200B | Fill        |        | If Task Pointer = 2 then go to step S0200                                                               |
| T0300B | Production  |        | If Task Pointer = 3 then go to step S0300                                                               |
| T0400B | Drain       |        | If Task Pointer = 4 then go to step S0400                                                               |
| T0500B | Drain Wait  |        | If Task Pointer = 5 then go to step S0500                                                               |
|        |             |        |                                                                                                         |
| S0100  | Standby     | A0010  | Closing and Disabling Conditional Logic for DEV2                                                        |
|        |             | A0020  | Closing and Disabling Conditional Logic for DEV3                                                        |
| T0200A | Fill        |        | If no actions pending in S0100 then go to step S0200                                                    |
|        |             |        |                                                                                                         |
| S0200  | Fill        | A0010  | Enabling Conditional Logic for DEV2                                                                     |
|        |             | A0020  | Closing and Disabling Conditional Logic for DEV3                                                        |
|        |             | A0040  | Set Message: Waiting for LSL_702 is not in alarm or LSH_702 is in alarm or LT_702 is in VHL alarm       |
|        |             | A0050  | Set Message: Scratch Time                                                                               |
| T0300A | Production  |        | If LSH_702 is in alarm or LT_702 is in VHL alarm or 5 min delay time (TP001) is over then go step S0300 |
|        |             |        |                                                                                                         |
| S0300  | Production  | A0010  | Enabling Conditional Logic for DEV2                                                                     |
|        |             | A0020  | Enabling Conditional Logic for DEV3                                                                     |
|        |             | A0030  | Check DEV1 is not owned by some other EM                                                                |
|        |             | A0040  | Acquire DEV1                                                                                            |
|        |             | A0050  | Start Heating command of DEV1                                                                           |
|        |             | A0060  | Start Production command of DEV1                                                                        |
|        |             | A0070  | Start DEV1 of DEV11                                                                                     |
| T0400A | Drain       |        | If DEOD_FA_CKT2 is in <Drain 802> then go to step S0400                                                 |
|        |             |        |                                                                                                         |
| S0400  | Drain       | A0010  | Closing and Disabling Conditional Logic for DEV2                                                        |
|        |             | A0020  | Closing and Disabling Conditional Logic for DEV3                                                        |
|        |             | A0040  | Set Message: LSL_702 is in alarm and LT_702 and FT_702 are in VL alarm                                  |
|        |             | A0050  | 10 sec Delay Time (TP002)                                                                               |
| T0500A | Drain Wait  |        | If 10 sec delay time (TP002) is over then go to step S0500                                              |
|        |             |        |                                                                                                         |
| S0500  | Drain Wait  | A0010  | Closing and Disabling Conditional Logic for DEV2                                                        |

| Step  | Description       | Action | Action Description                                         |
|-------|-------------------|--------|------------------------------------------------------------|
|       |                   | A0020  | Closing and Disabling Conditional Logic for DEV3           |
|       |                   | A0030  | Set Message: Checking Fatty acid tank 2 is in Drain 821A2  |
|       |                   | A0040  | Set Message: 10 sec Delay                                  |
| T0600 | Disable HM and SM |        | If 10 sec delay time (TP002) is over then go to step S0600 |
|       |                   |        |                                                            |
| S0600 | Disable HM        | A0010  | Disable HM 01 (Module Ownership Lost)                      |
| T0700 | Release Devices   |        | If no pending action in S0600 then go to step S0700        |
|       |                   |        |                                                            |
| S0700 | Release Devices   | A0010  | Release Devices Message                                    |
|       |                   | A0020  | Release Private Device                                     |
|       |                   | A0030  | Release Private Device                                     |
| T9900 | Complete          |        | If no pending action in S0700 then go to step S9900        |
|       |                   |        |                                                            |
| S9900 | Complete          | A0010  | Set Message: Production Command Complete                   |
|       |                   | A0020  | Self Release the EM & Reset First Pass                     |
| END1  | Command Complete  |        | True                                                       |

Table 13: Production Sequence

## 2.4

### Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

| Parameter            | Description                          | Type           | Units | Default Value/Reference |
|----------------------|--------------------------------------|----------------|-------|-------------------------|
| Operating Parameters |                                      |                |       |                         |
| OP001                | Selection button to switch ON 1-821X | Named Set      | NA    | OFF                     |
| Tuning Parameters    |                                      |                |       |                         |
| TP001                | Delay Timer 1                        | Floating Point | sec   | 300                     |
| TP002                | Delay Timer 2                        | Floating Point | sec   | 10                      |

Table 14: Operating and Tuning Parameters

## 2.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

| Parameter Name | Parameter Type     | Value/Reference            |
|----------------|--------------------|----------------------------|
| MY_TAG         |                    | RF1EQ_802                  |
| OPR_CMD_MASK   |                    | 3                          |
| LSH_702        | External Reference | RF1_LSH1702/DI1/DISC_ACT   |
| LT_702         | String             | RF1_LT1702                 |
| LSL_702        | External Reference | RF1_LSL1702/DI1/DISC_ACT   |
| FT_702         | External Reference | RF1_FT1702/PID1/LO_ACT     |
| RF_STARTUP     | String             | RF1EQ_STARTUP1             |
| DEOD_FA_CKT2   | String             | RF1EQ_822QE                |
| DY_REF_1       | Dynamic Reference  |                            |
| DY_REF_2       | Dynamic Reference  |                            |
| HOLD           | External Reference | RF1EQ_STARTUP1/OP006_VALUE |

**Table 15: Configurable Parameters**

## 2.6 Named Set

Following Named set shall be used for configuration of EM.

| Name            | Description                                 |
|-----------------|---------------------------------------------|
| NS_DEOD_FA_CKT1 | Refinery Area: Deodorising & Ice Condensing |

**Table 16: Named Set Details**

Details of states for NS\_DEOD\_FA\_CKT1 named set are as follows,

| Named Set Value | Named Set Name |
|-----------------|----------------|
| 00              | Hold           |
| 01              | Production     |

**Table 17: Name Set State Details**

### 3 EQUIPMENT MODULE CLASS DESIGN – EM\_DEOD\_FA\_CKT2 (1-822QE)

#### 3.1 Functional Description

EM\_DEOD\_FA\_CKT2 is a class based equipment module that performs filling and heating operation and after that drains the material. EM\_DEOD\_FA\_CKT2 is derived from the Command Driven equipment module class . Refer FDS for the functional design requirements including faceplate and detail display. Initially Operator shall be provided only with Hold and Heating. Aux Off, Production, Recirculation and Drain command shall be enabled during the execution of Heating State. Hold Request shall be enabled for all acquired devices at the end of HOLD command. Commands which require operator intervention shall be configured as Operator Prompts wherever possible.

#### 3.2 Instances

The following Table contains the module instances of the equipment module class.

| Device Name | Module Class     | Instance Name |
|-------------|------------------|---------------|
| EM          | EM_DEOD_FA_CKT2  | RF1EQ_822QE   |
| DEV1        | _MRSS11_130_CUR  | RF1_PU1702    |
| DEV2        | _V12_IP_130_SE   | RF1_V1712D    |
| DEV3        | _MVS11_IL_130_SO | Ignored       |
| DEV4        | _MRSS11_130_SE   | RF1_PU1722C1  |
| DEV5        | _MRSS11_130_SE   | RF1_PU1722C2  |
| DEV6        | _MRSS11_130_CUR  | RF1_PU17221   |
| DEV7        | _MRSS11_130_CUR  | RF1_PU17222   |
| DEV8        | _V12_IP_130_SE   | RF1_V1702D    |
| DEV9        | _V12_IP_130_SE   | RF1_V1702B    |
| DEV10       | _V12_IP_130_SE   | RF1_V1702BY   |
| DEV11       | _V12_IP_130_SE   | RF1_V1722QEA  |
| DEV12       | _V12_IP_130_SE   | RF1_V1722QEB  |
| DEV13       | _V12_IP_130_SE   | RF1_V1722QEC  |
| DEV14       | _V12_IP_130_SE   | RF1_V1722QED  |
| DEV15       | _V12_IP_130_SE   | RF1_V1722CD1  |
| DEV16       | _V12_IP_130_SE   | RF1_V1722CB   |
| DEV17       | _V12_IP_130_SE   | RF1_V1780BD   |
| DEV18       | _V12_IP_130_SE   | RF1_V1780BB   |
| DEV19       | _V12_IP_130_SE   | RF1_V1781AB   |
| DEV20       | _CSD_130_SE      | RF1_LT1780B   |

| Device Name | Module Class   | Instance Name |
|-------------|----------------|---------------|
| DEV21       | _CSD_130_SE    | RF1_FT1702    |
| DEV22       | _CSD_130_SE    | RF1_LT1722C   |
| DEV23       | _CSD_130_SE    | RF1_TT1781B   |
| DEV24       | _V12_IP_130_SE | RF1_V1721A1D  |
| DEV25       | _V12_IP_130_SE | RF1_V1721A2D  |
| DEV26       | _V12_IP_130_SE | RF1_V1722A1D  |
| DEV27       | _V12_IP_130_SE | RF1_V1722A2D  |
| DEV28       | _V12_IP_130_SE | RF1_V1722A3D  |
| DEV29       | _V12_IP_130_SE | RF1_V1722A4D  |
| DEV30       | _V12_IP_130_SE | RF1_V1722A5D  |
| DEV31       | _V12_IP_130_SE | RF1_V1722A6D  |
| DEV32       | _V12_IP_130_SE | RF1_V1780A1A  |
| DEV33       | _V12_IP_130_SE | RF1_V1780A1B  |
| DEV34       | _V12_IP_130_SE | RF1_V1780A2A  |
| DEV35       | _V12_IP_130_SE | RF1_V1780A2B  |

Table 18: Instances

### 3.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

#### 3.3.1 Command 00 – Hold

| Step  | Description                               | Action | Action Description                                |
|-------|-------------------------------------------|--------|---------------------------------------------------|
| H0010 | Command Hold Start                        | A0010  | Store Fail Message                                |
|       |                                           | A0020  | Reset Hold Request                                |
|       |                                           | A0030  | Save Previous Step Time                           |
| T1000 | Actions Complete or Time Out & Hold=True  |        | Go to step H1000                                  |
| T2000 | Actions Complete or Time Out & Hold=False |        | Go to Step H2000                                  |
|       |                                           |        |                                                   |
| H1000 | Hold Command 1                            | A0010  | Stopping and Disabling Conditional logic for DEV1 |
|       |                                           | A0020  | Stopping DEV3                                     |
|       |                                           | A0030  | Stopping and Disabling Conditional Logic for DEV4 |

| Step   | Description                  | Action | Action Description                                       |
|--------|------------------------------|--------|----------------------------------------------------------|
|        |                              | A0040  | Stopping and Disabling Conditional Logic for DEV5        |
|        |                              | A0050  | Stopping and Disabling Conditional Logic for DEV6        |
|        |                              | A0060  | Stopping and Disabling Conditional Logic for DEV7        |
| T9900A | Actions Complete or Time Out |        | If no pending actions in S1000 then go step H9900B       |
|        |                              |        |                                                          |
| H2000  | General Reset                | A0010  | Stopping and Disabling Conditional logic for DEV1        |
|        |                              | A0020  | Closing DEV2                                             |
|        |                              | A0030  | Stopping DEV3                                            |
|        |                              | A0040  | Stopping and Disabling Conditional Logic for DEV4        |
|        |                              | A0050  | Stopping and Disabling Conditional Logic for DEV5        |
|        |                              | A0060  | Stopping and Disabling Conditional Logic for DEV6        |
|        |                              | A0070  | Stopping and Disabling Conditional Logic for DEV7        |
|        |                              | A0080  | Closing DEV8                                             |
|        |                              | A0090  | Closing and Disabling Conditional Logic for DEV9         |
|        |                              | A0100  | Closing DEV10                                            |
|        |                              | A0110  | Closing and Disabling Conditional Logic for DEV11        |
|        |                              | A0120  | Closing DEV12                                            |
|        |                              | A0130  | Closing and Disabling Conditional Logic for DEV13        |
|        |                              | A0140  | Closing and Disabling Conditional Logic for DEV14        |
|        |                              | A0150  | Closing DEV15                                            |
|        |                              | A0160  | Closing and Disabling Conditional Logic for DEV16        |
|        |                              | A0170  | Closing DEV17                                            |
|        |                              | A0180  | Closing and Disabling Conditional Logic for DEV18        |
|        |                              | A0190  | Closing and Disabling Conditional Logic for DEV19        |
|        |                              | A0200  | Set DEV20 Mode to Remote Out                             |
|        |                              | A0210  | Set DEV20 output to Closed and Disable Conditional Logic |
|        |                              | A0220  | Set DEV21 Mode to Remote Out                             |
|        |                              | A0230  | Set DEV21 output to Closed and Disable Conditional Logic |
|        |                              | A0240  | Set DEV22 Mode to Remote Out                             |

| Step   | Description                  | Action | Action Description                                       |
|--------|------------------------------|--------|----------------------------------------------------------|
|        |                              | A0250  | Set DEV22 output to Closed and Disable Conditional Logic |
|        |                              | A0260  | Set DEV23 Mode to Remote Out                             |
|        |                              | A0270  | Set DEV23 output to Closed and Disable Conditional Logic |
|        |                              | A0280  | Closing and Disabling Conditional Logic for DEV24        |
|        |                              | A0290  | Closing and Disabling Conditional Logic for DEV25        |
|        |                              | A0300  | Closing and Disabling Conditional Logic for DEV26        |
|        |                              | A0310  | Closing and Disabling Conditional Logic for DEV27        |
|        |                              | A0320  | Closing and Disabling Conditional Logic for DEV28        |
|        |                              | A0330  | Closing and Disabling Conditional Logic for DEV29        |
|        |                              | A0340  | Closing and Disabling Conditional Logic for DEV30        |
|        |                              | A0350  | Closing and Disabling Conditional Logic for DEV31        |
|        |                              | A0360  | Closing and Disabling Conditional Logic for DEV32        |
|        |                              | A0370  | Closing and Disabling Conditional Logic for DEV33        |
|        |                              | A0380  | Closing and Disabling Conditional Logic for DEV34        |
|        |                              | A0390  | Closing and Disabling Conditional Logic for DEV35        |
|        |                              | A0400  | Set DEV36 Mode to Remote Out                             |
|        |                              | A0410  | Set DEV36 output to Closed and Disable Conditional Logic |
| T9900B | Actions Complete or Time Out |        | If no pending actions in S2000 then go step H9900B       |
|        |                              |        |                                                          |
| H9900A | Command Hold End             | A0010  | Completed                                                |
| END    | Command Complete             |        | True                                                     |
|        |                              |        |                                                          |
| H9900B | Release Devices              | A0010  | Release Devices Message                                  |
|        |                              | A0020  | Release Private Device                                   |
|        |                              | A0030  | Release Private Device                                   |
|        |                              | A0040  | Release Private Device                                   |
|        |                              | A0050  | Release Private Device                                   |
|        |                              | A0060  | Release Private Device                                   |
|        |                              | A0070  | Release Private Device                                   |

| Step | Description      | Action | Action Description     |
|------|------------------|--------|------------------------|
|      |                  | A0080  | Release Private Device |
|      |                  | A0090  | Release Private Device |
|      |                  | A0100  | Release Private Device |
|      |                  | A0110  | Release Private Device |
|      |                  | A0120  | Release Private Device |
|      |                  | A0130  | Release Private Device |
|      |                  | A0140  | Release Private Device |
|      |                  | A0150  | Release Private Device |
|      |                  | A0160  | Release Private Device |
|      |                  | A0170  | Release Private Device |
|      |                  | A0180  | Release Private Device |
|      |                  | A0190  | Release Private Device |
|      |                  | A0200  | Release Private Device |
|      |                  | A0210  | Release Private Device |
|      |                  | A0220  | Release Private Device |
|      |                  | A0230  | Release Private Device |
|      |                  | A0240  | Release Private Device |
|      |                  | A0250  | Release Private Device |
|      |                  | A0260  | Release Private Device |
|      |                  | A0270  | Release Private Device |
|      |                  | A0280  | Release Private Device |
|      |                  | A0290  | Release Private Device |
|      |                  | A0300  | Release Private Device |
|      |                  | A0310  | Release Private Device |
|      |                  | A0320  | Release Private Device |
|      |                  | A0330  | Release Private Device |
|      |                  | A0340  | Release Private Device |
|      |                  | A0350  | Release Private Device |
|      |                  | A0360  | Release Private Device |
|      |                  | A0370  | Reset Parameters       |
| END1 | Command Complete |        | True                   |

Table 19: Hold Sequence

## 3.3.2 Command 01 – Heating

| Step   | Description        | Action | Action Description                          |
|--------|--------------------|--------|---------------------------------------------|
| S0000  | Start              | A0010  | Set Message: Starting Heating Command       |
| T0010  | First Pass = False |        | If First Pass = False then go to step S0010 |
| T0020A | First Pass = True  |        | If First Pass = True then go to step S0020  |
|        |                    |        |                                             |

| Step   | Description      | Action | Action Description                                   |
|--------|------------------|--------|------------------------------------------------------|
| S0010  | Initializations  | A0010  | First Pass Logic                                     |
|        |                  | A0020  | First Pass = True                                    |
|        |                  | A0030  | Set Task Pointer                                     |
| T0020B | Actions Complete |        | If no pending actions in S0010 then go to step S0020 |
| S0020  | Acquire Devices  | A0010  | Acquire Devices Message                              |
|        |                  | A0020  | Acquire Private Device                               |
|        |                  | A0030  | Acquire Private Device                               |
|        |                  | A0040  | Acquire Private Device                               |
|        |                  | A0050  | Acquire Private Device                               |
|        |                  | A0060  | Acquire Private Device                               |
|        |                  | A0070  | Acquire Private Device                               |
|        |                  | A0080  | Acquire Private Device                               |
|        |                  | A0090  | Acquire Private Device                               |
|        |                  | A0100  | Acquire Private Device                               |
|        |                  | A0110  | Acquire Private Device                               |
|        |                  | A0120  | Acquire Private Device                               |
|        |                  | A0130  | Acquire Private Device                               |
|        |                  | A0140  | Acquire Private Device                               |
|        |                  | A0150  | Acquire Private Device                               |
|        |                  | A0160  | Acquire Private Device                               |
|        |                  | A0170  | Acquire Private Device                               |
|        |                  | A0180  | Acquire Private Device                               |
|        |                  | A0190  | Acquire Private Device                               |
|        |                  | A0200  | Acquire Private Device                               |
|        |                  | A0210  | Acquire Private Device                               |
|        |                  | A0220  | Acquire Private Device                               |
|        |                  | A0230  | Acquire Private Device                               |
|        |                  | A0240  | Acquire Private Device                               |
|        |                  | A0250  | Acquire Private Device                               |
|        |                  | A0260  | Acquire Private Device                               |
|        |                  | A0270  | Acquire Private Device                               |
|        |                  | A0280  | Acquire Private Device                               |
|        |                  | A0290  | Acquire Private Device                               |
|        |                  | A0300  | Acquire Private Device                               |
|        |                  | A0310  | Acquire Private Device                               |
|        |                  | A0320  | Acquire Private Device                               |
|        |                  | A0330  | Acquire Private Device                               |
|        |                  | A0340  | Acquire Private Device                               |
|        |                  | A0350  | Acquire Private Device                               |
|        |                  | A0360  | Acquire Private Device                               |

| Step   | Description | Action | Action Description                                       |
|--------|-------------|--------|----------------------------------------------------------|
|        |             | A0370  | Reset Hold Request                                       |
| T0100  | Standby     |        | If Task Pointer = 1 then go to step S0100                |
| T0200B | Fill        |        | If Task Pointer = 2 then go to step S0200                |
| T0300B | Heating     |        | If Task Pointer = 3 then go to step S0300                |
|        |             |        |                                                          |
| S0100  | Standby     | A0010  | Stopping and Disabling Conditional logic for DEV1        |
|        |             | A0020  | Closing DEV2                                             |
|        |             | A0030  | Stopping DEV3                                            |
|        |             | A0040  | Stopping and Disabling Conditional Logic for DEV4        |
|        |             | A0050  | Stopping and Disabling Conditional Logic for DEV5        |
|        |             | A0060  | Stopping and Disabling Conditional Logic for DEV6        |
|        |             | A0070  | Stopping and Disabling Conditional Logic for DEV7        |
|        |             | A0080  | Closing DEV8                                             |
|        |             | A0090  | Closing and Disabling Conditional Logic for DEV9         |
|        |             | A0100  | Closing DEV10                                            |
|        |             | A0110  | Closing and Disabling Conditional Logic for DEV11        |
|        |             | A0120  | Closing DEV12                                            |
|        |             | A0130  | Closing and Disabling Conditional Logic for DEV13        |
|        |             | A0140  | Closing and Disabling Conditional Logic for DEV14        |
|        |             | A0150  | Closing DEV15                                            |
|        |             | A0160  | Closing and Disabling Conditional Logic for DEV16        |
|        |             | A0170  | Closing DEV17                                            |
|        |             | A0180  | Closing and Disabling Conditional Logic for DEV18        |
|        |             | A0190  | Closing and Disabling Conditional Logic for DEV19        |
|        |             | A0200  | Set DEV20 Mode to Remote Out                             |
|        |             | A0210  | Set DEV20 output to Closed and Disable Conditional Logic |
|        |             | A0220  | Set DEV21 Mode to Remote Out                             |
|        |             | A0230  | Set DEV21 output to Closed and Disable Conditional Logic |
|        |             | A0240  | Set DEV22 Mode to Remote Out                             |
|        |             | A0250  | Set DEV22 output to Closed and Disable Conditional Logic |
|        |             | A0260  | Set DEV23 Mode to Remote Out                             |

| Step   | Description | Action | Action Description                                                                              |
|--------|-------------|--------|-------------------------------------------------------------------------------------------------|
|        |             | A0270  | Set DEV23 output to Closed and Disable Conditional Logic                                        |
|        |             | A0280  | Closing and Disabling Conditional Logic for DEV24                                               |
|        |             | A0290  | Closing and Disabling Conditional Logic for DEV25                                               |
|        |             | A0300  | Closing and Disabling Conditional Logic for DEV26                                               |
|        |             | A0310  | Closing and Disabling Conditional Logic for DEV27                                               |
|        |             | A0320  | Closing and Disabling Conditional Logic for DEV28                                               |
|        |             | A0330  | Closing and Disabling Conditional Logic for DEV29                                               |
|        |             | A0340  | Closing and Disabling Conditional Logic for DEV30                                               |
|        |             | A0350  | Closing and Disabling Conditional Logic for DEV31                                               |
|        |             | A0360  | Closing and Disabling Conditional Logic for DEV32                                               |
|        |             | A0370  | Closing and Disabling Conditional Logic for DEV33                                               |
|        |             | A0380  | Closing and Disabling Conditional Logic for DEV34                                               |
|        |             | A0390  | Closing and Disabling Conditional Logic for DEV35                                               |
|        |             | A0400  | Set DEV36 Mode to Remote Out                                                                    |
|        |             | A0410  | Set DEV36 output to Closed and Disable Conditional Logic                                        |
| T0200A | Fill        |        | If EM_DEOD_FA_CKT1 is in step <Production> & No Vacuum alarm (VACUUM_ALM) then go to step S0200 |
|        |             |        |                                                                                                 |
| S0200  | Fill        | A0010  | Enabling Conditional logic for DEV1                                                             |
|        |             | A0020  | Opening DEV2                                                                                    |
|        |             | A0030  | Stopping DEV3                                                                                   |
|        |             | A0040  | Stopping and Disabling Conditional Logic for DEV4                                               |
|        |             | A0050  | Stopping and Disabling Conditional Logic for DEV5                                               |
|        |             | A0060  | Stopping and Disabling Conditional Logic for DEV6                                               |
|        |             | A0070  | Stopping and Disabling Conditional Logic for DEV7                                               |
|        |             | A0080  | Opening DEV8                                                                                    |
|        |             | A0090  | Closing and Disabling Conditional Logic for DEV9                                                |
|        |             | A0100  | Closing DEV10                                                                                   |

| Step | Description | Action | Action Description                                                           |
|------|-------------|--------|------------------------------------------------------------------------------|
|      |             | A0110  | Closing and Disabling Conditional Logic for DEV11                            |
|      |             | A0120  | Opening DEV12                                                                |
|      |             | A0130  | Enabling Conditional Logic for DEV13                                         |
|      |             | A0140  | Enabling Conditional Logic for DEV14                                         |
|      |             | A0150  | Closing DEV15                                                                |
|      |             | A0160  | Closing and Disabling Conditional Logic for DEV16                            |
|      |             | A0170  | Closing DEV17                                                                |
|      |             | A0180  | Closing and Disabling Conditional Logic for DEV18                            |
|      |             | A0190  | Closing and Disabling Conditional Logic for DEV19                            |
|      |             | A0200  | Set DEV20 Mode to Remote Out                                                 |
|      |             | A0210  | Set DEV20 output to Closed and Disable Conditional Logic                     |
|      |             | A0220  | Set DEV21 Mode to Remote Cascade                                             |
|      |             | A0230  | Setting DEV21 setpoint to ( FT_702 Setpoint ) and Enabling Conditional Logic |
|      |             | A0240  | Set DEV22 Mode to Remote Out                                                 |
|      |             | A0250  | Set DEV22 output to Closed and Disable Conditional Logic                     |
|      |             | A0260  | Set DEV23 Mode to Remote Out                                                 |
|      |             | A0270  | Set DEV23 output to Closed and Disable Conditional Logic                     |
|      |             | A0280  | Closing and Disabling Conditional Logic for DEV24                            |
|      |             | A0290  | Closing and Disabling Conditional Logic for DEV25                            |
|      |             | A0300  | Closing and Disabling Conditional Logic for DEV26                            |
|      |             | A0310  | Closing and Disabling Conditional Logic for DEV27                            |
|      |             | A0320  | Closing and Disabling Conditional Logic for DEV28                            |
|      |             | A0330  | Closing and Disabling Conditional Logic for DEV29                            |
|      |             | A0340  | Closing and Disabling Conditional Logic for DEV30                            |
|      |             | A0350  | Closing and Disabling Conditional Logic for DEV31                            |
|      |             | A0360  | Closing and Disabling Conditional Logic for DEV32                            |
|      |             | A0370  | Closing and Disabling Conditional Logic for DEV33                            |
|      |             | A0380  | Closing and Disabling Conditional Logic for DEV34                            |

| Step   | Description | Action | Action Description                                        |
|--------|-------------|--------|-----------------------------------------------------------|
|        |             | A0390  | Closing and Disabling Conditional Logic for DEV35         |
|        |             | A0400  | Set DEV36 Mode to Remote Out                              |
|        |             | A0410  | Set DEV36 output to Open and Disable Conditional Logic    |
|        |             | A0420  | Set Message: LSM_721A alarm is not active                 |
|        |             | A0430  | 5 sec Delay Time (TP001)                                  |
| T0300A | Heating     |        | If 5 sec delay time (TP001) is over then go to step S0300 |
| S0300  | Heating     | A0010  | Stopping and Disabling Conditional Logic for DEV1         |
|        |             | A0020  | Opening DEV2                                              |
|        |             | A0030  | Stopping DEV3                                             |
|        |             | A0040  | Stopping and Disabling Conditional Logic for DEV4         |
|        |             | A0050  | Stopping and Disabling Conditional Logic for DEV5         |
|        |             | A0060  | Stopping and Disabling Conditional Logic for DEV6         |
|        |             | A0070  | Stopping and Disabling Conditional Logic for DEV7         |
|        |             | A0080  | Opening DEV8                                              |
|        |             | A0090  | Closing and Disabling Conditional Logic for DEV9          |
|        |             | A0100  | Closing DEV10                                             |
|        |             | A0110  | Closing and Disabling Conditional Logic for DEV11         |
|        |             | A0120  | Opening DEV12                                             |
|        |             | A0130  | Enabling Conditional Logic for DEV13                      |
|        |             | A0140  | Enabling Conditional Logic for DEV14                      |
|        |             | A0150  | Closing DEV15                                             |
|        |             | A0160  | Closing and Disabling Conditional Logic for DEV16         |
|        |             | A0170  | Closing DEV17                                             |
|        |             | A0180  | Closing and Disabling Conditional Logic for DEV18         |
|        |             | A0190  | Closing and Disabling Conditional Logic for DEV19         |
|        |             | A0200  | Set DEV20 Mode to Remote Out                              |
|        |             | A0210  | Set DEV20 output to Closed and Disable Conditional Logic  |
|        |             | A0220  | Set DEV21 Mode to Remote Out                              |
|        |             | A0230  | Set DEV21 output to Closed and Disable Conditional Logic  |
|        |             | A0240  | Set DEV22 Mode to Remote Out                              |

| Step  | Description      | Action | Action Description                                                   |
|-------|------------------|--------|----------------------------------------------------------------------|
|       |                  | A0250  | Set DEV22 output to Closed and Disable Conditional Logic             |
|       |                  | A0260  | Set DEV23 Mode to Remote Out                                         |
|       |                  | A0270  | Set DEV23 output to Closed and Disable Conditional Logic             |
|       |                  | A0280  | Closing and Disabling Conditional Logic for DEV24                    |
|       |                  | A0290  | Closing and Disabling Conditional Logic for DEV25                    |
|       |                  | A0300  | Closing and Disabling Conditional Logic for DEV26                    |
|       |                  | A0310  | Closing and Disabling Conditional Logic for DEV27                    |
|       |                  | A0320  | Closing and Disabling Conditional Logic for DEV28                    |
|       |                  | A0330  | Closing and Disabling Conditional Logic for DEV29                    |
|       |                  | A0340  | Closing and Disabling Conditional Logic for DEV30                    |
|       |                  | A0350  | Closing and Disabling Conditional Logic for DEV31                    |
|       |                  | A0360  | Closing and Disabling Conditional Logic for DEV32                    |
|       |                  | A0370  | Closing and Disabling Conditional Logic for DEV33                    |
|       |                  | A0380  | Closing and Disabling Conditional Logic for DEV34                    |
|       |                  | A0390  | Closing and Disabling Conditional Logic for DEV35                    |
|       |                  | A0400  | Set DEV36 Mode to Remote Out                                         |
|       |                  | A0410  | Set DEV36 output to Open and Disable Conditional Logic               |
|       |                  | A0420  | Set Message: Waiting for Deodorising Temperature Setpoint (TP011)    |
| T9900 | Complete         |        | If Deodorising Temp. set point (TP011) reached then go to step S9900 |
|       |                  |        |                                                                      |
| S9900 | Complete         | A0010  | Set Message: Heating Command Complete                                |
| END   | Command Complete |        | True                                                                 |

**Table 20: Heating Sequence****3.3.3****Command 02 – Production**

| Step  | Description    | Action | Action Description                        |
|-------|----------------|--------|-------------------------------------------|
| S0000 | Start          | A0010  | Set Message: Starting Production Command  |
| T0100 | Pre-Production |        | If Task Pointer = 3 then go to step S0100 |

| Step   | Description                | Action | Action Description                                                          |
|--------|----------------------------|--------|-----------------------------------------------------------------------------|
| T0200C | Confirm Production         |        | If Task Pointer = 4 then go to step S0200                                   |
| T0300B | Production                 |        | If Task Pointer = 5 then go to step S0300                                   |
| T0400B | Confirm Hot Recirculation  |        | If Task Pointer = 6 then go to step S0400                                   |
| T0500B | Hot Recirculation          |        | If Task Pointer = 7 then go to step S0500                                   |
| T0600B | Confirm Cold Recirculation |        | If Task Pointer = 8 then go to step S0600                                   |
| T0700B | Cold Recirculation         |        | If Task Pointer = 9 then go to step S0700                                   |
|        |                            |        |                                                                             |
| S0100  | Pre-Production             | A0010  | Enabling Conditional logic for DEV1                                         |
|        |                            | A0020  | Opening DEV2                                                                |
|        |                            | A0030  | Stopping DEV3                                                               |
|        |                            | A0040  | Enabling Conditional Logic for DEV4                                         |
|        |                            | A0050  | Enabling Conditional Logic for DEV5                                         |
|        |                            | A0060  | Enabling Conditional Logic for DEV6                                         |
|        |                            | A0070  | Enabling Conditional Logic for DEV7                                         |
|        |                            | A0080  | Opening DEV8                                                                |
|        |                            | A0090  | Closing and Disabling Conditional Logic for DEV9                            |
|        |                            | A0100  | Closing DEV10                                                               |
|        |                            | A0110  | Closing and Disabling Conditional Logic for DEV11                           |
|        |                            | A0120  | Opening DEV12                                                               |
|        |                            | A0130  | Enabling Conditional Logic for DEV13                                        |
|        |                            | A0140  | Enabling Conditional Logic for DEV14                                        |
|        |                            | A0150  | Opening DEV15                                                               |
|        |                            | A0160  | Closing and Disabling Conditional Logic for DEV16                           |
|        |                            | A0170  | Opening DEV17                                                               |
|        |                            | A0180  | Closing and Disabling Conditional Logic for DEV18                           |
|        |                            | A0190  | Closing and Disabling Conditional Logic for DEV19                           |
|        |                            | A0200  | Set DEV20 Mode to Remote Cascade                                            |
|        |                            | A0210  | Setting DEV20 setpoint to (LT_780B Setpoint) and Enabling Conditional Logic |
|        |                            | A0220  | Set DEV21 Mode to Remote Cascade                                            |
|        |                            | A0230  | Setting DEV21 setpoint to (FT_702 Setpoint) and Enabling Conditional Logic  |
|        |                            | A0240  | Set DEV22 Mode to Remote Cascade                                            |
|        |                            | A0250  | Setting DEV22 setpoint to (LT_722C Setpoint) and Enabling Conditional Logic |
|        |                            | A0260  | Set DEV23 Mode to Remote Cascade                                            |

| Step   | Description        | Action | Action Description                                                                                       |
|--------|--------------------|--------|----------------------------------------------------------------------------------------------------------|
|        |                    | A0270  | Setting DEV23 setpoint to (TT_781B Setpoint) and Enabling Conditional Logic                              |
|        |                    | A0280  | Closing and Disabling Conditional Logic for DEV24                                                        |
|        |                    | A0290  | Closing and Disabling Conditional Logic for DEV25                                                        |
|        |                    | A0300  | Closing and Disabling Conditional Logic for DEV26                                                        |
|        |                    | A0310  | Closing and Disabling Conditional Logic for DEV27                                                        |
|        |                    | A0320  | Closing and Disabling Conditional Logic for DEV28                                                        |
|        |                    | A0330  | Closing and Disabling Conditional Logic for DEV29                                                        |
|        |                    | A0340  | Closing and Disabling Conditional Logic for DEV30                                                        |
|        |                    | A0350  | Closing and Disabling Conditional Logic for DEV31                                                        |
|        |                    | A0360  | Closing and Disabling Conditional Logic for DEV32                                                        |
|        |                    | A0370  | Closing and Disabling Conditional Logic for DEV33                                                        |
|        |                    | A0380  | Closing and Disabling Conditional Logic for DEV34                                                        |
|        |                    | A0390  | Closing and Disabling Conditional Logic for DEV35                                                        |
|        |                    | A0400  | Set DEV36 Mode to Remote Out                                                                             |
|        |                    | A0410  | Set DEV36 output to Open and Disable Conditional Logic                                                   |
| T0200A | Confirm Production |        | If LSL_780B is not in alarm or LSH_780B is in alarm or LT_780B_VHL is in VHL alarm then go to step S0200 |
|        |                    |        |                                                                                                          |
| S0200  | Confirm Production | A0010  | Stopping and Disabling Conditional logic for DEV1                                                        |
|        |                    | A0020  | Opening DEV2                                                                                             |
|        |                    | A0030  | Stopping DEV3                                                                                            |
|        |                    | A0040  | Stopping and Disabling Conditional Logic for DEV4                                                        |
|        |                    | A0050  | Stopping and Disabling Conditional Logic for DEV5                                                        |
|        |                    | A0060  | Stopping and Disabling Conditional Logic for DEV6                                                        |
|        |                    | A0070  | Stopping and Disabling Conditional Logic for DEV7                                                        |
|        |                    | A0080  | Opening DEV8                                                                                             |
|        |                    | A0090  | Closing and Disabling Conditional Logic for DEV9                                                         |
|        |                    | A0100  | Closing DEV10                                                                                            |

| Step | Description | Action | Action Description                                       |
|------|-------------|--------|----------------------------------------------------------|
|      |             | A0110  | Closing and Disabling Conditional Logic for DEV11        |
|      |             | A0120  | Opening DEV12                                            |
|      |             | A0130  | Enabling Conditional Logic for DEV13                     |
|      |             | A0140  | Enabling Conditional Logic for DEV14                     |
|      |             | A0150  | Opening DEV15                                            |
|      |             | A0160  | Closing and Disabling Conditional Logic for DEV16        |
|      |             | A0170  | Opening DEV17                                            |
|      |             | A0180  | Closing and Disabling Conditional Logic for DEV18        |
|      |             | A0190  | Closing and Disabling Conditional Logic for DEV19        |
|      |             | A0200  | Set DEV20 Mode to Remote Out                             |
|      |             | A0210  | Set DEV20 output to Closed and Disable Conditional Logic |
|      |             | A0220  | Set DEV21 Mode to Remote Out                             |
|      |             | A0230  | Set DEV21 output to Closed and Disable Conditional Logic |
|      |             | A0240  | Set DEV22 Mode to Remote Out                             |
|      |             | A0250  | Set DEV22 output to Closed and Disable Conditional Logic |
|      |             | A0260  | Set DEV23 Mode to Remote Out                             |
|      |             | A0270  | Set DEV23 output to Closed and Disable Conditional Logic |
|      |             | A0280  | Closing and Disabling Conditional Logic for DEV24        |
|      |             | A0290  | Closing and Disabling Conditional Logic for DEV25        |
|      |             | A0300  | Closing and Disabling Conditional Logic for DEV26        |
|      |             | A0310  | Closing and Disabling Conditional Logic for DEV27        |
|      |             | A0320  | Closing and Disabling Conditional Logic for DEV28        |
|      |             | A0330  | Closing and Disabling Conditional Logic for DEV29        |
|      |             | A0340  | Closing and Disabling Conditional Logic for DEV30        |
|      |             | A0350  | Closing and Disabling Conditional Logic for DEV31        |
|      |             | A0360  | Closing and Disabling Conditional Logic for DEV32        |
|      |             | A0370  | Closing and Disabling Conditional Logic for DEV33        |
|      |             | A0380  | Closing and Disabling Conditional Logic for DEV34        |

| Step   | Description | Action | Action Description                                                          |
|--------|-------------|--------|-----------------------------------------------------------------------------|
|        |             | A0390  | Closing and Disabling Conditional Logic for DEV35                           |
|        |             | A0400  | Set DEV36 Mode to Remote Cascade                                            |
|        |             | A0410  | Setting DEV36 setpoint to (TT_722C Setpoint) and Enabling Conditional Logic |
|        |             | A0420  | Set Message: Confirm Production after closing/opening manual valves         |
|        |             | A0430  | Reset Hot Recirculation (OP005) Parameter                                   |
| T0300A | Production  |        | If Operator Confirm <Production> (OP004) step then go to step S0300         |
| S0300  | Production  |        |                                                                             |
|        |             | A0010  | Enabling Conditional logic for DEV1                                         |
|        |             | A0020  | Opening DEV2                                                                |
|        |             | A0030  | Stopping DEV3                                                               |
|        |             | A0040  | Enabling Conditional Logic for DEV4                                         |
|        |             | A0050  | Enabling Conditional Logic for DEV5                                         |
|        |             | A0060  | Enabling Conditional Logic for DEV6                                         |
|        |             | A0070  | Enabling Conditional Logic for DEV7                                         |
|        |             | A0080  | Opening DEV8                                                                |
|        |             | A0090  | Closing and Disabling Conditional Logic for DEV9                            |
|        |             | A0100  | Closing DEV10                                                               |
|        |             | A0110  | Closing and Disabling Conditional Logic for DEV11                           |
|        |             | A0120  | Opening DEV12                                                               |
|        |             | A0130  | Enabling Conditional Logic for DEV13                                        |
|        |             | A0140  | Enabling Conditional Logic for DEV14                                        |
|        |             | A0150  | Opening DEV15                                                               |
|        |             | A0160  | Closing and Disabling Conditional Logic for DEV16                           |
|        |             | A0170  | Opening DEV17                                                               |
|        |             | A0180  | Closing and Disabling Conditional Logic for DEV18                           |
|        |             | A0190  | Closing and Disabling Conditional Logic for DEV19                           |
|        |             | A0200  | Set DEV20 Mode to Remote Cascade                                            |
|        |             | A0210  | Setting DEV20 setpoint to (LT_780B Setpoint) and Enabling Conditional Logic |
|        |             | A0220  | Set DEV21 Mode to Remote Cascade                                            |
|        |             | A0230  | Setting DEV21 setpoint to (FT_702 Setpoint) and Enabling Conditional Logic  |
|        |             | A0240  | Set DEV22 Mode to Remote Cascade                                            |
|        |             | A0250  | Setting DEV22 setpoint to (LT_722C Setpoint) and Enabling Conditional Logic |
|        |             | A0260  | Set DEV23 Mode to Remote Cascade                                            |

| Step   | Description                | Action | Action Description                                                                |
|--------|----------------------------|--------|-----------------------------------------------------------------------------------|
|        |                            | A0270  | Setting DEV23 setpoint to (TT_781B Setpoint) and Enabling Conditional Logic       |
|        |                            | A0280  | Closing and Disabling Conditional Logic for DEV24                                 |
|        |                            | A0290  | Closing and Disabling Conditional Logic for DEV25                                 |
|        |                            | A0300  | Closing and Disabling Conditional Logic for DEV26                                 |
|        |                            | A0310  | Closing and Disabling Conditional Logic for DEV27                                 |
|        |                            | A0320  | Closing and Disabling Conditional Logic for DEV28                                 |
|        |                            | A0330  | Closing and Disabling Conditional Logic for DEV29                                 |
|        |                            | A0340  | Closing and Disabling Conditional Logic for DEV30                                 |
|        |                            | A0350  | Closing and Disabling Conditional Logic for DEV31                                 |
|        |                            | A0360  | Closing and Disabling Conditional Logic for DEV32                                 |
|        |                            | A0370  | Closing and Disabling Conditional Logic for DEV33                                 |
|        |                            | A0380  | Closing and Disabling Conditional Logic for DEV34                                 |
|        |                            | A0390  | Closing and Disabling Conditional Logic for DEV35                                 |
|        |                            | A0400  | Set DEV36 Mode to Remote Cascade                                                  |
|        |                            | A0410  | Setting DEV36 setpoint to (TT_722C Setpoint) and Enabling Conditional Logic       |
| T0400A | Confirm Hot Recirculation  |        | If Operator select <Confirm Hot Recirculation> (OP005) step then go to step S0400 |
| T0600A | Confirm Cold Recirculation |        | If Startup is in step <Drain> then go to step S0600                               |
|        |                            |        |                                                                                   |
| S0400  | Confirm Hot Recirculation  | A0010  | Stopping and Disabling Conditional logic for DEV1                                 |
|        |                            | A0020  | Opening DEV2                                                                      |
|        |                            | A0030  | Stopping DEV3                                                                     |
|        |                            | A0040  | Stopping and Disabling Conditional Logic for DEV4                                 |
|        |                            | A0050  | Stopping and Disabling Conditional Logic for DEV5                                 |
|        |                            | A0060  | Stopping and Disabling Conditional Logic for DEV6                                 |
|        |                            | A0070  | Stopping and Disabling Conditional Logic for DEV7                                 |
|        |                            | A0080  | Opening DEV8                                                                      |

| Step | Description | Action | Action Description                                       |
|------|-------------|--------|----------------------------------------------------------|
|      |             | A0090  | Closing and Disabling Conditional Logic for DEV9         |
|      |             | A0100  | Closing DEV10                                            |
|      |             | A0110  | Closing and Disabling Conditional Logic for DEV11        |
|      |             | A0120  | Opening DEV12                                            |
|      |             | A0130  | Enabling Conditional Logic for DEV13                     |
|      |             | A0140  | Enabling Conditional Logic for DEV14                     |
|      |             | A0150  | Opening DEV15                                            |
|      |             | A0160  | Closing and Disabling Conditional Logic for DEV16        |
|      |             | A0170  | Opening DEV17                                            |
|      |             | A0180  | Closing and Disabling Conditional Logic for DEV18        |
|      |             | A0190  | Closing and Disabling Conditional Logic for DEV19        |
|      |             | A0200  | Set DEV20 Mode to Remote Out                             |
|      |             | A0210  | Set DEV20 output to Closed and Disable Conditional Logic |
|      |             | A0220  | Set DEV21 Mode to Remote Out                             |
|      |             | A0230  | Set DEV21 output to Closed and Disable Conditional Logic |
|      |             | A0240  | Set DEV22 Mode to Remote Out                             |
|      |             | A0250  | Set DEV22 output to Closed and Disable Conditional Logic |
|      |             | A0260  | Set DEV23 Mode to Remote Out                             |
|      |             | A0270  | Set DEV23 output to Closed and Disable Conditional Logic |
|      |             | A0280  | Closing and Disabling Conditional Logic for DEV24        |
|      |             | A0290  | Closing and Disabling Conditional Logic for DEV25        |
|      |             | A0300  | Closing and Disabling Conditional Logic for DEV26        |
|      |             | A0310  | Closing and Disabling Conditional Logic for DEV27        |
|      |             | A0320  | Closing and Disabling Conditional Logic for DEV28        |
|      |             | A0330  | Closing and Disabling Conditional Logic for DEV29        |
|      |             | A0340  | Closing and Disabling Conditional Logic for DEV30        |
|      |             | A0350  | Closing and Disabling Conditional Logic for DEV31        |
|      |             | A0360  | Closing and Disabling Conditional Logic for DEV32        |
|      |             | A0370  | Closing and Disabling Conditional Logic for DEV33        |

| Step   | Description                        | Action | Action Description                                                          |
|--------|------------------------------------|--------|-----------------------------------------------------------------------------|
|        |                                    | A0380  | Closing and Disabling Conditional Logic for DEV34                           |
|        |                                    | A0390  | Closing and Disabling Conditional Logic for DEV35                           |
|        |                                    | A0400  | Set DEV36 Mode to Remote Cascade                                            |
|        |                                    | A0410  | Setting DEV36 setpoint to (TT_722C Setpoint) and Enabling Conditional Logic |
|        |                                    | A0420  | Set Message: Confirm Hot Recirculation after opening/closing manual valves  |
|        |                                    | A0430  | Reset Production (OP004) Parameter                                          |
| T0500A | Hot Recirculation                  |        | If Operator confirm <Hot Recirculation> (OP005) step then go to step S0500  |
| S0500  | Hot Recirculation (With Boiler On) | A0010  | Enabling Conditional logic for DEV1                                         |
|        |                                    | A0020  | Opening DEV2                                                                |
|        |                                    | A0030  | Stopping DEV3                                                               |
|        |                                    | A0040  | Enabling Conditional Logic for DEV4                                         |
|        |                                    | A0050  | Enabling Conditional Logic for DEV5                                         |
|        |                                    | A0060  | Enabling Conditional Logic for DEV6                                         |
|        |                                    | A0070  | Enabling Conditional Logic for DEV7                                         |
|        |                                    | A0080  | Opening DEV8                                                                |
|        |                                    | A0090  | Closing and Disabling Conditional Logic for DEV9                            |
|        |                                    | A0100  | Closing DEV10                                                               |
|        |                                    | A0110  | Closing and Disabling Conditional Logic for DEV11                           |
|        |                                    | A0120  | Opening DEV12                                                               |
|        |                                    | A0130  | Enabling Conditional Logic for DEV13                                        |
|        |                                    | A0140  | Enabling Conditional Logic for DEV14                                        |
|        |                                    | A0150  | Opening DEV15                                                               |
|        |                                    | A0160  | Closing and Disabling Conditional Logic for DEV16                           |
|        |                                    | A0170  | Opening DEV17                                                               |
|        |                                    | A0180  | Closing and Disabling Conditional Logic for DEV18                           |
|        |                                    | A0190  | Closing and Disabling Conditional Logic for DEV19                           |
|        |                                    | A0200  | Set DEV20 Mode to Remote Cascade                                            |
|        |                                    | A0210  | Setting DEV20 setpoint to (LT_780B Setpoint) and Enabling Conditional Logic |
|        |                                    | A0220  | Set DEV21 Mode to Remote Cascade                                            |
|        |                                    | A0230  | Setting DEV21 setpoint to (FT_702 Setpoint) and Enabling Conditional Logic  |
|        |                                    | A0240  | Set DEV22 Mode to Remote Cascade                                            |
|        |                                    | A0250  | Setting DEV22 setpoint to (LT_722C Setpoint) and Enabling Conditional Logic |

| Step   | Description                | Action | Action Description                                                          |
|--------|----------------------------|--------|-----------------------------------------------------------------------------|
|        |                            | A0260  | Set DEV23 Mode to Remote Cascade                                            |
|        |                            | A0270  | Setting DEV23 setpoint to (TT_781B Setpoint) and Enabling Conditional Logic |
|        |                            | A0280  | Closing and Disabling Conditional Logic for DEV24                           |
|        |                            | A0290  | Closing and Disabling Conditional Logic for DEV25                           |
|        |                            | A0300  | Closing and Disabling Conditional Logic for DEV26                           |
|        |                            | A0310  | Closing and Disabling Conditional Logic for DEV27                           |
|        |                            | A0320  | Closing and Disabling Conditional Logic for DEV28                           |
|        |                            | A0330  | Closing and Disabling Conditional Logic for DEV29                           |
|        |                            | A0340  | Closing and Disabling Conditional Logic for DEV30                           |
|        |                            | A0350  | Closing and Disabling Conditional Logic for DEV31                           |
|        |                            | A0360  | Closing and Disabling Conditional Logic for DEV32                           |
|        |                            | A0370  | Closing and Disabling Conditional Logic for DEV33                           |
|        |                            | A0380  | Closing and Disabling Conditional Logic for DEV34                           |
|        |                            | A0390  | Closing and Disabling Conditional Logic for DEV35                           |
|        |                            | A0400  | Set DEV36 Mode to Remote Cascade                                            |
|        |                            | A0410  | Setting DEV36 setpoint (TT_722C Setpoint) and Enabling Conditional Logic    |
|        |                            | A0420  | Set Preset Hot Recirculation time                                           |
|        |                            | A0430  | Start Preset Hot Recirculation Timer                                        |
|        |                            | A0440  | Preset Hot Recirculation time Message                                       |
| T0200B | Confirm Production         |        | If Preset Hot Recirculation time (TP013) is over then go to step S0200      |
| S0600  | Confirm Cold Recirculation | A0010  | Stopping and Disabling Conditional logic for DEV1                           |
|        |                            | A0020  | Opening DEV2                                                                |
|        |                            | A0030  | Stopping DEV3                                                               |
|        |                            | A0040  | Stopping and Disabling Conditional Logic for DEV4                           |
|        |                            | A0050  | Stopping and Disabling Conditional Logic for DEV5                           |
|        |                            | A0060  | Stopping and Disabling Conditional Logic for DEV6                           |
|        |                            | A0070  | Stopping and Disabling Conditional Logic for DEV7                           |

| Step | Description | Action | Action Description                                       |
|------|-------------|--------|----------------------------------------------------------|
|      |             | A0080  | Opening DEV8                                             |
|      |             | A0090  | Closing and Disabling Conditional Logic for DEV9         |
|      |             | A0100  | Closing DEV10                                            |
|      |             | A0110  | Closing and Disabling Conditional Logic for DEV11        |
|      |             | A0120  | Opening DEV12                                            |
|      |             | A0130  | Enabling Conditional Logic for DEV13                     |
|      |             | A0140  | Enabling Conditional Logic for DEV14                     |
|      |             | A0150  | Opening DEV15                                            |
|      |             | A0160  | Closing and Disabling Conditional Logic for DEV16        |
|      |             | A0170  | Opening DEV17                                            |
|      |             | A0180  | Closing and Disabling Conditional Logic for DEV18        |
|      |             | A0190  | Closing and Disabling Conditional Logic for DEV19        |
|      |             | A0200  | Set DEV20 Mode to Remote Out                             |
|      |             | A0210  | Set DEV20 output to Closed and Disable Conditional Logic |
|      |             | A0220  | Set DEV21 Mode to Remote Out                             |
|      |             | A0230  | Set DEV21 output to Closed and Disable Conditional Logic |
|      |             | A0240  | Set DEV22 Mode to Remote Out                             |
|      |             | A0250  | Set DEV22 output to Closed and Disable Conditional Logic |
|      |             | A0260  | Set DEV23 Mode to Remote Out                             |
|      |             | A0270  | Set DEV23 output to Closed and Disable Conditional Logic |
|      |             | A0280  | Closing and Disabling Conditional Logic for DEV24        |
|      |             | A0290  | Closing and Disabling Conditional Logic for DEV25        |
|      |             | A0300  | Closing and Disabling Conditional Logic for DEV26        |
|      |             | A0310  | Closing and Disabling Conditional Logic for DEV27        |
|      |             | A0320  | Closing and Disabling Conditional Logic for DEV28        |
|      |             | A0330  | Closing and Disabling Conditional Logic for DEV29        |
|      |             | A0340  | Closing and Disabling Conditional Logic for DEV30        |
|      |             | A0350  | Closing and Disabling Conditional Logic for DEV31        |
|      |             | A0360  | Closing and Disabling Conditional Logic for DEV32        |

| Step   | Description                          | Action | Action Description                                                          |
|--------|--------------------------------------|--------|-----------------------------------------------------------------------------|
|        |                                      | A0370  | Closing and Disabling Conditional Logic for DEV33                           |
|        |                                      | A0380  | Closing and Disabling Conditional Logic for DEV34                           |
|        |                                      | A0390  | Closing and Disabling Conditional Logic for DEV35                           |
|        |                                      | A0400  | Set DEV36 Mode to Remote Cascade                                            |
|        |                                      | A0410  | Setting DEV36 setpoint to (TT_722C Setpoint) and Enabling Conditional Logic |
|        |                                      | A0420  | Set Message: Confirm Cold Recirculation after opening/closing manual valves |
| T0700A | Cold Recirculation                   |        | If Opeator confirm <Cold Recirculation> step (OP006) then go to step S0700  |
| S0700  | Cold Recirculation (With Boiler Off) | A0010  | Enabling Conditional logic (TP02) for DEV1                                  |
|        |                                      | A0020  | Opening DEV2                                                                |
|        |                                      | A0030  | Stopping DEV3                                                               |
|        |                                      | A0040  | Enabling Conditional Logic for DEV4                                         |
|        |                                      | A0050  | Enabling Conditional Logic for DEV5                                         |
|        |                                      | A0060  | Enabling Conditional Logic for DEV6                                         |
|        |                                      | A0070  | Enabling Conditional Logic for DEV7                                         |
|        |                                      | A0080  | Opening DEV8                                                                |
|        |                                      | A0090  | Closing and Disabling Conditional Logic for DEV9                            |
|        |                                      | A0100  | Closing DEV10                                                               |
|        |                                      | A0110  | Closing and Disabling Conditional Logic for DEV11                           |
|        |                                      | A0120  | Opening DEV12                                                               |
|        |                                      | A0130  | Enabling Conditional Logic for DEV13                                        |
|        |                                      | A0140  | Enabling Conditional Logic for DEV14                                        |
|        |                                      | A0150  | Opening DEV15                                                               |
|        |                                      | A0160  | Closing and Disabling Conditional Logic for DEV16                           |
|        |                                      | A0170  | Opening DEV17                                                               |
|        |                                      | A0180  | Closing and Disabling Conditional Logic for DEV18                           |
|        |                                      | A0190  | Closing and Disabling Conditional Logic for DEV19                           |
|        |                                      | A0200  | Set DEV20 Mode to Remote Cascade                                            |
|        |                                      | A0210  | Setting DEV20 setpoint to (LT_780B Setpoint) and Enabling Conditional Logic |
|        |                                      | A0220  | Set DEV21 Mode to Remote Cascade                                            |
|        |                                      | A0230  | Setting DEV21 setpoint to (FT_702 Setpoint) and Enabling Conditional Logic  |
|        |                                      | A0240  | Set DEV22 Mode to Remote Cascade                                            |

| Step  | Description      | Action | Action Description                                                                                                          |
|-------|------------------|--------|-----------------------------------------------------------------------------------------------------------------------------|
|       |                  | A0250  | Setting DEV22 setpoint to (LT_722C Setpoint) and Enabling Conditional Logic                                                 |
|       |                  | A0260  | Set DEV23 Mode to Remote Cascade                                                                                            |
|       |                  | A0270  | Setting DEV23 setpoint (TT_781B Setpoint) and Enabling Conditional Logic                                                    |
|       |                  | A0280  | Closing and Disabling Conditional Logic for DEV24                                                                           |
|       |                  | A0290  | Closing and Disabling Conditional Logic for DEV25                                                                           |
|       |                  | A0300  | Closing and Disabling Conditional Logic for DEV26                                                                           |
|       |                  | A0310  | Closing and Disabling Conditional Logic for DEV27                                                                           |
|       |                  | A0320  | Closing and Disabling Conditional Logic for DEV28                                                                           |
|       |                  | A0330  | Closing and Disabling Conditional Logic for DEV29                                                                           |
|       |                  | A0340  | Closing and Disabling Conditional Logic for DEV30                                                                           |
|       |                  | A0350  | Closing and Disabling Conditional Logic for DEV31                                                                           |
|       |                  | A0360  | Closing and Disabling Conditional Logic for DEV32                                                                           |
|       |                  | A0370  | Closing and Disabling Conditional Logic for DEV33                                                                           |
|       |                  | A0380  | Closing and Disabling Conditional Logic for DEV34                                                                           |
|       |                  | A0390  | Closing and Disabling Conditional Logic for DEV35                                                                           |
|       |                  | A0400  | Set DEV36 Mode to Remote Cascade                                                                                            |
|       |                  | A0410  | Setting DEV36 setpoint (TT_722C Setpoint) and Enabling Conditional Logic                                                    |
|       |                  | A0420  | Set Message: Starting Cold Recirculation Operation                                                                          |
| T9900 | Complete         |        | If TT_721A2 < the Set Cold Recirculation temp (TP014) and Operator confirm Proceed to <Drain> (OP007) then go to step S9900 |
|       |                  |        |                                                                                                                             |
| S9900 | Complete         | A0010  | Set Message: Production Command Complete                                                                                    |
|       |                  | A0020  | Reset Cold Recirculation (OP006) Parameter                                                                                  |
| END   | Command Complete |        | True                                                                                                                        |

Table 21: Production Sequence

## 3.3.4 Command 03 – Recirculation

| Step   | Description             | Action | Action Description                                |
|--------|-------------------------|--------|---------------------------------------------------|
| S0000  | Start                   | A0010  | Set Message: Starting Recirculation Command       |
| T0010  | Drain 802               |        | If Task Pointer = 7 then go to step S0010         |
| T0100B | Blow 802                |        | If Task Pointer = 8 then go to step S0100         |
| T0200B | Drain 821A1 & Blow Coil |        | If Task Pointer = 9 then go to step S0200         |
| T0300B | Drain 821A2             |        | If Task Pointer = 10 then go to step S0300        |
| T0400B | Drain 822C              |        | If Task Pointer = 11 then go to step S0400        |
| T0500B | Drain 822A1 & Blow 822C |        | If Task Pointer = 12 then go to step S0500        |
| T0600B | Drain 822A2             |        | If Task Pointer = 13 then go to step S0600        |
| T0700B | Drain 822A3             |        | If Task Pointer = 14 then go to step S0700        |
| T0800B | Drain 822A4             |        | If Task Pointer = 15 then go to step S0800        |
| T0900B | Drain 822A5             |        | If Task Pointer = 16 then go to step S0900        |
| T1000B | Drain 822A6             |        | If Task Pointer = 17 then go to step S1000        |
| T1100B | Drain 880A1             |        | If Task Pointer = 18 then go to step S1100        |
| T1200B | Drain 880A2             |        | If Task Pointer = 19 then go to step S1200        |
| T1300B | Drain 880B              |        | If Task Pointer = 20 then go to step S1300        |
| T1400B | Blow 880B               |        | If Task Pointer = 21 then go to step S1400        |
| T1500B | Blow to Storage         |        | If Task Pointer = 22 then go to step S1500        |
|        |                         |        |                                                   |
|        |                         | A0010  | Enabling Conditional logic for DEV1               |
|        |                         | A0020  | Opening DEV2                                      |
|        |                         | A0030  | Stopping DEV3                                     |
|        |                         | A0040  | Enabling Conditional Logic for DEV4               |
|        |                         | A0050  | Enabling Conditional Logic for DEV5               |
|        |                         | A0060  | Enabling Conditional Logic for DEV6               |
|        |                         | A0070  | Enabling Conditional Logic for DEV7               |
|        |                         | A0080  | Opening DEV8                                      |
|        |                         | A0090  | Closing and Disabling Conditional Logic for DEV9  |
|        |                         | A0100  | Closing DEV10                                     |
|        |                         | A0110  | Closing and Disabling Conditional Logic for DEV11 |
|        |                         | A0120  | Opening DEV12                                     |
|        |                         | A0130  | Enabling Conditional Logic for DEV13              |
|        |                         | A0140  | Enabling Conditional Logic for DEV14              |
|        |                         | A0150  | Opening DEV15                                     |
|        |                         | A0160  | Closing and Disabling Conditional Logic for DEV16 |
| S0010  | Drain 802               | A0170  | Opening DEV17                                     |

| Step   | Description | Action | Action Description                                                         |
|--------|-------------|--------|----------------------------------------------------------------------------|
|        |             | A0180  | Closing and Disabling Conditional Logic for DEV18                          |
|        |             | A0190  | Closing and Disabling Conditional Logic for DEV19                          |
|        |             | A0200  | Set DEV20 Mode to Remote Cascade                                           |
|        |             | A0210  | Setting DEV20 setpoint (LT_780B Setpoint) and Enabling Conditional Logic   |
|        |             | A0220  | Set DEV21 Mode to Remote Cascade                                           |
|        |             | A0230  | Setting DEV21 setpoint to (FT_702 Setpoint) and Enabling Conditional Logic |
|        |             | A0240  | Set DEV22 Mode to Remote Cascade                                           |
|        |             | A0250  | Setting DEV22 setpoint (LT_722C Setpoint) and Enabling Conditional Logic   |
|        |             | A0260  | Set DEV23 Mode to Remote Cascade                                           |
|        |             | A0270  | Setting DEV23 setpoint (TT_781B Setpoint) and Enabling Conditional Logic   |
|        |             | A0280  | Closing and Disabling Conditional Logic for DEV24                          |
|        |             | A0290  | Closing and Disabling Conditional Logic for DEV25                          |
|        |             | A0300  | Closing and Disabling Conditional Logic for DEV26                          |
|        |             | A0310  | Closing and Disabling Conditional Logic for DEV27                          |
|        |             | A0320  | Closing and Disabling Conditional Logic for DEV28                          |
|        |             | A0330  | Closing and Disabling Conditional Logic for DEV29                          |
|        |             | A0340  | Closing and Disabling Conditional Logic for DEV30                          |
|        |             | A0350  | Closing and Disabling Conditional Logic for DEV31                          |
|        |             | A0360  | Closing and Disabling Conditional Logic for DEV32                          |
|        |             | A0370  | Closing and Disabling Conditional Logic for DEV33                          |
|        |             | A0380  | Closing and Disabling Conditional Logic for DEV34                          |
|        |             | A0390  | Closing and Disabling Conditional Logic for DEV35                          |
|        |             | A0400  | Set DEV36 Mode to Remote Out                                               |
|        |             | A0410  | Set DEV36 output to Open and Disable Conditional Logic                     |
|        |             | A0430  | Reset Cold Recirculation Button (OP006)                                    |
| T0100A | Blow 802    |        | If EM_DEOD_FA_CKT1 is in step <Drain Wait> then go to step S0100           |
|        |             |        |                                                                            |

| Step  | Description | Action | Action Description                                                       |
|-------|-------------|--------|--------------------------------------------------------------------------|
| S0100 | Blow 802    | A0010  | Stopping and Disabling Conditional logic for DEV1                        |
|       |             | A0020  | Closing DEV2                                                             |
|       |             | A0030  | Stopping DEV3                                                            |
|       |             | A0040  | Enabling Conditional Logic for DEV4                                      |
|       |             | A0050  | Enabling Conditional Logic for DEV5                                      |
|       |             | A0060  | Enabling Conditional Logic for DEV6                                      |
|       |             | A0070  | Enabling Conditional Logic for DEV7                                      |
|       |             | A0080  | Opening DEV8                                                             |
|       |             | A0090  | Enabling Conditional Logic for DEV9                                      |
|       |             | A0100  | Closing DEV10                                                            |
|       |             | A0110  | Closing and Disabling Conditional Logic for DEV11                        |
|       |             | A0120  | Opening DEV12                                                            |
|       |             | A0130  | Enabling Conditional Logic for DEV13                                     |
|       |             | A0140  | Enabling Conditional Logic for DEV14                                     |
|       |             | A0150  | Opening DEV15                                                            |
|       |             | A0160  | Closing and Disabling Conditional Logic for DEV16                        |
|       |             | A0170  | Opening DEV17                                                            |
|       |             | A0180  | Closing and Disabling Conditional Logic for DEV18                        |
|       |             | A0190  | Closing and Disabling Conditional Logic for DEV19                        |
|       |             | A0200  | Set DEV20 Mode to Remote Cascade                                         |
|       |             | A0210  | Setting DEV20 setpoint (LT_780B Setpoint) and Enabling Conditional Logic |
|       |             | A0220  | Set DEV21 Mode to Remote Out and Disabling Conditional Logic             |
|       |             | A0230  | Set DEV21 output to Open                                                 |
|       |             | A0240  | Set DEV22 Mode to Remote Cascade                                         |
|       |             | A0250  | Setting DEV22 setpoint (LT_722C Setpoint) and Enabling Conditional Logic |
|       |             | A0260  | Set DEV23 Mode to Remote Cascade                                         |
|       |             | A0270  | Setting DEV23 setpoint (TT1781B Setpoint) and Enabling Conditional Logic |
|       |             | A0280  | Closing and Disabling Conditional Logic for DEV24                        |
|       |             | A0290  | Closing and Disabling Conditional Logic for DEV25                        |
|       |             | A0300  | Closing and Disabling Conditional Logic for DEV26                        |
|       |             | A0310  | Closing and Disabling Conditional Logic for DEV27                        |
|       |             | A0320  | Closing and Disabling Conditional Logic for DEV28                        |

| Step   | Description             | Action | Action Description                                               |
|--------|-------------------------|--------|------------------------------------------------------------------|
|        |                         | A0330  | Closing and Disabling Conditional Logic for DEV29                |
|        |                         | A0340  | Closing and Disabling Conditional Logic for DEV30                |
|        |                         | A0350  | Closing and Disabling Conditional Logic for DEV31                |
|        |                         | A0360  | Closing and Disabling Conditional Logic for DEV32                |
|        |                         | A0370  | Closing and Disabling Conditional Logic for DEV33                |
|        |                         | A0380  | Closing and Disabling Conditional Logic for DEV34                |
|        |                         | A0390  | Closing and Disabling Conditional Logic for DEV35                |
|        |                         | A0400  | Set DEV36 Mode to Remote Out                                     |
|        |                         | A0410  | Set DEV36 output to Open and Disable Conditional Logic           |
|        |                         | A0420  | Set Message: Setting Preset blow time                            |
|        |                         | A0430  | Start Preset Blow Timer                                          |
|        |                         | A0440  | Preset Blow time Message                                         |
| T0200A | Drain 821A1 & Blow Coil |        | If Preset (DEV9) Blow time (TP006) is over then go to step S0200 |
|        |                         |        |                                                                  |
|        |                         | A0010  | Stopping and Disabling Conditional logic for DEV1                |
|        |                         | A0020  | Closing DEV2                                                     |
|        |                         | A0030  | Stopping DEV3                                                    |
|        |                         | A0040  | Enabling Conditional Logic for DEV4                              |
|        |                         | A0050  | Enabling Conditional Logic for DEV5                              |
|        |                         | A0060  | Enabling Conditional Logic for DEV6                              |
|        |                         | A0070  | Enabling Conditional Logic for DEV7                              |
|        |                         | A0080  | Opening DEV8                                                     |
|        |                         | A0090  | Closing and Disabling Conditional Logic for DEV9                 |
|        |                         | A0100  | Opening DEV10                                                    |
|        |                         | A0110  | Enabling Conditional Logic for DEV11                             |
|        |                         | A0120  | Closing DEV12                                                    |
|        |                         | A0130  | Enabling Conditional Logic for DEV13                             |
|        |                         | A0140  | Enabling Conditional Logic for DEV14                             |
|        |                         | A0150  | Opening DEV15                                                    |
|        |                         | A0160  | Closing and Disabling Conditional Logic for DEV16                |
|        |                         | A0170  | Opening DEV17                                                    |
|        |                         | A0180  | Closing and Disabling Conditional Logic for DEV18                |

| Step   | Description             | Action | Action Description                                                                       |
|--------|-------------------------|--------|------------------------------------------------------------------------------------------|
| S0200  | Drain 821A1 & Blow Coil | A0190  | Closing and Disabling Conditional Logic for DEV19                                        |
|        |                         | A0200  | Set DEV20 Mode to Remote Cascade                                                         |
|        |                         | A0210  | Setting DEV20 setpoint (LT_780B Setpoint) and Enabling Conditional Logic                 |
|        |                         | A0220  | Set DEV21 Mode to Remote Out                                                             |
|        |                         | A0230  | Set DEV21 output to Closed and Disable Conditional Logic                                 |
|        |                         | A0240  | Set DEV22 Mode to Remote Cascade                                                         |
|        |                         | A0250  | Setting DEV22 setpoint (LT_722C Setpoint) and Enabling Conditional Logic                 |
|        |                         | A0260  | Set DEV23 Mode to Remote Cascade                                                         |
|        |                         | A0270  | Setting DEV23 setpoint (TT_781B Setpoint) and Enabling Conditional Logic                 |
|        |                         | A0280  | Enabling Conditional Logic for DEV24                                                     |
|        |                         | A0290  | Closing and Disabling Conditional Logic for DEV25                                        |
|        |                         | A0300  | Closing and Disabling Conditional Logic for DEV26                                        |
|        |                         | A0310  | Closing and Disabling Conditional Logic for DEV27                                        |
|        |                         | A0320  | Closing and Disabling Conditional Logic for DEV28                                        |
|        |                         | A0330  | Closing and Disabling Conditional Logic for DEV29                                        |
|        |                         | A0340  | Closing and Disabling Conditional Logic for DEV30                                        |
|        |                         | A0350  | Closing and Disabling Conditional Logic for DEV31                                        |
|        |                         | A0360  | Closing and Disabling Conditional Logic for DEV32                                        |
|        |                         | A0370  | Closing and Disabling Conditional Logic for DEV33                                        |
|        |                         | A0380  | Closing and Disabling Conditional Logic for DEV34                                        |
|        |                         | A0390  | Closing and Disabling Conditional Logic for DEV35                                        |
|        |                         | A0400  | Set DEV36 Mode to Remote Out                                                             |
|        |                         | A0410  | Set DEV36 output to Open and Disable Conditional Logic                                   |
|        |                         | A0420  | Set Preset Drain time and Blow Time                                                      |
|        |                         | A0430  | Start Preset Drain and Blow Timer                                                        |
|        |                         | A0440  | Preset Drain and Blow time Message                                                       |
| T0300A | Drain 821A2             |        | If Preset Drain time (TP016) over and DEV11 Blow time over (TP018) then go to step S0300 |
|        |                         |        |                                                                                          |

| Step  | Description | Action | Action Description                                                       |
|-------|-------------|--------|--------------------------------------------------------------------------|
| S0300 | Drain 821A2 | A0010  | Stopping and Disabling Conditional logic for DEV1                        |
|       |             | A0020  | Closing DEV2                                                             |
|       |             | A0030  | Stopping DEV3                                                            |
|       |             | A0040  | Enabling Conditional Logic for DEV4                                      |
|       |             | A0050  | Enabling Conditional Logic for DEV5                                      |
|       |             | A0060  | Enabling Conditional Logic for DEV6                                      |
|       |             | A0070  | Enabling Conditional Logic for DEV7                                      |
|       |             | A0080  | Closing DEV8                                                             |
|       |             | A0090  | Closing and Disabling Conditional Logic for DEV9                         |
|       |             | A0100  | Closing DEV10                                                            |
|       |             | A0110  | Closing and Disabling Conditional Logic for DEV11                        |
|       |             | A0120  | Closing DEV12                                                            |
|       |             | A0130  | Closing and Disabling Conditional Logic for DEV13                        |
|       |             | A0140  | Closing and Disabling Conditional Logic for DEV14                        |
|       |             | A0150  | Opening DEV15                                                            |
|       |             | A0160  | Closing and Disabling Conditional Logic for DEV16                        |
|       |             | A0170  | Opening DEV17                                                            |
|       |             | A0180  | Closing and Disabling Conditional Logic for DEV18                        |
|       |             | A0190  | Closing and Disabling Conditional Logic for DEV19                        |
|       |             | A0200  | Set DEV20 Mode to Remote Cascade                                         |
|       |             | A0210  | Setting DEV20 setpoint (LT_780B Setpoint) and Enabling Conditional Logic |
|       |             | A0220  | Set DEV21 Mode to Remote Out                                             |
|       |             | A0230  | Set DEV21 output to Closed and Disable Conditional Logic                 |
|       |             | A0240  | Set DEV22 Mode to Remote Cascade                                         |
|       |             | A0250  | Setting DEV22 setpoint (LT_722C Setpoint) and Enabling Conditional Logic |
|       |             | A0260  | Set DEV23 Mode to Remote Cascade                                         |
|       |             | A0270  | Setting DEV23 setpoint (TT_781B Setpoint) and Enabling Conditional Logic |
|       |             | A0280  | Opening and Disabling Conditional Logic for DEV24                        |
|       |             | A0290  | Enabling Conditional Logic for DEV25                                     |
|       |             | A0300  | Closing and Disabling Conditional Logic for DEV26                        |
|       |             | A0310  | Closing and Disabling Conditional Logic for DEV27                        |

| Step   | Description | Action | Action Description                                         |
|--------|-------------|--------|------------------------------------------------------------|
|        |             | A0320  | Closing and Disabling Conditional Logic for DEV28          |
|        |             | A0330  | Closing and Disabling Conditional Logic for DEV29          |
|        |             | A0340  | Closing and Disabling Conditional Logic for DEV30          |
|        |             | A0350  | Closing and Disabling Conditional Logic for DEV31          |
|        |             | A0360  | Closing and Disabling Conditional Logic for DEV32          |
|        |             | A0370  | Closing and Disabling Conditional Logic for DEV33          |
|        |             | A0380  | Closing and Disabling Conditional Logic for DEV34          |
|        |             | A0390  | Closing and Disabling Conditional Logic for DEV35          |
|        |             | A0400  | Set DEV36 Mode to Remote Out                               |
|        |             | A0410  | Set DEV36 output to Closed and Disable Conditional Logic   |
|        |             | A0420  | Set Preset Drain time                                      |
|        |             | A0430  | Start Preset Drain Timer                                   |
|        |             | A0440  | Preset Drain time Message                                  |
| T0400A | Drain 822C  |        | If Preset Drain time (TP016) is over then go to step S0400 |
|        |             |        |                                                            |
|        |             | A0010  | Stopping and Disabling Conditional logic for DEV1          |
|        |             | A0020  | Closing DEV2                                               |
|        |             | A0030  | Stopping DEV3                                              |
|        |             | A0040  | Enabling Conditional Logic for DEV4                        |
|        |             | A0050  | Enabling Conditional Logic for DEV5                        |
|        |             | A0060  | Enabling Conditional Logic for DEV6                        |
|        |             | A0070  | Enabling Conditional Logic for DEV7                        |
|        |             | A0080  | Closing DEV8                                               |
|        |             | A0090  | Closing and Disabling Conditional Logic for DEV9           |
|        |             | A0100  | Closing DEV10                                              |
|        |             | A0110  | Closing and Disabling Conditional Logic for DEV11          |
|        |             | A0120  | Closing DEV12                                              |
|        |             | A0130  | Closing and Disabling Conditional Logic for DEV13          |
|        |             | A0140  | Closing and Disabling Conditional Logic for DEV14          |
|        |             | A0150  | Opening DEV15                                              |
|        |             | A0160  | Closing and Disabling Conditional Logic for DEV16          |

| Step   | Description             | Action | Action Description                                                            |
|--------|-------------------------|--------|-------------------------------------------------------------------------------|
| S0400  | Drain 822C              | A0170  | Opening DEV17                                                                 |
|        |                         | A0180  | Closing and Disabling Conditional Logic for DEV18                             |
|        |                         | A0190  | Closing and Disabling Conditional Logic for DEV19                             |
|        |                         | A0200  | Set DEV20 Mode to Remote Cascade                                              |
|        |                         | A0210  | Setting DEV20 setpoint (LT_780B Setpoint) and Enabling Conditional Logic      |
|        |                         | A0220  | Set DEV21 Mode to Remote Out                                                  |
|        |                         | A0230  | Set DEV21 output to Closed and Disable Conditional Logic                      |
|        |                         | A0240  | Set DEV22 Mode to Remote Out                                                  |
|        |                         | A0250  | Set DEV22 output to Open and Enable Conditional logic                         |
|        |                         | A0260  | Set DEV23 Mode to Remote Cascade                                              |
|        |                         | A0270  | Setting DEV23 setpoint (TT_781B Setpoint) and Enabling Conditional Logic      |
|        |                         | A0280  | Opening and Disabling Conditional Logic for DEV24                             |
|        |                         | A0290  | Opening and Disabling Conditional Logic for DEV25                             |
|        |                         | A0300  | Closing and Disabling Conditional Logic for DEV26                             |
|        |                         | A0310  | Closing and Disabling Conditional Logic for DEV27                             |
|        |                         | A0320  | Closing and Disabling Conditional Logic for DEV28                             |
|        |                         | A0330  | Closing and Disabling Conditional Logic for DEV29                             |
|        |                         | A0340  | Closing and Disabling Conditional Logic for DEV30                             |
|        |                         | A0350  | Closing and Disabling Conditional Logic for DEV31                             |
|        |                         | A0360  | Closing and Disabling Conditional Logic for DEV32                             |
|        |                         | A0370  | Closing and Disabling Conditional Logic for DEV33                             |
|        |                         | A0380  | Closing and Disabling Conditional Logic for DEV34                             |
|        |                         | A0390  | Closing and Disabling Conditional Logic for DEV35                             |
|        |                         | A0400  | Set DEV36 Mode to Remote Out                                                  |
|        |                         | A0410  | Set DEV36 output to Closed and Disable Conditional Logic                      |
| T0500A | Drain 822A1 & Blow 822C |        | If LSL_722C is in alarm and LT_722C_VLL is in VLL alarm then go to step S0500 |
|        |                         |        |                                                                               |

| Step  | Description                | Action | Action Description                                                         |
|-------|----------------------------|--------|----------------------------------------------------------------------------|
| S0500 | Drain 822A1<br>& Blow 822C | A0010  | Stopping and Disabling Conditional logic for DEV1                          |
|       |                            | A0020  | Closing DEV2                                                               |
|       |                            | A0030  | Stopping DEV3                                                              |
|       |                            | A0040  | Stopping and Disabling Conditional Logic for DEV4                          |
|       |                            | A0050  | Stopping and Disabling Conditional Logic for DEV5                          |
|       |                            | A0060  | Enabling Conditional Logic for DEV6                                        |
|       |                            | A0070  | Enabling Conditional Logic for DEV7                                        |
|       |                            | A0080  | Closing DEV8                                                               |
|       |                            | A0090  | Closing and Disabling Conditional Logic for DEV9                           |
|       |                            | A0100  | Closing DEV10                                                              |
|       |                            | A0110  | Closing and Disabling Conditional Logic for DEV11                          |
|       |                            | A0120  | Closing DEV12                                                              |
|       |                            | A0130  | Closing and Disabling Conditional Logic for DEV13                          |
|       |                            | A0140  | Closing and Disabling Conditional Logic for DEV14                          |
|       |                            | A0150  | Closing DEV15                                                              |
|       |                            | A0160  | Enabling Conditional Logic for DEV16                                       |
|       |                            | A0170  | Opening DEV17                                                              |
|       |                            | A0180  | Closing and Disabling Conditional Logic for DEV18                          |
|       |                            | A0190  | Closing and Disabling Conditional Logic for DEV19                          |
|       |                            | A0200  | Set DEV20 Mode to Remote Cascade                                           |
|       |                            | A0210  | Setting DEV20 setpoint (LT_780B Setpoint) and Enabling Conditional Logic   |
|       |                            | A0220  | Set DEV21 Mode to Remote Out                                               |
|       |                            | A0230  | Set DEV21 output to Closed and Disable Conditional Logic                   |
|       |                            | A0240  | Set DEV22 Mode to Remote Out                                               |
|       |                            | A0250  | Set DEV22 output to Open and Disable Conditional logic                     |
|       |                            | A0260  | Set DEV23 Mode to Remote Cascade                                           |
|       |                            | A0270  | Setting DEV23 setpoint ( TT_781B Setpoint ) and Enabling Conditional Logic |
|       |                            | A0280  | Opening and Disabling Conditional Logic for DEV24                          |
|       |                            | A0290  | Opening and Disabling Conditional Logic for DEV25                          |
|       |                            | A0300  | Enabling Conditional Logic for DEV26                                       |
|       |                            | A0310  | Closing and Disabling Conditional Logic for DEV27                          |

| Step   | Description | Action | Action Description                                                                               |
|--------|-------------|--------|--------------------------------------------------------------------------------------------------|
|        |             | A0320  | Closing and Disabling Conditional Logic for DEV28                                                |
|        |             | A0330  | Closing and Disabling Conditional Logic for DEV29                                                |
|        |             | A0340  | Closing and Disabling Conditional Logic for DEV30                                                |
|        |             | A0350  | Closing and Disabling Conditional Logic for DEV31                                                |
|        |             | A0360  | Closing and Disabling Conditional Logic for DEV32                                                |
|        |             | A0370  | Closing and Disabling Conditional Logic for DEV33                                                |
|        |             | A0380  | Closing and Disabling Conditional Logic for DEV34                                                |
|        |             | A0390  | Closing and Disabling Conditional Logic for DEV35                                                |
|        |             | A0400  | Set DEV36 Mode to Remote Out                                                                     |
|        |             | A0410  | Set DEV36 output to Closed and Disable Conditional Logic                                         |
|        |             | A0420  | Set Preset Drain time and Blow Time                                                              |
|        |             | A0430  | Start Preset Drain and Blow Timer                                                                |
|        |             | A0440  | Preset Drain and Blow time Message                                                               |
| T0600A | Drain 822A2 |        | If Preset Drain time (TP016) is over and (DEV16) Blow time (TP007) is over then go to step S0600 |
| S0600  | Drain 822A2 | A0010  | Stopping and Disabling Conditional logic for DEV1                                                |
|        |             | A0020  | Closing DEV2                                                                                     |
|        |             | A0030  | Stopping DEV3                                                                                    |
|        |             | A0040  | Stopping and Disabling Conditional Logic for DEV4                                                |
|        |             | A0050  | Stopping and Disabling Conditional Logic for DEV5                                                |
|        |             | A0060  | Enabling Conditional Logic for DEV6                                                              |
|        |             | A0070  | Enabling Conditional Logic for DEV7                                                              |
|        |             | A0080  | Closing DEV8                                                                                     |
|        |             | A0090  | Closing and Disabling Conditional Logic for DEV9                                                 |
|        |             | A0100  | Closing DEV10                                                                                    |
|        |             | A0110  | Closing and Disabling Conditional Logic for DEV11                                                |
|        |             | A0120  | Closing DEV12                                                                                    |
|        |             | A0130  | Closing and Disabling Conditional Logic for DEV13                                                |
|        |             | A0140  | Closing and Disabling Conditional Logic for DEV14                                                |

| Step | Description | Action | Action Description                                                       |
|------|-------------|--------|--------------------------------------------------------------------------|
|      |             | A0150  | Closing DEV15                                                            |
|      |             | A0160  | Closing and Disabling Conditional Logic for DEV16                        |
|      |             | A0170  | Opening DEV17                                                            |
|      |             | A0180  | Closing and Disabling Conditional Logic for DEV18                        |
|      |             | A0190  | Closing and Disabling Conditional Logic for DEV19                        |
|      |             | A0200  | Set DEV20 Mode to Remote Cascade                                         |
|      |             | A0210  | Setting DEV20 setpoint (LT_780B Setpoint) and Enabling Conditional Logic |
|      |             | A0220  | Set DEV21 Mode to Remote Out                                             |
|      |             | A0230  | Set DEV21 output to Closed and Disable Conditional Logic                 |
|      |             | A0240  | Set DEV22 Mode to Remote Out                                             |
|      |             | A0250  | Set DEV22 output to Closed and Disable Conditional Logic                 |
|      |             | A0260  | Set DEV23 Mode to Remote Cascade                                         |
|      |             | A0270  | Setting DEV23 setpoint (TT_781B Setpoint) and Enabling Conditional Logic |
|      |             | A0280  | Opening and Disabling Conditional Logic for DEV24                        |
|      |             | A0290  | Opening and Disabling Conditional Logic for DEV25                        |
|      |             | A0300  | Opening and Disabling Conditional Logic for DEV26                        |
|      |             | A0310  | Enabling Conditional Logic for DEV27                                     |
|      |             | A0320  | Closing and Disabling Conditional Logic for DEV28                        |
|      |             | A0330  | Closing and Disabling Conditional Logic for DEV29                        |
|      |             | A0340  | Closing and Disabling Conditional Logic for DEV30                        |
|      |             | A0350  | Closing and Disabling Conditional Logic for DEV31                        |
|      |             | A0360  | Closing and Disabling Conditional Logic for DEV32                        |
|      |             | A0370  | Closing and Disabling Conditional Logic for DEV33                        |
|      |             | A0380  | Closing and Disabling Conditional Logic for DEV34                        |
|      |             | A0390  | Closing and Disabling Conditional Logic for DEV35                        |
|      |             | A0400  | Set DEV36 Mode to Remote Out                                             |
|      |             | A0410  | Set DEV36 output to Closed and Disable Conditional Logic                 |
|      |             | A0420  | Set Preset Drain time                                                    |
|      |             | A0430  | Start Preset Drain Timer                                                 |

| Step   | Description | Action | Action Description                                                       |
|--------|-------------|--------|--------------------------------------------------------------------------|
|        |             | A0440  | Preset Drain time Message                                                |
| T0700A | Drain 822A3 |        | If Preset Drain time (TP016) is over then go to step S0700               |
|        |             |        |                                                                          |
| S0700  | Drain 822A3 | A0010  | Stopping and Disabling Conditional logic for DEV1                        |
|        |             | A0020  | Closing DEV2                                                             |
|        |             | A0030  | Stopping DEV3                                                            |
|        |             | A0040  | Stopping and Disabling Conditional Logic for DEV4                        |
|        |             | A0050  | Stopping and Disabling Conditional Logic for DEV5                        |
|        |             | A0060  | Enabling Conditional Logic for DEV6                                      |
|        |             | A0070  | Enabling Conditional Logic for DEV7                                      |
|        |             | A0080  | Closing DEV8                                                             |
|        |             | A0090  | Closing and Disabling Conditional Logic for DEV9                         |
|        |             | A0100  | Closing DEV10                                                            |
|        |             | A0110  | Closing and Disabling Conditional Logic for DEV11                        |
|        |             | A0120  | Closing DEV12                                                            |
|        |             | A0130  | Closing and Disabling Conditional Logic for DEV13                        |
|        |             | A0140  | Closing and Disabling Conditional Logic for DEV14                        |
|        |             | A0150  | Closing DEV15                                                            |
|        |             | A0160  | Closing and Disabling Conditional Logic for DEV16                        |
|        |             | A0170  | Opening DEV17                                                            |
|        |             | A0180  | Closing and Disabling Conditional Logic for DEV18                        |
|        |             | A0190  | Closing and Disabling Conditional Logic for DEV19                        |
|        |             | A0200  | Set DEV20 Mode to Remote Cascade                                         |
|        |             | A0210  | Setting DEV20 setpoint (LT_780B Setpoint) and Enabling Conditional Logic |
|        |             | A0220  | Set DEV21 Mode to Remote Out                                             |
|        |             | A0230  | Set DEV21 output to Closed and Disable Conditional Logic                 |
|        |             | A0240  | Set DEV22 Mode to Remote Out                                             |
|        |             | A0250  | Set DEV22 output to Closed and Disable Conditional Logic                 |
|        |             | A0260  | Set DEV23 Mode to Remote Cascade                                         |
|        |             | A0270  | Setting DEV23 setpoint (TT_781B Setpoint) and Enabling Conditional Logic |
|        |             | A0280  | Opening and Disabling Conditional Logic for DEV24                        |

| Step   | Description | Action | Action Description                                         |
|--------|-------------|--------|------------------------------------------------------------|
|        |             | A0290  | Opening and Disabling Conditional Logic for DEV25          |
|        |             | A0300  | Opening and Disabling Conditional Logic for DEV26          |
|        |             | A0310  | Opening and Disabling Conditional Logic for DEV27          |
|        |             | A0320  | Enabling Conditional Logic for DEV28                       |
|        |             | A0330  | Closing and Disabling Conditional Logic for DEV29          |
|        |             | A0340  | Closing and Disabling Conditional Logic for DEV30          |
|        |             | A0350  | Closing and Disabling Conditional Logic for DEV31          |
|        |             | A0360  | Closing and Disabling Conditional Logic for DEV32          |
|        |             | A0370  | Closing and Disabling Conditional Logic for DEV33          |
|        |             | A0380  | Closing and Disabling Conditional Logic for DEV34          |
|        |             | A0390  | Closing and Disabling Conditional Logic for DEV35          |
|        |             | A0400  | Set DEV36 Mode to Remote Out                               |
|        |             | A0410  | Set DEV36 output to Closed and Disable Conditional Logic   |
|        |             | A0420  | Set Preset Drain time                                      |
|        |             | A0430  | Start Preset Drain Timer                                   |
|        |             | A0440  | Preset Drain time Message                                  |
| T0800A | Drain 822A4 |        | If Preset Drain time (TP016) is over then go to step S0800 |
| S0800  | Drain 822A4 | A0010  | Stopping and Disabling Conditional logic for DEV1          |
|        |             | A0020  | Closing DEV2                                               |
|        |             | A0030  | Stopping DEV3                                              |
|        |             | A0040  | Stopping and Disabling Conditional Logic for DEV4          |
|        |             | A0050  | Stopping and Disabling Conditional Logic for DEV5          |
|        |             | A0060  | Enabling Conditional Logic for DEV6                        |
|        |             | A0070  | Enabling Conditional Logic for DEV7                        |
|        |             | A0080  | Closing DEV8                                               |
|        |             | A0090  | Closing and Disabling Conditional Logic for DEV9           |
|        |             | A0100  | Closing DEV10                                              |
|        |             | A0110  | Closing and Disabling Conditional Logic for DEV11          |
|        |             | A0120  | Closing DEV12                                              |

| Step | Description | Action | Action Description                                                       |
|------|-------------|--------|--------------------------------------------------------------------------|
|      |             | A0130  | Closing and Disabling Conditional Logic for DEV13                        |
|      |             | A0140  | Closing and Disabling Conditional Logic for DEV14                        |
|      |             | A0150  | Closing DEV15                                                            |
|      |             | A0160  | Closing and Disabling Conditional Logic for DEV16                        |
|      |             | A0170  | Opening DEV17                                                            |
|      |             | A0180  | Closing and Disabling Conditional Logic for DEV18                        |
|      |             | A0190  | Closing and Disabling Conditional Logic for DEV19                        |
|      |             | A0200  | Set DEV20 Mode to Remote Cascade                                         |
|      |             | A0210  | Setting DEV20 setpoint (LT_780B Setpoint) and Enabling Conditional Logic |
|      |             | A0220  | Set DEV21 Mode to Remote Out                                             |
|      |             | A0230  | Set DEV21 output to Closed and Disable Conditional Logic                 |
|      |             | A0240  | Set DEV22 Mode to Remote Out                                             |
|      |             | A0250  | Set DEV22 output to Closed and Disable Conditional Logic                 |
|      |             | A0260  | Set DEV23 Mode to Remote Cascade                                         |
|      |             | A0270  | Setting DEV23 setpoint (TT_781B Setpoint) and Enabling Conditional Logic |
|      |             | A0280  | Opening and Disabling Conditional Logic for DEV24                        |
|      |             | A0290  | Opening and Disabling Conditional Logic for DEV25                        |
|      |             | A0300  | Opening and Disabling Conditional Logic for DEV26                        |
|      |             | A0310  | Opening and Disabling Conditional Logic for DEV27                        |
|      |             | A0320  | Opening and Disabling Conditional Logic for DEV28                        |
|      |             | A0330  | Enabling Conditional Logic for DEV29                                     |
|      |             | A0340  | Closing and Disabling Conditional Logic for DEV30                        |
|      |             | A0350  | Closing and Disabling Conditional Logic for DEV31                        |
|      |             | A0360  | Closing and Disabling Conditional Logic for DEV32                        |
|      |             | A0370  | Closing and Disabling Conditional Logic for DEV33                        |
|      |             | A0380  | Closing and Disabling Conditional Logic for DEV34                        |
|      |             | A0390  | Closing and Disabling Conditional Logic for DEV35                        |
|      |             | A0400  | Set DEV36 Mode to Remote Out                                             |

| Step   | Description | Action | Action Description                                                       |
|--------|-------------|--------|--------------------------------------------------------------------------|
|        |             | A0410  | Set DEV36 output to Closed and Disable Conditional Logic                 |
|        |             | A0420  | Set Preset Drain time                                                    |
|        |             | A0430  | Start Preset Drain Timer                                                 |
|        |             | A0440  | Preset Drain time Message                                                |
| T0900A | Drain 822A5 |        | If Preset Drain time (TP016) is over then go to step S0900               |
| S0900  | Drain 822A5 | A0010  | Stopping and Disabling Conditional logic for DEV1                        |
|        |             | A0020  | Closing DEV2                                                             |
|        |             | A0030  | Stopping DEV3                                                            |
|        |             | A0040  | Stopping and Disabling Conditional Logic for DEV4                        |
|        |             | A0050  | Stopping and Disabling Conditional Logic for DEV5                        |
|        |             | A0060  | Enabling Conditional Logic for DEV6                                      |
|        |             | A0070  | Enabling Conditional Logic for DEV7                                      |
|        |             | A0080  | Closing DEV8                                                             |
|        |             | A0090  | Closing and Disabling Conditional Logic for DEV9                         |
|        |             | A0100  | Closing DEV10                                                            |
|        |             | A0110  | Closing and Disabling Conditional Logic for DEV11                        |
|        |             | A0120  | Closing DEV12                                                            |
|        |             | A0130  | Closing and Disabling Conditional Logic for DEV13                        |
|        |             | A0140  | Closing and Disabling Conditional Logic for DEV14                        |
|        |             | A0150  | Closing DEV15                                                            |
|        |             | A0160  | Closing and Disabling Conditional Logic for DEV16                        |
|        |             | A0170  | Opening DEV17                                                            |
|        |             | A0180  | Closing and Disabling Conditional Logic for DEV18                        |
|        |             | A0190  | Closing and Disabling Conditional Logic for DEV19                        |
|        |             | A0200  | Set DEV20 Mode to Remote Cascade                                         |
|        |             | A0210  | Setting DEV20 setpoint (LT_780B Setpoint) and Enabling Conditional Logic |
|        |             | A0220  | Set DEV21 Mode to Remote Out                                             |
|        |             | A0230  | Set DEV21 output to Closed and Disable Conditional Logic                 |
|        |             | A0240  | Set DEV22 Mode to Remote Out                                             |
|        |             | A0250  | Set DEV22 output to Closed and Disable Conditional Logic                 |

| Step   | Description | Action | Action Description                                                       |
|--------|-------------|--------|--------------------------------------------------------------------------|
|        |             | A0260  | Set DEV23 Mode to Remote Cascade                                         |
|        |             | A0270  | Setting DEV23 setpoint (TT1781B Setpoint) and Enabling Conditional Logic |
|        |             | A0280  | Opening and Disabling Conditional Logic for DEV24                        |
|        |             | A0290  | Opening and Disabling Conditional Logic for DEV25                        |
|        |             | A0300  | Opening and Disabling Conditional Logic for DEV26                        |
|        |             | A0310  | Opening and Disabling Conditional Logic for DEV27                        |
|        |             | A0320  | Opening and Disabling Conditional Logic for DEV28                        |
|        |             | A0330  | Opening and Disabling Conditional Logic for DEV29                        |
|        |             | A0340  | Enabling Conditional Logic for DEV30                                     |
|        |             | A0350  | Closing and Disabling Conditional Logic for DEV31                        |
|        |             | A0360  | Closing and Disabling Conditional Logic for DEV32                        |
|        |             | A0370  | Closing and Disabling Conditional Logic for DEV33                        |
|        |             | A0380  | Closing and Disabling Conditional Logic for DEV34                        |
|        |             | A0390  | Closing and Disabling Conditional Logic for DEV35                        |
|        |             | A0400  | Set DEV36 Mode to Remote Out                                             |
|        |             | A0410  | Set DEV36 output to Closed and Disable Conditional Logic                 |
|        |             | A0420  | Set Preset Drain time                                                    |
|        |             | A0430  | Start Preset Drain Timer                                                 |
|        |             | A0440  | Preset Drain time Message                                                |
| T1000A | Drain 822A6 |        | If Preset Drain time (TP016) is over then go to step S1000               |
| S1000  | Drain 822A6 | A0010  | Stopping and Disabling Conditional logic for DEV1                        |
|        |             | A0020  | Closing DEV2                                                             |
|        |             | A0030  | Stopping DEV3                                                            |
|        |             | A0040  | Stopping and Disabling Conditional Logic for DEV4                        |
|        |             | A0050  | Stopping and Disabling Conditional Logic for DEV5                        |
|        |             | A0060  | Enabling Conditional Logic for DEV6                                      |
|        |             | A0070  | Enabling Conditional Logic for DEV7                                      |
|        |             | A0080  | Closing DEV8                                                             |

| Step | Description | Action | Action Description                                                       |
|------|-------------|--------|--------------------------------------------------------------------------|
|      |             | A0090  | Closing and Disabling Conditional Logic for DEV9                         |
|      |             | A0100  | Closing DEV10                                                            |
|      |             | A0110  | Closing and Disabling Conditional Logic for DEV11                        |
|      |             | A0120  | Closing DEV12                                                            |
|      |             | A0130  | Closing and Disabling Conditional Logic for DEV13                        |
|      |             | A0140  | Closing and Disabling Conditional Logic for DEV14                        |
|      |             | A0150  | Closing DEV15                                                            |
|      |             | A0160  | Closing and Disabling Conditional Logic for DEV16                        |
|      |             | A0170  | Opening DEV17                                                            |
|      |             | A0180  | Closing and Disabling Conditional Logic for DEV18                        |
|      |             | A0190  | Closing and Disabling Conditional Logic for DEV19                        |
|      |             | A0200  | Set DEV20 Mode to Remote Cascade                                         |
|      |             | A0210  | Setting DEV20 setpoint (LT_780B Setpoint) and Enabling Conditional Logic |
|      |             | A0220  | Set DEV21 Mode to Remote Out                                             |
|      |             | A0230  | Set DEV21 output to Closed and Disable Conditional Logic                 |
|      |             | A0240  | Set DEV22 Mode to Remote Out                                             |
|      |             | A0250  | Set DEV22 output to Closed and Disable Conditional Logic                 |
|      |             | A0260  | Set DEV23 Mode to Remote Cascade                                         |
|      |             | A0270  | Setting DEV23 setpoint (TT_781B Setpoint) and Enabling Conditional Logic |
|      |             | A0280  | Opening and Disabling Conditional Logic for DEV24                        |
|      |             | A0290  | Opening and Disabling Conditional Logic for DEV25                        |
|      |             | A0300  | Opening and Disabling Conditional Logic for DEV26                        |
|      |             | A0310  | Opening and Disabling Conditional Logic for DEV27                        |
|      |             | A0320  | Opening and Disabling Conditional Logic for DEV28                        |
|      |             | A0330  | Opening and Disabling Conditional Logic for DEV29                        |
|      |             | A0340  | Opening and Disabling Conditional Logic for DEV30                        |
|      |             | A0350  | Enabling Conditional Logic for DEV31                                     |
|      |             | A0360  | Closing and Disabling Conditional Logic for DEV32                        |

| Step   | Description | Action | Action Description                                                       |
|--------|-------------|--------|--------------------------------------------------------------------------|
|        |             | A0370  | Closing and Disabling Conditional Logic for DEV33                        |
|        |             | A0380  | Closing and Disabling Conditional Logic for DEV34                        |
|        |             | A0390  | Closing and Disabling Conditional Logic for DEV35                        |
|        |             | A0400  | Set DEV36 Mode to Remote Out                                             |
|        |             | A0410  | Set DEV36 output to Closed and Disable Conditional Logic                 |
|        |             | A0420  | Set Preset Drain time                                                    |
|        |             | A0430  | Start Preset Drain Timer                                                 |
|        |             | A0440  | Preset Drain time Message                                                |
| T1100A | Drain 880A1 |        | If Preset Drain time (TP016) is over then go to step S1100               |
| S1100  | Drain 880A1 | A0010  | Stopping and Disabling Conditional logic for DEV1                        |
|        |             | A0020  | Closing DEV2                                                             |
|        |             | A0030  | Stopping DEV3                                                            |
|        |             | A0040  | Stopping and Disabling Conditional Logic for DEV4                        |
|        |             | A0050  | Stopping and Disabling Conditional Logic for DEV5                        |
|        |             | A0060  | Enabling Conditional Logic for DEV6                                      |
|        |             | A0070  | Enabling Conditional Logic for DEV7                                      |
|        |             | A0080  | Closing DEV8                                                             |
|        |             | A0090  | Closing and Disabling Conditional Logic for DEV9                         |
|        |             | A0100  | Closing DEV10                                                            |
|        |             | A0110  | Closing and Disabling Conditional Logic for DEV11                        |
|        |             | A0120  | Closing DEV12                                                            |
|        |             | A0130  | Closing and Disabling Conditional Logic for DEV13                        |
|        |             | A0140  | Closing and Disabling Conditional Logic for DEV14                        |
|        |             | A0150  | Closing DEV15                                                            |
|        |             | A0160  | Closing and Disabling Conditional Logic for DEV16                        |
|        |             | A0170  | Opening DEV17                                                            |
|        |             | A0180  | Closing and Disabling Conditional Logic for DEV18                        |
|        |             | A0190  | Closing and Disabling Conditional Logic for DEV19                        |
|        |             | A0200  | Set DEV20 Mode to Remote Cascade                                         |
|        |             | A0210  | Setting DEV20 setpoint (LT_780B Setpoint) and Enabling Conditional Logic |

| Step   | Description | Action | Action Description                                                       |
|--------|-------------|--------|--------------------------------------------------------------------------|
|        |             | A0220  | Set DEV21 Mode to Remote Out                                             |
|        |             | A0230  | Set DEV21 output to Closed and Disable Conditional Logic                 |
|        |             | A0240  | Set DEV22 Mode to Remote Out                                             |
|        |             | A0250  | Set DEV22 output to Closed and Disable Conditional Logic                 |
|        |             | A0260  | Set DEV23 Mode to Remote Cascade                                         |
|        |             | A0270  | Setting DEV23 setpoint (TT_781B Setpoint) and Enabling Conditional Logic |
|        |             | A0280  | Opening and Disabling Conditional Logic for DEV24                        |
|        |             | A0290  | Opening and Disabling Conditional Logic for DEV25                        |
|        |             | A0300  | Opening and Disabling Conditional Logic for DEV26                        |
|        |             | A0310  | Opening and Disabling Conditional Logic for DEV27                        |
|        |             | A0320  | Opening and Disabling Conditional Logic for DEV28                        |
|        |             | A0330  | Opening and Disabling Conditional Logic for DEV29                        |
|        |             | A0340  | Opening and Disabling Conditional Logic for DEV30                        |
|        |             | A0350  | Opening and Disabling Conditional Logic for DEV31                        |
|        |             | A0360  | Enabling Conditional Logic for DEV32                                     |
|        |             | A0370  | Enabling Conditional Logic for DEV33                                     |
|        |             | A0380  | Closing and Disabling Conditional Logic for DEV34                        |
|        |             | A0390  | Closing and Disabling Conditional Logic for DEV35                        |
|        |             | A0400  | Set DEV36 Mode to Remote Out                                             |
|        |             | A0410  | Set DEV36 output to Closed and Disable Conditional Logic                 |
|        |             | A0420  | Set Preset Drain time                                                    |
|        |             | A0430  | Start Preset Drain Timer                                                 |
|        |             | A0440  | Preset Drain time Message                                                |
| T1200A | Drain 880A2 |        | If Preset Drain time (TP016) is over then go to step S1200               |
|        |             |        |                                                                          |
| S1200  | Drain 880A2 | A0010  | Stopping and Disabling Conditional logic for DEV1                        |
|        |             | A0020  | Closing DEV2                                                             |
|        |             | A0030  | Stopping DEV3                                                            |
|        |             | A0040  | Stopping and Disabling Conditional Logic for DEV4                        |

| Step | Description | Action | Action Description                                                          |
|------|-------------|--------|-----------------------------------------------------------------------------|
|      |             | A0050  | Stopping and Disabling Conditional Logic for DEV5                           |
|      |             | A0060  | Enabling Conditional Logic for DEV6                                         |
|      |             | A0070  | Enabling Conditional Logic for DEV7                                         |
|      |             | A0080  | Closing DEV8                                                                |
|      |             | A0090  | Closing and Disabling Conditional Logic for DEV9                            |
|      |             | A0100  | Closing DEV10                                                               |
|      |             | A0110  | Closing and Disabling Conditional Logic for DEV11                           |
|      |             | A0120  | Closing DEV12                                                               |
|      |             | A0130  | Closing and Disabling Conditional Logic for DEV13                           |
|      |             | A0140  | Closing and Disabling Conditional Logic for DEV14                           |
|      |             | A0150  | Closing DEV15                                                               |
|      |             | A0160  | Closing and Disabling Conditional Logic for DEV16                           |
|      |             | A0170  | Opening DEV17                                                               |
|      |             | A0180  | Closing and Disabling Conditional Logic for DEV18                           |
|      |             | A0190  | Closing and Disabling Conditional Logic for DEV19                           |
|      |             | A0200  | Set DEV20 Mode to Remote Cascade                                            |
|      |             | A0210  | Setting DEV20 setpoint (LT_780B Setpoint) and Enabling Conditional Logic    |
|      |             | A0220  | Set DEV21 Mode to Remote Out                                                |
|      |             | A0230  | Set DEV21 output to Closed and Disable Conditional Logic                    |
|      |             | A0240  | Set DEV22 Mode to Remote Out                                                |
|      |             | A0250  | Set DEV22 output to Closed and Disable Conditional Logic                    |
|      |             | A0260  | Set DEV23 Mode to Remote Cascade                                            |
|      |             | A0270  | Setting DEV23 setpoint to (TT_781B Setpoint) and Enabling Conditional Logic |
|      |             | A0280  | Opening and Disabling Conditional Logic for DEV24                           |
|      |             | A0290  | Opening and Disabling Conditional Logic for DEV25                           |
|      |             | A0300  | Opening and Disabling Conditional Logic for DEV26                           |
|      |             | A0310  | Opening and Disabling Conditional Logic for DEV27                           |
|      |             | A0320  | Opening and Disabling Conditional Logic for DEV28                           |
|      |             | A0330  | Opening and Disabling Conditional Logic for DEV29                           |

| Step   | Description | Action | Action Description                                         |
|--------|-------------|--------|------------------------------------------------------------|
|        |             | A0340  | Opening and Disabling Conditional Logic for DEV30          |
|        |             | A0350  | Opening and Disabling Conditional Logic for DEV31          |
|        |             | A0360  | Opening and Disabling Conditional Logic for DEV32          |
|        |             | A0370  | Opening and Disabling Conditional Logic for DEV33          |
|        |             | A0380  | Enabling Conditional Logic for DEV34                       |
|        |             | A0390  | Enabling Conditional Logic for DEV35                       |
|        |             | A0400  | Set DEV36 Mode to Remote Out                               |
|        |             | A0410  | Set DEV36 output to Closed and Disable Conditional Logic   |
|        |             | A0420  | Set Preset Drain time                                      |
|        |             | A0430  | Start Preset Drain Timer                                   |
|        |             | A0440  | Preset Drain time Message                                  |
| T1300A | Drain 880B  |        | If Preset Drain time (TP016) is over then go to step S1300 |
| S1300  | Drain 880B  | A0010  | Stopping and Disabling Conditional logic for DEV1          |
|        |             | A0020  | Closing DEV2                                               |
|        |             | A0030  | Stopping DEV3                                              |
|        |             | A0040  | Stopping and Disabling Conditional Logic for DEV4          |
|        |             | A0050  | Stopping and Disabling Conditional Logic for DEV5          |
|        |             | A0060  | Enabling Conditional Logic for DEV6                        |
|        |             | A0070  | Enabling Conditional Logic for DEV7                        |
|        |             | A0080  | Closing DEV8                                               |
|        |             | A0090  | Closing and Disabling Conditional Logic for DEV9           |
|        |             | A0100  | Closing DEV10                                              |
|        |             | A0110  | Closing and Disabling Conditional Logic for DEV11          |
|        |             | A0120  | Closing DEV12                                              |
|        |             | A0130  | Closing and Disabling Conditional Logic for DEV13          |
|        |             | A0140  | Closing and Disabling Conditional Logic for DEV14          |
|        |             | A0150  | Closing DEV15                                              |
|        |             | A0160  | Closing and Disabling Conditional Logic for DEV16          |
|        |             | A0170  | Opening DEV17                                              |
|        |             | A0180  | Closing and Disabling Conditional Logic for DEV18          |

| Step   | Description | Action | Action Description                                                            |
|--------|-------------|--------|-------------------------------------------------------------------------------|
|        |             | A0190  | Closing and Disabling Conditional Logic for DEV19                             |
|        |             | A0200  | Set DEV20 Mode to Remote Out                                                  |
|        |             | A0210  | Set DEV20 output to Open and Enable Conditional Logic                         |
|        |             | A0220  | Set DEV21 Mode to Remote Out                                                  |
|        |             | A0230  | Set DEV21 output to Closed and Disable Conditional Logic                      |
|        |             | A0240  | Set DEV22 Mode to Remote Out                                                  |
|        |             | A0250  | Set DEV22 output to Closed and Disable Conditional Logic                      |
|        |             | A0260  | Set DEV23 Mode to Remote Cascade                                              |
|        |             | A0270  | Setting DEV23 setpoint (TT_781B Setpoint) and Enabling Conditional Logic      |
|        |             | A0280  | Opening and Disabling Conditional Logic for DEV24                             |
|        |             | A0290  | Opening and Disabling Conditional Logic for DEV25                             |
|        |             | A0300  | Opening and Disabling Conditional Logic for DEV26                             |
|        |             | A0310  | Opening and Disabling Conditional Logic for DEV27                             |
|        |             | A0320  | Opening and Disabling Conditional Logic for DEV28                             |
|        |             | A0330  | Opening and Disabling Conditional Logic for DEV29                             |
|        |             | A0340  | Opening and Disabling Conditional Logic for DEV30                             |
|        |             | A0350  | Opening and Disabling Conditional Logic for DEV31                             |
|        |             | A0360  | Opening and Disabling Conditional Logic for DEV32                             |
|        |             | A0370  | Opening and Disabling Conditional Logic for DEV33                             |
|        |             | A0380  | Opening and Disabling Conditional Logic for DEV34                             |
|        |             | A0390  | Opening and Disabling Conditional Logic for DEV35                             |
|        |             | A0400  | Set DEV36 Mode to Remote Out                                                  |
|        |             | A0410  | Set DEV36 output to Closed and Disable Conditional Logic                      |
| T1400A | Blow 880B   |        | If LSL_780B is in alarm and LT_780B_VLL is in VLL alarm then go to step S1400 |
|        |             |        |                                                                               |
| S1400  | Blow 880B   | A0010  | Stopping and Disabling Conditional logic for DEV1                             |
|        |             | A0020  | Closing DEV2                                                                  |
|        |             | A0030  | Stopping DEV3                                                                 |

| Step | Description | Action | Action Description                                       |
|------|-------------|--------|----------------------------------------------------------|
|      |             | A0040  | Stopping and Disabling Conditional Logic for DEV4        |
|      |             | A0050  | Stopping and Disabling Conditional Logic for DEV5        |
|      |             | A0060  | Stopping and Disabling Conditional Logic for DEV6        |
|      |             | A0070  | Stopping and Disabling Conditional Logic for DEV7        |
|      |             | A0080  | Closing DEV8                                             |
|      |             | A0090  | Closing and Disabling Conditional Logic for DEV9         |
|      |             | A0100  | Closing DEV10                                            |
|      |             | A0110  | Closing and Disabling Conditional Logic for DEV11        |
|      |             | A0120  | Closing DEV12                                            |
|      |             | A0130  | Closing and Disabling Conditional Logic for DEV13        |
|      |             | A0140  | Closing and Disabling Conditional Logic for DEV14        |
|      |             | A0150  | Closing DEV15                                            |
|      |             | A0160  | Closing and Disabling Conditional Logic for DEV16        |
|      |             | A0170  | Closing DEV17                                            |
|      |             | A0180  | Enabling Conditional Logic for DEV18                     |
|      |             | A0190  | Closing and Disabling Conditional Logic for DEV19        |
|      |             | A0200  | Set DEV20 Mode to Remote Out                             |
|      |             | A0210  | Set DEV20 output to Open and Disable Conditional Logic   |
|      |             | A0220  | Set DEV21 Mode to Remote Out                             |
|      |             | A0230  | Set DEV21 output to Closed and Disable Conditional Logic |
|      |             | A0240  | Set DEV22 Mode to Remote Out                             |
|      |             | A0250  | Set DEV22 output to Closed and Disable Conditional Logic |
|      |             | A0260  | Set DEV23 Mode to Remote Out                             |
|      |             | A0270  | Set DEV23 output to Closed and Disable Conditional Logic |
|      |             | A0280  | Opening and Disabling Conditional Logic for DEV24        |
|      |             | A0290  | Opening and Disabling Conditional Logic for DEV25        |
|      |             | A0300  | Opening and Disabling Conditional Logic for DEV26        |
|      |             | A0310  | Opening and Disabling Conditional Logic for DEV27        |
|      |             | A0320  | Opening and Disabling Conditional Logic for DEV28        |

| Step   | Description     | Action | Action Description                                       |
|--------|-----------------|--------|----------------------------------------------------------|
|        |                 | A0330  | Opening and Disabling Conditional Logic for DEV29        |
|        |                 | A0340  | Opening and Disabling Conditional Logic for DEV30        |
|        |                 | A0350  | Opening and Disabling Conditional Logic for DEV31        |
|        |                 | A0360  | Opening and Disabling Conditional Logic for DEV32        |
|        |                 | A0370  | Opening and Disabling Conditional Logic for DEV33        |
|        |                 | A0380  | Opening and Disabling Conditional Logic for DEV34        |
|        |                 | A0390  | Opening and Disabling Conditional Logic for DEV35        |
|        |                 | A0400  | Set DEV36 Mode to Remote Out                             |
|        |                 | A0410  | Set DEV36 output to Closed and Disable Conditional Logic |
|        |                 | A0420  | Set Message: Setting Preset blow time                    |
|        |                 | A0430  | Start Preset Blow Timer                                  |
|        |                 | A0440  | Preset Blow time Message                                 |
| T1500A | Blow to Storage |        | If DEV18 blow time (TP008) is over then go to step S1500 |
|        |                 |        |                                                          |
| S1500  | Blow to Storage | A0010  | Stopping and Disabling Conditional logic for DEV1        |
|        |                 | A0020  | Closing DEV2                                             |
|        |                 | A0030  | Stopping DEV3                                            |
|        |                 | A0040  | Stopping and Disabling Conditional Logic for DEV4        |
|        |                 | A0050  | Stopping and Disabling Conditional Logic for DEV5        |
|        |                 | A0060  | Stopping and Disabling Conditional Logic for DEV6        |
|        |                 | A0070  | Stopping and Disabling Conditional Logic for DEV7        |
|        |                 | A0080  | Closing DEV8                                             |
|        |                 | A0090  | Closing and Disabling Conditional Logic for DEV9         |
|        |                 | A0100  | Closing DEV10                                            |
|        |                 | A0110  | Closing and Disabling Conditional Logic for DEV11        |
|        |                 | A0120  | Closing DEV12                                            |
|        |                 | A0130  | Closing and Disabling Conditional Logic for DEV13        |
|        |                 | A0140  | Closing and Disabling Conditional Logic for DEV14        |
|        |                 | A0150  | Closing DEV15                                            |

| Step | Description | Action | Action Description                                       |
|------|-------------|--------|----------------------------------------------------------|
|      |             | A0160  | Closing and Disabling Conditional Logic for DEV16        |
|      |             | A0170  | Closing DEV17                                            |
|      |             | A0180  | Closing and Disabling Conditional Logic for DEV18        |
|      |             | A0190  | Enabling Conditional Logic for DEV19                     |
|      |             | A0200  | Set DEV20 Mode to Remote Out                             |
|      |             | A0210  | Set DEV20 output to Closed and Disable Conditional Logic |
|      |             | A0220  | Set DEV21 Mode to Remote Out                             |
|      |             | A0230  | Set DEV21 output to Closed and Disable Conditional Logic |
|      |             | A0240  | Set DEV22 Mode to Remote Out                             |
|      |             | A0250  | Set DEV22 output to Closed and Disable Conditional Logic |
|      |             | A0260  | Set DEV23 Mode to Remote Out                             |
|      |             | A0270  | Set DEV23 output to Closed and Disable Conditional Logic |
|      |             | A0280  | Opening and Disabling Conditional Logic for DEV24        |
|      |             | A0290  | Opening and Disabling Conditional Logic for DEV25        |
|      |             | A0300  | Opening and Disabling Conditional Logic for DEV26        |
|      |             | A0310  | Opening and Disabling Conditional Logic for DEV27        |
|      |             | A0320  | Opening and Disabling Conditional Logic for DEV28        |
|      |             | A0330  | Opening and Disabling Conditional Logic for DEV29        |
|      |             | A0340  | Opening and Disabling Conditional Logic for DEV30        |
|      |             | A0350  | Opening and Disabling Conditional Logic for DEV31        |
|      |             | A0360  | Opening and Disabling Conditional Logic for DEV32        |
|      |             | A0370  | Opening and Disabling Conditional Logic for DEV33        |
|      |             | A0380  | Opening and Disabling Conditional Logic for DEV34        |
|      |             | A0390  | Opening and Disabling Conditional Logic for DEV35        |
|      |             | A0400  | Set DEV36 Mode to Remote Out                             |
|      |             | A0410  | Set DEV36 output to Closed and Disable Conditional Logic |
|      |             | A0420  | Set Message: Setting Preset blow time                    |
|      |             | A0430  | Start Preset Blow Timer                                  |
|      |             | A0440  | Preset Blow time Message                                 |

| Step  | Description   | Action | Action Description                                       |
|-------|---------------|--------|----------------------------------------------------------|
|       |               | A0450  | Reset Cold Recirculation Button                          |
| T1600 | Close Devices |        | If DEV19 blow time (TP009) is over then go to step S1600 |
|       |               |        |                                                          |
| S1600 | Close Devices | A0010  | Stopping and Disabling Conditional logic for DEV1        |
|       |               | A0020  | Closing DEV2                                             |
|       |               | A0030  | Stopping DEV3                                            |
|       |               | A0040  | Stopping and Disabling Conditional Logic for DEV4        |
|       |               | A0050  | Stopping and Disabling Conditional Logic for DEV5        |
|       |               | A0060  | Stopping and Disabling Conditional Logic for DEV6        |
|       |               | A0070  | Stopping and Disabling Conditional Logic for DEV7        |
|       |               | A0080  | Closing DEV8                                             |
|       |               | A0090  | Closing and Disabling Conditional Logic for DEV9         |
|       |               | A0100  | Closing DEV10                                            |
|       |               | A0110  | Closing and Disabling Conditional Logic for DEV11        |
|       |               | A0120  | Closing DEV12                                            |
|       |               | A0130  | Closing and Disabling Conditional Logic for DEV13        |
|       |               | A0140  | Closing and Disabling Conditional Logic for DEV14        |
|       |               | A0150  | Closing DEV15                                            |
|       |               | A0160  | Closing and Disabling Conditional Logic for DEV16        |
|       |               | A0170  | Closing DEV17                                            |
|       |               | A0180  | Closing and Disabling Conditional Logic for DEV18        |
|       |               | A0190  | Closing and Disabling Conditional Logic for DEV19        |
|       |               | A0200  | Set DEV20 Mode to Remote Out                             |
|       |               | A0210  | Set DEV20 output to Closed and Disable Conditional Logic |
|       |               | A0220  | Set DEV21 Mode to Remote Out                             |
|       |               | A0230  | Set DEV21 output to Closed and Disable Conditional Logic |
|       |               | A0240  | Set DEV22 Mode to Remote Out                             |
|       |               | A0250  | Set DEV22 output to Closed and Disable Conditional Logic |
|       |               | A0260  | Set DEV23 Mode to Remote Out                             |
|       |               | A0270  | Set DEV23 output to Closed and Disable Conditional Logic |

| Step  | Description     | Action | Action Description                                       |
|-------|-----------------|--------|----------------------------------------------------------|
|       |                 | A0280  | Closing and Disabling Conditional Logic for DEV24        |
|       |                 | A0290  | Closing and Disabling Conditional Logic for DEV25        |
|       |                 | A0300  | Closing and Disabling Conditional Logic for DEV26        |
|       |                 | A0310  | Closing and Disabling Conditional Logic for DEV27        |
|       |                 | A0320  | Closing and Disabling Conditional Logic for DEV28        |
|       |                 | A0330  | Closing and Disabling Conditional Logic for DEV29        |
|       |                 | A0340  | Closing and Disabling Conditional Logic for DEV30        |
|       |                 | A0350  | Closing and Disabling Conditional Logic for DEV31        |
|       |                 | A0360  | Closing and Disabling Conditional Logic for DEV32        |
|       |                 | A0370  | Closing and Disabling Conditional Logic for DEV33        |
|       |                 | A0380  | Closing and Disabling Conditional Logic for DEV34        |
|       |                 | A0390  | Closing and Disabling Conditional Logic for DEV35        |
|       |                 | A0400  | Set DEV36 Mode to Remote Out                             |
|       |                 | A0410  | Set DEV36 output to Closed and Disable Conditional Logic |
| T1700 | Disable HM      |        | If no pending actions in S1600 then go to step S1700     |
|       |                 |        |                                                          |
| S1700 | Disable HM      | A0010  | Disable HM 01 (Module Ownership Lost)                    |
| T1800 | Release Devices |        | If no pending actions in S1700 then go to step S1800     |
|       |                 |        |                                                          |
| S1800 | Release Devices | A0010  | Release Devices Message                                  |
|       |                 | A0020  | Release Private Device                                   |
|       |                 | A0030  | Release Private Device                                   |
|       |                 | A0040  | Release Private Device                                   |
|       |                 | A0050  | Release Private Device                                   |
|       |                 | A0060  | Release Private Device                                   |
|       |                 | A0070  | Release Private Device                                   |
|       |                 | A0080  | Release Private Device                                   |
|       |                 | A0090  | Release Private Device                                   |
|       |                 | A0100  | Release Private Device                                   |
|       |                 | A0110  | Release Private Device                                   |
|       |                 | A0120  | Release Private Device                                   |

| Step  | Description      | Action | Action Description                                   |
|-------|------------------|--------|------------------------------------------------------|
|       |                  | A0130  | Release Private Device                               |
|       |                  | A0140  | Release Private Device                               |
|       |                  | A0150  | Release Private Device                               |
|       |                  | A0160  | Release Private Device                               |
|       |                  | A0170  | Release Private Device                               |
|       |                  | A0180  | Release Private Device                               |
|       |                  | A0190  | Release Private Device                               |
|       |                  | A0200  | Release Private Device                               |
|       |                  | A0210  | Release Private Device                               |
|       |                  | A0220  | Release Private Device                               |
|       |                  | A0230  | Release Private Device                               |
|       |                  | A0240  | Release Private Device                               |
|       |                  | A0250  | Release Private Device                               |
|       |                  | A0260  | Release Private Device                               |
|       |                  | A0270  | Release Private Device                               |
|       |                  | A0280  | Release Private Device                               |
|       |                  | A0290  | Release Private Device                               |
|       |                  | A0300  | Release Private Device                               |
|       |                  | A0310  | Release Private Device                               |
|       |                  | A0320  | Release Private Device                               |
|       |                  | A0330  | Release Private Device                               |
|       |                  | A0340  | Release Private Device                               |
|       |                  | A0350  | Release Private Device                               |
|       |                  | A0360  | Release Private Device                               |
| T9900 | Complete         |        | If no pending actions in S1800 then go to step S9900 |
|       |                  |        |                                                      |
| S9900 | Complete         | A0010  | Set Message: Drain Command Complete                  |
|       |                  | A0020  | Self Release the EM                                  |
| END   | Command Complete |        | True                                                 |

Table 22: Recirculation Sequence

### 3.4

### Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

| Parameter            | Description        | Type      | Units | Default Value/Reference |
|----------------------|--------------------|-----------|-------|-------------------------|
| Operating Parameters |                    |           |       |                         |
| OP001                | Deodorising 1 Pump | Named Set | NA    | RF1_PU1722C1            |

| Parameter         | Description                     | Type               | Units | Default Value/Reference |
|-------------------|---------------------------------|--------------------|-------|-------------------------|
| OP002             | Deodorising 2 Pump              | Named Set          | NA    | RF1_PU1722              |
| OP003             | Quality Recirculation           | Boolean            | NA    | False                   |
| OP004             | Production                      | Boolean            | NA    | False                   |
| OP005             | Hot Oil Recirculation (HOR)     | Boolean            | NA    | False                   |
| OP006             | Cold Recirculation (CR)         | Boolean            | NA    | False                   |
| OP007             | Start Drain Sequence - After CR | String             | NA    | 0                       |
| OP008             | Bypass 1-822QE2 Heating Coil    | String             | NA    | 0                       |
| OP009             | Bypass 1-822QE2 Heating Coil    | Boolean            | NA    | False                   |
| OP010             | Manual Valve Confirmation       | Boolean            | NA    | False                   |
| Tuning Parameters |                                 |                    |       |                         |
| TP001             | Delay Timer                     | Floating Point     | sec   | 5                       |
| TP002             | Remote Setpoint FT_702          | Floating Point     | kg/h  | 62500                   |
| TP003             | Remote Setpoint LT_722C         | Floating Point     | %     | 70                      |
| TP004             | Remote Setpoint LT_780B         | Floating Point     | %     | 40                      |
| TP005             | Remote Setpoint TT_781B         | Floating Point     | °C    | 60                      |
| TP006             | XV_802-B Blowing                | Floating Point     | sec   | 60                      |
| TP007             | XV_822C-B Blowing               | Floating Point     | sec   | 60                      |
| TP008             | XV_880B-B Blowing               | Floating Point     | sec   | 60                      |
| TP009             | XV_881A-B Blowing               | Floating Point     | sec   | 60                      |
| TP010             | Vacuum Setpoint                 | External Reference | mbar  | RF1_PT1722AB/HI_HI_LIM  |
| TP011             | Deodorising Temperature         | External Reference | °C    | RF1_TT1721A2/LO_LO_LIM  |
| TP012             | Max P822 outlet Temperature     | External Reference | °C    | RF1_TT1781B/HI_HI_LIM   |
| TP013             | Hot recirculation Preset Time   | Floating Point     | min   | 60                      |
| TP014             | Cold Recirculation Temperature  | Floating Point     | °C    | 60                      |
| TP015             | Blow Coil preset time           | Floating Point     | sec   | 400                     |

| Parameter | Description                  | Type           | Units | Default Value/Reference |
|-----------|------------------------------|----------------|-------|-------------------------|
| TP016     | Drain preset Time            | Floating Point | sec   | 600                     |
| TP017     | Bypass 1-822QE2 Heating Coil | Named Set      | NA    | NO                      |
| TP018     | XV_822QE1-B Blowing          | Floating Point | sec   | 0                       |
| TP019     | RSP for TT_722C              | Floating Point | NA    | 180                     |

Table 23: Operating and Tuning Parameters

### 3.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

| Parameter Name  | Parameter Type     | Value                      |
|-----------------|--------------------|----------------------------|
| MY_TAG          |                    | RF1EQ_822QE                |
| OPR_CMD_MASK    |                    | 31                         |
| LSM_721A        | External Reference | RF1_LSM1721A/DI1/DISC_ACT  |
| TT_721A2        | External Reference | RF1_TT1721A2/AI1/PV        |
| LSL_780B        | External Reference | RF1_LSL1780B/DI1/DISC_ACT  |
| LSH_780B        | External Reference | RF1_LSH1780B/DI1/DISC_ACT  |
| LT_780B_VLL     | External Reference | RF1_LT1780B/PID1/LO_LO_ACT |
| LSL_722C        | External Reference | RF1_LSL1722C/DI1/DISC_ACT  |
| LT_722C_VLL     | External Reference | RF1_LT1722C/PID1/LO_LO_ACT |
| LSH_722C        | External Reference | RF1_LSH1722C/DI1/DISC_ACT  |
| LSL_702         | External Reference | RF1_LSL1702/DI1/DISC_ACT   |
| LT_722C_VHL     | External Reference | RF1_LT1722C/PID1/HI_HI_ACT |
| LT_780B_VHL     | External Reference | RF1_LT1780B/PID1/HI_HI_ACT |
| BYPASS          | Boolean            | False                      |
| STARTUP         | String             | RF1EQ_STARTUP1             |
| EM_DEOD_FA_CKT1 | String             | RF1EQ_802                  |
| DY_REF_1        | Dynamic Reference  |                            |
| VACUUM_ALM      | External Reference | RF1_PT1722AB/AI1/HI_HI_ACT |
| HOLD            | External Reference | RF1EQ_STARTUP1/OP006_VALUE |
| STEP_TIME       | Floating point     | 0                          |
| COMPLETED_TIME  | Floating point     | 0                          |
| RESET_COMP_TM   | Boolean            | False                      |

Table 24: Configurable Parameters

### 3.6 Named Set

Following Named set shall be used for configuration of EM.

| Name            | Description               |
|-----------------|---------------------------|
| NS_DEOD_FA_CKT2 | Refinery Area: Startup EM |

**Table 25: Named Set Details**

Details of states for NS\_DEOD\_FA\_CKT2 named set are as follows,

| Named Set Value | Named Set Name |
|-----------------|----------------|
| 00              | Hold           |
| 01              | Heating        |
| 02              | Production     |
| 03              | Recirculation  |

**Table 26: Name Set State Details**

## 4 EQUIPMENT MODULE CLASS DESIGN – EM\_DEOD\_FA\_CKT3 (1-814/23)

### 4.1 Functional Description

EM\_DEOD\_FA\_CKT3 is a class based equipment module that driven by EM\_RF\_STARTUP1 sequence. EM\_DEOD\_FA\_CKT3 is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. Initially Operator shall be provided only with Hold and Production command. All devices used in EM shall be acquired in the Production command and released in Stop Command. Hold Request shall be enabled for all acquired devices at the end of Hold command.

### 4.2 Instances

The following Table contains the module instances of the equipment module class.

| Device Name | Module Class    | Instance Name |
|-------------|-----------------|---------------|
| EM          | EM_DEOD_FA_CKT3 | RF1EQ_814_23  |
| DEV1        | _MRSS11_130_CUR | RF1_PU1714AG  |
| DEV2        | _V12_IP_130_SE  | RF1_V1714AGB  |
| DEV3        | _MRSS11_130_SE  | RF1_PU1750    |
| DEV4        | _CSD_130_SE     | RF1_TT1781AA  |
| DEV5        | _CSD_130_SE     | RF1_LT1714    |
| DEV6        | _CMT_130_SE     | RF1_TT1781AB  |

**Table 27: Instances**

### 4.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

#### 4.3.1 Command 00 – Hold

| Step  | Description                              | Action | Action Description      |
|-------|------------------------------------------|--------|-------------------------|
| H0010 | Command Hold Start                       | A0010  | Store Fail Message      |
|       |                                          | A0020  | Reset Hold Request      |
|       |                                          | A0030  | Save Previous Step Time |
| T1000 | Actions Complete or Time Out & Hold=True |        | Go to step T1000        |
| T2000 | Actions Complete or                      |        | Go to step T2000        |

| Step   | Description                  | Action | Action Description                                      |
|--------|------------------------------|--------|---------------------------------------------------------|
|        | Time Out & Hold=False        |        |                                                         |
|        |                              |        |                                                         |
| H1000  | Hold Command 1               | A0010  | Stopping and Disabling Conditional Logic for DEV3       |
| T9900A | Actions Complete or Time Out |        | Go to step H9900A                                       |
|        |                              |        |                                                         |
| H2000  | General Reset                | A0010  | Stopping DEV1                                           |
|        |                              | A0020  | Closing and Disabling Conditional Logic for DEV2        |
|        |                              | A0030  | Stopping and Disabling Conditional Logic for DEV3       |
|        |                              | A0040  | Set DEV4 Mode to Remote Output                          |
|        |                              | A0050  | Set DEV4 output to Closed and Disable Conditional Logic |
|        |                              | A0060  | Set DEV5 Mode to Remote Out                             |
|        |                              | A0070  | Set DEV5 output to Closed and Disable Conditional Logic |
| T9900B | Actions Complete or Time Out |        | Go to step H9900A                                       |
|        |                              |        |                                                         |
| H9900A | Command Hold End             | A0010  | Completed                                               |
| END    | Command Complete             |        | True                                                    |
|        |                              |        |                                                         |
| H9900B | Release Devices              | A0010  | Release Devices Message                                 |
|        |                              | A0020  | Release Private Devices                                 |
|        |                              | A0030  | Release Private Devices                                 |
|        |                              | A0040  | Release Private Devices                                 |
|        |                              | A0050  | Release Private Devices                                 |
|        |                              | A0060  | Release Private Devices                                 |
|        |                              | A0070  | Release Private Devices                                 |
|        |                              | A0080  | Reset Parameters                                        |
| END1   | Command Complete             |        | True                                                    |

Table 28: Hold Sequence

## 4.3.2 Command 01 – Production

| Step   | Description        | Action | Action Description                                      |
|--------|--------------------|--------|---------------------------------------------------------|
| S0000  | Start              | A0010  | Set Message: Starting Production Command                |
| T0010  | First Pass = False |        | If First Pass = False then go to step S0010             |
| T0020A | First Pass = True  |        | If First Pass = True then go to step S0020              |
|        |                    |        |                                                         |
| S0010  | Initializations    | A0010  | First Pass Logic                                        |
|        |                    | A0020  | First Pass = True                                       |
|        |                    | A0030  | Set Task Pointer                                        |
| T0020B | Actions Complete   |        | If no pending actions in S0010 then go to step S0020    |
|        |                    |        |                                                         |
| S0020  | Acquire Devices    | A0010  | Acquire Devices Message                                 |
|        |                    | A0020  | Acquire Private Devices                                 |
|        |                    | A0030  | Acquire Private Devices                                 |
|        |                    | A0040  | Acquire Private Devices                                 |
|        |                    | A0050  | Acquire Private Devices                                 |
|        |                    | A0060  | Acquire Private Devices                                 |
|        |                    | A0070  | Acquire Private Devices                                 |
|        |                    | A0080  | Reset Hold Request                                      |
| T0100B | Standby            |        | If Task Pointer = 1 then go to step S0100               |
| T0200B | Fill               |        | If Task Pointer = 2 then go to step S0200               |
| T0300B | Blow Inlet         |        | If Task Pointer = 3 then go to step S0300               |
|        |                    |        |                                                         |
| S0100  | Standby            | A0010  | Stopping DEV1                                           |
|        |                    | A0020  | Closing and Disabling Conditional Logic for DEV2        |
|        |                    | A0030  | Stopping and Disabling Conditional Logic for DEV3       |
|        |                    | A0040  | Set DEV4 Mode to Remote Output                          |
|        |                    | A0050  | Set DEV4 output to Closed and Disable Conditional Logic |
|        |                    | A0060  | Set DEV5 Mode to Remote Out                             |
|        |                    | A0070  | Set DEV5 output to Closed and Disable Conditional Logic |
| T0200A | Fill               |        | If no pending actions in S0100 then go to step S0200    |
|        |                    |        |                                                         |
| S0200  | Fill               | A0010  | Stopping DEV1                                           |
|        |                    | A0020  | Closing and Disabling Conditional Logic for DEV2        |
|        |                    | A0030  | Stopping and Disabling Conditional Logic for DEV3       |

| Step   | Description      | Action | Action Description                                                        |
|--------|------------------|--------|---------------------------------------------------------------------------|
|        |                  | A0040  | Set DEV4 Mode to Remote Output                                            |
|        |                  | A0050  | Set DEV4 output to Closed and Disable Conditional Logic                   |
|        |                  | A0060  | Set DEV5 Mode to Remote Out                                               |
|        |                  | A0070  | Set DEV5 output to Closed and Disable Conditional Logic                   |
|        |                  | A0080  | Waiting for LSL_714 to not be in alarm                                    |
| T0300A | Production       |        | If LSL_714 is not in alarm then go to step S0300                          |
| S0300  | Production       | A0010  | Starting DEV1                                                             |
|        |                  | A0020  | Closing and Disabling Conditional Logic for DEV2                          |
|        |                  | A0030  | Enabling Conditional Logic for DEV3                                       |
|        |                  | A0040  | Set DEV4 Mode to Cascade and DEV6 to Remote Cascade                       |
|        |                  | A0050  | Setting DEV6 setpoint (TT_781AG Setpoint) and Enabling Conditional Logic  |
|        |                  | A0060  | Set DEV5 Mode to Remote Cascade                                           |
|        |                  | A0070  | Setting DEV5 setpoint (LT_P714AG Setpoint) and Enabling Conditional Logic |
| T9900  | Complete         |        | If no pending actions in S0300 then go to step S9900                      |
| S9900  | Complete         | A0010  | Set Message: Production Command Complete                                  |
| END    | Command Complete |        | True                                                                      |

**Table 29: Production Sequence**

## 4.3.3

## Command 02 –Drain

| Step  | Description | Action | Action Description                                                       |
|-------|-------------|--------|--------------------------------------------------------------------------|
| S0000 | Start       | A0010  | Set Message: Starting Drain Command                                      |
|       |             | A0020  | Reset HOLD_REQ                                                           |
| T0100 | Drain       |        | If Drain Command (OP001) selected by operator then go to step S0100      |
| S0100 | Drain       | A0010  | Starting DEV1                                                            |
|       |             | A0020  | Closing and Disabling Conditional Logic for DEV2                         |
|       |             | A0030  | Enabling Conditional Logic for DEV3                                      |
|       |             | A0040  | Set DEV4 Mode to Cascade and DEV6 to Remote Cascade                      |
|       |             | A0050  | Setting DEV6 setpoint (TT_781AG Setpoint) and Enabling Conditional Logic |
|       |             | A0060  | Set DEV5 Mode to Remote Out                                              |

| Step  | Description      | Action | Action Description                                    |
|-------|------------------|--------|-------------------------------------------------------|
|       |                  | A0070  | Set DEV5 output to Open and Disable Conditional Logic |
|       |                  | A0080  | Setting Auto Drain Timer (TMR1)                       |
| T9900 | Complete         |        | If no pending actions in S0100 the go to step S9900   |
| S9900 | Complete         | A0010  | Set Message: Drain Command Complete                   |
| END   | Command Complete |        | True                                                  |

Table 30: Drain Sequence

## 4.3.4 Command 03 – Stop

| Step  | Description   | Action | Action Description                                                                                                     |
|-------|---------------|--------|------------------------------------------------------------------------------------------------------------------------|
| S0000 | Start         | A0010  | Set Message: Starting Stop Drain Command                                                                               |
|       |               | A0020  | Reset HOLD_REQ                                                                                                         |
| T0100 | Blow          |        | If LSL_714 is in alarm and FT_714AG is in VLL or <Stop> command (OP002) by operator or Auto drain time finished (TMR1) |
| S0100 | Blow          | A0010  | Stopping DEV1                                                                                                          |
|       |               | A0020  | Enabling Conditional Logic for DEV2                                                                                    |
|       |               | A0030  | Stopping and Disabling Conditional Logic for DEV3                                                                      |
|       |               | A0040  | Set DEV4 Mode to Remote Output                                                                                         |
|       |               | A0050  | Set DEV4 output to Closed and Disable Conditional Logic                                                                |
|       |               | A0060  | Set DEV5 Mode to Remote Out                                                                                            |
|       |               | A0070  | Set DEV5 output to Closed and Disable Conditional Logic                                                                |
|       |               | A0080  | Set Message: Setting Preset blow time                                                                                  |
|       |               | A0090  | Start Preset Blow Timer                                                                                                |
|       |               | A0100  | Preset Blow time Message                                                                                               |
| T0200 | Close Devices |        | If Preset Blow time (TP001) is over then go to step S0200                                                              |
| S0200 | Close Devices | A0010  | Stopping DEV1                                                                                                          |
|       |               | A0020  | Closing and Disabling Conditional Logic for DEV2                                                                       |
|       |               | A0030  | Stopping and Disabling Conditional Logic for DEV3                                                                      |
|       |               | A0040  | Set DEV4 Mode to Remote Output                                                                                         |
|       |               | A0050  | Set DEV4 output to Closed and Disable Conditional Logic                                                                |
|       |               | A0060  | Set DEV5 Mode to Remote Out                                                                                            |

| Step  | Description       | Action | Action Description                                      |
|-------|-------------------|--------|---------------------------------------------------------|
|       |                   | A0070  | Set DEV5 output to Closed and Disable Conditional Logic |
| T0300 | Disable HM and SM |        | If no pending actions in S0200 the go to step S0300     |
|       |                   |        |                                                         |
| S0300 | Disable HM        | A0010  | Disable HM 01 (Module Ownership Lost)                   |
| T0400 | Release Devices   |        | If no pending actions in S0300 the go to step S0400     |
|       |                   |        |                                                         |
| S0400 | Release Devices   | A0010  | Release Devices Message                                 |
|       |                   | A0020  | Release Private Devices                                 |
|       |                   | A0030  | Release Private Devices                                 |
|       |                   | A0040  | Release Private Devices                                 |
|       |                   | A0050  | Release Private Devices                                 |
|       |                   | A0060  | Release Private Devices                                 |
|       |                   | A0070  | Release Private Devices                                 |
| T9900 | Complete          |        | If no pending actions in S0400 the go to step S9900     |
|       |                   |        |                                                         |
| S9900 | Complete          | A0010  | Set Message: Stop Drain Command Complete                |
|       |                   | A0020  | Self Release the EM                                     |
| END   | Command Complete  |        | True                                                    |

Table 31: Stop Sequence

## 4.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

| Parameter            | Description                  | Type               | Units | Default Value/Reference     |
|----------------------|------------------------------|--------------------|-------|-----------------------------|
| Operating Parameters |                              |                    |       |                             |
| OP001                | DRAIN                        | External Reference | NA    | RF1EQ_STARTUP1 /OP005_VALUE |
| OP002                | STOP                         | Named Set          | NA    | OFF                         |
| Tuning Parameters    |                              |                    |       |                             |
| TP001                | Preset Blow Coil Time        | Floating Point     | sec   | 400                         |
| TP002                | Remote Setpoint for TT_781AG | Floating Point     | NA    | 55                          |
| TP003                | Remote Setpoint for LT_714   | Floating Point     | NA    | 40                          |

| Parameter | Description     | Type           | Units | Default Value/Reference |
|-----------|-----------------|----------------|-------|-------------------------|
| TP004     | Auto Drain Time | Floating Point | sec   | 100                     |

**Table 32: Operating and Tuning Parameters**

## 4.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

| Parameter Name | Parameter Type     | Value/Reference            |
|----------------|--------------------|----------------------------|
| MY_TAG         |                    | RF1EQ_814_23               |
| OPR_CMD_MASK   |                    | 15                         |
| LSL_714        | External Reference | RF1_LSL1714/DI1/DISC_ACT   |
| FT_714AG       | External Reference | RF1_FT1714AG/AI1/LO_LO_ACT |
| RF_STARTUP     | String             | RF1EQ_STARTUP1             |
| DY_REF_1       | Dynamic Reference  |                            |
| HOLD           | External Reference | RF1EQ_STARTUP1/OP006_VALUE |
| EM_DEOD_CKT2   |                    | RF1EQ_822QE/A_PV           |
| STEP_TIME      | Floating point     | 0                          |
| COMPLETED_TIME | Floating point     | 0                          |
| RESET_COMP_TM  | Boolean            | False                      |

**Table 33: Configurable Parameters**

## 4.6 Named Set

Following Named set shall be used for configuration of EM.

| Name            | Description                                 |
|-----------------|---------------------------------------------|
| NS_DEOD_FA_CKT3 | Refinery Area: Deodorising & Ice Condensing |

**Table 34: Named Set Detail**

Details of states for NS\_DEOD\_FA\_CKT3 named set are as follows,

| Named Set Value | Named Set Name |
|-----------------|----------------|
| 00              | Hold           |
| 01              | Production     |
| 02              | Drain          |
| 03              | Stop           |

**Table 35: Name Set State Details**

## 5 EQUIPMENT MODULE CLASS DESIGN – EM\_IC\_VAC\_PROD1 (1-819IC1/C2)

### 5.1 Functional Description

EM\_IC\_VAC\_PROD1 is a class based equipment module that communicates with EM\_RF1\_STARTUP1 sequence to complete fluid process that includes cooling, icing, heating, melting operation. Refer FDS for the functional design requirements including faceplate and detail display. Initially Operator shall be provided only with Hold and Process Fluid command. All devices used in EM shall be acquired and released in Process Fluid Command. Hold Request shall be enabled for all acquired devices at the end of Hold command.

### 5.2 Instances

The following Table contains the module instances of the equipment module class.

| Device Name | Module Class    | Instance Name |              |
|-------------|-----------------|---------------|--------------|
| EM          | EM_IC_VAC_PROD1 | RF1EQ_819IC1  | RF1EQ_819IC2 |
| DEV1        | _V12_IP_130_SE  | RF1_V1719ICG  | RF1_V1719ICM |
| DEV2        | _V12_IP_130_SE  | RF1_V1719ICH  | RF1_V1719ICN |
| DEV3        | _V12_IP_130_SE  | RF1_V1719ICF  | RF1_V1719ICL |
| DEV4        | _V12_IP_130_SE  | RF1_V1719ICC  | RF1_V1719ICJ |
| DEV5        | _V12_IP_130_SE  | RF1_V1719ICD  | RF1_V1719ICI |
| DEV6        | _V12_IP_130_SE  | RF1_V1719ICE  | RF1_V1719ICK |

**Table 36: Instances**

### 5.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

#### 5.3.1 Command 00 – Hold

| Step  | Description         | Action | Action Description |
|-------|---------------------|--------|--------------------|
| H0010 | Command Hold Start  | A0010  | Store Fail Message |
|       |                     | A0020  | Set Hold Request   |
|       |                     | A0030  | Stop Cooling Timer |
|       |                     | A0040  | Stop Icing Timer   |
|       |                     | A0050  | Stop Melting Timer |
|       |                     | A0060  | Stop Drain Timer   |
| T1000 | Actions Complete or |        | Go to step H1000   |

| Step   | Description                               | Action | Action Description                               |
|--------|-------------------------------------------|--------|--------------------------------------------------|
|        | Time Out & Hold=True                      |        |                                                  |
| T2000  | Actions Complete or Time Out & Hold=False |        | Go to step H2000                                 |
|        |                                           |        |                                                  |
| H1000  | Hold Command 1                            | A0010  | Set Message: Executing Hold Command              |
| T9900A | Actions Complete or Time Out              |        | Go to step H9900A                                |
|        |                                           |        |                                                  |
| H2000  | General Reset                             | A0010  | Closing and Disabling Conditional Logic for DEV1 |
|        |                                           | A0020  | Closing and Disabling Conditional Logic for DEV2 |
|        |                                           | A0030  | Closing and Disabling Conditional Logic for DEV3 |
|        |                                           | A0040  | Closing and Disabling Conditional Logic for DEV4 |
|        |                                           | A0050  | Closing and Disabling Conditional Logic for DEV5 |
|        |                                           | A0060  | Closing and Disabling Conditional Logic for DEV6 |
| T9900B | Actions Complete or Time Out              |        | Go to step H9900B                                |
|        |                                           |        |                                                  |
| H9900A | Command Hold End                          | A0010  | Completed                                        |
| END    | Command Complete                          |        | True                                             |
|        |                                           |        |                                                  |
| H9900B | Release Devices                           | A0010  | Release Devices Message                          |
|        |                                           | A0020  | Release Private Devices                          |
|        |                                           | A0030  | Release Private Devices                          |
|        |                                           | A0040  | Release Private Devices                          |
|        |                                           | A0050  | Release Private Devices                          |
|        |                                           | A0060  | Release Private Devices                          |
|        |                                           | A0070  | Release Private Devices                          |
|        |                                           | A0080  | Reset Parameters                                 |
| END1   | Command Complete                          |        | True                                             |

Table 37: Hold Sequence

## 5.3.2 Command 01 – Process Fluid

| Step   | Description        | Action | Action Description                                   |
|--------|--------------------|--------|------------------------------------------------------|
| S0000  | Start              | A0010  | Set Message: Starting Process Fluid Command          |
| T0010  | First Pass = False |        | If First Pass = False then go to step S0010          |
| T0020A | First Pass = True  |        | If First Pass = True then go to step S0020           |
|        |                    |        |                                                      |
| S0010  | Initializations    | A0010  | First Pass Logic                                     |
|        |                    | A0020  | First Pass = True                                    |
|        |                    | A0030  | Stop timers(Cooling, Icing, Melting, Drain)          |
|        |                    | A0040  | Reset timers(Cooling, Icing, Melting, Drain)         |
|        |                    | A0050  | Set SP & Description for Cooling timer               |
|        |                    | A0060  | Set SP & Description for Icing timer                 |
|        |                    | A0070  | Set SP & Description for Melting timer               |
|        |                    | A0080  | Set SP & Description for Drain timer                 |
|        |                    | A0090  | Set Task Pointer                                     |
| T0020B | Acquire Devices    |        | If no pending actions in S0010 then go to step S0020 |
|        |                    |        |                                                      |
| S0020  | Acquire Devices    | A0010  | Acquire Devices Message                              |
|        |                    | A0020  | Acquire Private Devices                              |
|        |                    | A0030  | Acquire Private Devices                              |
|        |                    | A0040  | Acquire Private Devices                              |
|        |                    | A0050  | Acquire Private Devices                              |
|        |                    | A0060  | Acquire Private Devices                              |
|        |                    | A0070  | Acquire Private Devices                              |
|        |                    | A0090  | Reset Hold Request                                   |
| T0100A | Start              |        | If Task Pointer = 1 then go to step S0100            |
| T0200B | Wait               |        | If Task Pointer = 2 then go to step S0200            |
| T0300B | Cooling            |        | If Task Pointer = 3 then go to step S0300            |
| T0400B | Ready              |        | If Task Pointer = 4 then go to step S0400            |
| T0500B | Icing              |        | If Task Pointer = 5 then go to step S0500            |
| T0600B | Melting            |        | If Task Pointer = 6 then go to step S0600            |
| T0700B | Drain              |        | If Task Pointer = 7 then go to step S0700            |
|        |                    |        |                                                      |
| S0100  | Standby            | A0010  | Closing and Disabling Conditional Logic for DEV1     |
|        |                    | A0020  | Closing and Disabling Conditional Logic for DEV2     |
|        |                    | A0030  | Closing and Disabling Conditional Logic for DEV3     |
|        |                    | A0040  | Closing and Disabling Conditional Logic for DEV4     |
|        |                    | A0050  | Closing and Disabling Conditional Logic for DEV5     |
|        |                    | A0060  | Closing and Disabling Conditional Logic for DEV6     |
|        |                    | A0070  | Waiting for Other 819ICX                             |



| Step   | Description | Action | Action Description                                                                                                                                                                        |
|--------|-------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T0200A | Wait        |        | If 1-819ICX is enabled (OP001 for 1-819IC1 and OP002 for 1-819IC2) and Chiller (CHILLER_ALM) is not in alarm and other 1-819ICX is not in Step <Wait> or <Cooling> then go to step S0200. |
|        |             |        |                                                                                                                                                                                           |
| S0200  | Wait        | A0010  | Closing and Disabling Conditional Logic for DEV1                                                                                                                                          |
|        |             | A0020  | Closing and Disabling Conditional Logic for DEV2                                                                                                                                          |
|        |             | A0030  | Closing and Disabling Conditional Logic for DEV3                                                                                                                                          |
|        |             | A0040  | Closing and Disabling Conditional Logic for DEV4                                                                                                                                          |
|        |             | A0050  | Closing and Disabling Conditional Logic for DEV5                                                                                                                                          |
|        |             | A0060  | Closing and Disabling Conditional Logic for DEV6                                                                                                                                          |
|        |             | A0070  | Waiting for Chiller is not in alarm and Chiller is ready                                                                                                                                  |
| T0100B | Standby     |        | If 1-819ICX is disabled (OP001 for 1-819IC1 and OP002 for 1-819IC2) by operator then go to step S0100                                                                                     |
| T0300A | Cooling     |        | If Chiller is not in alarm (CHILLER_ALM) and Chiller ready (CHILLER_READY) then go to step S0300                                                                                          |
|        |             |        |                                                                                                                                                                                           |
| S0300  | Cooling     | A0010  | Closing and Disabling Conditional Logic for DEV1                                                                                                                                          |
|        |             | A0020  | Closing and Disabling Conditional Logic for DEV2                                                                                                                                          |
|        |             | A0030  | Closing and Disabling Conditional Logic for DEV3                                                                                                                                          |
|        |             | A0040  | Closing and Disabling Conditional Logic for DEV4                                                                                                                                          |
|        |             | A0050  | Closing and Disabling Conditional Logic for DEV5                                                                                                                                          |
|        |             | A0060  | Closing and Disabling Conditional Logic for DEV6                                                                                                                                          |
|        |             | A0070  | Set cooling timer (TMR1=TP001)                                                                                                                                                            |
|        |             | A0080  | Starting cooling timer                                                                                                                                                                    |
|        |             | A0090  | Waiting Cooling Timer                                                                                                                                                                     |
| T0100C | Standby     |        | If 1-819ICX is disabled (OP001 for 1-819IC1 and OP002 for 1-819IC2) by operator then go to step S0100                                                                                     |
| T0400A | Ready       |        | If Cooling timer is over (TMR1) and Chiller is not in alarm (CHILLER_ALM) then go to step S0400                                                                                           |
|        |             |        |                                                                                                                                                                                           |
| S0400  | Ready       | A0010  | Closing and Disabling Conditional Logic for DEV1                                                                                                                                          |
|        |             | A0020  | Closing and Disabling Conditional Logic for DEV2                                                                                                                                          |
|        |             | A0030  | Closing and Disabling Conditional Logic for DEV3                                                                                                                                          |
|        |             | A0040  | Closing and Disabling Conditional Logic for DEV4                                                                                                                                          |
|        |             | A0050  | Closing and Disabling Conditional Logic for DEV5                                                                                                                                          |
|        |             | A0060  | Closing and Disabling Conditional Logic for DEV6                                                                                                                                          |
|        |             | A0070  | Set Dynamic Reference and stop Cooing Timer                                                                                                                                               |
|        |             | A0080  | Delayed Massege                                                                                                                                                                           |
|        |             | A0090  | Reset Cooling Timer (TMR1)                                                                                                                                                                |

| Step   | Description | Action | Action Description                                                                                                                                                                                                                                                                  |
|--------|-------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T0100D | Standby     |        | If 1-819ICX is disabled (OP001 for 819IC1 and OP002 for 819IC2) by operator then go to step S0100                                                                                                                                                                                   |
| T0500A | Icing       |        | If (Chiller is not in alarm and other 1-819ICX Icing time over (other 1-819ICX is in <Icing> step) and TT_778 > RSP_TT_778) or (other 819ICX is not in <Icing> step) then go to step S0500                                                                                          |
| S0500  | Icing       | A0010  | Enabling Conditional Logic for DEV1                                                                                                                                                                                                                                                 |
|        |             | A0020  | Enabling Conditional Logic for DEV2                                                                                                                                                                                                                                                 |
|        |             | A0030  | Closing and Disabling Conditional Logic for DEV3                                                                                                                                                                                                                                    |
|        |             | A0040  | Enabling Conditional Logic for DEV4                                                                                                                                                                                                                                                 |
|        |             | A0050  | Enabling Conditional Logic for DEV5                                                                                                                                                                                                                                                 |
|        |             | A0060  | Closing and Disabling Conditional Logic for DEV6                                                                                                                                                                                                                                    |
|        |             | A0070  | Set Icing Timer (TMR2=TP002)                                                                                                                                                                                                                                                        |
|        |             | A0080  | Starting Icing timer                                                                                                                                                                                                                                                                |
| T0600A | Melting     |        | If (Other 1-819IC2 is in <Icing> and Remain Time of other 1-819ICX > Remain Time 1-819ICX) or (Operator select 1-819ICX is Disabled or Startup step in <Standby> or <Aux off>) and TT 1-TT_778 > RSP_TT_778 and other 1-819ICX is not in <Melting> or <Draining> then go step S0600 |
| S0600  | Melting     | A0010  | Closing and Disabling Conditional Logic for DEV1                                                                                                                                                                                                                                    |
|        |             | A0020  | Closing and Disabling Conditional Logic for DEV2                                                                                                                                                                                                                                    |
|        |             | A0030  | Enabling Conditional Logic for DEV3                                                                                                                                                                                                                                                 |
|        |             | A0040  | Closing and Disabling Conditional Logic for DEV4                                                                                                                                                                                                                                    |
|        |             | A0050  | Closing and Disabling Conditional Logic for DEV5                                                                                                                                                                                                                                    |
|        |             | A0060  | Enabling Conditional Logic for DEV6                                                                                                                                                                                                                                                 |
|        |             | A0070  | Stop Icing Timer                                                                                                                                                                                                                                                                    |
|        |             | A0080  | Start Melting of 878/32IC                                                                                                                                                                                                                                                           |
|        |             | A0090  | Setting Melting timer (TMR3=TP003)                                                                                                                                                                                                                                                  |
|        |             | A0100  | Start Melting timer                                                                                                                                                                                                                                                                 |
| T0700A | Drain       |        | If 1-878/32IC is in step <Melting> and 800 (RF1_STARTUP1) <HOLD> command is not enabled and Melting time (TP003) is over then go to step S0700                                                                                                                                      |
| S0700  | Drain       | A0010  | Closing and Disabling Conditional Logic for DEV1                                                                                                                                                                                                                                    |
|        |             | A0020  | Closing and Disabling Conditional Logic for DEV2                                                                                                                                                                                                                                    |
|        |             | A0030  | Closing and Disabling Conditional Logic for DEV3                                                                                                                                                                                                                                    |
|        |             | A0040  | Closing and Disabling Conditional Logic for DEV4                                                                                                                                                                                                                                    |
|        |             | A0050  | Closing and Disabling Conditional Logic for DEV5                                                                                                                                                                                                                                    |
|        |             | A0060  | Opening and Disabling Conditional Logic for DEV6                                                                                                                                                                                                                                    |

| Step   | Description       | Action | Action Description                                                                                                                |
|--------|-------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------|
|        |                   | A0070  | Stopping Melting Timer                                                                                                            |
|        |                   | A0080  | Set Drain Timer (TMR4=TP004)                                                                                                      |
|        |                   | A0090  | Start Drain Timer                                                                                                                 |
|        |                   | A0100  | Start Heating of 878/32IC                                                                                                         |
|        |                   | A0110  | Reset Melting Timer                                                                                                               |
| T0800A | Close Devices     |        | If Drain time (TP004) is over and LSL_719A is in alarm and 800 (RF1_STARTUP1) <HOLD> command is not enabled then go to step S0800 |
| S0800  | Close Devices     | A0010  | Closing and Disabling Conditional Logic for DEV1                                                                                  |
|        |                   | A0020  | Closing and Disabling Conditional Logic for DEV2                                                                                  |
|        |                   | A0030  | Closing and Disabling Conditional Logic for DEV3                                                                                  |
|        |                   | A0040  | Closing and Disabling Conditional Logic for DEV4                                                                                  |
|        |                   | A0050  | Closing and Disabling Conditional Logic for DEV5                                                                                  |
|        |                   | A0060  | Closing and Disabling Conditional Logic for DEV6                                                                                  |
|        |                   | A0070  | Stopping Drain Timer                                                                                                              |
| T0900  | Disable HM and SM |        | If no pending actions in S0800 then go to step S0900                                                                              |
| S0900  | Disable HM        | A0010  | Disable HM 01 (Module Ownership Lost)                                                                                             |
| T1000  | Release Devices   |        | If no pending actions in S0900 then go to step S1000                                                                              |
| S1000  | Release Devices   | A0010  | Release Devices Message                                                                                                           |
|        |                   | A0020  | Release Private Devices                                                                                                           |
|        |                   | A0030  | Release Private Devices                                                                                                           |
|        |                   | A0040  | Release Private Devices                                                                                                           |
|        |                   | A0050  | Release Private Devices                                                                                                           |
|        |                   | A0060  | Release Private Devices                                                                                                           |
|        |                   | A0070  | Release Private Devices                                                                                                           |
| T9900  | Complete          |        | If no pending actions in S1000 then go to step S9900                                                                              |
| S9900  | Complete          | A0010  | Set Message: Process Fluid Command Completed                                                                                      |
|        |                   | A0020  | Self Release the EM & Reset First Pass                                                                                            |
| END    | Command Complete  |        | True                                                                                                                              |

Table 38: Process Fluid Sequence

## 5.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

| Parameter            | Description     | Type           | Units | Default Value/Reference |
|----------------------|-----------------|----------------|-------|-------------------------|
| Operating Parameters |                 |                |       |                         |
| OP001                | Enable 1-819IC1 | Name Set       | NA    | NO                      |
| OP002                | Enable 1-819IC2 | Name Set       | NA    | NO                      |
| OP003                | Plant           | Name Set       | NA    | MNA1                    |
| Tuning Parameters    |                 |                |       |                         |
| TP001                | Cooling Time    | Floating Point | min   | 50                      |
| TP002                | Icing Time      | Floating Point | min   | 90                      |
| TP003                | Melting Time    | Floating Point | min   | 30                      |
| TP004                | Drain Time      | Floating Point | min   | 30                      |
| TP005                | Wait            | Floating Point | min   | 5                       |
| TP006                | RSP TCV 1-821IC | Floating Point | °C    | 5                       |
| TP007                | Delay Timer     | Floating Point | sec   | 0                       |

Table 39: Operating and Tuning Parameters

## 5.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

| Parameter Name | Parameter Type     | Value/Reference           |                           |
|----------------|--------------------|---------------------------|---------------------------|
| MY_TAG         |                    | RF1EQ_819IC1              | RF1EQ_819IC2              |
| OPR_CMD_MASK   |                    | 3                         | 3                         |
| LSL_719A       | External reference | RF1_LSL1719A/DI1/DISC_ACT | RF1_LSL1719B/DI1/DISC_ACT |
| DY_REF_1       | Dynamic Reference  |                           |                           |
| DY_REF_2       | Dynamic Reference  |                           |                           |
| DY_REF_3       | Dynamic Reference  |                           |                           |
| DY_REF_4       | Dynamic Reference  |                           |                           |
| DY_REF_5       | Dynamic Reference  |                           |                           |
| RF1_STARTUP1   | Sting              | RF1EQ_STARTUP1            |                           |
| RF1EQ_819IC    | Sting              | RF1EQ_819IC2              | RF1EQ_819IC1              |
| PU_781IB       | External reference | RF1_PU1781IB              |                           |
| TT_778         | External reference | RF1_TT1778/PV             |                           |
| 878_32IC       | Sting              | RF1EQ_878_32IC            |                           |
| PT_719I        | External reference | RF1_PT1719IA/AI1/LO_ACT   |                           |
| CHILLER_ALM    | Boolean            | False                     |                           |
| CHILLER_READY  | Boolean            | False                     |                           |
| READY_MELT     | Boolean            | False                     |                           |

| Parameter Name | Parameter Type     | Value/Reference            |
|----------------|--------------------|----------------------------|
| HOLD           | External reference | RF1EQ_STARTUP1/OP006_VALUE |
| RSP_TT_778     | External reference | RF1EQ_878_32IC/TP001_VALUE |
| RF_IC_VAC_PROD | Sting              | RF1EQ_878_32IC             |

**Table 40: Configurable Parameters**

## 5.6 Named Set

Following Named set shall be used for configuration of EM.

| Name            | Description                                 |
|-----------------|---------------------------------------------|
| NS_IC_VAC_PROD1 | Refinery Area: Deodorising & Ice Condensing |

**Table 41: Named Set Details**

Details of states for NS\_IC\_VAC\_PROD1 named set are as follows,

| Named Set Value | Named Set Name |
|-----------------|----------------|
| 00              | Hold           |
| 01              | Fluid Process  |

**Table 42: Name Set State Details**

## 6 EQUIPMENT MODULE CLASS DESIGN – EM\_IC\_VAC\_PROD2 (1-878/32IC)

### 6.1 Functional Description

EM\_IC\_VAC\_PROD2 is a class based equipment module that communicates with EM\_IC\_VAC\_PROD1 sequence to complete fluid process that includes heating & melting operation. Refer FDS for the functional design requirements including faceplate and detail display. Initially Operator shall be provided only with Hold, Heating, Melting and Stop command. All devices used in EM shall be acquired in Heating command and released in Stop Command. Hold Request shall be enabled for all acquired devices at the end of Hold command.

### 6.2 Instances

The following Table contains the module instances of the equipment module class.

| Device Name | Module Class    | Instance Name  |
|-------------|-----------------|----------------|
| EM          | EM_IC_VAC_PROD2 | RF1EQ_878_32IC |
| DEV1        | _MRSS11_130_SE  | RF1_PU1781IB   |
| DEV2        | _V12_IP_130_SE  | RF1_V1778      |
| DEV3        | _CSD_130_SE     | RF1_TT1721IC   |

**Table 43: Instances**

### 6.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

#### 6.3.1 Command 00 – Hold

| Step  | Description                               | Action | Action Description |
|-------|-------------------------------------------|--------|--------------------|
| H0010 | Common Hold Start                         | A0010  | Store Fail Message |
|       |                                           | A0020  | Set Hold Request   |
| T1000 | Actions Complete or Time Out & Hold=True  |        | Go to step H1000   |
| T2000 | Actions Complete or Time Out & Hold=False |        | Go to step H2000   |
|       |                                           |        |                    |
| H1000 | Hold Command 1                            | A0010  | Set Message        |

| Step   | Description                  | Action | Action Description                                      |
|--------|------------------------------|--------|---------------------------------------------------------|
| T9900A | Actions Complete or Time Out |        | Go to step H9900A                                       |
|        |                              |        |                                                         |
| H2000  | General Reset                | A0010  | Stopping and Disabling Conditional Logic for DEV1       |
|        |                              | A0020  | Closing DEV2                                            |
|        |                              | A0030  | Set DEV3 Mode to Remote Out                             |
|        |                              | A0040  | Set DEV3 output to Closed and Disable Conditional Logic |
| T9900B | Actions Complete or Time Out |        | Go to step H9900B                                       |
|        |                              |        |                                                         |
| H9900A | Command Hold End             | A0010  | Completed                                               |
| END    | Command Complete             |        | True                                                    |
|        |                              |        |                                                         |
| H9900B | Release Devices              | A0010  | Release Devices Message                                 |
|        |                              | A0020  | Release Private Devices                                 |
|        |                              | A0030  | Release Private Devices                                 |
|        |                              | A0040  | Release Private Devices                                 |
|        |                              | A0050  | Reset Parameters                                        |
| END1   | Command Complete             |        | True                                                    |

Table 44: Hold Sequence

## 6.3.2 Command 01 – Heating

| Step   | Description        | Action | Action Description                                   |
|--------|--------------------|--------|------------------------------------------------------|
| S0000  | Start              | A0010  | Set Message: Starting Heating Command                |
| T0010  | First Pass = False |        | If First Pass = False then go to step S0010          |
| T0020A | First Pass = True  |        | If First Pass = True then go to step S0020           |
|        |                    |        |                                                      |
| S0010  | Initializations    | A0010  | First Pass Logic                                     |
|        |                    | A0020  | First Pass = True                                    |
|        |                    | A0030  | Set Task Pointer                                     |
| T0020B | Acquire Devices    |        | If no pending actions in S0010 then go to step S0020 |
|        |                    |        |                                                      |
| S0020  | Acquire Devices    | A0010  | Acquire Devices Message                              |
|        |                    | A0020  | Acquire Private Devices                              |
|        |                    | A0030  | Acquire Private Devices                              |

| Step   | Description      | Action | Action Description                                                       |
|--------|------------------|--------|--------------------------------------------------------------------------|
|        |                  | A0040  | Acquire Private Devices                                                  |
|        |                  | A0050  | Reset Hold Request                                                       |
| T0100  | Standby          |        | If Task Pointer = 1 then go to step S0100                                |
| T0200B | Heating          |        | If Task Pointer = 2 then go to step S0200                                |
|        |                  |        |                                                                          |
| S0100  | Standby          | A0010  | Stopping and Disabling Conditional Logic for DEV1                        |
|        |                  | A0020  | Closing DEV2                                                             |
|        |                  | A0030  | Set DEV3 Mode to Remote Out                                              |
|        |                  | A0040  | Set DEV3 output to Closed and Disable Conditional Logic                  |
| T0200A | Complete         |        | If 1-819IC1 or 1-819IC2 is not in step <Standby> then go to step S0200   |
|        |                  |        |                                                                          |
| S0200  | Heating          | A0010  | Enabling Conditional Logic for DEV1                                      |
|        |                  | A0020  | Opening DEV2                                                             |
|        |                  | A0030  | Set DEV3 Mode to Remote Cascade                                          |
|        |                  | A0040  | Setting DEV3 setpoint (TT_721IC Setpoint) and Enabling Conditional Logic |
|        |                  | A0050  | Set Task Pointer                                                         |
| T9900  | Complete         |        | If no pending actions in S0200 then go to step S9900                     |
|        |                  |        |                                                                          |
| S9900  | Complete         | A0010  | Set Message: Heating Command Complete                                    |
| END    | Command Complete |        | True                                                                     |

**Table 45: Heating Sequence**

### 6.3.3 Command 02 – Melting

| Step   | Description | Action | Action Description                                                                    |
|--------|-------------|--------|---------------------------------------------------------------------------------------|
| S0000  | Start       | A0010  | Set Message: Starting Melting Command                                                 |
| T0100  | Heating     |        | If Task Pointer = 2 then go to step S0100                                             |
| T0200B | Melting     |        | If Task Pointer = 3 then go to step S0200                                             |
|        |             |        |                                                                                       |
| S0100  | Heating     | A0010  | Set Message: Check RF1EQ_819IC1 or RF1EQ_819IC2 are in step Melting                   |
| T0200A | Melting     |        | If 1-819IC1 or 1-819IC2 is in step <Melting> and TT_778 > TP001 then go to step S0200 |
|        |             |        |                                                                                       |
| S0200  | Melting     | A0010  | Enabling Conditional Logic for DEV1                                                   |
|        |             | A0020  | Closing DEV2                                                                          |
|        |             | A0030  | Set DEV3 Mode to Remote Cascade                                                       |
|        |             | A0040  | Setting DEV3 setpoint (TT_721IC Setpoint) and Enabling Conditional Logic              |

| Step  | Description      | Action | Action Description                                                              |
|-------|------------------|--------|---------------------------------------------------------------------------------|
|       |                  | A0050  | Set Task Pointer                                                                |
| T9900 | Complete         |        | If 1-819IC1 and 1-819IC2 are not in Melting and after 10 sec then go step S9900 |
|       |                  |        |                                                                                 |
| S9900 | Complete         | A0010  | Set Message: Melting Command Complete                                           |
| END   | Command Complete |        | True                                                                            |

**Table 46: Melting Sequence**

## 6.3.4

## Command 03 – Stop

| Step  | Description      | Action | Action Description                                      |
|-------|------------------|--------|---------------------------------------------------------|
| S0000 | Start            | A0010  | Set Message: Starting Stop Command                      |
|       |                  | A0020  | Reset HOLD_REQ                                          |
| T0100 | Close Devices    |        | If no pending actions in S0000 then go to step S0100    |
|       |                  |        |                                                         |
| S0100 | Close Devices    | A0010  | Stopping and Disabling Conditional Logic for DEV1       |
|       |                  | A0020  | Closing DEV2                                            |
|       |                  | A0030  | Set DEV3 Mode to Remote Out                             |
|       |                  | A0040  | Set DEV3 output to Closed and Disable Conditional Logic |
| T0200 | Disable HM & SM  |        | If no pending actions in S0100 then go to step S0200    |
|       |                  |        |                                                         |
| S0200 | Disable HM       | A0010  | Disable HM 01 (Module Ownership Lost)                   |
| T0300 | Release Devices  |        | If no pending actions in S0200 then go to step S0300    |
|       |                  |        |                                                         |
| S0300 | Release Devices  | A0010  | Release Devices Message                                 |
|       |                  | A0020  | Release Private Devices                                 |
|       |                  | A0030  | Release Private Devices                                 |
|       |                  | A0040  | Release Private Devices                                 |
| T9900 | Complete         |        | If no pending actions in S0300 then go to step S9900    |
|       |                  |        |                                                         |
| S9900 | Complete         | A0010  | Set Message: Stop Command Complete                      |
|       |                  | A0020  | Self Release of EM                                      |
| END   | Command Complete |        | True                                                    |

**Table 47: Stop Sequence**

## 6.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

| Parameter            | Description    | Type           | Units | Default Value / Reference |
|----------------------|----------------|----------------|-------|---------------------------|
| Operating Parameters |                |                |       |                           |
| NA                   | NA             | NA             | NA    | NA                        |
| Tuning Parameters    |                |                |       |                           |
| TP001                | RSP for TT_778 | Floating Point | °C    | 60                        |
| TP002                | Delay Timer    | Floating Point | sec   | 10                        |

**Table 48: Operating and Tuning Parameters**

## 6.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

| Parameter Name | Parameter Type     | Value/Reference                          |
|----------------|--------------------|------------------------------------------|
| MY_TAG         |                    | RF1EQ_878_32IC                           |
| OPR_CMD_MASK   |                    | 15                                       |
| TT_778         | External reference | RF1_TT1778/PV                            |
| 1-819IC2       | External reference | RF1EQ_819IC2/COMMAND_0000<br>1/ACT_STEPS |
| 1-819IC1       | External reference | RF1EQ_819IC1/COMMAND_0000<br>1/ACT_STEPS |
| HOLD           | External reference | RF1EQ_STARTUP1/OP006_VAL<br>UE           |

**Table 49: Configurable Parameters**

## 6.6 Named Set

Following Named set shall be used for configuration of EM.

| Name            | Description                                 |
|-----------------|---------------------------------------------|
| NS_IC_VAC_PROD2 | Refinery Area: Deodorising & Ice Condensing |

**Table 50: Named Set Details**

Details of states for NS\_IC\_VAC\_PROD2 named set are as follows,

| Named Set Value | Named Set Name |
|-----------------|----------------|
| 00              | Hold           |

| Named Set Value | Named Set Name |
|-----------------|----------------|
| 01              | Heating        |
| 02              | Melting        |
| 03              | Stop           |

**Table 51: Name Set State Details**

## 7 EQUIPMENT MODULE CLASS DESIGN – EM\_RF\_STARTUP3 (STARTUP)

### 7.1 Functional Description

EM\_RF\_STARTUP3 is a class based equipment module that performs filling operation to produce the material and afterwards drains the material. EM\_RF\_STARTUP3 is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. Initially Operator shall be provided only with Hold and Aux On. Production, Drain and Aux off command shall be enabled during the execution of Aux On State. Hold Request shall be enabled for all acquired devices at the end of Hold command.

### 7.2 Instances

The following Table contains the module instances of the equipment module class.

| Device Name | Module Class      | Instance Name  |
|-------------|-------------------|----------------|
| EM          | EM_RF_OF_STARTUP3 | RF1EQ_STARTUP3 |
| DEV1        | _MRSS11_130_CUR   | RF1_PU1641X1   |
| DEV2        | _MRSS11_130_CUR   | RF1_PU1641X2   |
| DEV3        | EM_RF_OF_BLEACH2  | RF1EQ_T506     |

**Table 52: Instances**

### 7.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

#### 7.3.1 Command 00 – Hold

| Step   | Description                              | Action | Action Description |
|--------|------------------------------------------|--------|--------------------|
| H0000  | Command Hold Start                       | A0010  | Store Fail Message |
|        |                                          | A0020  | Set Hold Request   |
| T0010  | Actions Complete or Time Out             |        | Go to step H1000   |
|        |                                          |        |                    |
| H1000  | Hold Command 1                           | A0010  | Stopping DEV1      |
|        |                                          | A0020  | Stopping DEV2      |
| T9900A | Actions Complete or Time Out & Hold=True |        | Go to step H9900A  |

| Step   | Description                               | Action | Action Description      |
|--------|-------------------------------------------|--------|-------------------------|
| T9900B | Actions Complete or Time Out & Hold=False |        | Go to step H9900B       |
| H9900A | Command Hold End                          | A0010  | Completed               |
| END    | Command Complete                          |        | True                    |
| H9900B | Release Device                            | A0010  | Release Private Devices |
|        |                                           | A0020  | Release Private Devices |
|        |                                           | A0030  | Rest Parameters         |
| END1   | Command Complete                          |        | True                    |

**Table 53: Hold Sequence**

## 7.3.2 Command 01 – Aux On

| Step   | Description        | Action | Action Description                                         |
|--------|--------------------|--------|------------------------------------------------------------|
| S0000  | Start              | A0010  | Set Message: Starting Aux On Command                       |
| T0010  | First Pass = False |        | If First Pass = False then go to step S0010                |
| T0020A | First Pass = True  |        | If First Pass = True then go to step S0020                 |
| S0010  | Initializations    | A0010  | First Pass Logic                                           |
|        |                    | A0020  | First Pass = True                                          |
| T0020B | Acquire Devices    |        | If no pending actions in S0010 then go to step S0020       |
| S0020  | Acquire Devices    | A0010  | Acquire Devices Message                                    |
|        |                    | A0020  | Acquire Private Devices                                    |
|        |                    | A0030  | Acquire Private Devices                                    |
|        |                    | A0040  | Reset Hold Request                                         |
| T0100  | Start Aux Ckt      |        | If Task Pointer = 1 then go to step S0100                  |
| T0200B | Ready              |        | If Task Pointer = 2 then go to step S0200                  |
| S0100  | Start Aux Ckt      | A0010  | Set Message: Starting pump as per operator selection       |
|        |                    | A0020  | Starting DEV1 or DEV2                                      |
|        |                    | A0030  | Set Message: Checking PT_622< the set Vacuum set point     |
| T0200A | Ready              |        | If PT_622 < Vacuum set point (TP002) then go to step S0200 |

| Step  | Description      | Action | Action Description                                                        |
|-------|------------------|--------|---------------------------------------------------------------------------|
|       |                  |        |                                                                           |
| S0200 | Ready            | A0010  | Set Message: Starting pump as per operator selection and Set Task Pointer |
|       |                  | A0020  | Starting DEV1 or DEV2                                                     |
| T9900 | Complete         |        | If no pending actions in S0200 then go to step S9900                      |
|       |                  |        |                                                                           |
| S9900 | Complete         | A0010  | Set Message: Aux On Command Complete.                                     |
| END   | Command Complete |        | True                                                                      |

**Table 54: Aux On Sequence**

## 7.3.3 Command 02 – Production

| Step   | Description      | Action | Action Description                                                  |
|--------|------------------|--------|---------------------------------------------------------------------|
| S0000  | Start            | A0010  | Set Message: Starting Production Command                            |
| T0100  | Startup          |        | If Task Pointer = 2 then go to step S0100                           |
| T0200B | Production       |        | If Task Pointer = 3 then go to step S0200                           |
|        |                  |        |                                                                     |
| S0100  | Startup          | A0010  | Set Message: Checking EM_RF_OF_BLEACH2 is Not Owned.                |
|        |                  | A0020  | Acquire DEV3.                                                       |
|        |                  | A0030  | Provide Production command to EM_RF_OF_BLEACH2.                     |
|        |                  | A0040  | Set Message: Starting pump as per operator selection                |
|        |                  | A0050  | Starting DEV1 or DEV2                                               |
|        |                  | A0060  | Set Message: Waiting for Bleach5 to be in Production step           |
| T0200A | Production       |        | If 622/21 (EM_622_21) is in Step <Production> then go to step S0200 |
|        |                  |        |                                                                     |
| S0200  | Production       | A0010  | Set Message: Starting pump as per operator selection                |
|        |                  | A0020  | Starting DEV1 or DEV2                                               |
|        |                  | A0030  | Set Task Pointer                                                    |
| T9900  | Complete         |        | If no pending actions in S0200 then go to step S9900                |
|        |                  |        |                                                                     |
| S9900  | Complete         | A0010  | Set Message: Production Command Complete                            |
| END    | Actions Complete |        | True                                                                |

**Table 55: Production Sequence**

## 7.3.4 Command 03 – Drain

| Step  | Description      | Action | Action Description                                                                                                                                                   |
|-------|------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| S0000 | Start            | A0010  | Set Message: Starting Drain Command                                                                                                                                  |
| T0100 | Drain            |        | Go to step S0100                                                                                                                                                     |
|       |                  |        |                                                                                                                                                                      |
| S0100 | Drain            | A0010  | Set Message: Starting pump as per operator selection                                                                                                                 |
|       |                  | A0020  | Starting DEV1 or DEV2                                                                                                                                                |
|       |                  | A0030  | Reset Production Command                                                                                                                                             |
| T9900 | Complete         |        | If T506 (EM_T506), T503 (EM_T503), 622/21 (EM_622_21), 616A1-5 (EM_616A1/A2/A3/A4/A5), 682B (EM_682B) and 616S (EM_616S) are in step <Standby> then go to step S9900 |
|       |                  |        |                                                                                                                                                                      |
| S9900 | Complete         | A0010  | Set Message: Drain Command Complete                                                                                                                                  |
|       |                  | A0020  | Reset Drain Command                                                                                                                                                  |
| END   | Actions Complete |        | True                                                                                                                                                                 |

Table 56: Drain Sequence

## 7.3.5 Command 04 –Aux Off

| Step  | Description      | Action | Action Description                                   |
|-------|------------------|--------|------------------------------------------------------|
| S0000 | Start            | A0010  | Set Message: Starting Aux Off Command                |
| T0100 | Close Device     |        | If no pending actions in S0000 then go to step S0100 |
|       |                  |        |                                                      |
| S0100 | Close Device     | A0010  | Stopping DEV1 or DEV2                                |
|       |                  | A0020  | AUX ON command                                       |
| T0200 | Disable HM & SM  |        | If no pending actions in S0000 then go to step S0100 |
|       |                  |        |                                                      |
| S0200 | Disable HM       | A0010  | Disable HM 01 (Module Ownership Lost)                |
| T0300 | Release Device   |        | If no pending actions in S0200 then go to step S0300 |
|       |                  |        |                                                      |
| S0300 | Release Device   | A0010  | Release Private Devices                              |
|       |                  | A0020  | Release Private Devices                              |
| T9900 | Complete         |        | If no pending actions in S0300 then go to step S9900 |
|       |                  |        |                                                      |
| S9900 | Complete         | A0010  | Set Message: Aux Off Command Complete                |
| END   | Command Complete |        | True                                                 |



**Table 57: Aux Off Sequence**

## 7.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

| Parameter            | Description                  | Type           | Units | Default Value / Reference |
|----------------------|------------------------------|----------------|-------|---------------------------|
| Operating Parameters |                              |                |       |                           |
| OP001                | Pump Selection               | Named Set      | NA    | PUMPX1                    |
| OP002                | AUX                          | Named Set      | NA    | OFF                       |
| OP003                | RUN                          | Named Set      | NA    | OFF                       |
| OP004                | DRAIN                        | Named Set      | NA    | OFF                       |
| OP005                | HOLD                         | Named Set      | NA    | OFF                       |
| Tuning Parameters    |                              |                |       |                           |
| TP001                | Delay Timer                  | Floating Point | sec   | 5                         |
| TP002                | Vacuum Setpoint<br>PT_622/21 | Floating Point | mbar  | 150                       |

**Table 58: Operating and Tuning Parameters**

## 7.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

| Parameter Name | Parameter Type     | Value / Reference    |
|----------------|--------------------|----------------------|
| MY_TAG         |                    | RF1EQ_STARTUP3       |
| OPR_CMD_MASK   |                    | 31                   |
| PT_622         | External Reference | RF1_PT1622/PV        |
| EM_622_21      | String             | RF1EQ_622_21         |
| EM_T506        | External Reference | RF1EQ_T506/OWNER_ID  |
| EM_T503        | External Reference | RF1EQ_T503/OWNER_ID  |
| EM_616A1       | External Reference | RF1EQ_616A1/OWNER_ID |
| EM_616A2       | External Reference | RF1EQ_616A2/OWNER_ID |
| EM_616A3       | External Reference | RF1EQ_616A3/OWNER_ID |
| EM_616A4       | External Reference | RF1EQ_616A4/OWNER_ID |
| EM_616A5       | External Reference | RF1EQ_616A5/OWNER_ID |
| EM_682B        | External Reference | RF1EQ_682B/OWNER_ID  |
| EM_616S        | External Reference | RF1EQ_616S/OWNER_ID  |
| DY_REF_1       | Dynamic Reference  |                      |

**Table 59: Configurable Parameters**

## 7.6 Named Set

Following Named set shall be used for configuration of EM.

| Name           | Description                                     |
|----------------|-------------------------------------------------|
| NS_RF_STARTUP3 | Refinery Area: Filtration and Vacuum Production |

**Table 60: Named Set Details**

Details of states for NS\_RF\_STARTUP3 named set are as follows,

| Named Set Value | Named Set Name |
|-----------------|----------------|
| 00              | Hold           |
| 01              | Aux On         |
| 02              | Production     |
| 03              | Drain          |
| 04              | Aux Off        |

**Table 61: Name Set State Details**

## 8 EQUIPMENT MODULE CLASS DESIGN – EM\_RF\_OF\_BLEACH1 (1-630A OR 1-630B)

### 8.1 Functional Description

EM\_RF\_OF\_BLEACH1 is a class based equipment module that drives two different instances which performs filling operation using vacuum pump/fan. EM\_RF\_OF\_BLEACH1 is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. Initially Operator shall be provided only with Hold and Fill. Hold Request shall be enabled for all acquired devices at the end of Hold command. Commands which require operator intervention shall be configured as Operator Prompts wherever possible.

### 8.2 Instances

The following Table contains the module instances of the equipment module class.

| Device Name | Module Class     | Instance Name |             |
|-------------|------------------|---------------|-------------|
| EM          | EM_RF_OF_BLEACH1 | RF1EQ_630A    | RF1EQ_630B  |
| DEV1        | _MRSS11_130_SE   | RF1_BW1609A   | RF1_BW1609B |
| DEV2        | _DO_130_SE       | RF1_SV1609A   | RF1_SV1609B |
| DEV3        | _MRSS11_130_SE   | RF1_RL1610A   | RF1_RL1610B |

**Table 62: Instances**

### 8.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

#### 8.3.1 Command 00 – Hold

| Step  | Description                  | Action | Action Description      |
|-------|------------------------------|--------|-------------------------|
| H0010 | Start                        | A0010  | Store Fail Message      |
|       |                              | A0020  | Reset HOLD_REQ          |
|       |                              | A0030  | Save Previous Step Time |
| T1000 | Actions Complete or Time Out |        | Go to step H1000        |
|       |                              |        |                         |
| H1000 | Hold Command 1               | A0010  | Stopping DEV1           |
|       |                              | A0020  | Turning Off DEV2        |
|       |                              | A0030  | Stopping DEV3           |

| Step   | Description                                 | Action | Action Description      |
|--------|---------------------------------------------|--------|-------------------------|
| T9900A | Actions Complete or Time Out and Hold=True  |        | Go to step H9900B       |
| T9900B | Actions Complete or Time Out and Hold=False |        | Go to step H9900B       |
|        |                                             |        |                         |
| H9900A | Command Hold End                            | A0010  | Completed               |
| END    | Command Complete                            |        | True                    |
|        |                                             |        |                         |
| H9900B | Release Devices                             | A0010  | Release Devices Message |
|        |                                             | A0020  | Release Private Device  |
|        |                                             | A0030  | Release Private Device  |
|        |                                             | A0040  | Release Private Device  |
|        |                                             | A0050  | Reset Parameters        |
| END1   | Command Complete                            |        | True                    |

Table 63: Hold Sequence

## 8.3.2

## Command 01 – Start Fill

| Step   | Description        | Action | Action Description                                   |
|--------|--------------------|--------|------------------------------------------------------|
| S0000  | Start              | A0010  | Set Message: Starting Start Fill Command             |
| T0010  | First Pass = False |        | If First Pass = False then go to step S0010          |
| T0020A | First Pass = True  |        | If First Pass = True then go to step S0020           |
|        |                    |        |                                                      |
| S0010  | Initializations    | A0010  | First Pass Logic                                     |
|        |                    | A0020  | First Pass = True                                    |
|        |                    | A0030  | Set Task Pointer                                     |
| T0020B | Actions Complete   |        | If no pending actions in S0010 then go to step S0020 |
|        |                    |        |                                                      |
| S0020  | Acquire Device     | A0010  | Acquire Devices Message                              |
|        |                    | A0020  | Acquire Private Device                               |
|        |                    | A0030  | Acquire Private Device                               |
|        |                    | A0040  | Acquire Private Device                               |
|        |                    | A0060  | Reset Hold Request                                   |
| T0100  | Standby            |        | If Task Pointer = 1 then go to step S0100            |

| Step   | Description         | Action | Action Description                                                               |
|--------|---------------------|--------|----------------------------------------------------------------------------------|
| T0200B | Start Vac. Pump/Fan |        | If Task Pointer = 2 then go to step S0200                                        |
| TP300B | Filling             |        | If Task Pointer = 3 then go to step S0300                                        |
|        |                     |        |                                                                                  |
| S0100  | Standby             | A0010  | Stopping DEV1                                                                    |
|        |                     | A0020  | Turning Off DEV2                                                                 |
|        |                     | A0030  | Stopping DEV3                                                                    |
| T0200A | Standby             |        | If LSL_630 is in alarm then go step S0200                                        |
|        |                     |        |                                                                                  |
| S0200  | Start Vac. Pump/Fan | A0010  | Starting DEV1                                                                    |
|        |                     | A0020  | Turning On DEV2                                                                  |
|        |                     | A0030  | Stopping DEV3                                                                    |
|        |                     | A0040  | Set Start Fan time                                                               |
|        |                     | A0050  | Start Fan Timer                                                                  |
|        |                     | A0060  | Start Fan time Message                                                           |
| T0300B | Filling             |        | If DEV1 is running and Start Fan Timer (TP002) elapsed then go step S0300        |
|        |                     |        |                                                                                  |
| S0300  | Filling             | A0010  | Starting DEV1                                                                    |
|        |                     | A0020  | Turning On DEV2                                                                  |
|        |                     | A0030  | Starting DEV3                                                                    |
| T9900  | Complete            |        | If Stop button (OP001) is triggered or LSH_630 is in alarm then go to step S9900 |
|        |                     |        |                                                                                  |
| S9900  | Complete            | A0010  | Set Message: Start Fill Command Complete                                         |
|        |                     | A0020  | Give A_CMD to next Command on LSH_630 is in alarm                                |
| END    | Command Complete    |        | True                                                                             |

Table 64: Fill Sequence

## 8.3.3 Command 01 – Stop Fill

| Step  | Description        | Action | Action Description                        |
|-------|--------------------|--------|-------------------------------------------|
| S0000 | Start              | A0010  | Set Message: Starting Stop Fill Command   |
|       |                    | A0020  | Check if LSH_630A is active               |
| T0100 | Stop Vac. Pump/Fan |        | If Task Pointer = 3 then go to step S0100 |
|       |                    |        |                                           |
| S0100 | Stop Vac. Pump/Fan | A0010  | Starting DEV1                             |
|       |                    | A0020  | Turning On DEV2                           |
|       |                    | A0030  | Stopping DEV3                             |
|       |                    | A0040  | Set Stop Fan time                         |
|       |                    | A0050  | Stop Fan Timer                            |

| Step  | Description                                   | Action | Action Description                                      |
|-------|-----------------------------------------------|--------|---------------------------------------------------------|
|       |                                               | A0060  | Stop Fan time Message                                   |
| T0200 | Step Time > TP003_VALUE (Stop Fan Time (sec)) |        | If Stop Fan Timer elapsed (TP003) then go to step S0200 |
|       |                                               |        |                                                         |
| S0200 | Close Devices                                 | A0010  | Stopping DEV1                                           |
|       |                                               | A0020  | Turning Off DEV2                                        |
|       |                                               | A0030  | Stopping DEV3                                           |
|       |                                               | A0040  | Reset Filling Selection Button (OP001)                  |
| T0300 | Disable HM and SM                             |        | If no pending actions S0200 then go to step S0300       |
|       |                                               |        |                                                         |
| S0300 | Disable HM                                    | A0010  | Disable HM 01 (Module Ownership Lost)                   |
| T0400 | Release Devices                               |        | If no pending actions S0300 then go to step S0400       |
|       |                                               |        |                                                         |
| S0400 | Release Devices                               | A0010  | Release Devices Message                                 |
|       |                                               | A0020  | Release Private Device                                  |
|       |                                               | A0030  | Release Private Device                                  |
|       |                                               | A0040  | Release Private Device                                  |
| T9900 | Complete                                      |        | If no pending actions S0400 then go to step S9900       |
|       |                                               |        |                                                         |
| S9900 | Complete                                      | A0010  | Set Message: Stop Fill Command Complete                 |
|       |                                               | A0020  | Self release the EM                                     |
| END   | Command Complete                              |        | True                                                    |

Table 65: Fill Sequence

## 8.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

| Parameter            | Description    | Type           | Units | Default Value/Reference |
|----------------------|----------------|----------------|-------|-------------------------|
| Operating Parameters |                |                |       |                         |
| OP001                | Start Fill     | Boolean        | NA    | False                   |
| OP002                | Stop Fill      | Boolean        | NA    | False                   |
| Tuning Parameters    |                |                |       |                         |
| TP001                | Delay Timer    | Floating Point | sec   | 5                       |
| TP002                | Start Fan Time | Floating Point | sec   | 600                     |

| Parameter | Description   | Type           | Units | Default Value/Reference |
|-----------|---------------|----------------|-------|-------------------------|
| TP003     | Stop Fan Time | Floating Point | sec   | 600                     |

**Table 66: Operating and Tuning Parameters**

## 8.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

| Parameter Name | Parameter Type     | Value/Reference            |                           |
|----------------|--------------------|----------------------------|---------------------------|
| MY_TAG         |                    | RF1EQ_630A                 | RF1EQ_630B                |
| OPR_CMD_MASK   |                    | 7                          | 7                         |
| LSH_630        | External Reference | RF1_LSH1630A/DI1/DISC_ACT  | RF1_LSH1630B/DI1/DISC_ACT |
| LSL_630        | External Reference | RF1_LSL1630A/DI1/DISC_ACT  | RF1_LSL1630B/DI1/DISC_ACT |
| HOLD           | External Reference | RF1EQ_STARTUP3/OP005_VALUE |                           |
| STEP_TIME      | Floating Point     | 0                          |                           |
| COMPLETED_TIME | Floating Point     | 0                          |                           |
| RESET_COMP_TM  | Boolean            | False                      |                           |

**Table 67: Configurable Parameters**

## 8.6 Named Set

Following Named set shall be used for configuration of EM.

| Name             | Description                                     |
|------------------|-------------------------------------------------|
| NS_RF_OF_BLEACH1 | Refinery Area: Filtration and Vacuum Production |

**Table 68: Named Set Details**

Details of states for NS\_RF\_OF\_BLEACH1 named set are as follows,

| Named Set Value | Named Set Name |
|-----------------|----------------|
| 00              | Hold           |
| 01              | Start Fill     |
| 02              | Stop Fill      |

**Table 69: Name Set State Details**

## 9 EQUIPMENT MODULE CLASS DESIGN – EM\_RF\_OF\_BLEACH2 (1-T506)

### 9.1 Functional Description

EM\_RF\_OF\_BLEACH2 is a class based equipment module that performs filling operation to produce the material and afterwards drains the material. EM\_RF\_OF\_BLEACH2 is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. Initially Operator shall be provided only with Hold and Production. Hold Request shall be enabled for all acquired devices at the end of Hold command.

### 9.2 Instances

The following Table contains the module instances of the equipment module class.

| Device Name | Module Class     | Instance Name |
|-------------|------------------|---------------|
| EM          | EM_RF_OF_BLEACH2 | RF1EQ_T506    |
| DEV1        | _MRSS11_130_SE   | RF1_PU1006    |
| DEV2        | _MRSS11_130_CUR  | RF1_MX1004AC  |
| DEV3        | _MVS11_IL_130_SO | RF1_PU1044CA  |
| DEV4        | _MVS11_IL_130_SO | RF1_PU1034PA  |
| DEV5        | _V12_IP_130_SE   | RF1_V1006D    |
| DEV6        | _V12_IP_130_SE   | RF1_V1024F    |
| DEV7        | _V12_IP_130_SE   | RF1_V1024B    |
| DEV8        | _V12_IP_130_SE   | RF1_V1006B    |
| DEV9        | _CSD_130_SE      | RF1_FT1011    |
| DEV10       | _CSD_130_SE      | RF1_TT1021B   |
| DEV11       | _CSD_130_SE      | RF1_LT1006    |

**Table 70: Instances**

### 9.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

#### 9.3.1 Command 00 – Hold

| Step  | Description        | Action | Action Description |
|-------|--------------------|--------|--------------------|
| H0010 | Command Hold Start | A0010  | Store Fail Message |
|       |                    | A0020  | Reset Hold Request |

| Step   | Description                                 | Action | Action Description                                       |
|--------|---------------------------------------------|--------|----------------------------------------------------------|
| T1000  | Actions Complete or Time Out                |        | Go to step H1000                                         |
| H1000  | Hold Command 1                              | A0010  | Stopping and Disabling Conditional Logic for DEV1        |
|        |                                             | A0020  | Stopping and Disabling Conditional Logic for DEV2        |
|        |                                             | A0030  | Stopping and Disabling Conditional Logic for DEV3        |
|        |                                             | A0040  | Stopping and Disabling Conditional Logic for DEV4        |
|        |                                             | A0050  | Closing DEV5                                             |
|        |                                             | A0060  | Closing and Disabling Conditional Logic for DEV6         |
|        |                                             | A0070  | Closing DEV7                                             |
|        |                                             | A0080  | Closing DEV8                                             |
|        |                                             | A0090  | Set DEV9 Mode to Remote Out                              |
|        |                                             | A0100  | Set DEV9 output to Closed and Disable Conditional Logic  |
|        |                                             | A0110  | Set DEV10 Mode to Remote Out                             |
|        |                                             | A0120  | Set DEV10 output to Closed and Disable Conditional Logic |
|        |                                             | A0130  | Set DEV11 Mode to Remote Out                             |
|        |                                             | A0140  | Set DEV11 output to Closed and Disable Conditional Logic |
| T9900A | Actions Complete or Time Out and Hold=True  |        | Go to step H9900A                                        |
| T9900B | Actions Complete or Time Out and Hold=False |        | Go to step H9900B                                        |
| H9900A | Commom Hold End                             | A0010  | Completed                                                |
| END    | Command Complete                            |        | True                                                     |
| H9900B | Release Devices                             | A0010  | Release Devices Message                                  |
|        |                                             | A0020  | Release Private Device                                   |
|        |                                             | A0030  | Release Private Device                                   |
|        |                                             | A0040  | Release Private Device                                   |
|        |                                             | A0050  | Release Private Device                                   |

| Step | Description      | Action | Action Description     |
|------|------------------|--------|------------------------|
|      |                  | A0060  | Release Private Device |
|      |                  | A0070  | Release Private Device |
|      |                  | A0080  | Release Private Device |
|      |                  | A0090  | Release Private Device |
|      |                  | A0100  | Release Private Device |
|      |                  | A0110  | Release Private Device |
|      |                  | A0120  | Release Private Device |
|      |                  | A0130  | Reset Parameters       |
| END1 | Command Complete |        | True                   |

**Table 71: Hold Sequence**

## 9.3.2 Command 01 – Production

| Step   | Description        | Action | Action Description                                   |
|--------|--------------------|--------|------------------------------------------------------|
| S0000  | Start              | A0010  | Set Message: Starting Production Command             |
| T0010  | First Pass = False |        | If First Pass = False then go to step S0010          |
| T0020A | First Pass = True  |        | If First Pass = True then go to step S0020           |
| S0010  | Initializations    | A0010  | First Pass Logic                                     |
|        |                    | A0020  | First Pass = True                                    |
|        |                    | A0030  | Set Task Pointer                                     |
| T0020B | Actions Complete   |        | If no pending actions in S0010 then go to step S0020 |
| S0020  | Acquire Devices    | A0010  | Acquire Devices Message                              |
|        |                    | A0020  | Acquire Private Device                               |
|        |                    | A0030  | Acquire Private Device                               |
|        |                    | A0040  | Acquire Private Device                               |
|        |                    | A0050  | Acquire Private Device                               |
|        |                    | A0060  | Acquire Private Device                               |
|        |                    | A0070  | Acquire Private Device                               |
|        |                    | A0080  | Acquire Private Device                               |
|        |                    | A0090  | Acquire Private Device                               |
|        |                    | A0100  | Acquire Private Device                               |
|        |                    | A0110  | Acquire Private Device                               |
|        |                    | A0120  | Acquire Private Device                               |
|        |                    | A0130  | Reset Hold Request                                   |
| T0100  | Standby            |        | If Task Pointer = 1 then go to step S0100            |
| T0200B | Fill               |        | If Task Pointer = 2 then go to step S0200            |
| T0300B | Production         |        | If Task Pointer = 3 then go to step S0300            |



| Step   | Description | Action | Action Description                                                                       |
|--------|-------------|--------|------------------------------------------------------------------------------------------|
| T0400B | Drain       |        | If Task Pointer = 4 then go to step S0400                                                |
| T0500B | Drain Wait  |        | If Task Pointer = 5 then go to step S0500                                                |
|        |             |        |                                                                                          |
| S0100  | Standby     | A0010  | Stopping and Disabling Conditional Logic for DEV1                                        |
|        |             | A0020  | Stopping and Disabling Conditional Logic for DEV2                                        |
|        |             | A0030  | Stopping and Disabling Conditional Logic for DEV3                                        |
|        |             | A0040  | Stopping and Disabling Conditional Logic for DEV4                                        |
|        |             | A0050  | Closing DEV5                                                                             |
|        |             | A0060  | Closing and Disabling Conditional Logic for DEV6                                         |
|        |             | A0070  | Closing DEV7                                                                             |
|        |             | A0080  | Closing DEV8                                                                             |
|        |             | A0090  | Set DEV9 Mode to Remote Out                                                              |
|        |             | A0100  | Set DEV9 output to Closed and Disable Conditional Logic                                  |
|        |             | A0110  | Set DEV10 Mode to Remote Out                                                             |
|        |             | A0120  | Set DEV10 output to Closed and Disable Conditional Logic                                 |
|        |             | A0130  | Set DEV11 Mode to Remote Out                                                             |
|        |             | A0140  | Set DEV11 output to Closed and Disable Conditional Logic                                 |
| T0200A | Fill        |        | If Start up (EM_STARTUP3) is in step in <Start-Up> or <Production> then go to step S0200 |
|        |             |        |                                                                                          |
| S0200  | Fill        | A0010  | Stopping and Disabling Conditional Logic for DEV1                                        |
|        |             | A0020  | Stopping and Disabling Conditional Logic for DEV2                                        |
|        |             | A0030  | Stopping and Disabling Conditional Logic for DEV3                                        |
|        |             | A0040  | Stopping and Disabling Conditional Logic for DEV4                                        |
|        |             | A0050  | Closing DEV5                                                                             |
|        |             | A0060  | Enabling Conditional Logic for DEV6                                                      |
|        |             | A0070  | Closing DEV7                                                                             |
|        |             | A0080  | Closing DEV8                                                                             |
|        |             | A0090  | Set DEV9 Mode to Remote Cascade                                                          |
|        |             | A0100  | Setting DEV9 setpoint (FT_011 Setpoint) and Enabling Conditional Logic                   |
|        |             | A0110  | Set DEV10 Mode to Remote Cascade                                                         |
|        |             | A0120  | Set DEV10 Setpoint (TT_021B Setpoint) and Enable Conditional Logic                       |

| Step   | Description | Action | Action Description                                                                                     |
|--------|-------------|--------|--------------------------------------------------------------------------------------------------------|
|        |             | A0130  | Set DEV11 Mode to Remote Out                                                                           |
|        |             | A0140  | Set DEV11 output to Closed and Disable Conditional Logic                                               |
|        |             | A0150  | Set Message: Waiting for EM_RF_OF_BLEACH3 to be in Standby                                             |
| T0300A | Production  |        | If DEV11 $\geq$ LT_006 level reached (TP003) and EM_BLEACH3 is in step <Standby> then go to step S0300 |
| S0300  | Production  | A0010  | Enabling Conditional Logic for DEV1 (TP01)                                                             |
|        |             | A0020  | Enabling Conditional Logic for DEV2                                                                    |
|        |             | A0030  | Enabling Conditional Logic for DEV3                                                                    |
|        |             | A0040  | Enabling Conditional Logic for DEV4                                                                    |
|        |             | A0050  | Opening DEV5                                                                                           |
|        |             | A0060  | Enabling Conditional Logic for DEV6                                                                    |
|        |             | A0070  | Closing DEV7                                                                                           |
|        |             | A0080  | Closing DEV8                                                                                           |
|        |             | A0090  | Set DEV9 Mode to Remote Cascade                                                                        |
|        |             | A0100  | Setting DEV9 setpoint (FT_011 Setpoint) and Enabling Conditional Logic                                 |
|        |             | A0110  | Set DEV10 Mode to Remote Cascade                                                                       |
|        |             | A0120  | Set DEV10 Setpoint (TT_021B Setpoint) and Enable Conditional Logic                                     |
|        |             | A0130  | Set DEV11 Mode to Remote Cascade                                                                       |
|        |             | A0140  | Setting DEV11 setpoint (LT_006 Setpoint) and Enabling Conditional Logic                                |
|        |             | A0150  | Acquire EM_RF_OF_BLEACH3                                                                               |
| T0400A | Drain       | A0170  | Starting production command for EM_RF_OF_BLEACH3                                                       |
|        |             | A0180  | Set Message: Waiting for Startup3 to be in Step Drain                                                  |
|        |             |        | If Start up (EM_STARTUP3) is in step <Drain> then go to step S0400                                     |
| S0400  | Drain       | A0010  | Enabling Conditional Logic for DEV1 (TP02)                                                             |
|        |             | A0020  | Enabling Conditional Logic for DEV2                                                                    |
|        |             | A0030  | Enabling Conditional Logic for DEV3                                                                    |
|        |             | A0040  | Enabling Conditional Logic for DEV4                                                                    |
|        |             | A0050  | Opening DEV5                                                                                           |
|        |             | A0060  | Closing and Disabling Conditional Logic for DEV6                                                       |
|        |             | A0070  | Opening DEV7                                                                                           |
|        |             | A0080  | Closing DEV8                                                                                           |
|        |             | A0090  | Set DEV9 Mode to Remote Out                                                                            |

| Step   | Description   | Action | Action Description                                                        |
|--------|---------------|--------|---------------------------------------------------------------------------|
|        |               | A0100  | Set DEV9 output to Open and Disable Conditional Logic                     |
|        |               | A0110  | Set DEV10 Mode to Remote Out                                              |
|        |               | A0120  | Set DEV10 output to Closed and Disable Conditional Logic                  |
|        |               | A0130  | Set DEV11 Mode to Remote Out                                              |
|        |               | A0140  | Set DEV11 output to Open and Enable Conditional Logic                     |
| T0500A | Drain Wait    |        | If LAL_T506 is in alarm and LT_T506 is in VLL alarm then go to step S0500 |
|        |               |        |                                                                           |
| S0500  | Drain Wait    | A0010  | Stopping and Disabling Conditional Logic for DEV1                         |
|        |               | A0020  | Stopping and Disabling Conditional Logic for DEV2                         |
|        |               | A0030  | Stopping and Disabling Conditional Logic for DEV3                         |
|        |               | A0040  | Stopping and Disabling Conditional Logic for DEV4                         |
|        |               | A0050  | Closing DEV5                                                              |
|        |               | A0060  | Closing and Disabling Conditional Logic for DEV6                          |
|        |               | A0070  | Closing DEV7                                                              |
|        |               | A0080  | Opening DEV8                                                              |
|        |               | A0090  | Set DEV9 Mode to Remote Out                                               |
|        |               | A0100  | Set DEV9 output to Closed and Disable Conditional Logic                   |
|        |               | A0110  | Set DEV10 Mode to Remote Out                                              |
|        |               | A0120  | Set DEV10 output to Closed and Disable Conditional Logic                  |
|        |               | A0130  | Set DEV11 Mode to Remote Out                                              |
|        |               | A0140  | Set DEV11 output to Open and Disable Conditional Logic                    |
|        |               | A0150  | Set Message: Waiting for EM_RF_OF_BLEACH3 to be in Step Drain             |
| T0600  | Close Devices |        | If EM_BLEACH3 is in step <Drain> then go to step S0600                    |
|        |               |        |                                                                           |
| S0600  | Close Devices | A0010  | Stopping and Disabling Conditional Logic for DEV1                         |
|        |               | A0020  | Stopping and Disabling Conditional Logic for DEV2                         |
|        |               | A0030  | Stopping and Disabling Conditional Logic for DEV3                         |
|        |               | A0040  | Stopping and Disabling Conditional Logic for DEV4                         |
|        |               | A0050  | Closing DEV5                                                              |

| Step  | Description       | Action | Action Description                                       |
|-------|-------------------|--------|----------------------------------------------------------|
|       |                   | A0060  | Closing and Disabling Conditional Logic for DEV6         |
|       |                   | A0070  | Closing DEV7                                             |
|       |                   | A0080  | Closing DEV8                                             |
|       |                   | A0090  | Set DEV9 Mode to Remote Out                              |
|       |                   | A0100  | Set DEV9 output to Closed and Disable Conditional Logic  |
|       |                   | A0110  | Set DEV10 Mode to Remote Out                             |
|       |                   | A0120  | Set DEV10 output to Closed and Disable Conditional Logic |
|       |                   | A0130  | Set DEV11 Mode to Remote Out                             |
|       |                   | A0140  | Set DEV11 output to Closed and Disable Conditional Logic |
|       |                   | A0150  | Reset Acid Dosing Selection Button                       |
| T0700 | Disable HM and SM |        | If no pending actions in S0600 then go to step S0700     |
|       |                   |        |                                                          |
| S0700 | Disable HM        | A0010  | Disable HM 01 (Module Ownership Lost)                    |
| T0800 | Release Devices   |        | If no pending actions in S0700 then go to step S0800     |
|       |                   |        |                                                          |
| S0800 | Release Devices   | A0010  | Release Devices Message                                  |
|       |                   | A0020  | Release Private Device                                   |
|       |                   | A0030  | Release Private Device                                   |
|       |                   | A0040  | Release Private Device                                   |
|       |                   | A0050  | Release Private Device                                   |
|       |                   | A0060  | Release Private Device                                   |
|       |                   | A0070  | Release Private Device                                   |
|       |                   | A0080  | Release Private Device                                   |
|       |                   | A0090  | Release Private Device                                   |
|       |                   | A0100  | Release Private Device                                   |
|       |                   | A0110  | Release Private Device                                   |
|       |                   | A0120  | Release Private Device                                   |
| T9900 | Complete          |        | If no pending actions in S0800 then go to step S9900     |
|       |                   |        |                                                          |
| S9900 | Complete          | A0010  | Set Message: Production Command Complete                 |
|       |                   | A0020  | Self release the EM                                      |
| END   | Command Complete  |        | True                                                     |

Table 72: Production Sequence

## 9.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

| Parameter            | Description                  | Type           | Units | Default Value/Reference |
|----------------------|------------------------------|----------------|-------|-------------------------|
| Operating Parameters |                              |                |       |                         |
| OP001                | Citric Acid Selection        | Boolean        | NA    | False                   |
| OP002                | Phosphoric Acid Selection    | Boolean        | NA    | False                   |
| Tuning Parameters    |                              |                |       |                         |
| TP001                | Delay Timer                  | Floating Point | sec   | 5                       |
| TP002                | Remote set point for FT_011  | Floating Point | Kg/h  | 80                      |
| TP003                | Remote set point for LT_006  | Floating Point | %     | 80                      |
| TP004                | Remote set point for TT_021B | Floating Point | °C    | 0                       |

**Table 73: Operating and Tuning Parameters**

## 9.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

| Parameter Name | Parameter Type     | Value/Reference                       |
|----------------|--------------------|---------------------------------------|
| MY_TAG         |                    | RF1EQ_T506                            |
| OPR_CMD_MASK   |                    | 3                                     |
| EM_BLEACH3     | String             | RF1EQ_T503                            |
| EM_STARTUP3    | String             | RF1EQ_STARTUP3/COMMAND_0002/ACT_STEPS |
| DY_REF_1       | Dynamic Reference  |                                       |
| DY_REF_2       | Dynamic Reference  |                                       |
| LAL_T506       | External Reference | RF1_LSL1006/DI1/DISC_ACT              |
| LAH_622_21     | External Reference | RF1_LSH1622/DI1/DISC_ACT              |
| LAH_T506       | External Reference | RF1_LSH1006/DI1/DISC_ACT              |
| LT_T534CA      | External Reference | RF1_LT1044CA/AI1/LO_LO_ACT            |
| LT_T534PA      | External Reference | RF1_LT1034PA/AI1/LO_LO_ACT            |
| LT_T506        | External Reference | RF1_LT1006/DI1/DISC_ACT               |
| BLEACH3        | External Reference | RF1EQ_T503/OWNER_ID                   |

| Parameter Name | Parameter Type     | Value/Reference                |
|----------------|--------------------|--------------------------------|
| HOLD           | External Reference | RF1EQ_STARTUP3/OP005_VALU<br>E |
| STARTUP3       | External Reference | RF1EQ_STARTUP3                 |

**Table 74: Configurable Parameters**

## 9.6

### Named Set

Following Named set shall be used for configuration of EM.

| Name             | Description                                     |
|------------------|-------------------------------------------------|
| NS_RF_OF_BLEACH2 | Refinery Area: Filtration and Vacuum Production |

**Table 75: Named Set Details**

Details of states for NS\_RF\_OF\_BLEACH2 named set are as follows,

| Named Set Value | Named Set Name |
|-----------------|----------------|
| 00              | Hold           |
| 01              | Production     |

**Table 76: Name Set State Details**

## 10 EQUIPMENT MODULE CLASS DESIGN – EM\_RF\_OF\_BLEACH3 (1-T503)

### 10.1 Functional Description

EM\_RF\_OF\_BLEACH3 is a class based equipment module that performs filling operation to produce the material and afterwards drains the material. EM\_RF\_OF\_BLEACH3 is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. Initially Operator shall be provided only with Hold and Production. Hold Request shall be enabled for all acquired devices at the end of Hold command.

### 10.2 Instances

The following Table contains the module instances of the equipment module class.

| Device Name | Module Class     | Instance Name |
|-------------|------------------|---------------|
| EM          | EM_RF_OF_BLEACH3 | RF1EQ_T503    |
| DEV1        | _MRSS11_130_SE   | RF1_AG1003    |
| DEV2        | _V12_IP_130_SE   | RF1_V1003F    |
| DEV3        | _V12_IP_130_SE   | RF1_V1003BY   |
| DEV4        | EM_RF_OF_BLEACH6 | RF1EQ_635     |

Table 77: Instances

### 10.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

#### 10.3.1 Command 00 – Hold

| Step   | Description                  | Action | Action Description                                |
|--------|------------------------------|--------|---------------------------------------------------|
| H0010  | Command Hold Start           | A0010  | Store Fail Message                                |
|        |                              | A0020  | Reset Hold Request                                |
|        |                              | A0030  | Save Previous Step Time                           |
| T1000  | Actions Complete or Time Out |        | Go to step H1000                                  |
|        |                              |        |                                                   |
| H1000  | Hold Command 1               | A0010  | Stopping and Disabling Conditional Logic for DEV1 |
|        |                              | A0020  | Closing DEV2                                      |
|        |                              | A0030  | Closing DEV3                                      |
| T9900A | Actions Complete or          |        | Go to step H9900A                                 |

| Step   | Description                                 | Action | Action Description      |
|--------|---------------------------------------------|--------|-------------------------|
|        | Time Out and Hold=True                      |        |                         |
| T9900B | Actions Complete or Time Out and Hold=False |        | Go to step H9900B       |
|        |                                             |        |                         |
| H9900A | Command Hold End                            | A0010  | Completed               |
| END    | Command Complete                            |        | True                    |
|        |                                             |        |                         |
| H9900B | Release Devices                             | A0010  | Release Devices Message |
|        |                                             | A0020  | Release Private Device  |
|        |                                             | A0030  | Release Private Device  |
|        |                                             | A0040  | Release Private Device  |
|        |                                             | A0050  | Reset Parameters        |
| END1   | Command Complete                            |        | True                    |

**Table 78: Hold Sequence**

## 10.3.2

## Command 01 – Production

| Step   | Description        | Action | Action Description                                   |
|--------|--------------------|--------|------------------------------------------------------|
| S0000  | Start              | A0010  | Set Message: Starting Production Command             |
| T0010  | First Pass = False |        | If First Pass = False then go to step S0010          |
| T0020A | First Pass = True  |        | If First Pass = True then go to step S0020           |
|        |                    |        |                                                      |
| S0010  | Initializations    | A0010  | First Pass Logic                                     |
|        |                    | A0020  | First Pass = True                                    |
|        |                    | A0030  | Set Task Pointer                                     |
| T0020B | Actions Complete   |        | If no pending actions in S0010 then go to step S0020 |
|        |                    |        |                                                      |
| S0020  | Acquire Devices    | A0010  | Acquire Devices Message                              |
|        |                    | A0020  | Acquire Private Device                               |
|        |                    | A0030  | Acquire Private Device                               |
|        |                    | A0040  | Acquire Private Device                               |
|        |                    | A0050  | Reset Hold Request                                   |
| T0100  | Standby            |        | If Task Pointer = 1 then go to step S0100            |
| T0200B | Fill               |        | If Task Pointer = 2 then go to step S0400            |
| T0300B | Production         |        | If Task Pointer = 3 then go to step S0500            |
| T0400B | Drain              |        | If Task Pointer = 4 then go to step S0600            |



| Step   | Description | Action | Action Description                                                                                     |
|--------|-------------|--------|--------------------------------------------------------------------------------------------------------|
| T0500B | Drain Wait  |        | If Task Pointer = 5 then go to step S0700                                                              |
|        |             |        |                                                                                                        |
| S0100  | Standby     | A0010  | Stopping and Disabling Conditional Logic for DEV1                                                      |
|        |             | A0020  | Closing DEV2                                                                                           |
|        |             | A0030  | Closing DEV3                                                                                           |
|        |             | A0040  | Set Message: Waiting for EM_RF_OF_BLEACH2 to be in Step Production                                     |
| T0200A | Fill        |        | If EM_BLEACH2 is in step <Production> then go to step S0200                                            |
|        |             |        |                                                                                                        |
| S0200  | Fill        | A0010  | Enabling Conditional Logic for DEV1 (TP01)                                                             |
|        |             | A0020  | Opening DEV2                                                                                           |
|        |             | A0030  | Closing DEV3                                                                                           |
|        |             | A0040  | Set Message                                                                                            |
| T0300A | Production  |        | If LSL_T503 is not in alarm then go to step S0300                                                      |
|        |             |        |                                                                                                        |
| S0300  | Production  | A0010  | Enabling Conditional Logic for DEV1 (TP02)                                                             |
|        |             | A0020  | Opening DEV2                                                                                           |
|        |             | A0030  | Closing DEV3                                                                                           |
|        |             | A0040  | Acquire EM_RF_OF_BLEACH5                                                                               |
|        |             | A0060  | Starting Production command for EM_RF_OF_BLEACH5                                                       |
|        |             | A0070  | Acquire DEV6                                                                                           |
|        |             | A0075  | Start DEV6                                                                                             |
|        |             | A0080  | Set Blow time                                                                                          |
|        |             | A0090  | Start Blow Timer                                                                                       |
|        |             | A0100  | Blow time Message                                                                                      |
| T0400A | Drain       |        | If EM_T506 is in step <Drain Wait> and XV 1-T506-B blowing timer (TP001) elapsed then go to step S0400 |
|        |             |        |                                                                                                        |
| S0400  | Drain       | A0010  | Enabling Conditional Logic for DEV1 (TP02)                                                             |
|        |             | A0020  | Opening DEV2                                                                                           |
|        |             | A0030  | Opening DEV3                                                                                           |
|        |             | A0040  | Set Message: Waiting for LSL_003 to be in alarm                                                        |
|        |             | A0050  | Set Scratch Time                                                                                       |
| T0500A | Drain Wait  |        | If 30 secs delay time completed then go to step S0500                                                  |
|        |             |        |                                                                                                        |
| S0500  | Drain Wait  | A0010  | Stopping and Disabling Conditional Logic for DEV1                                                      |

| Step  | Description      | Action | Action Description                                              |
|-------|------------------|--------|-----------------------------------------------------------------|
|       |                  | A0020  | Closing DEV2                                                    |
|       |                  | A0030  | Closing DEV3                                                    |
|       |                  | A0040  | Set Drain 635A & 635B preset time                               |
|       |                  | A0050  | Start Drain 635A & 635B preset Timer                            |
|       |                  | A0060  | Blow time Message                                               |
| T0600 | Disable HM       |        | If Drain 635 A&B timer (TP002) is elapsed then go to step S0600 |
|       |                  |        |                                                                 |
| S0600 | Disable HM       | A0010  | Disable HM 01 (Module Ownership Lost)                           |
| T0700 | Release Devices  |        | If no pending actions in S0600 then go to step S0700            |
|       |                  |        |                                                                 |
| S0700 | Release Devices  | A0010  | Release Devices Message                                         |
|       |                  | A0020  | Release Private Device                                          |
|       |                  | A0030  | Release Private Device                                          |
|       |                  | A0040  | Release Private Device                                          |
| T9900 | Complete         |        | If no pending actions in S0700 then go to step S9900            |
|       |                  |        |                                                                 |
| S9900 | Complete         | A0010  | Set Message: Production Command Complete                        |
|       |                  | A0020  | Self release the EM                                             |
| END   | Command Complete |        | True                                                            |

Table 79: Production Sequence

## 10.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

| Parameter            | Description                         | Type           | Units | Default Value/Reference |
|----------------------|-------------------------------------|----------------|-------|-------------------------|
| Operating Parameters |                                     |                |       |                         |
| NA                   |                                     |                |       |                         |
| Tuning Parameters    |                                     |                |       |                         |
| TP001                | Blowing Preset Time                 | Floating Point | sec   | 5                       |
| TP002                | Drain 635A & 635B preset Time (sec) | Floating Point | sec   | 20                      |

Table 80: Operating and Tuning Parameters

## 10.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

| Parameter Name   | Parameter Type     | Value/Reference                       |
|------------------|--------------------|---------------------------------------|
| MY_TAG           |                    | RF1EQ_T503                            |
| OPR_CMD_MASK     |                    | 3                                     |
| LSL_T503         | String             | RF1_LSL1003/DI1/DISC_ACT              |
| MX_004AC         | String             | RF1_MX1004AC/PV_D                     |
| LSL_T503_CM_NAME | String             | RF1_LSL1003                           |
| DY_REF_1         | Dynamic Reference  |                                       |
| HOLD             | External Reference | RF1EQ_STARTUP3/OP005_VALUE            |
| EM_BLEACH2       | String             | RF1EQ_T506                            |
| EM_BLEACH5       | String             | RF1EQ_622_21                          |
| BLEACH5          | External Reference | RF1EQ_622_21/OWNER_ID                 |
| STEP_TIME        | Floating Point     | 0                                     |
| COMPLETED_TIME   | Floating Point     | 0                                     |
| RESET_COMP_TM    | Boolean            | False                                 |
| EM_T506          | External Reference | RF1EQ_T506/COMMAND_00001/S0500/ACTIVE |

**Table 81: Configurable Parameters**

## 10.6 Named Set

Following Named set shall be used for configuration of EM.

| Name             | Description                                     |
|------------------|-------------------------------------------------|
| NS_RF_OF_BLEACH3 | Refinery Area: Filtration and Vacuum Production |

**Table 82: Named Set Details**

Details of states for NS\_RF\_OF\_BLEACH3 named set are as follows,

| Named Set Value | Named Set Name |
|-----------------|----------------|
| 00              | Hold           |
| 01              | Production     |

**Table 83: Name Set State Details**

## 11 EQUIPMENT MODULE CLASS DESIGN – EM\_RF\_OF\_BLEACH4 (1-606A1~A6 & 1-606B1~B2)

### 11.1 Functional Description

EM\_RF\_OF\_BLEACH4 is a class based equipment module that performs filling operation after evacuation and then discharge the material. EM\_RF\_OF\_BLEACH4 is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. Initially Operator shall be provided only with Hold and Fill commands. Discharge command will enable while execution of filling operation. Hold Request shall be enabled for all acquired devices at the end of Hold command.

### 11.2 Instances

The following Table contains the module instances of the equipment module class.

| Device Name | Module Class     | Instance Name         |                       |                       |                       |                       |                       |                       |                       |
|-------------|------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| EM          | EM_RF_OF_BLEACH4 | RF1<br>EQ_606<br>A1   | RF1<br>EQ_606<br>A2   | RF1<br>EQ_606<br>A3   | RF1<br>EQ_606<br>A4   | RF1<br>EQ_606<br>A5   | RF1<br>EQ_606<br>A6   | RF1<br>EQ_606<br>B1   | RF1<br>EQ_606<br>B2   |
| DEV1        | _V12_IP_130_SE   | RF1<br>_V1_606<br>A1F | RF1<br>_V1_606<br>A2F | RF1<br>_V1_606<br>A3F | RF1<br>_V1_606<br>A4F | RF1<br>_V1_606<br>A5F | RF1<br>_V1_606<br>A6F | RF1<br>_V1_606<br>B1F | RF1<br>_V1_606<br>B2F |
| DEV2        | _V12_IP_130_SE   | RF1<br>_V1_606<br>A1D | RF1<br>_V1_606<br>A2D | RF1<br>_V1_606<br>A3D | RF1<br>_V1_606<br>A4D | RF1<br>_V1_606<br>A5D | RF1<br>_V1_606<br>A6D | RF1<br>_V1_606<br>B1D | RF1<br>_V1_606<br>B2D |
| DEV3        | _V12_IP_130_SE   | RF1_V1630A12          |                       |                       |                       |                       |                       | RF1_V1630B            |                       |
| DEV4        | _V12_IP_130_SE   | RF1_V1630A34          |                       |                       |                       |                       |                       | Ignored               |                       |

**Table 84: Instances**

### 11.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

#### 11.3.1 Command 00 – Hold

| Step  | Description | Action | Action Description |
|-------|-------------|--------|--------------------|
| H0010 | Start       | A0010  | Store Fail Message |
|       |             | A0020  | Reset Hold Request |

| Step   | Description                               | Action | Action Description                               |
|--------|-------------------------------------------|--------|--------------------------------------------------|
|        |                                           | A0030  | Save Previous Step Time                          |
| T1000  | Actions Complete or Time Out              |        | Go to step H1000                                 |
|        |                                           |        |                                                  |
| H1000  | Hold Command 1                            | A0010  | Closing and Disabling Conditional Logic for DEV1 |
|        |                                           | A0020  | Closing and Disabling Conditional Logic for DEV2 |
|        |                                           | A0030  | Closing and Disabling Conditional Logic for DEV3 |
|        |                                           | A0040  | Closing and Disabling Conditional Logic for DEV4 |
| T9900A | Actions Complete or Time Out & Hold=True  |        | Go to step H9900A                                |
| T9900B | Actions Complete or Time Out & Hold=False |        | Go to step H9900B                                |
|        |                                           |        |                                                  |
| H9900A | Command Hold End                          | A0010  | Completed                                        |
| END    | Command Complete                          |        | True                                             |
|        |                                           |        |                                                  |
| H9900B | Release Devices                           | A0010  | Release Devices Message                          |
|        |                                           | A0020  | Release Private Device                           |
|        |                                           | A0030  | Release Private Device                           |
|        |                                           | A0040  | Release Shared Device                            |
|        |                                           | A0050  | Release Shared Device                            |
| END1   | Command Complete                          |        | True                                             |

**Table 85: Hold Sequence**

## 11.3.2 Command 01 – Fill

| Step   | Description        | Action | Action Description                          |
|--------|--------------------|--------|---------------------------------------------|
| S0000  | Start              | A0010  | Set Message: Starting Fill Command          |
|        |                    | A0020  | Set OWNER_ID                                |
| T0010  | First Pass = False |        | If First Pass = False then go to step S0010 |
| T0020A | First Pass = True  |        | If First Pass = True then go to step S0020  |
|        |                    |        |                                             |
| S0010  | Initializations    | A0010  | First Pass Logic                            |

| Step   | Description      | Action | Action Description                                            |
|--------|------------------|--------|---------------------------------------------------------------|
| T0020B | Actions Complete | A0020  | First Pass = True                                             |
|        |                  | A0030  | Set Task Pointer                                              |
|        |                  |        | If no pending actions in S0010 then go to step S0020          |
| S0020  | Acquire Devices  | A0010  | Acquire Devices Message                                       |
|        |                  | A0020  | Acquire Private Device                                        |
|        |                  | A0030  | Acquire Private Device                                        |
|        |                  | A0040  | Acquire Private Device                                        |
|        |                  | A0050  | Acquire Private Device                                        |
|        |                  | A0060  | Reset Hold Request                                            |
| T0100  | Standby          |        | If Task Pointer = 1 then go to step S0100                     |
| T0200B | Evacuate         |        | If Task Pointer = 2 then go to step S0200                     |
| T0300B | Fill             |        | If Task Pointer = 3 then go to step S0300                     |
| S0100  | Standby          | A0010  | Closing and Disabling Conditional Logic for DEV1              |
|        |                  | A0020  | Closing and Disabling Conditional Logic for DEV2              |
|        |                  | A0030  | Closing and Disabling Conditional Logic for DEV3              |
|        |                  | A0040  | Closing and Disabling Conditional Logic for DEV4              |
| T0200A | Evacuate         |        | If no pending actions in S0100 then go to step S0200          |
| S0200  | Evacuate         | A0010  | Enabling Conditional Logic for DEV2                           |
|        |                  | A0020  | Integrate Quantity Dosed                                      |
|        |                  | A0030  | Set Preset Evacuate time                                      |
|        |                  | A0040  | Start Preset Evacuate Timer                                   |
|        |                  | A0050  | Preset Evacuate time Message                                  |
|        |                  | A0060  | Increment Arms in Discharge Counter by 1                      |
| T0300A | Fill             |        | If Preset Evacuate time (TP002) is over then go to step S0300 |
| S0300  | Fill             | A0010  | Closing and Disabling Conditional Logic for DEV2              |
|        |                  | A0020  | Enabling Conditional Logic for DEV1                           |
|        |                  | A0030  | Enabling Conditional Logic for DEV4                           |
|        |                  | A0040  | Enabling Conditional Logic for DEV3                           |
|        |                  | A0050  | Set Preset Filling time                                       |
|        |                  | A0060  | Start Preset Filling Timer                                    |
|        |                  | A0070  | Preset Waiting Fast time Message                              |

| Step   | Description      | Action | Action Description                                        |
|--------|------------------|--------|-----------------------------------------------------------|
| T09900 | Complete         |        | If Preset Fill time (TP003) is over then go to step S9900 |
|        |                  |        |                                                           |
| S9900  | Complete         | A0010  | Set Message: Fill Command Complete                        |
| END    | Command Complete |        | True                                                      |

**Table 86: Fill Sequence**

## 11.3.3 Command 02 – Discharge

| Step   | Description   | Action | Action Description                                                                  |
|--------|---------------|--------|-------------------------------------------------------------------------------------|
| S0000  | Start         | A0010  | Set Message: Starting Discharge Command                                             |
|        |               | A0020  | Reset Hold Request                                                                  |
| T0100  | Wait          |        | If Task Pointer = 3 then go to step S0100                                           |
| T0200B | Discharge     |        | If Task Pointer = 4 then go to step S0200                                           |
|        |               |        |                                                                                     |
| S0100  | Wait          | A0010  | Closing and Disabling Conditional Logic for DEV1                                    |
|        |               | A0020  | Closing and Disabling Conditional Logic for DEV2                                    |
|        |               | A0030  | Closing and Disabling Conditional Logic for DEV3                                    |
|        |               | A0040  | Closing and Disabling Conditional Logic for DEV4                                    |
|        |               | A0050  | Set Preset Waiting time                                                             |
|        |               | A0060  | Start Preset Waiting Timer                                                          |
|        |               | A0070  | Preset Waiting Fast time Message                                                    |
| T0200A | Discharge     |        | If Preset Wait time (TP005) is over and DEMAND_DOSING > TP008 then go to step S0200 |
|        |               |        |                                                                                     |
| S0200  | Discharge     | A0010  | Closing and Disabling Conditional Logic for DEV1                                    |
|        |               | A0020  | Enabling Conditional Logic for DEV2                                                 |
|        |               | A0030  | Closing and Disabling Conditional Logic for DEV3                                    |
|        |               | A0040  | Closing and Disabling Conditional Logic for DEV4                                    |
|        |               | A0050  | Integrate Quantity Dosed                                                            |
|        |               | A0060  | Set Preset Discharge time                                                           |
|        |               | A0070  | Start Preset Discharge Timer                                                        |
|        |               | A0080  | Preset Discharge time Message                                                       |
|        |               | A0090  | Decrement Arms in Discharge Counter by 1                                            |
| T0300  | Close Devices |        | If Preset Discharge time (TP006) is over then go to step S0300                      |

| Step  | Description      | Action | Action Description                                   |
|-------|------------------|--------|------------------------------------------------------|
| S0300 | Close Devices    | A0010  | Closing and Disabling Conditional Logic for DEV1     |
|       |                  | A0020  | Closing and Disabling Conditional Logic for DEV2     |
|       |                  | A0030  | Closing and Disabling Conditional Logic for DEV3     |
|       |                  | A0040  | Closing and Disabling Conditional Logic for DEV4     |
| S0400 | Disable HM       | A0010  | Disable HM 01 (Module Ownership Lost)                |
| T0500 | Close Devices    |        | If no pending actions in S0400 then go to step S0500 |
| S0500 | Release Devices  | A0010  | Release Devices Message                              |
|       |                  | A0020  | Release Private Device                               |
|       |                  | A0030  | Release Private Device                               |
|       |                  | A0040  | Release Shared Device                                |
|       |                  | A0050  | Release Shared Device                                |
| T9900 | Close Devices    |        | If no pending actions in S0500 then go to step S9900 |
| S9900 | Complete         | A0010  | Set Message: Discharge Command Complete              |
|       |                  | A0020  | Self Release the EM                                  |
| END   | Command Complete |        | True                                                 |

Table 87: Discharge Sequence

## 11.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

| Parameter            | Description      | Type           | Units | Default Value/Reference |
|----------------------|------------------|----------------|-------|-------------------------|
| Operating Parameters |                  |                |       |                         |
| OP001                | Flow             | Floating Point | kg/h  | 62500                   |
| OP002                | Dosing aid       | Floating Point | %     | False                   |
| OP003                | Dosing Arm       | String         | NA    | 606A                    |
| OP004                | Mode             | Boolean        | NA    | MAN = 0<br>(AUTO = 1)   |
| OP005                | Manual Set Point | Floating Point | kg/h  | 0                       |
| OP006                | Actual Step      | Floating Point |       | 0                       |
| OP007                | Remain Time      | Floating Point | sec   | 0                       |

| Parameter         | Description              | Type           | Units | Default Value/Reference |
|-------------------|--------------------------|----------------|-------|-------------------------|
| OP008             | Dosed Quantity           | Floating Point | kg    | 100                     |
| OP009             | Reset dosed quantity     | Named Set      | NA    | NO                      |
| Tuning Parameters |                          |                |       |                         |
| TP001             | Delay Timer              | Floating Point | sec   | 5                       |
| TP002             | Preset Evacuate Time     | Floating Point | sec   | 5                       |
| TP003             | Preset Filling Time      | Floating Point | sec   | 10                      |
| TP004             | Preset Waiting Time Slow | Floating Point | sec   | 5                       |
| TP005             | Preset Waiting Time Fast | Floating Point | sec   | 2                       |
| TP006             | Discharge Time           | Floating Point | sec   | 10                      |
| TP007             | Dosing (Ax)              | Floating Point | %     | 0.8                     |
| TP008             | Delta Demand             | Floating Point | kg    | 5                       |
| TP009             | Volume 606A              | Floating Point | kg    | 4                       |
| TP010             | Call Demand/hour         | Floating Point | kg/h  | 500                     |
| TP011             | Call Demand              | Floating Point | kg    | 0.2                     |
| TP012             | Total Dosed              | Floating Point | kg    | 234                     |
| TP013             | Volume 606Bx             | Floating Point | kg    | 0                       |
| TP014             | Demand Limit             | Floating Point | NA    | 0                       |

Table 88: Operating and Tuning Parameters

## 11.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

| Parameter Name   | Parameter Type | Value/Reference                 |                                 |                                 |                                 |                                 |                                 |                                 |                                 |  |  |
|------------------|----------------|---------------------------------|---------------------------------|---------------------------------|---------------------------------|---------------------------------|---------------------------------|---------------------------------|---------------------------------|--|--|
| MY_TAG           |                | RF<br>1E<br>Q_<br>60<br>6A<br>1 | RF<br>1E<br>Q_<br>60<br>6A<br>2 | RF<br>1E<br>Q_<br>60<br>6A<br>3 | RF<br>1E<br>Q_<br>60<br>6A<br>4 | RF<br>1E<br>Q_<br>60<br>6A<br>5 | RF<br>1E<br>Q_<br>60<br>6A<br>6 | RF<br>1E<br>Q_<br>60<br>6B<br>1 | RF<br>1E<br>Q_<br>60<br>6B<br>2 |  |  |
| OPR_CMD_MASK     |                | 7                               |                                 |                                 |                                 |                                 |                                 |                                 |                                 |  |  |
| DEMAND_DOSIN_G   | Floating Point |                                 |                                 |                                 |                                 |                                 |                                 |                                 |                                 |  |  |
| DEMAND_IN_KG_SEC | Floating Point |                                 |                                 |                                 |                                 |                                 |                                 |                                 |                                 |  |  |

| Parameter Name | Parameter Type     | Value/Reference                    |                                   |
|----------------|--------------------|------------------------------------|-----------------------------------|
| HOLD           |                    | RF1EQ_STARTUP3/OP005_VALUE         |                                   |
| STEP_TIME      | Floating Point     | 0                                  |                                   |
| COMPLETED_TIME | Floating Point     | 0                                  |                                   |
| RESET_COMP_TM  | Boolean            | False                              |                                   |
| EM_BLEACH3     | External Reference | RF1EQ_T503/COMMAND_00001/ACT_STEPS | RF1EQ_635/COMMAND_00001/ACT_STEPS |
| OWNER          | String             | RF1EQ_T503                         | RF1EQ_635                         |

**Table 89: Configurable Parameters**

## 11.6 Named Set

Following Named set shall be used for configuration of EM.

| Name             | Description                                     |
|------------------|-------------------------------------------------|
| NS_RF_OF_BLEACH4 | Refinery Area: Filtration and Vacuum Production |

**Table 90: Named Set Details**

Details of states for NS\_RF\_OF\_BLEACH4 named set are as follows,

| Named Set Value | Named Set Name |
|-----------------|----------------|
| 00              | Hold           |
| 01              | Fill           |
| 02              | Discharge      |

**Table 91: Name Set State Details**

## 12 EQUIPMENT MODULE CLASS DESIGN – EM\_RF\_OF\_BLEACH5 (1-622/21)

### 12.1 Functional Description

EM\_RF\_OF\_BLEACH5 is a class based equipment module that performs filling operation to produce the material and afterwards drains the material. EM\_RF\_OF\_BLEACH5 is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. Initially Operator shall be provided only with Hold and Production. Drain command shall be enabled during the execution of Production State. Hold Request shall be enabled for all acquired devices at the end of Hold command.

### 12.2 Instances

The following Table contains the module instances of the equipment module class.

| Device Name | Module Class     | Instance Name |
|-------------|------------------|---------------|
| EM          | EM_RF_OF_BLEACH5 | RF1EQ_622_21  |
| DEV1        | EM_RF_FILTRATION | RF1EQ_616A1   |
| DEV2        | EM_RF_FILTRATION | RF1EQ_616A2   |
| DEV3        | EM_RF_FILTRATION | RF1EQ_616A3   |
| DEV4        | EM_RF_FILTRATION | RF1EQ_616A4   |
| DEV5        | EM_RF_FILTRATION | RF1EQ_616A5   |
| DEV6        | _V12_IP_130_SE   | RF1_V1616A15  |
| DEV7        | _V12_IP_130_SE   | RF1_V1622C    |
| DEV8        | _V12_IP_130_SE   | RF1_V1622B    |
| DEV9        | _V12_IP_130_SE   | RF1_V1622A    |
| DEV10       | _CSD_130_SE      | RF1_TT1622A   |

**Table 92: Instances**

### 12.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

#### 12.3.1 Command 00 – Hold

| Step  | Description        | Action | Action Description      |
|-------|--------------------|--------|-------------------------|
| H0010 | Command Hold Start | A0010  | Store Fail Message      |
|       |                    | A0020  | Reset Hold Request      |
|       |                    | A0030  | Save Previous Step Time |

| Step   | Description                                 | Action | Action Description                               |
|--------|---------------------------------------------|--------|--------------------------------------------------|
| T1000  | Actions Complete or Time Out                |        | Go to step H1000                                 |
|        |                                             |        |                                                  |
| H1000  | Hold Command 1                              | A0010  | Closing DEV6                                     |
|        |                                             | A0020  | Closing DEV7                                     |
|        |                                             | A0030  | Closing and Disabling Conditional Logic for DEV8 |
|        |                                             | A0040  | Closing DEV9                                     |
|        |                                             | A0050  | Set DEV10 Mode to Remote Out                     |
|        |                                             | A0060  | Set DEV10 output to Closed                       |
|        |                                             | A0070  | Step Name                                        |
| T9900A | Actions Complete or Time Out and Hold=True  |        | Go to step H9900A                                |
| T9900B | Actions Complete or Time Out and Hold=False |        | Go to step H9900B                                |
|        |                                             |        |                                                  |
| H9900A | Command Hold End                            | A0010  | Completed                                        |
| END    | Command Complete                            |        | True                                             |
|        |                                             |        |                                                  |
| H9900B | Release Devices                             | A0010  | Release Devices Message                          |
|        |                                             | A0020  | Release Private Device                           |
|        |                                             | A0030  | Release Private Device                           |
|        |                                             | A0040  | Release Private Device                           |
|        |                                             | A0050  | Release Private Device                           |
|        |                                             | A0060  | Release Private Device                           |
|        |                                             | A0070  | Release Private Device                           |
|        |                                             | A0080  | Release Private Device                           |
|        |                                             | A0090  | Release Private Device                           |
|        |                                             | A0100  | Release Private Device                           |
|        |                                             | A0110  | Release Private Device                           |
|        |                                             | A0120  | Reset Parameters                                 |
| END1   | Command Complete                            |        | True                                             |

Table 93: Hold Sequence

## 12.3.2 Command 01 – Production

| Step   | Description         | Action | Action Description                                   |
|--------|---------------------|--------|------------------------------------------------------|
| S0000  | Start               | A0010  | Set Message: Starting Production Command             |
| T0010  | First Pass = False  |        | If First Pass = False then go to step S0010          |
| T0020A | First Pass = True   |        | If First Pass = True then go to step S0020           |
|        |                     |        |                                                      |
| S0010  | Initializations     | A0010  | First Pass Logic                                     |
|        |                     | A0020  | First Pass = True                                    |
|        |                     | A0030  | Set Task Pointer                                     |
|        |                     | A0040  | Step Name                                            |
| T0020B | Actions Complete    |        | If no pending actions in S0010 then go to step S0020 |
|        |                     |        |                                                      |
| S0020  | Acquire Devices     | A0010  | Acquire Devices Message                              |
|        |                     | A0020  | Acquire Private Device                               |
|        |                     | A0030  | Acquire Private Device                               |
|        |                     | A0040  | Acquire Private Device                               |
|        |                     | A0050  | Acquire Private Device                               |
|        |                     | A0060  | Acquire Private Device                               |
|        |                     | A0070  | Reset Hold Request                                   |
| T0100  | Standby             |        | If Task Pointer = 1 then go to step S0100            |
| T0200B | Fill                |        | If Task Pointer = 2 then go to step S0200            |
| T0300B | Production          |        | If Task Pointer = 3 then go to step S0300            |
| T0400B | Pre Drain           |        | If Task Pointer = 4 then go to step S0400            |
| T0500B | Final Recirculation |        | If Task Pointer = 5 then go to step S0500            |
|        |                     |        |                                                      |
| S0100  | Standby             | A0010  | Closing DEV6                                         |
|        |                     | A0020  | Closing DEV7                                         |
|        |                     | A0030  | Closing and Disabling Conditional Logic for DEV8     |
|        |                     | A0040  | Closing DEV9                                         |
|        |                     | A0050  | Set DEV10 Mode to Remote Out                         |
|        |                     | A0060  | Set DEV10 output to Closed                           |
| T0200A | Fill                |        | If no pending actions in S0100 then go to step S0200 |
|        |                     |        |                                                      |
| S0200  | Fill                | A0010  | Closing DEV6                                         |
|        |                     | A0020  | Closing DEV7                                         |
|        |                     | A0030  | Closing and Disabling Conditional Logic for DEV8     |
|        |                     | A0040  | Closing DEV9                                         |

| Step   | Description         | Action | Action Description                                                                                                                                                                               |
|--------|---------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        |                     | A0050  | Set DEV10 Mode to Remote Cascade                                                                                                                                                                 |
|        |                     | A0060  | Set DEV10 setpoint to Remote Setpoint for TT_622A                                                                                                                                                |
|        |                     | A0070  | Step Name                                                                                                                                                                                        |
| T0300A | Production          |        | If LSH_622_21 is in alarm and any one 616A Filter in step Standby ON (STANDBY_C > 0) then go to step S0300                                                                                       |
| S0300  | Production          |        |                                                                                                                                                                                                  |
|        |                     | A0010  | Closing DEV6                                                                                                                                                                                     |
|        |                     | A0020  | Opening DEV7                                                                                                                                                                                     |
|        |                     | A0030  | Enabling Conditional Logic for DEV8                                                                                                                                                              |
|        |                     | A0040  | Closing DEV9                                                                                                                                                                                     |
|        |                     | A0050  | Set DEV10 Mode to Remote Cascade                                                                                                                                                                 |
|        |                     | A0060  | Set DEV10 setpoint to Remote Setpoint for TT_622A                                                                                                                                                |
|        |                     | A0070  | Acquire Filters 1                                                                                                                                                                                |
|        |                     | A0080  | Starting Fill command for DEV1                                                                                                                                                                   |
|        |                     | A0090  | Acquire Filters 2                                                                                                                                                                                |
|        |                     | A0100  | Starting Fill command for DEV2                                                                                                                                                                   |
|        |                     | A0110  | Acquire Filters 3                                                                                                                                                                                |
|        |                     | A0120  | Starting Fill command for DEV3                                                                                                                                                                   |
|        |                     | A0130  | Acquire Filters 4                                                                                                                                                                                |
|        |                     | A0140  | Starting Fill command for DEV4                                                                                                                                                                   |
|        |                     | A0150  | Acquire Filters 5                                                                                                                                                                                |
|        |                     | A0160  | Starting Fill command for DEV5                                                                                                                                                                   |
|        |                     | A0170  | Set Message                                                                                                                                                                                      |
|        |                     | A0180  | Step Name                                                                                                                                                                                        |
| T0400A | Pre Drain           |        | If EM_BLEACH3 is in step <Drain Wait> and any one filter is in <Filtration> (FILTRATION_C > 0) then go to step S0400                                                                             |
| S0400  | Pre Drain           |        |                                                                                                                                                                                                  |
|        |                     | A0010  | Closing DEV6                                                                                                                                                                                     |
|        |                     | A0020  | Opening DEV7                                                                                                                                                                                     |
|        |                     | A0030  | Enabling Conditional Logic for DEV8                                                                                                                                                              |
|        |                     | A0040  | Closing DEV9                                                                                                                                                                                     |
|        |                     | A0050  | Set DEV10 Mode to Remote Out                                                                                                                                                                     |
|        |                     | A0060  | Set DEV10 output to Closed                                                                                                                                                                       |
|        |                     | A0070  | Step Name                                                                                                                                                                                        |
| T0500A | Final Recirculation |        | If Final Recirculation Level Reached (LT_622_PV >= TP003) and no filter is in step <DRAIN TO 622> (DRAIN_TO_622C) & number of filters is in <Filtration> (FILTRATION_C) is not > 1 and number of |

| Step  | Description         | Action | Action Description                                                                                                   |
|-------|---------------------|--------|----------------------------------------------------------------------------------------------------------------------|
|       |                     |        | filters is in <Fill> (FILL_C) = 0 then go to step S0500                                                              |
|       |                     |        |                                                                                                                      |
| S0500 | Final Recirculation | A0010  | Closing DEV6                                                                                                         |
|       |                     | A0020  | Opening DEV7                                                                                                         |
|       |                     | A0030  | Opening and Disabling Conditional Logic for DEV8                                                                     |
|       |                     | A0040  | Closing DEV9                                                                                                         |
|       |                     | A0050  | Set DEV10 Mode to Remote Out                                                                                         |
|       |                     | A0060  | Set DEV10 output to Closed                                                                                           |
|       |                     | A0070  | Step Name                                                                                                            |
| T9900 | Complete            |        | If no filter is in step <Final Recirculation> and any one filter is in step <Final Filtration> then go to step S9900 |
|       |                     |        |                                                                                                                      |
| S9900 | Complete            | A0010  | Set Message: Production Command Complete                                                                             |
| END   | Command Complete    |        | True                                                                                                                 |

**Table 94: Production Sequence**

## 12.3.3

## Command 02 – Drain

| Step   | Description            | Action | Action Description                               |
|--------|------------------------|--------|--------------------------------------------------|
| S0000  | Start                  | A0010  | Set Message: Starting Drain Command              |
| T0100  | Drain                  |        | If Task Pointer = 5 then go to step S0100        |
| T0200B | Fill from Filter 616Ax |        | If Task Pointer = 6 then go to step S000         |
| T0300B | Empty                  |        | If Task Pointer = 7 then go to step S0500        |
| T0400B | Blow                   |        | If Task Pointer = 8 then go to step S0600        |
| T0500B | Drain Wait             |        | If Task Pointer = 9 then go to step S0700        |
|        |                        |        |                                                  |
| S0100  | Drain                  | A0010  | Closing DEV6                                     |
|        |                        | A0020  | Opening DEV7                                     |
|        |                        | A0030  | Opening and Disabling Conditional Logic for DEV8 |
|        |                        | A0040  | Closing DEV9                                     |
|        |                        | A0050  | Set DEV10 Mode to Remote Out                     |
|        |                        | A0060  | Set DEV10 output to Closed                       |
|        |                        | A0070  | Check if LT_622 is active                        |
|        |                        | A0080  | Set Message: Adding 10 secs delay                |
| T0200A | Fill from Filter 616Ax |        | If 10 seconds completed then go to step S0200    |
|        |                        |        |                                                  |

| Step   | Description            | Action | Action Description                                                                                                                                                                                                              |
|--------|------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| S0200  | Fill from Filter 616Ax | A0010  | Closing DEV6                                                                                                                                                                                                                    |
|        |                        | A0020  | Opening DEV7                                                                                                                                                                                                                    |
|        |                        | A0030  | Opening and Disabling Conditional Logic for DEV8                                                                                                                                                                                |
|        |                        | A0040  | Closing DEV9                                                                                                                                                                                                                    |
|        |                        | A0050  | Set DEV10 Mode to Remote Out                                                                                                                                                                                                    |
|        |                        | A0060  | Set DEV10 output to Closed                                                                                                                                                                                                      |
| T0300A | Empty                  |        | If no filter is in step <DRAIN TO 622> (DRAIN_TO_622C) or <Drying> (DRYING_C) <Sec. Drain> (SEC_DRAIN_C) or <Depress.> (DEPRESS_C) and 682B in <Standby> and Empty Confirmation Button (OP001) is Pressed then go to step S0300 |
|        |                        |        |                                                                                                                                                                                                                                 |
| S0300  | Empty                  | A0010  | Opening DEV6                                                                                                                                                                                                                    |
|        |                        | A0020  | Opening DEV7                                                                                                                                                                                                                    |
|        |                        | A0030  | Opening and Disabling Conditional Logic for DEV8                                                                                                                                                                                |
|        |                        | A0040  | Closing DEV9                                                                                                                                                                                                                    |
|        |                        | A0050  | Set DEV10 Mode to Remote Out                                                                                                                                                                                                    |
|        |                        | A0060  | Set DEV10 output to Closed                                                                                                                                                                                                      |
|        |                        | A0070  | Check if LSL_622 is active                                                                                                                                                                                                      |
|        |                        | A0080  | Set Message: Adding 10 secs delay                                                                                                                                                                                               |
|        |                        | A0090  | Step Name                                                                                                                                                                                                                       |
|        |                        | A0100  | Enabling Conditional Logic for PU_6225                                                                                                                                                                                          |
| T0400A | Blow                   |        | If no action pending in S0300 then go to step S0400                                                                                                                                                                             |
|        |                        |        |                                                                                                                                                                                                                                 |
| S0400  | Blow                   | A0010  | Opening DEV6                                                                                                                                                                                                                    |
|        |                        | A0020  | Closing DEV7                                                                                                                                                                                                                    |
|        |                        | A0030  | Opening and Disabling Conditional Logic for DEV8                                                                                                                                                                                |
|        |                        | A0040  | Opening DEV9                                                                                                                                                                                                                    |
|        |                        | A0050  | Set DEV10 Mode to Remote Out                                                                                                                                                                                                    |
|        |                        | A0060  | Set DEV10 output to Closed                                                                                                                                                                                                      |
|        |                        | A0070  | Step Name                                                                                                                                                                                                                       |
|        |                        | A0080  | Set Preset Blow time                                                                                                                                                                                                            |
|        |                        | A0090  | Start Preset Blow Timer                                                                                                                                                                                                         |
|        |                        | A0100  | Preset Blow time Message                                                                                                                                                                                                        |
| T0500A | Drain Wait             |        | If XV 1-622/21-BS Blowing timer elapsed (TP005) and 5 secs completed then go to step S0500                                                                                                                                      |
|        |                        |        |                                                                                                                                                                                                                                 |
| S0500  | Drain Wait             | A0010  | Opening DEV6                                                                                                                                                                                                                    |

| Step  | Description      | Action | Action Description                                   |
|-------|------------------|--------|------------------------------------------------------|
|       |                  | A0020  | Closing DEV7                                         |
|       |                  | A0030  | Opening and Disabling Conditional Logic for DEV8     |
|       |                  | A0040  | Closing DEV9                                         |
|       |                  | A0050  | Set DEV10 Mode to Remote Out                         |
|       |                  | A0060  | Set DEV10 output to Closed                           |
|       |                  | A0070  | Disable VPRD1                                        |
| T0600 | Close Devices    |        | If 5 secs (TP001) completed then go to step S0600    |
| S0600 | Close Devices    | A0010  | Closing DEV6                                         |
|       |                  | A0020  | Closing DEV7                                         |
|       |                  | A0030  | Closing and Disabling Conditional Logic for DEV8     |
|       |                  | A0040  | Closing DEV9                                         |
|       |                  | A0050  | Set DEV10 Mode to Remote Out                         |
|       |                  | A0060  | Set DEV10 output to Closed                           |
| T0700 | Disable HM       |        | If no pending actions in S0600 then go to step S0700 |
| S0700 | Disable HM       | A0010  | Disable HM 01 (Module Ownership Lost)                |
| T0800 | Release Devices  |        | If no pending actions in S0700 then go to step S0800 |
| S0800 | Release Devices  | A0010  | Release Devices Message                              |
|       |                  | A0020  | Release Private Device                               |
|       |                  | A0030  | Release Private Device                               |
|       |                  | A0040  | Release Private Device                               |
|       |                  | A0050  | Release Private Device                               |
|       |                  | A0060  | Release Private Device                               |
|       |                  | A0070  | Release Private Device                               |
|       |                  | A0080  | Release Private Device                               |
|       |                  | A0090  | Release Private Device                               |
|       |                  | A0100  | Release Private Device                               |
|       |                  | A0110  | Release Private Device                               |
| T9900 | Complete         |        | If no pending actions in S0800 then go to step S9900 |
| S9900 | Complete         | A0010  | Set Message: Drain Command Complete                  |
|       |                  | A0020  | Self release the EM                                  |
| END   | Command Complete |        | True                                                 |

Table 95: Drain Sequence

## 12.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

| Parameter            | Description                    | Type           | Units | Default Value/Reference |
|----------------------|--------------------------------|----------------|-------|-------------------------|
| Operating Parameters |                                |                |       |                         |
| OP001                | Empty Confirmation Button      | Named Set      | NA    | NO                      |
| Tuning Parameters    |                                |                |       |                         |
| TP001                | Delay Timer                    | Floating Point | sec   | 5                       |
| TP002                | Safe Drain level               | Floating Point | sec   | 70                      |
| TP003                | Final Recirculation level      | Floating Point | %     | 5                       |
| TP004                | Vacuum                         | Floating Point | mbar  | 100                     |
| TP005                | Blow Time for XV_622/21-BS     | Floating Point | sec   | 0                       |
| TP006                | Remote Setpoint for TCV_622/21 | Floating Point | °C    | 105                     |

**Table 96: Operating and Tuning Parameters**

## 12.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

| Parameter Name | Type               | Value/Reference            |
|----------------|--------------------|----------------------------|
| MY_TAG         | String             | RF1EQ_622_21               |
| OPR_CMD_MASK   |                    | 7                          |
| LSH_622_21     | External Reference | RF1_LSH1622/DI1/DISC_ACT   |
| LT_622_PV      | External Reference | RF1_LT1622/LO_LO_ALM       |
| LT_622_ALM     | External Reference | RF1_LT1622/PV              |
| LSLL_622       | External Reference | RF1_LSLL1622/DI1/DISC_ACT  |
| EM_BLEACH3     | String             | RF1EQ_T503                 |
| EM_VPRD2       | External Reference | RF1EQ_682B/OWNER_ID        |
| HOLD           | External Reference | RF1EQ_STARTUP3/OP005_VALUE |
| STEP_NAME      | String             |                            |
| STANDBY_C      | External Reference | RF1_MISC/STANDBY_C         |
| FILTRATION_C   | External Reference | RF1_MISC/FILTRATION_C      |
| DRAIN_TO_622C  | External Reference | RF1_MISC/DRAIN_TO_622_C    |
| FILL_C         | External Reference | RF1_MISC/FILL_C            |

| Parameter Name  | Type               | Value/Reference          |
|-----------------|--------------------|--------------------------|
| F_FILTRATION_C  | External Reference | RF1_MISC/F_FILTRATION_C  |
| F_RECIRCULATN_C | External Reference | RF1_MISC/F_RECIRCULATN_C |
| DRYING_C        | External Reference | RF1_MISC/DRYING_C        |
| SEC_DRAIN_C     | External Reference | RF1_MISC/SEC_DRAIN_C     |
| DEPRESS_C       | External Reference | RF1_MISC/DEPRESS_C       |
| STEP_TIME       | Floating Point     | 0                        |
| COMPLETED_TIME  | Floating Point     | 0                        |
| RESET_COMP_TM   | Boolean            | False                    |
| PU_6225         | External Reference |                          |

**Table 97: Configurable Parameters**

## 12.6 Named Set

Following Named set shall be used for configuration of EM.

| Name             | Description                                     |
|------------------|-------------------------------------------------|
| NS_RF_OF_BLEACH5 | Refinery Area: Filtration and Vacuum Production |

**Table 98: Named Set Details**

Details of states for NS\_RF\_OF\_BLEACH5 named set are as follows,

| Named Set Value | Named Set Name |
|-----------------|----------------|
| 00              | Hold           |
| 01              | Production     |
| 02              | Drain          |

**Table 99: Name Set State Details**

## 13 EQUIPMENT MODULE CLASS DESIGN – EM\_RF\_FILTRATION (1-616A1-5)

### 13.1 Functional Description

EM\_RF\_FILTRATION is a class based equipment module that performs filtration and final recirculation and afterwards discharge the material. EM\_RF\_FILTRATION is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. Initially Operator shall be provided only with Hold and Fill. Filtration and Discharge command shall be enabled during the execution of Fill State. Hold Request shall be enabled for all acquired devices at the end of Hold command. Commands which require operator intervention shall be configured as Operator Prompts wherever possible.

### 13.2 Instances

The following Table contains the module instances of the equipment module class.

| Device Name | Module Class       | Instance Name    |                  |                  |                  |                  |
|-------------|--------------------|------------------|------------------|------------------|------------------|------------------|
| EM          | EM_RF_FILTRATION   | RF1EQ_6<br>16A1  | RF1EQ_6<br>16A2  | RF1EQ_6<br>16A3  | RF1EQ_6<br>16A4  | RF1EQ_6<br>16A5  |
| DEV1        | _MRSS1<br>1_130_SE | RF1_PU1<br>6221  | RF1_PU1<br>6222  | RF1_PU1<br>6223  | RF1_PU1<br>6224  | RF1_PU1<br>6225  |
| DEV2        | _V12_IP_<br>130_SE | RF1_V16<br>16A1A | RF1_V16<br>16A2A | RF1_V16<br>16A3A | RF1_V16<br>16A4A | RF1_V16<br>16A5A |
| DEV3        | _V12_IP_<br>130_SE | RF1_V16<br>16A1C | RF1_V16<br>16A2C | RF1_V16<br>16A3C | RF1_V16<br>16A4C | RF1_V16<br>16A5C |
| DEV4        | _V12_IP_<br>130_SE | RF1_V16<br>16A1D | RF1_V16<br>16A2D | RF1_V16<br>16A3D | RF1_V16<br>16A4D | RF1_V16<br>16A5D |
| DEV5        | _V12_IP_<br>130_SE | RF1_V16<br>16A1F | RF1_V16<br>16A2F | RF1_V16<br>16A3F | RF1_V16<br>16A4F | RF1_V16<br>16A5F |
| DEV6        | _V12_IP_<br>130_SE | Ignored          | Ignored          | Ignored          | Ignored          | Ignored          |
| DEV7        | _V12_IP_<br>130_SE | RF1_V16<br>16A1G | RF1_V16<br>16A2G | RF1_V16<br>16A3G | RF1_V16<br>16A4G | RF1_V16<br>16A5G |
| DEV8        | _V12_IP_<br>130_SE | RF1_V16<br>16A1H | RF1_V16<br>16A2H | RF1_V16<br>16A3H | RF1_V16<br>16A4H | RF1_V16<br>16A5H |
| DEV9        | _V12_IP_<br>130_SE | RF1_V16<br>16A1I | RF1_V16<br>16A2I | RF1_V16<br>16A3I | RF1_V16<br>16A4I | RF1_V16<br>16A5I |
| DEV10       | _V12_IP_<br>130_SE | RF1_V16<br>16A1J | RF1_V16<br>16A2J | RF1_V16<br>16A3J | RF1_V16<br>16A4J | RF1_V16<br>16A5J |
| DEV11       | _V12_IP_<br>130_SE | RF1_V16<br>16A1K | RF1_V16<br>16A2L | RF1_V16<br>16A3L | RF1_V16<br>16A4K | RF1_V16<br>16A5L |
| DEV12       | _V12_IP_<br>130_SE | RF1_V16<br>16A1L | RF1_V16<br>16A2M | RF1_V16<br>16A3M | RF1_V16<br>16A4L | RF1_V16<br>16A5M |
| DEV13       | _V12_IP_<br>130_SE | RF1_V16<br>16A1M | RF1_V16<br>16A2K | RF1_V16<br>16A3K | RF1_V16<br>16A4M | RF1_V16<br>16A5K |



| Device Name | Module Class   | Instance Name |              |              |              |              |
|-------------|----------------|---------------|--------------|--------------|--------------|--------------|
| DEV14       | _V12_IP_130_SE | RF1_V1616A1B  | RF1_V1616A2B | RF1_V1616A3B | RF1_V1616A4B | RF1_V1616A5B |
| DEV15       | _V12_IP_130_SE | RF1_V1616A1E  | RF1_V1616A2E | RF1_V1616A3E | RF1_V1616A4E | RF1_V1616A5E |
| DEV16       | _MRSS11_130_SE | Ignored       |              |              |              |              |
| DEV17       | _V12_IP_130_SE | RF1_V1623     |              |              |              |              |

Table 100: Instances

### 13.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

#### 13.3.1 Command 00 – Hold

| Step  | Description                  | Action | Action Description                                |
|-------|------------------------------|--------|---------------------------------------------------|
| H0010 | Start                        | A0010  | Store Fail Message                                |
|       |                              | A0020  | Reset Hold Request                                |
| T1000 | Actions Complete or Time Out |        | Go to step H1000                                  |
| H1000 | Hold Command 1               | A0010  | Stopping and Disabling Conditional Logic for DEV1 |
|       |                              | A0020  | Closing and Disabling Conditional Logic for DEV2  |
|       |                              | A0030  | Closing DEV3                                      |
|       |                              | A0040  | Closing and Disabling Conditional Logic for DEV4  |
|       |                              | A0050  | Closing and Disabling Conditional Logic for DEV5  |
|       |                              | A0060  | Closing and Disabling Conditional Logic for DEV6  |
|       |                              | A0070  | Closing and Disabling Conditional Logic for DEV7  |
|       |                              | A0080  | Closing and Disabling Conditional Logic for DEV8  |
|       |                              | A0090  | Closing DEV9                                      |
|       |                              | A0100  | Closing and Disabling Conditional Logic for DEV10 |
|       |                              | A0110  | Closing and Disabling Conditional Logic for DEV11 |

| Step   | Description                               | Action | Action Description                                 |
|--------|-------------------------------------------|--------|----------------------------------------------------|
|        |                                           | A0120  | Closing and Disabling Conditional Logic for DEV12  |
|        |                                           | A0130  | Closing and Disabling Conditional Logic for DEV13  |
|        |                                           | A0140  | Closing and Disabling Conditional Logic for DEV14  |
|        |                                           | A0150  | Closing and Disabling Conditional Logic for DEV15  |
|        |                                           | A0160  | Stopping and Disabling Conditional Logic for DEV16 |
|        |                                           | A0170  | Closing DEV17                                      |
| T9900A | Actions Complete or Time Out & Hold=True  |        | Go to step H9900A                                  |
| T9900B | Actions Complete or Time Out & Hold=False |        | Go to step H9900B                                  |
|        |                                           |        |                                                    |
| H9900A | Command Hold End                          | A0010  | Completed                                          |
| END    | Command Complete                          |        | True                                               |
|        |                                           |        |                                                    |
| H9900B | Release Devices                           | A0010  | Release Devices Message                            |
|        |                                           | A0020  | Release Private Devices                            |
|        |                                           | A0030  | Release Private Devices                            |
|        |                                           | A0040  | Release Private Devices                            |
|        |                                           | A0050  | Release Private Devices                            |
|        |                                           | A0060  | Release Private Devices                            |
|        |                                           | A0070  | Release Private Devices                            |
|        |                                           | A0080  | Release Private Devices                            |
|        |                                           | A0090  | Release Private Devices                            |
|        |                                           | A0100  | Release Private Devices                            |
|        |                                           | A0110  | Release Private Devices                            |
|        |                                           | A0120  | Release Private Devices                            |
|        |                                           | A0130  | Release Private Devices                            |
|        |                                           | A0140  | Release Private Devices                            |
|        |                                           | A0150  | Release Private Devices                            |
|        |                                           | A0160  | Release Private Devices                            |
|        |                                           | A0170  | Reset Parameters                                   |
| END1   | Command Complete                          |        | True                                               |

Table 101: Hold Sequence

## 13.3.2 Command 01 – Fill

| Step   | Description        | Action | Action Description                                   |
|--------|--------------------|--------|------------------------------------------------------|
| S0000  | Start              | A0010  | Set Message: Starting Fill Command                   |
| T0010  | First Pass = False |        | If First Pass = False then go to step S0010          |
| T0020A | First Pass = True  |        | If First Pass = True then go to step S0020           |
|        |                    |        |                                                      |
| S0010  | Initializations    | A0010  | First Pass Logic                                     |
|        |                    | A0020  | First Pass = True                                    |
|        |                    | A0030  | Set Task Pointer                                     |
| T0020B | Actions Complete   |        | If no pending actions in S0010 then go to step S0020 |
|        |                    |        |                                                      |
| S0020  | Acquire Devices    | A0010  | Acquire Devices Message                              |
|        |                    | A0020  | Acquire Private Devices                              |
|        |                    | A0030  | Acquire Private Devices                              |
|        |                    | A0040  | Acquire Private Devices                              |
|        |                    | A0050  | Acquire Private Devices                              |
|        |                    | A0060  | Acquire Private Devices                              |
|        |                    | A0070  | Acquire Private Devices                              |
|        |                    | A0080  | Acquire Private Devices                              |
|        |                    | A0090  | Acquire Private Devices                              |
|        |                    | A0100  | Acquire Private Devices                              |
|        |                    | A0110  | Acquire Private Devices                              |
|        |                    | A0120  | Acquire Private Devices                              |
|        |                    | A0130  | Acquire Private Devices                              |
|        |                    | A0140  | Acquire Private Devices                              |
|        |                    | A0150  | Acquire Private Devices                              |
|        |                    | A0160  | Acquire Private Devices                              |
|        |                    | A0170  | Reset Hold Request                                   |
| T0100B | Standby Off        |        | If Task Pointer = 1 then go to step S0100            |
| T0200B | Standby On         |        | If Task Pointer = 2 then go to step S0200            |
| T0300B | Close V9           |        | If Task Pointer = 3 then go to step S0300            |
| T0400B | Close V9           |        | If Task Pointer = 4 then go to step S0400            |
| T0500B | Filling            |        | If Task Pointer = 5 then go to step S0500            |
| T0600B | Fillinf from 604C  |        | If Task Pointer = 6 then go to step S0600            |
| S0100  | Standby Off        | A0010  | Stopping and Disabling Conditional Logic for DEV1    |
|        |                    | A0020  | Closing and Disabling Conditional Logic for DEV2     |
|        |                    | A0030  | Closing DEV3                                         |

| Step   | Description | Action | Action Description                                                                       |
|--------|-------------|--------|------------------------------------------------------------------------------------------|
|        |             | A0040  | Closing and Disabling Conditional Logic for DEV4                                         |
|        |             | A0050  | Closing and Disabling Conditional Logic for DEV5                                         |
|        |             | A0060  | Closing and Disabling Conditional Logic for DEV6                                         |
|        |             | A0070  | Closing and Disabling Conditional Logic for DEV7                                         |
|        |             | A0080  | Closing and Disabling Conditional Logic for DEV8                                         |
|        |             | A0090  | Closing DEV9                                                                             |
|        |             | A0100  | Closing and Disabling Conditional Logic for DEV10                                        |
|        |             | A0110  | Closing and Disabling Conditional Logic for DEV11                                        |
|        |             | A0120  | Closing and Disabling Conditional Logic for DEV12                                        |
|        |             | A0130  | Closing and Disabling Conditional Logic for DEV13                                        |
|        |             | A0140  | Closing and Disabling Conditional Logic for DEV14                                        |
|        |             | A0150  | Closing and Disabling Conditional Logic for DEV15                                        |
|        |             | A0160  | Stopping and Disabling Conditional Logic for DEV16                                       |
|        |             | A0170  | Closing DEV17                                                                            |
|        |             | A0180  | Operator Prompt                                                                          |
| T0200A | Standby On  |        | If (Auto Mode selected) or (Manual Mode and <Standby On> selected) then go to step S0200 |
| S0200  | Standby On  | A0010  | Stopping and Disabling Conditional Logic for DEV1                                        |
|        |             | A0020  | Closing and Disabling Conditional Logic for DEV2                                         |
|        |             | A0030  | Opening DEV3                                                                             |
|        |             | A0040  | Closing and Disabling Conditional Logic for DEV4                                         |
|        |             | A0050  | Closing and Disabling Conditional Logic for DEV5                                         |
|        |             | A0060  | Closing and Disabling Conditional Logic for DEV6                                         |
|        |             | A0070  | Closing and Disabling Conditional Logic for DEV7                                         |
|        |             | A0080  | Closing and Disabling Conditional Logic for DEV8                                         |
|        |             | A0090  | Closing DEV9                                                                             |
|        |             | A0100  | Closing and Disabling Conditional Logic for DEV10                                        |

| Step   | Description | Action | Action Description                                                                                                                                                                                                                                       |
|--------|-------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        |             | A0110  | Closing and Disabling Conditional Logic for DEV11                                                                                                                                                                                                        |
|        |             | A0120  | Closing and Disabling Conditional Logic for DEV12                                                                                                                                                                                                        |
|        |             | A0130  | Closing and Disabling Conditional Logic for DEV13                                                                                                                                                                                                        |
|        |             | A0140  | Closing and Disabling Conditional Logic for DEV14                                                                                                                                                                                                        |
|        |             | A0150  | Closing and Disabling Conditional Logic for DEV15                                                                                                                                                                                                        |
|        |             | A0160  | Stopping and Disabling Conditional Logic for DEV16                                                                                                                                                                                                       |
|        |             | A0170  | Closing DEV17                                                                                                                                                                                                                                            |
|        |             | A0180  | Operator Prompt                                                                                                                                                                                                                                          |
| T0100B | Standby Off |        | If Manual mode (OP001) & step <Standby Off> selected then go to step S0100                                                                                                                                                                               |
| T0300A | Close V9    |        | If Auto mode (OP001) selected and no. of filters in step <Fill> = 0 (FILL_C ) and no. of filters in step <Filtration> < 4 (FILTRATION_C) and EM_BLEACH5 is in step <Production> and Fill From 1-604C button (OP002) is not pressed then go to step S0300 |
| T0400A | Close V9    |        | If Auto mode (OP001) selected and no. of filters in step <Fill> = 0 (FILL_C ) and no. of filters in step <Filtration> < 4 (FILTRATION_C) and EM_BLEACH5 is in step <Production> and Fill From 1-604C button (OP002) is pressed then go to step S0400     |
|        |             |        |                                                                                                                                                                                                                                                          |
| S0300  | Close V9    | A0010  | Stopping and Disabling Conditional Logic for DEV1                                                                                                                                                                                                        |
|        |             | A0020  | Closing and Disabling Conditional Logic for DEV2                                                                                                                                                                                                         |
|        |             | A0030  | Opening DEV3                                                                                                                                                                                                                                             |
|        |             | A0040  | Closing and Disabling Conditional Logic for DEV4                                                                                                                                                                                                         |
|        |             | A0050  | Closing and Disabling Conditional Logic for DEV5                                                                                                                                                                                                         |
|        |             | A0060  | Closing and Disabling Conditional Logic for DEV6                                                                                                                                                                                                         |
|        |             | A0070  | Closing and Disabling Conditional Logic for DEV7                                                                                                                                                                                                         |
|        |             | A0080  | Closing and Disabling Conditional Logic for DEV8                                                                                                                                                                                                         |
|        |             | A0090  | Closing DEV9                                                                                                                                                                                                                                             |
|        |             | A0100  | Closing and Disabling Conditional Logic for DEV10                                                                                                                                                                                                        |

| Step   | Description | Action | Action Description                                 |
|--------|-------------|--------|----------------------------------------------------|
|        |             | A0110  | Closing and Disabling Conditional Logic for DEV11  |
|        |             | A0120  | Closing and Disabling Conditional Logic for DEV12  |
|        |             | A0130  | Closing and Disabling Conditional Logic for DEV13  |
|        |             | A0140  | Closing and Disabling Conditional Logic for DEV14  |
|        |             | A0150  | Closing and Disabling Conditional Logic for DEV15  |
|        |             | A0160  | Stopping and Disabling Conditional Logic for DEV16 |
|        |             | A0170  | Closing DEV17                                      |
|        |             | A0180  | Setting Delay Time                                 |
| T0500A | Filling     |        | If 10 secs completed (TP002) then go to step S0500 |
| S0400  | Close V9    | A0010  | Stopping and Disabling Conditional Logic for DEV1  |
|        |             | A0020  | Closing and Disabling Conditional Logic for DEV2   |
|        |             | A0030  | Opening DEV3                                       |
|        |             | A0040  | Closing and Disabling Conditional Logic for DEV4   |
|        |             | A0050  | Closing and Disabling Conditional Logic for DEV5   |
|        |             | A0060  | Closing and Disabling Conditional Logic for DEV6   |
|        |             | A0070  | Closing and Disabling Conditional Logic for DEV7   |
|        |             | A0080  | Closing and Disabling Conditional Logic for DEV8   |
|        |             | A0090  | Closing DEV9                                       |
|        |             | A0100  | Closing and Disabling Conditional Logic for DEV10  |
|        |             | A0110  | Closing and Disabling Conditional Logic for DEV11  |
|        |             | A0120  | Closing and Disabling Conditional Logic for DEV12  |
|        |             | A0130  | Closing and Disabling Conditional Logic for DEV13  |
|        |             | A0140  | Closing and Disabling Conditional Logic for DEV14  |
|        |             | A0150  | Closing and Disabling Conditional Logic for DEV15  |
|        |             | A0160  | Stopping and Disabling Conditional Logic for DEV16 |
|        |             | A0170  | Closing DEV17                                      |

| Step   | Description       | Action | Action Description                                                                                                                                         |
|--------|-------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        |                   | A0180  | Setting Delay Time                                                                                                                                         |
|        |                   | A0190  | Acquire Shared Device                                                                                                                                      |
| T0600A | Filling           |        | If 10 secs completed (TP002) then go to step S0600                                                                                                         |
| S0500  | Filling           | A0010  | Enabling Conditional Logic for DEV1                                                                                                                        |
|        |                   | A0020  | Opening and Disabling Conditional Logic for DEV2                                                                                                           |
|        |                   | A0030  | Opening DEV3                                                                                                                                               |
|        |                   | A0040  | Closing and Disabling Conditional Logic for DEV4                                                                                                           |
|        |                   | A0050  | Closing and Disabling Conditional Logic for DEV5                                                                                                           |
|        |                   | A0060  | Closing and Disabling Conditional Logic for DEV6                                                                                                           |
|        |                   | A0070  | Closing and Disabling Conditional Logic for DEV7                                                                                                           |
|        |                   | A0080  | Closing and Disabling Conditional Logic for DEV8                                                                                                           |
|        |                   | A0090  | Closing DEV9                                                                                                                                               |
|        |                   | A0100  | Closing and Disabling Conditional Logic for DEV10                                                                                                          |
|        |                   | A0110  | Closing and Disabling Conditional Logic for DEV11                                                                                                          |
|        |                   | A0120  | Closing and Disabling Conditional Logic for DEV12                                                                                                          |
|        |                   | A0130  | Closing and Disabling Conditional Logic for DEV13                                                                                                          |
|        |                   | A0140  | Closing and Disabling Conditional Logic for DEV14                                                                                                          |
|        |                   | A0150  | Closing and Disabling Conditional Logic for DEV15                                                                                                          |
|        |                   | A0160  | Stopping and Disabling Conditional Logic for DEV16                                                                                                         |
|        |                   | A0170  | Closing DEV17                                                                                                                                              |
|        |                   | A0180  | Operator Prompt                                                                                                                                            |
|        |                   | A0190  | Setting Delay                                                                                                                                              |
| T9900A | Actions Complete  |        | If (Auto mode (OP001) selected & LSH_616AX is in alarm and 5 secs (TP001))<br>Or (Manual mode (OP001) selected & <Precoat> selected) then go to step S9900 |
| S0600  | Filling from 604C | A0010  | Stopping and Disabling Conditional Logic for DEV1                                                                                                          |
|        |                   | A0020  | Closing and Disabling Conditional Logic for DEV2                                                                                                           |
|        |                   | A0030  | Opening DEV3                                                                                                                                               |

| Step   | Description      | Action | Action Description                                                                                                                                                  |
|--------|------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        |                  | A0040  | Closing and Disabling Conditional Logic for DEV4                                                                                                                    |
|        |                  | A0050  | Closing and Disabling Conditional Logic for DEV5                                                                                                                    |
|        |                  | A0060  | Closing and Disabling Conditional Logic for DEV6                                                                                                                    |
|        |                  | A0070  | Closing and Disabling Conditional Logic for DEV7                                                                                                                    |
|        |                  | A0080  | Closing and Disabling Conditional Logic for DEV8                                                                                                                    |
|        |                  | A0090  | Closing DEV9                                                                                                                                                        |
|        |                  | A0100  | Closing and Disabling Conditional Logic for DEV10                                                                                                                   |
|        |                  | A0110  | Closing and Disabling Conditional Logic for DEV11                                                                                                                   |
|        |                  | A0120  | Closing and Disabling Conditional Logic for DEV12                                                                                                                   |
|        |                  | A0130  | Closing and Disabling Conditional Logic for DEV13                                                                                                                   |
|        |                  | A0140  | Opening and Disabling Conditional Logic for DEV14                                                                                                                   |
|        |                  | A0150  | Closing and Disabling Conditional Logic for DEV15                                                                                                                   |
|        |                  | A0160  | Enabling Conditional Logic for DEV16                                                                                                                                |
|        |                  | A0170  | Closing DEV17                                                                                                                                                       |
|        |                  | A0180  | Operator Prompt                                                                                                                                                     |
|        |                  | A0190  | Setting Delay                                                                                                                                                       |
| T9900B | Actions Complete |        | If (Auto mode (OP001) selected & LSH_616AX is in alarm and 5 secs (TP001)) Or (Manual mode (OP001) selected & <Precoat From 1-604C> selected) then go to step S9900 |
| S9900  | Complete         | A0010  | Set Message: Fill Command Complete                                                                                                                                  |
| END    | Command Complete |        | True                                                                                                                                                                |

Table 102: Fill Sequence

## 13.3.3 Command 02 – Filtration

| Step   | Description       | Action | Action Description                                   |
|--------|-------------------|--------|------------------------------------------------------|
| S0000  | Start             | A0010  | Set Message: Starting Filtration Command             |
| T0100  | Initializations   |        | If no pending actions in S0000 then go to step S0100 |
| T0200C | Precoat           |        | If Task Pointer = 7 then go to step S0200            |
| T0300B | Precoat from 604C |        | If Task Pointer = 8 then go to step S0300            |

| Step   | Description         | Action | Action Description                                 |
|--------|---------------------|--------|----------------------------------------------------|
| T0400C | Xfer1               |        | If Task Pointer = 9 then go to step S0400          |
| T0500B | Xfer1               |        | If Task Pointer = 10 then go to step S0500         |
| T0600B | Xfer 2              |        | If Task Pointer = 11 then go to step S0600         |
| T0700B | Xfer 2              |        | If Task Pointer = 12 then go to step S0700         |
| T0800C | Filtration          |        | If Task Pointer = 13 then go to step S0800         |
| TP900B | Final Filtration    |        | If Task Pointer = 14 then go to step S0900         |
| T1000B | Empty 604C          |        | If Task Pointer = 15 then go to step S1000         |
| T1100B | Final Recirculation |        | If Task Pointer = 16 then go to step S1100         |
| T1200E | Drain to 622        |        | If Task Pointer = 17 then go to step S1200         |
| T1300D | Drying              |        | If Task Pointer = 18 then go to step S1300         |
| T1400C | Sec. Drain          |        | If Task Pointer = 19 then go to step S1400         |
| T1500B | Depressurization    |        | If Task Pointer = 20 then go to step S1500         |
|        |                     |        |                                                    |
| S0100  | Initializations     |        | Empty                                              |
| T0200A | Precoat             |        | If Task Pointer = 5 then go to step S0200          |
| T0300A | Precoat from 604C   |        | If Task Pointer = 6 then go to step S0300          |
|        |                     |        |                                                    |
| S0200  | Precoat             | A0010  | Enabling Conditional Logic for DEV1                |
|        |                     | A0020  | Enabling Conditional Logic for DEV2                |
|        |                     | A0030  | Closing DEV3                                       |
|        |                     | A0040  | Enabling Conditional Logic for DEV4                |
|        |                     | A0050  | Closing and Disabling Conditional Logic for DEV5   |
|        |                     | A0060  | Closing and Disabling Conditional Logic for DEV6   |
|        |                     | A0070  | Closing and Disabling Conditional Logic for DEV7   |
|        |                     | A0080  | Closing and Disabling Conditional Logic for DEV8   |
|        |                     | A0090  | Closing DEV9                                       |
|        |                     | A0100  | Closing and Disabling Conditional Logic for DEV10  |
|        |                     | A0110  | Closing and Disabling Conditional Logic for DEV11  |
|        |                     | A0120  | Closing and Disabling Conditional Logic for DEV12  |
|        |                     | A0130  | Enabling Conditional Logic for DEV13               |
|        |                     | A0140  | Closing and Disabling Conditional Logic for DEV14  |
|        |                     | A0150  | Closing and Disabling Conditional Logic for DEV15  |
|        |                     | A0160  | Stopping and Disabling Conditional Logic for DEV16 |

| Step   | Description       | Action | Action Description                                                                                                                                |
|--------|-------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------|
|        |                   | A0170  | Closing DEV17                                                                                                                                     |
|        |                   | A0180  | Operator Prompt                                                                                                                                   |
|        |                   | A0190  | Set Precoat time                                                                                                                                  |
|        |                   | A0200  | Start Precoat Timer                                                                                                                               |
|        |                   | A0210  | Precoat time Message                                                                                                                              |
|        |                   | A0220  | Set bit "From Precoat"                                                                                                                            |
| T0400A | Xfer1             |        | If (Auto mode (OP001) and Precoat time elapsed (TP003)) Or (Manual mode (OP001) and <Filtration> step selected by operator) then go to step S0400 |
| S0300  | Precoat from 604C | A0010  | Stopping and Disabling Conditional Logic for DEV1                                                                                                 |
|        |                   | A0020  | Closing and Disabling Conditional Logic for DEV2                                                                                                  |
|        |                   | A0030  | Closing DEV3                                                                                                                                      |
|        |                   | A0040  | Closing and Disabling Conditional Logic for DEV4                                                                                                  |
|        |                   | A0050  | Closing and Disabling Conditional Logic for DEV5                                                                                                  |
|        |                   | A0060  | Closing and Disabling Conditional Logic for DEV6                                                                                                  |
|        |                   | A0070  | Closing and Disabling Conditional Logic for DEV7                                                                                                  |
|        |                   | A0080  | Closing and Disabling Conditional Logic for DEV8                                                                                                  |
|        |                   | A0090  | Closing DEV9                                                                                                                                      |
|        |                   | A0100  | Closing and Disabling Conditional Logic for DEV10                                                                                                 |
|        |                   | A0110  | Closing and Disabling Conditional Logic for DEV11                                                                                                 |
|        |                   | A0120  | Closing and Disabling Conditional Logic for DEV12                                                                                                 |
|        |                   | A0130  | Enabling Conditional Logic for DEV13                                                                                                              |
|        |                   | A0140  | Enabling Conditional Logic for DEV14                                                                                                              |
|        |                   | A0150  | Enabling Conditional Logic for DEV15                                                                                                              |
|        |                   | A0160  | Enabling Conditional Logic for DEV16                                                                                                              |
|        |                   | A0170  | Closing DEV17                                                                                                                                     |
|        |                   | A0180  | Operator Prompt                                                                                                                                   |
|        |                   | A0190  | Set Precoat time                                                                                                                                  |
|        |                   | A0200  | Start Precoat Timer                                                                                                                               |
|        |                   | A0210  | Precoat time Message                                                                                                                              |
|        |                   | A0220  | Set bit "From Precoat From 1-604C"                                                                                                                |
| T0500A | Xfer1             |        | If (Auto Mode (OP001) and Precoat time elapsed (TP013)) Or (Manual mode (OP001)                                                                   |

| Step   | Description | Action | Action Description                                                |
|--------|-------------|--------|-------------------------------------------------------------------|
|        |             |        | and <Filtration> step selected by operator) then go to step S0500 |
|        |             |        |                                                                   |
| S0400  | Xfer 1      | A0010  | Enabling Conditional Logic for DEV1                               |
|        |             | A0020  | Opening and Disabling Conditional Logic for DEV2                  |
|        |             | A0030  | Closing DEV3                                                      |
|        |             | A0040  | Opening and Disabling Conditional Logic for DEV4                  |
|        |             | A0050  | Closing and Disabling Conditional Logic for DEV5                  |
|        |             | A0060  | Closing and Disabling Conditional Logic for DEV6                  |
|        |             | A0070  | Closing and Disabling Conditional Logic for DEV7                  |
|        |             | A0080  | Closing and Disabling Conditional Logic for DEV8                  |
|        |             | A0090  | Closing DEV9                                                      |
|        |             | A0100  | Closing and Disabling Conditional Logic for DEV10                 |
|        |             | A0110  | Closing and Disabling Conditional Logic for DEV11                 |
|        |             | A0120  | Closing and Disabling Conditional Logic for DEV12                 |
|        |             | A0130  | Opening and Disabling Conditional Logic for DEV13                 |
|        |             | A0140  | Closing and Disabling Conditional Logic for DEV14                 |
|        |             | A0150  | Closing and Disabling Conditional Logic for DEV15                 |
|        |             | A0160  | Stopping and Disabling Conditional Logic for DEV16                |
|        |             | A0170  | Closing DEV17                                                     |
|        |             | A0180  | Setting Delay Time                                                |
| T0600A | Xfer 2      |        | If 10 secs (TP002) completed then go to step S0600                |
|        |             |        |                                                                   |
| S0500  | Xfer 1      | A0010  | Stopping and Disabling Conditional Logic for DEV1                 |
|        |             | A0020  | Closing and Disabling Conditional Logic for DEV2                  |
|        |             | A0030  | Closing DEV3                                                      |
|        |             | A0040  | Closing and Disabling Conditional Logic for DEV4                  |
|        |             | A0050  | Closing and Disabling Conditional Logic for DEV5                  |
|        |             | A0060  | Closing and Disabling Conditional Logic for DEV6                  |

| Step   | Description | Action | Action Description                                 |
|--------|-------------|--------|----------------------------------------------------|
|        |             | A0070  | Closing and Disabling Conditional Logic for DEV7   |
|        |             | A0080  | Closing and Disabling Conditional Logic for DEV8   |
|        |             | A0090  | Closing DEV9                                       |
|        |             | A0100  | Closing and Disabling Conditional Logic for DEV10  |
|        |             | A0110  | Closing and Disabling Conditional Logic for DEV11  |
|        |             | A0120  | Closing and Disabling Conditional Logic for DEV12  |
|        |             | A0130  | Opening and Disabling Conditional Logic for DEV13  |
|        |             | A0140  | Opening and Disabling Conditional Logic for DEV14  |
|        |             | A0150  | Opening and Disabling Conditional Logic for DEV15  |
|        |             | A0160  | Enabling Conditional Logic for DEV16               |
|        |             | A0170  | Closing DEV17                                      |
|        |             | A0180  | Setting Delay Time                                 |
| T0700A | Xfer 2      |        | If 10 secs (TP002) completed then go to step S0700 |
|        |             |        |                                                    |
| S0600  | Xfer 2      | A0010  | Enabling Conditional Logic for DEV1                |
|        |             | A0020  | Opening and Disabling Conditional Logic for DEV2   |
|        |             | A0030  | Closing DEV3                                       |
|        |             | A0040  | Closing and Disabling Conditional Logic for DEV4   |
|        |             | A0050  | Opening and Disabling Conditional Logic for DEV5   |
|        |             | A0060  | Closing and Disabling Conditional Logic for DEV6   |
|        |             | A0070  | Closing and Disabling Conditional Logic for DEV7   |
|        |             | A0080  | Closing and Disabling Conditional Logic for DEV8   |
|        |             | A0090  | Closing DEV9                                       |
|        |             | A0100  | Closing and Disabling Conditional Logic for DEV10  |
|        |             | A0110  | Closing and Disabling Conditional Logic for DEV11  |
|        |             | A0120  | Closing and Disabling Conditional Logic for DEV12  |
|        |             | A0130  | Opening and Disabling Conditional Logic for DEV13  |
|        |             | A0140  | Closing and Disabling Conditional Logic for DEV14  |

| Step   | Description      | Action | Action Description                                                                                               |
|--------|------------------|--------|------------------------------------------------------------------------------------------------------------------|
|        |                  | A0150  | Closing and Disabling Conditional Logic for DEV15                                                                |
|        |                  | A0160  | Stopping and Disabling Conditional Logic for DEV16                                                               |
|        |                  | A0170  | Closing DEV17                                                                                                    |
|        |                  | A0180  | Setting Delay Time                                                                                               |
| T0800A | Filtration       |        | If bit "From Precoat" (FROM_PRECOAT) is set and 10 secs (TP002) completed then go to step S0800                  |
| T0900A | Final Filtration |        | If bit "From Final Recirculation" (FROM_FNL_RECIRC) is set and 10 secs (TP002) completed then go to step S0900   |
| S0700  | Xfer 2           | A0010  | Stopping and Disabling Conditional Logic for DEV1                                                                |
|        |                  | A0020  | Closing and Disabling Conditional Logic for DEV2                                                                 |
|        |                  | A0030  | Closing DEV3                                                                                                     |
|        |                  | A0040  | Closing and Disabling Conditional Logic for DEV4                                                                 |
|        |                  | A0050  | Opening and Disabling Conditional Logic for DEV5                                                                 |
|        |                  | A0060  | Closing and Disabling Conditional Logic for DEV6                                                                 |
|        |                  | A0070  | Closing and Disabling Conditional Logic for DEV7                                                                 |
|        |                  | A0080  | Closing and Disabling Conditional Logic for DEV8                                                                 |
|        |                  | A0090  | Closing DEV9                                                                                                     |
|        |                  | A0100  | Closing and Disabling Conditional Logic for DEV10                                                                |
|        |                  | A0110  | Closing and Disabling Conditional Logic for DEV11                                                                |
|        |                  | A0120  | Closing and Disabling Conditional Logic for DEV12                                                                |
|        |                  | A0130  | Opening and Disabling Conditional Logic for DEV13                                                                |
|        |                  | A0140  | Opening and Disabling Conditional Logic for DEV14                                                                |
|        |                  | A0150  | Closing and Disabling Conditional Logic for DEV15                                                                |
|        |                  | A0160  | Enabling Conditional Logic for DEV16                                                                             |
|        |                  | A0170  | Closing DEV17                                                                                                    |
|        |                  | A0180  | Setting Delay Time                                                                                               |
| T0900A | Empty 604C       |        | If bit " From Precoat from 1-604C" (FROM_PRECOAT604C) is set and 10 secs (TP002) completed then go to step S0900 |
|        |                  |        |                                                                                                                  |

| Step   | Description         | Action | Action Description                                                                                                                                                                                                                                                                                                                     |
|--------|---------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| S0800  | Filtration          | A0010  | Enabling Conditional Logic for DEV1                                                                                                                                                                                                                                                                                                    |
|        |                     | A0020  | Enabling Conditional Logic for DEV2                                                                                                                                                                                                                                                                                                    |
|        |                     | A0030  | Closing DEV3                                                                                                                                                                                                                                                                                                                           |
|        |                     | A0040  | Closing and Disabling Conditional Logic for DEV4                                                                                                                                                                                                                                                                                       |
|        |                     | A0050  | Enabling Conditional Logic for DEV5                                                                                                                                                                                                                                                                                                    |
|        |                     | A0060  | Closing and Disabling Conditional Logic for DEV6                                                                                                                                                                                                                                                                                       |
|        |                     | A0070  | Closing and Disabling Conditional Logic for DEV7                                                                                                                                                                                                                                                                                       |
|        |                     | A0080  | Closing and Disabling Conditional Logic for DEV8                                                                                                                                                                                                                                                                                       |
|        |                     | A0090  | Closing DEV9                                                                                                                                                                                                                                                                                                                           |
|        |                     | A0100  | Closing and Disabling Conditional Logic for DEV10                                                                                                                                                                                                                                                                                      |
|        |                     | A0110  | Closing and Disabling Conditional Logic for DEV11                                                                                                                                                                                                                                                                                      |
|        |                     | A0120  | Closing and Disabling Conditional Logic for DEV12                                                                                                                                                                                                                                                                                      |
|        |                     | A0130  | Enabling Conditional Logic for DEV13                                                                                                                                                                                                                                                                                                   |
|        |                     | A0140  | Closing and Disabling Conditional Logic for DEV14                                                                                                                                                                                                                                                                                      |
|        |                     | A0150  | Closing and Disabling Conditional Logic for DEV15                                                                                                                                                                                                                                                                                      |
|        |                     | A0160  | Stopping and Disabling Conditional Logic for DEV16                                                                                                                                                                                                                                                                                     |
|        |                     | A0170  | Closing DEV17                                                                                                                                                                                                                                                                                                                          |
|        |                     | A0180  | Operator Prompt                                                                                                                                                                                                                                                                                                                        |
|        |                     | A0190  | Reset bit "From Precoat"<br>(FROM_PRECOAT)                                                                                                                                                                                                                                                                                             |
|        |                     | A0200  | Release Shared Device                                                                                                                                                                                                                                                                                                                  |
| T0200B | Precoat             |        | If Manual mode (OP001) selected and operator select <Precoat> and EM_BLEACH5 is in step <Production> or <Pre Drain> then go to step S0200                                                                                                                                                                                              |
| T1100A | Final Recirculation |        | If Auto mode (OP001) selected and EM_BLEACH5 is in step <Final Recirculation> then go to step S1100                                                                                                                                                                                                                                    |
| T1200A | Drain to 622        |        | If (Auto mode (OP001) selected and Max. BE Qty (MAX_QTY_FILTER) in filter or Max Pressure (PT_616A >= TP012) in filter and EM_BLEACH5 is in step <Production> and LI_622_21 < Safe Level to drain (TP002 of 622/21) and No other filter in step <DRAIN TO 622> (DRAIN_TO_622))<br>or<br>(Manual mode and step <Drain to 622> selected) |

| Step   | Description      | Action | Action Description                                                                                                                                                                                                                                                    |
|--------|------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        |                  |        | or<br>(Auto mode (OP001) selected and EM_BLEACH5 is in step <Pre Drain> and no. of filters in Filtration (FILTRATION_C) is > 1 and LI_622_21 < Safe Level to drain (TP002 of 622/21) and no other filter in step <Drain to 622> (DRAIN_TO_622)) then go to step S1200 |
|        |                  |        |                                                                                                                                                                                                                                                                       |
| S0900  | Final Filtration | A0010  | Enabling Conditional Logic for DEV1                                                                                                                                                                                                                                   |
|        |                  | A0020  | Enabling Conditional Logic for DEV2                                                                                                                                                                                                                                   |
|        |                  | A0030  | Closing DEV3                                                                                                                                                                                                                                                          |
|        |                  | A0040  | Closing and Disabling Conditional Logic for DEV4                                                                                                                                                                                                                      |
|        |                  | A0050  | Enabling Conditional Logic for DEV5                                                                                                                                                                                                                                   |
|        |                  | A0060  | Closing and Disabling Conditional Logic for DEV6                                                                                                                                                                                                                      |
|        |                  | A0070  | Closing and Disabling Conditional Logic for DEV7                                                                                                                                                                                                                      |
|        |                  | A0080  | Closing and Disabling Conditional Logic for DEV8                                                                                                                                                                                                                      |
|        |                  | A0090  | Closing DEV9                                                                                                                                                                                                                                                          |
|        |                  | A0100  | Closing and Disabling Conditional Logic for DEV10                                                                                                                                                                                                                     |
|        |                  | A0110  | Closing and Disabling Conditional Logic for DEV11                                                                                                                                                                                                                     |
|        |                  | A0120  | Closing and Disabling Conditional Logic for DEV12                                                                                                                                                                                                                     |
|        |                  | A0130  | Enabling Conditional Logic for DEV13                                                                                                                                                                                                                                  |
|        |                  | A0140  | Closing and Disabling Conditional Logic for DEV14                                                                                                                                                                                                                     |
|        |                  | A0150  | Closing and Disabling Conditional Logic for DEV15                                                                                                                                                                                                                     |
|        |                  | A0160  | Stopping and Disabling Conditional Logic for DEV16                                                                                                                                                                                                                    |
|        |                  | A0170  | Closing DEV17                                                                                                                                                                                                                                                         |
|        |                  | A0190  | Starting Drain command for EM_RF_OF_BLEACH5                                                                                                                                                                                                                           |
|        |                  | A0210  | Reset bit "From Final Recirculation" (FROM_FNL_RECIRC)                                                                                                                                                                                                                |
| T1200B | Drain to 622     |        | If Auto Mode (OP001) and STR_BLEACH5 is in step <Fill From Filter 616Ax> then go to step S1200                                                                                                                                                                        |
|        |                  |        |                                                                                                                                                                                                                                                                       |
| S1000  | Empty 604C       | A0010  | Stopping and Disabling Conditional Logic for DEV1                                                                                                                                                                                                                     |
|        |                  | A0020  | Closing and Disabling Conditional Logic for DEV2                                                                                                                                                                                                                      |

| Step   | Description         | Action | Action Description                                                        |
|--------|---------------------|--------|---------------------------------------------------------------------------|
|        |                     | A0030  | Closing DEV3                                                              |
|        |                     | A0040  | Closing and Disabling Conditional Logic for DEV4                          |
|        |                     | A0050  | Opening and Disabling Conditional Logic for DEV5                          |
|        |                     | A0060  | Closing and Disabling Conditional Logic for DEV6                          |
|        |                     | A0070  | Closing and Disabling Conditional Logic for DEV7                          |
|        |                     | A0080  | Closing and Disabling Conditional Logic for DEV8                          |
|        |                     | A0090  | Closing DEV9                                                              |
|        |                     | A0100  | Closing and Disabling Conditional Logic for DEV10                         |
|        |                     | A0110  | Closing and Disabling Conditional Logic for DEV11                         |
|        |                     | A0120  | Closing and Disabling Conditional Logic for DEV12                         |
|        |                     | A0130  | Closing and Disabling Conditional Logic for DEV13                         |
|        |                     | A0140  | Opening and Disabling Conditional Logic for DEV14                         |
|        |                     | A0150  | Closing and Disabling Conditional Logic for DEV15                         |
|        |                     | A0160  | Enabling Conditional Logic for DEV16                                      |
|        |                     | A0170  | Closing DEV17                                                             |
|        |                     | A0180  | Set Message: LSL_604C alarm is active and 5 Sec delay.                    |
| T0800B | Filtration          |        | If LSL_604C is in alarm and 5 sec (TP001) completed then go to step S0800 |
| S1100  | Final Recirculation | A0010  | Enabling Conditional Logic for DEV1                                       |
|        |                     | A0020  | Enabling Conditional Logic for DEV2                                       |
|        |                     | A0030  | Closing DEV3                                                              |
|        |                     | A0040  | Enabling Conditional Logic for DEV4                                       |
|        |                     | A0050  | Closing and Disabling Conditional Logic for DEV5                          |
|        |                     | A0060  | Closing and Disabling Conditional Logic for DEV6                          |
|        |                     | A0070  | Closing and Disabling Conditional Logic for DEV7                          |
|        |                     | A0080  | Closing and Disabling Conditional Logic for DEV8                          |
|        |                     | A0090  | Closing DEV9                                                              |
|        |                     | A0100  | Closing and Disabling Conditional Logic for DEV10                         |
|        |                     | A0110  | Closing and Disabling Conditional Logic for DEV11                         |

| Step   | Description  | Action | Action Description                                                                                                                                                            |
|--------|--------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        |              | A0120  | Closing and Disabling Conditional Logic for DEV12                                                                                                                             |
|        |              | A0130  | Enabling Conditional Logic for DEV13                                                                                                                                          |
|        |              | A0140  | Closing and Disabling Conditional Logic for DEV14                                                                                                                             |
|        |              | A0150  | Closing and Disabling Conditional Logic for DEV15                                                                                                                             |
|        |              | A0160  | Stopping and Disabling Conditional Logic for DEV16                                                                                                                            |
|        |              | A0170  | Closing DEV17                                                                                                                                                                 |
|        |              | A0180  | Prompt: Acknowledge prompt to proceed with Final Filtration.                                                                                                                  |
|        |              | A0190  | Set Final Recirculation time                                                                                                                                                  |
|        |              | A0200  | Start Final Recirculation Timer                                                                                                                                               |
|        |              | A0210  | Final Recirculation time Message                                                                                                                                              |
|        |              | A0220  | Set bit "From Final Recirculation" (FROM FNL RECIRC)                                                                                                                          |
| T0400B | Xfer 1       |        | If (Auto mode (OP001) selected and after Final recirculation preset time (TP004) elapsed) Or (Manual mode (OP001) and <Final Filtration> step selected) then go to step S0400 |
| S1200  | Drain to 622 | A0010  | Stopping and Disabling Conditional Logic for DEV1                                                                                                                             |
|        |              | A0020  | Closing and Disabling Conditional Logic for DEV2                                                                                                                              |
|        |              | A0030  | Closing DEV3                                                                                                                                                                  |
|        |              | A0040  | Closing and Disabling Conditional Logic for DEV4                                                                                                                              |
|        |              | A0050  | Closing and Disabling Conditional Logic for DEV5                                                                                                                              |
|        |              | A0060  | Opening and Disabling Conditional Logic for DEV6                                                                                                                              |
|        |              | A0070  | Enabling Conditional Logic for DEV7 (TP01)                                                                                                                                    |
|        |              | A0080  | Opening and Disabling Conditional Logic for DEV8                                                                                                                              |
|        |              | A0090  | Closing DEV9                                                                                                                                                                  |
|        |              | A0100  | Closing and Disabling Conditional Logic for DEV10                                                                                                                             |
|        |              | A0110  | Closing and Disabling Conditional Logic for DEV11                                                                                                                             |
|        |              | A0120  | Closing and Disabling Conditional Logic for DEV12                                                                                                                             |
|        |              | A0130  | Opening and Disabling Conditional Logic for DEV13                                                                                                                             |
|        |              | A0140  | Closing and Disabling Conditional Logic for DEV14                                                                                                                             |

| Step   | Description  | Action | Action Description                                                                                      |
|--------|--------------|--------|---------------------------------------------------------------------------------------------------------|
|        |              | A0150  | Closing and Disabling Conditional Logic for DEV15                                                       |
|        |              | A0160  | Stopping and Disabling Conditional Logic for DEV16                                                      |
|        |              | A0170  | Closing DEV17                                                                                           |
|        |              | A0180  | Prompt: Acknowledge prompt to proceed with Drying process.                                              |
|        |              | A0190  | Acquire Shared Device                                                                                   |
| Drying | T1300A       |        | If (Auto Mode Or Manual Mode and <Drying> step selected and LSL_616A is in alarm) then go to step S1300 |
| S1300  | Drying       | A0010  | Stopping and Disabling Conditional Logic for DEV1                                                       |
|        |              | A0020  | Closing and Disabling Conditional Logic for DEV2                                                        |
|        |              | A0030  | Closing DEV3                                                                                            |
|        |              | A0040  | Closing and Disabling Conditional Logic for DEV4                                                        |
|        |              | A0050  | Closing and Disabling Conditional Logic for DEV5                                                        |
|        |              | A0060  | Closing and Disabling Conditional Logic for DEV6                                                        |
|        |              | A0070  | Enabling Conditional Logic for DEV7 (TP02)                                                              |
|        |              | A0080  | Closing and Disabling Conditional Logic for DEV8                                                        |
|        |              | A0090  | Opening DEV9                                                                                            |
|        |              | A0100  | Closing and Disabling Conditional Logic for DEV10                                                       |
|        |              | A0110  | Closing and Disabling Conditional Logic for DEV11                                                       |
|        |              | A0120  | Closing and Disabling Conditional Logic for DEV12                                                       |
|        |              | A0130  | Closing and Disabling Conditional Logic for DEV13                                                       |
|        |              | A0140  | Closing and Disabling Conditional Logic for DEV14                                                       |
|        |              | A0150  | Closing and Disabling Conditional Logic for DEV15                                                       |
|        |              | A0160  | Stopping and Disabling Conditional Logic for DEV16                                                      |
|        |              | A0170  | Opening DEV17                                                                                           |
|        |              | A0180  | Operator Prompt                                                                                         |
|        |              | A0190  | Set Drying time                                                                                         |
|        |              | A0200  | Start Drying Timer                                                                                      |
|        |              | A0210  | Drying time Message                                                                                     |
| T1200C | Drain to 622 |        | If Manual Mode (OP001) and step <Drain to 622> selected and EM_BLEACH5 is in step                       |

| Step   | Description  | Action | Action Description                                                                                                                                                          |
|--------|--------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        |              |        | <Production> and LI_622_21< Safe Level to drain (TP002 of 622/21) and No other filter in step <DRAIN TO 622> (DRAIN_TO_622) then go to step S1200                           |
| T1400A | Sec. Drain   |        | If (Auto Mode (OP001) and Drying time (TP006) elapsed) Or (Manual mode (OP001) and step <Sec. Drain> selected) then go to step S1400                                        |
|        |              |        |                                                                                                                                                                             |
| S1400  | Sec. Drain   | A0010  | Stopping and Disabling Conditional Logic for DEV1                                                                                                                           |
|        |              | A0020  | Closing and Disabling Conditional Logic for DEV2                                                                                                                            |
|        |              | A0030  | Closing DEV3                                                                                                                                                                |
|        |              | A0040  | Closing and Disabling Conditional Logic for DEV4                                                                                                                            |
|        |              | A0050  | Closing and Disabling Conditional Logic for DEV5                                                                                                                            |
|        |              | A0060  | Enabling Conditional Logic for DEV6                                                                                                                                         |
|        |              | A0070  | Closing and Disabling Conditional Logic for DEV7                                                                                                                            |
|        |              | A0080  | Enabling Conditional Logic for DEV8                                                                                                                                         |
|        |              | A0090  | Closing DEV9                                                                                                                                                                |
|        |              | A0100  | Closing and Disabling Conditional Logic for DEV10                                                                                                                           |
|        |              | A0110  | Closing and Disabling Conditional Logic for DEV11                                                                                                                           |
|        |              | A0120  | Closing and Disabling Conditional Logic for DEV12                                                                                                                           |
|        |              | A0130  | Opening and Disabling Conditional Logic for DEV13                                                                                                                           |
|        |              | A0140  | Closing and Disabling Conditional Logic for DEV14                                                                                                                           |
|        |              | A0150  | Closing and Disabling Conditional Logic for DEV15                                                                                                                           |
|        |              | A0160  | Stopping and Disabling Conditional Logic for DEV16                                                                                                                          |
|        |              | A0170  | Closing DEV17                                                                                                                                                               |
|        |              | A0180  | Operator Prompt                                                                                                                                                             |
|        |              | A0190  | Set Security Drain time                                                                                                                                                     |
|        |              | A0200  | Start Security Drain Timer                                                                                                                                                  |
|        |              | A0210  | Security Drain time Message                                                                                                                                                 |
|        |              | A0220  | Release Shared Device                                                                                                                                                       |
| T1200D | Drain to 622 |        | If Manual Mode (OP001) and step <Drain to 622> selected and EM_BLEACH5 is in step <Production> and LI_622_21 < Safe Level to drain (TP002 of 622/21) and no other filter in |

| Step   | Description      | Action | Action Description                                                                                                                                                               |
|--------|------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        |                  |        | step "DRAIN TO 622" (DRAIN_TO_622) then go to step S1200                                                                                                                         |
| T1300B | Drying           |        | If Manual mode (OP001) and step <Drying> selected then go to step S1300                                                                                                          |
| T1500A | Depressurisation |        | If (Auto Mode (OP001) selected and Sec. Drain time (TP007) elapsed and LSL_616A is in alarm) Or (Manual mode (OP001) and step <Depressurisation> selected) then go to step S1500 |
|        |                  |        |                                                                                                                                                                                  |
| S1500  | Depressurisation | A0010  | Stopping and Disabling Conditional Logic for DEV1                                                                                                                                |
|        |                  | A0020  | Closing and Disabling Conditional Logic for DEV2                                                                                                                                 |
|        |                  | A0030  | Opening DEV3                                                                                                                                                                     |
|        |                  | A0040  | Closing and Disabling Conditional Logic for DEV4                                                                                                                                 |
|        |                  | A0050  | Closing and Disabling Conditional Logic for DEV5                                                                                                                                 |
|        |                  | A0060  | Closing and Disabling Conditional Logic for DEV6                                                                                                                                 |
|        |                  | A0070  | Closing and Disabling Conditional Logic for DEV7                                                                                                                                 |
|        |                  | A0080  | Closing and Disabling Conditional Logic for DEV8                                                                                                                                 |
|        |                  | A0090  | Closing DEV9                                                                                                                                                                     |
|        |                  | A0100  | Closing and Disabling Conditional Logic for DEV10                                                                                                                                |
|        |                  | A0110  | Closing and Disabling Conditional Logic for DEV11                                                                                                                                |
|        |                  | A0120  | Closing and Disabling Conditional Logic for DEV12                                                                                                                                |
|        |                  | A0130  | Closing and Disabling Conditional Logic for DEV13                                                                                                                                |
|        |                  | A0140  | Closing and Disabling Conditional Logic for DEV14                                                                                                                                |
|        |                  | A0150  | Closing and Disabling Conditional Logic for DEV15                                                                                                                                |
|        |                  | A0160  | Stopping and Disabling Conditional Logic for DEV16                                                                                                                               |
|        |                  | A0170  | Closing DEV17                                                                                                                                                                    |
|        |                  | A0180  | Operator Prompt                                                                                                                                                                  |
|        |                  | A0190  | Set Depressurisation time                                                                                                                                                        |
|        |                  | A0200  | Start Depressurisation Timer                                                                                                                                                     |
|        |                  | A0210  | Depressurization time Message                                                                                                                                                    |
| T1300C | Drying           |        | If Manual mode (OP001) and step <Drying> selected then go to step S1300                                                                                                          |

| Step   | Description      | Action | Action Description                                                                                                                                                                 |
|--------|------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1400B | Sec. Drain       |        | If Manual mode (OP001) and step <Sec. Drain> selected then go to step S1400                                                                                                        |
| T9900  | Complete         |        | If (Auto Mode and Depress. time (TP008) elapsed and confirmation button For Open Filter is pressed (OP003)) or (Manual mode and step <Open Filter> selected) then go to step T9900 |
|        |                  |        |                                                                                                                                                                                    |
| S9900  | Complete         | A0010  | Set Message: Filtration Command Complete                                                                                                                                           |
| END    | Command Complete |        | True                                                                                                                                                                               |

**Table 103: Filtration Sequence**

## 13.3.4 Command 03 – Discharge

| Step   | Description  | Action | Action Description                                |
|--------|--------------|--------|---------------------------------------------------|
| S0000  | Start        | A0010  | Set Message: Starting Discharge Command           |
|        |              | A0020  | Reset Open Filter Button (OP003)                  |
| T0100  | Open Filter  |        | If Task Pointer = 20 then go to step S0100        |
| T0200B | Discharge    |        | If Task Pointer = 21 then go to step S0200        |
| T0300B | Standby Open |        | If Task Pointer = 22 then go to step S0300        |
| T0400B | Close Filter |        | If Task Pointer = 23 then go to step S0400        |
|        |              |        |                                                   |
| S0100  | Open Filter  | A0010  | Stopping and Disabling Conditional Logic for DEV1 |
|        |              | A0020  | Closing and Disabling Conditional Logic for DEV2  |
|        |              | A0030  | Opening DEV3                                      |
|        |              | A0040  | Closing and Disabling Conditional Logic for DEV4  |
|        |              | A0050  | Closing and Disabling Conditional Logic for DEV5  |
|        |              | A0060  | Closing and Disabling Conditional Logic for DEV6  |
|        |              | A0070  | Closing and Disabling Conditional Logic for DEV7  |
|        |              | A0080  | Closing and Disabling Conditional Logic for DEV8  |
|        |              | A0090  | Closing DEV9                                      |
|        |              | A0100  | Enabling Conditional Logic for DEV10              |
|        |              | A0110  | Closing and Disabling Conditional Logic for DEV11 |
|        |              | A0120  | Closing and Disabling Conditional Logic for DEV12 |
|        |              | A0130  | Closing and Disabling Conditional Logic for DEV13 |

| Step   | Description | Action | Action Description                                                                                                                       |
|--------|-------------|--------|------------------------------------------------------------------------------------------------------------------------------------------|
|        |             | A0140  | Closing and Disabling Conditional Logic for DEV14                                                                                        |
|        |             | A0150  | Closing and Disabling Conditional Logic for DEV15                                                                                        |
|        |             | A0160  | Stopping and Disabling Conditional Logic for DEV16                                                                                       |
|        |             | A0170  | Closing DEV17                                                                                                                            |
|        |             | A0180  | Operator Prompt                                                                                                                          |
| T0200A | Discharge   |        | If Auto Mode or Manual mode (OP001) is selected and step <Discharge> selected and DEV10 open feedback is available then go to step S0200 |
| S0200  | Discharge   | A0010  | Stopping and Disabling Conditional Logic for DEV1                                                                                        |
|        |             | A0020  | Closing and Disabling Conditional Logic for DEV2                                                                                         |
|        |             | A0030  | Opening DEV3                                                                                                                             |
|        |             | A0040  | Closing and Disabling Conditional Logic for DEV4                                                                                         |
|        |             | A0050  | Closing and Disabling Conditional Logic for DEV5                                                                                         |
|        |             | A0060  | Closing and Disabling Conditional Logic for DEV6                                                                                         |
|        |             | A0070  | Closing and Disabling Conditional Logic for DEV7                                                                                         |
|        |             | A0080  | Closing and Disabling Conditional Logic for DEV8                                                                                         |
|        |             | A0090  | Closing DEV9                                                                                                                             |
|        |             | A0100  | Opening and Disabling Conditional Logic for DEV10                                                                                        |
|        |             | A0110  | Enabling Conditional Logic for DEV11                                                                                                     |
|        |             | A0120  | Enabling Conditional Logic for DEV12                                                                                                     |
|        |             | A0130  | Closing and Disabling Conditional Logic for DEV13                                                                                        |
|        |             | A0140  | Closing and Disabling Conditional Logic for DEV14                                                                                        |
|        |             | A0150  | Closing and Disabling Conditional Logic for DEV15                                                                                        |
|        |             | A0160  | Stopping and Disabling Conditional Logic for DEV16                                                                                       |
|        |             | A0170  | Closing DEV17                                                                                                                            |
|        |             | A0180  | Operator Prompt                                                                                                                          |
|        |             | A0190  | Set Discharge time                                                                                                                       |
|        |             | A0200  | Start Discharge Timer                                                                                                                    |
|        |             | A0210  | Discharge time Message                                                                                                                   |

| Step   | Description  | Action | Action Description                                                                                                                           |
|--------|--------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------|
| T0300A | Standby Open |        | If (Auto Mode and Discharge time (TP009) elapsed) or (Manual mode (OP001) and step <Standby Open> selected) then go to step S0300            |
|        |              |        |                                                                                                                                              |
| S0300  | Standby Open | A0010  | Stopping and Disabling Conditional Logic for DEV1                                                                                            |
|        |              | A0020  | Closing and Disabling Conditional Logic for DEV2                                                                                             |
|        |              | A0030  | Opening DEV3                                                                                                                                 |
|        |              | A0040  | Closing and Disabling Conditional Logic for DEV4                                                                                             |
|        |              | A0050  | Closing and Disabling Conditional Logic for DEV5                                                                                             |
|        |              | A0060  | Closing and Disabling Conditional Logic for DEV6                                                                                             |
|        |              | A0070  | Closing and Disabling Conditional Logic for DEV7                                                                                             |
|        |              | A0080  | Closing and Disabling Conditional Logic for DEV8                                                                                             |
|        |              | A0090  | Closing DEV9                                                                                                                                 |
|        |              | A0100  | Opening and Disabling Conditional Logic for DEV10                                                                                            |
|        |              | A0110  | Closing and Disabling Conditional Logic for DEV11                                                                                            |
|        |              | A0120  | Closing and Disabling Conditional Logic for DEV12                                                                                            |
|        |              | A0130  | Closing and Disabling Conditional Logic for DEV13                                                                                            |
|        |              | A0140  | Closing and Disabling Conditional Logic for DEV14                                                                                            |
|        |              | A0150  | Closing and Disabling Conditional Logic for DEV15                                                                                            |
|        |              | A0160  | Stopping and Disabling Conditional Logic for DEV16                                                                                           |
|        |              | A0170  | Closing DEV17                                                                                                                                |
|        |              | A0180  | Operator Prompt                                                                                                                              |
|        |              | A0190  | Set Standby Open time                                                                                                                        |
|        |              | A0200  | Start Standby Open Timer                                                                                                                     |
|        |              | A0210  | Standby Open time Message                                                                                                                    |
| T0400A | Close Filter |        | If (Auto Mode (OP001) and Standby Open time (TP010) elapsed) or (Manual mode (OP001) and step <Close Filter> selected) then go to step S0400 |
|        |              |        |                                                                                                                                              |
| S0400  | Close Filter | A0010  | Stopping and Disabling Conditional Logic for DEV1                                                                                            |

| Step  | Description   | Action | Action Description                                                                                                                        |
|-------|---------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------|
|       |               | A0020  | Closing and Disabling Conditional Logic for DEV2                                                                                          |
|       |               | A0030  | Opening DEV3                                                                                                                              |
|       |               | A0040  | Closing and Disabling Conditional Logic for DEV4                                                                                          |
|       |               | A0050  | Closing and Disabling Conditional Logic for DEV5                                                                                          |
|       |               | A0060  | Closing and Disabling Conditional Logic for DEV6                                                                                          |
|       |               | A0070  | Closing and Disabling Conditional Logic for DEV7                                                                                          |
|       |               | A0080  | Closing and Disabling Conditional Logic for DEV8                                                                                          |
|       |               | A0090  | Closing DEV9                                                                                                                              |
|       |               | A0100  | Closing and Disabling Conditional Logic for DEV10                                                                                         |
|       |               | A0110  | Closing and Disabling Conditional Logic for DEV11                                                                                         |
|       |               | A0120  | Closing and Disabling Conditional Logic for DEV12                                                                                         |
|       |               | A0130  | Closing and Disabling Conditional Logic for DEV13                                                                                         |
|       |               | A0140  | Closing and Disabling Conditional Logic for DEV14                                                                                         |
|       |               | A0150  | Closing and Disabling Conditional Logic for DEV15                                                                                         |
|       |               | A0160  | Stopping and Disabling Conditional Logic for DEV16                                                                                        |
|       |               | A0170  | Closing DEV17                                                                                                                             |
|       |               | A0180  | Operator Prompt                                                                                                                           |
| T0500 | Close Devices |        | If Auto Mode or Manual mode (OP001) is selected and step <Standby Off> selected and DEV10 closed feedback available then go to step S0500 |
|       |               |        |                                                                                                                                           |
| S0500 | Close Devices | A0010  | Stopping and Disabling Conditional Logic for DEV1                                                                                         |
|       |               | A0020  | Closing and Disabling Conditional Logic for DEV2                                                                                          |
|       |               | A0030  | Closing DEV3                                                                                                                              |
|       |               | A0040  | Closing and Disabling Conditional Logic for DEV4                                                                                          |
|       |               | A0050  | Closing and Disabling Conditional Logic for DEV5                                                                                          |
|       |               | A0060  | Closing and Disabling Conditional Logic for DEV6                                                                                          |
|       |               | A0070  | Closing and Disabling Conditional Logic for DEV7                                                                                          |

| Step  | Description     | Action | Action Description                                   |
|-------|-----------------|--------|------------------------------------------------------|
|       |                 | A0080  | Closing and Disabling Conditional Logic for DEV8     |
|       |                 | A0090  | Closing DEV9                                         |
|       |                 | A0100  | Closing and Disabling Conditional Logic for DEV10    |
|       |                 | A0110  | Closing and Disabling Conditional Logic for DEV11    |
|       |                 | A0120  | Closing and Disabling Conditional Logic for DEV12    |
|       |                 | A0130  | Closing and Disabling Conditional Logic for DEV13    |
|       |                 | A0140  | Closing and Disabling Conditional Logic for DEV14    |
|       |                 | A0150  | Closing and Disabling Conditional Logic for DEV15    |
|       |                 | A0160  | Stopping and Disabling Conditional Logic for DEV16   |
|       |                 | A0170  | Closing DEV17                                        |
| T0600 | Disable HM      |        | If no pending actions in S0500 then go to step S0600 |
|       |                 |        |                                                      |
| S0600 | Disable HM      | A0010  | Disable HM 01 (Module Ownership Lost)                |
| T0700 | Action Complete |        | If no pending actions in S0600 then go to step S0700 |
|       |                 |        |                                                      |
| S0700 | Release Devices | A0010  | Release Devices Message                              |
|       |                 | A0020  | Release Private Devices                              |
|       |                 | A0030  | Release Private Devices                              |
|       |                 | A0040  | Release Private Devices                              |
|       |                 | A0050  | Release Private Devices                              |
|       |                 | A0060  | Release Private Devices                              |
|       |                 | A0070  | Release Private Devices                              |
|       |                 | A0080  | Release Private Devices                              |
|       |                 | A0090  | Release Private Devices                              |
|       |                 | A0100  | Release Private Devices                              |
|       |                 | A0110  | Release Private Devices                              |
|       |                 | A0120  | Release Private Devices                              |
|       |                 | A0130  | Release Private Devices                              |
|       |                 | A0140  | Release Private Devices                              |
|       |                 | A0150  | Release Private Devices                              |
|       |                 | A0160  | Release Private Devices                              |
| T9900 | Complete        |        | If no pending actions in S0700 then go to step S9900 |
|       |                 |        |                                                      |

| Step  | Description      | Action | Action Description                      |
|-------|------------------|--------|-----------------------------------------|
| S9900 | Complete         | A0010  | Set Message: Discharge Command Complete |
|       |                  | A0020  | Self Release the EM                     |
| END   | Command Complete |        | True                                    |

**Table 104: Discharge Sequence**

## 13.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

| Parameter            | Description                        | Type           | Units | Default Value/Reference |
|----------------------|------------------------------------|----------------|-------|-------------------------|
| Operating Parameters |                                    |                |       |                         |
| OP001                | MODE                               | Boolean        | NA    | MAN = 0<br>(AUTO = 1)   |
| OP002                | FILL FROM 604C                     | Floating Point | NA    | 0                       |
| OP003                | Confirm Open Filter                | Named Set      | NA    | NO                      |
| Tuning Parameters    |                                    |                |       |                         |
| TP001                | Delay Timer                        | Floating Point | Sec   | 5                       |
| TP002                | Delay Timer                        | Floating Point | Sec   | 10                      |
| TP003                | Preset Precoat Time                | Floating Point | Sec   | 400                     |
| TP004                | Final Recirculation Preset Time    | Floating Point | Sec   | 400                     |
| TP005                | Delay Timer                        | Floating Point | Sec   | 3                       |
| TP006                | Drying Preset Time                 | Floating Point | Sec   | 300                     |
| TP007                | Security Drain Preset Time         | Floating Point | Sec   | 300                     |
| TP008                | Depressurization Preset Time       | Floating Point | Sec   | 200                     |
| TP009                | Discharge Preset Time              | Floating Point | Sec   | 250                     |
| TP010                | Standby Open Preset Time           | Floating Point | Sec   | 250                     |
| TP011                | Maximum Bleaching Earth            | Floating Point | kg    | 250                     |
| TP012                | PI_616A1-5                         | Floating Point | barg  | 3.5                     |
| TP013                | Precoating from 1-604C Preset Time | Floating Point | Sec   | 400                     |

**Table 105: Operating and Tuning Parameters**

## 13.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

| Parameter Name    | Parameter Type     | Value/Reference                      |                            |                            |                            |                            |
|-------------------|--------------------|--------------------------------------|----------------------------|----------------------------|----------------------------|----------------------------|
| MY_TAG            |                    | RF1EQ_616A1                          | RF1EQ_616A2                | RF1EQ_616A3                | RF1EQ_616A4                | RF1EQ_616A5                |
| OPR_CMD_MASK      |                    | 15                                   |                            |                            |                            |                            |
| LSH_616AX         | External Reference | RF1_LS H1616A/DI1/DISC_ACT           | RF1_LS H1616B/DI1/DISC_ACT | RF1_LS H1616C/DI1/DISC_ACT | RF1_LS H1616D/DI1/DISC_ACT | RF1_LS H1616E/DI1/DISC_ACT |
| STR_BLEACH5       | String             | RF1EQ_622_21                         |                            |                            |                            |                            |
| LSLL_622          | External Reference | RF1_LSLL1622/DI1/DISC_ACT            |                            |                            |                            |                            |
| LSL_604C          | External Reference | RF1_LSL1604C/DI1/DISC_ACT            |                            |                            |                            |                            |
| PT_616A           | External Reference | RF1_PT 1616A1/AI1/PV                 | RF1_PT 1616A2/AI1/PV       | RF1_PT 1616A3/AI1/PV       | RF1_PT 1616A4/AI1/PV       | RF1_PT 1616A5/AI1/PV       |
| LSL_616A          | External Reference | RF1_LS L1616A/DI1/DISC_ACT           | RF1_LS L1616B/DI1/DISC_ACT | RF1_LS L1616C/DI1/DISC_ACT | RF1_LS L1616D/DI1/DISC_ACT | RF1_LS L1616E/DI1/DISC_ACT |
| LI_622_21         | External Reference | RF1_LT1622/AI1/PV                    |                            |                            |                            |                            |
| EM_RF_FLT_VPRD6   | External Reference | RF1EQ_604C/COMMAND_00001/ACT_STEPS   |                            |                            |                            |                            |
| EM_BLEACH5        | External Reference | RF1EQ_622_21/COMMAND_00001/ACT_STEPS |                            |                            |                            |                            |
| HOLD              | External Reference | RF1EQ_STARTUP3/OP005_VALUE           |                            |                            |                            |                            |
| FROM_PRE_COAT     | Boolean            | False                                |                            |                            |                            |                            |
| FROM_FNL_RECIRC   | Boolean            | False                                |                            |                            |                            |                            |
| FROM_PRE_COAT604C | Boolean            | False                                |                            |                            |                            |                            |
| DRAIN_TO_622      | External Reference | RF1_MISC/DRAIN_TO_622_C              |                            |                            |                            |                            |
| STEP_TIME         | Floating Point     | 0                                    |                            |                            |                            |                            |



| Parameter Name  | Parameter Type     | Value/Reference       |
|-----------------|--------------------|-----------------------|
| COMPLETE_D_TIME | Floating Point     | 0                     |
| RESET_COMP_TM   | Boolean            | False                 |
| FILTRATION_C    | External Reference | RF1_MISC/FILTRATION_C |
| FILL_C          | External Reference | RF1_MISC/FILL_C       |
| DY_REF_1        | Dynamic Reference  |                       |
| DY_REF_2        | Dynamic Reference  |                       |
| DY_REF_3        | Dynamic Reference  |                       |
| MAX_QTY_FILTER  | External Reference | RF1_MISC/MAX_BE_QTY   |

**Table 106: Configurable Parameters**

## 13.6 Named Set

Following Named set shall be used for configuration of EM.

| Name             | Description                                     |
|------------------|-------------------------------------------------|
| NS_RF_FILTRATION | Refinery Area: Filtration and Vacuum Production |

**Table 107: Named Set Details**

Details of states for NS\_RF\_FILTRATION named set are as follows,

| Named Set Value | Named Set Name |
|-----------------|----------------|
| 00              | Hold           |
| 01              | Fill           |
| 02              | Filtration     |
| 03              | Discharge      |

**Table 108: Name Set State Details**

## 14 EQUIPMENT MODULE CLASS DESIGN – EM\_RF\_FLT\_VPRD1 (1-616S)

### 14.1 Functional Description

EM\_RF\_FLT\_VPRD1 is a class based equipment module that performs filtration and vacuum production. This EM performs Filling and Filtration Operation and drains the material based on the Operator command. EM\_RF\_FLT\_VPRD1 is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. Initially Operator shall be provided with Hold and Filtration Command. All devices used in EM shall be acquired in Filtration Command and should be released on Completion of Filtration Command. Hold Request shall be enabled for all acquired devices at the end of Hold command.

### 14.2 Instances

The following Table contains the module instances of the equipment module class.

| Device Name | Module Class    | Instance Name |
|-------------|-----------------|---------------|
| EM          | EM_RF_FLT_VPRD1 | RF1EQ_616S    |
| DEV1        | _V12_IP_130_SE  | RF1_V1611S1   |
| DEV2        | _V12_IP_130_SE  | RF1_V1612S1   |
| DEV3        | _V12_IP_130_SE  | RF1_V1614S1   |
| DEV4        | _V12_IP_130_SE  | Ignored       |
| DEV5        | _V12_IP_130_SE  | RF1_V1616S1   |
| DEV6        | _V12_IP_130_SE  | RF1_V1619S1   |
| DEV7        | _V12_IP_130_SE  | RF1_V1610S1   |
| DEV8        | _V12_IP_130_SE  | RF1_V1682BBY  |

**Table 109: Instances**

### 14.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

#### 14.3.1 Command 00 – Hold

| Step  | Description                  | Action | Action Description      |
|-------|------------------------------|--------|-------------------------|
| H0010 | Command Hold Start           | A0010  | Store Fail Message      |
|       |                              | A0020  | Reset Hold Request      |
|       |                              | A0030  | Save Previous Step Time |
| T1000 | Actions Complete or Time Out |        | Go to step H1000        |
|       |                              |        |                         |

| Step   | Description                               | Action | Action Description      |
|--------|-------------------------------------------|--------|-------------------------|
| H1000  | Hold Command 1                            | A0010  | Closing DEV1            |
|        |                                           | A0020  | Closing DEV2            |
|        |                                           | A0030  | Closing DEV3            |
|        |                                           | A0040  | Closing DEV4            |
|        |                                           | A0050  | Closing DEV5            |
|        |                                           | A0060  | Closing DEV6            |
|        |                                           | A0070  | Closing DEV7            |
|        |                                           | A0080  | Closing DEV8            |
| T9900A | Actions Complete or Time Out & Hold=True  |        | Go to step H9900A       |
| T9900B | Actions Complete or Time Out & Hold=False |        | Go to step H9900B       |
|        |                                           |        |                         |
| H9900A | Command Hold End                          | A0010  | Completed               |
| END    | Command Complete                          |        | True                    |
|        |                                           |        |                         |
| H9900B | Release Devices                           | A0010  | Release Private Devices |
|        |                                           | A0020  | Release Private Devices |
|        |                                           | A0030  | Release Private Devices |
|        |                                           | A0040  | Release Private Devices |
|        |                                           | A0050  | Release Private Devices |
|        |                                           | A0060  | Release Private Devices |
|        |                                           | A0070  | Release Private Devices |
|        |                                           | A0080  | Release Private Devices |
|        |                                           | A0090  | Reset Parameters        |
| END1   | Command Complete                          |        | True                    |

Table 110: Hold Sequence

## 14.3.2 Command 01 – Filtration

| Step   | Description        | Action | Action Description                          |
|--------|--------------------|--------|---------------------------------------------|
| S0000  | Start              | A0010  | Set Message: Starting Filtration Command    |
| T0010  | First Pass = False |        | If First Pass = False then go to step S0010 |
| T0020A | First Pass = True  |        | If First Pass = True then go to step S0020  |
|        |                    |        |                                             |
| S0010  | Initializations    | A0010  | First Pass Logic                            |

| Step   | Description      | Action | Action Description                                                                                                        |
|--------|------------------|--------|---------------------------------------------------------------------------------------------------------------------------|
|        |                  | A0020  | First Pass = True                                                                                                         |
| T0020B | Actions Complete |        | If no pending actions in S0010 then go to step S0020                                                                      |
|        |                  |        |                                                                                                                           |
| S0020  | Acquire Devices  | A0010  | Acquire Devices Message                                                                                                   |
|        |                  | A0020  | Acquire Private Devices                                                                                                   |
|        |                  | A0030  | Acquire Private Devices                                                                                                   |
|        |                  | A0040  | Acquire Private Devices                                                                                                   |
|        |                  | A0050  | Acquire Private Devices                                                                                                   |
|        |                  | A0060  | Acquire Private Devices                                                                                                   |
|        |                  | A0070  | Acquire Private Devices                                                                                                   |
|        |                  | A0080  | Acquire Private Devices                                                                                                   |
|        |                  | A0090  | Acquire Private Devices                                                                                                   |
|        |                  | A0110  | Reset HOLD_REQ                                                                                                            |
| T0100  | Fill             |        | If Task Pointer = 1 then go to step S0100                                                                                 |
| T0200A | Filtration       |        | If Task Pointer = 2 then go to step S0200A                                                                                |
| T0300A | Drain            |        | If Task Pointer = 3 then go to step S0300A                                                                                |
| T0400A | Drain Wait       |        | If Task Pointer = 4 then go to step S0400A                                                                                |
|        |                  |        |                                                                                                                           |
| S0100  | Fill             | A0010  | Opening DEV1                                                                                                              |
|        |                  | A0020  | Opening DEV2                                                                                                              |
|        |                  | A0030  | Closing DEV3                                                                                                              |
|        |                  | A0040  | Closing DEV4                                                                                                              |
|        |                  | A0050  | Closing DEV5                                                                                                              |
|        |                  | A0060  | Closing DEV6                                                                                                              |
|        |                  | A0070  | Closing DEV7                                                                                                              |
|        |                  | A0080  | Closing DEV8                                                                                                              |
| T0200  | Filtration       |        | If LSH_616S is in alarm then go to step S0200                                                                             |
|        |                  |        |                                                                                                                           |
| S0200  | Filtration       | A0010  | Opening DEV1                                                                                                              |
|        |                  | A0020  | Closing DEV2                                                                                                              |
|        |                  | A0030  | Opening DEV3                                                                                                              |
|        |                  | A0040  | Opening DEV4                                                                                                              |
|        |                  | A0050  | Closing DEV5                                                                                                              |
|        |                  | A0060  | Closing DEV6                                                                                                              |
|        |                  | A0070  | Closing DEV7                                                                                                              |
|        |                  | A0080  | Closing DEV8                                                                                                              |
|        |                  | A0090  | Checking other EM states                                                                                                  |
| T0300  | Drain            |        | If PT_616S > SV (TP004 ) or VPRD2_OWNER is in step <Standby> and BLEACH5_OWNER is in step <Standby> then go to step S0300 |
|        |                  |        |                                                                                                                           |

| Step   | Description   | Action | Action Description                                                                                                                                                |
|--------|---------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| S0300  | Drain         | A0010  | Closing DEV1                                                                                                                                                      |
|        |               | A0020  | Closing DEV2                                                                                                                                                      |
|        |               | A0030  | Closing DEV3                                                                                                                                                      |
|        |               | A0040  | Opening DEV4                                                                                                                                                      |
|        |               | A0050  | Enabling Conditional Logic for DEV5                                                                                                                               |
|        |               | A0060  | Opening DEV6                                                                                                                                                      |
|        |               | A0070  | Opening DEV7                                                                                                                                                      |
|        |               | A0080  | Opening DEV8                                                                                                                                                      |
|        |               | A0090  | Set Message                                                                                                                                                       |
|        |               | A0100  | Set Drain Wait Recirculation time                                                                                                                                 |
|        |               | A0110  | Start Drain Wait Timer                                                                                                                                            |
|        |               | A0120  | Preset Drain time Message                                                                                                                                         |
| T0400  | Drain Wait    |        | If LSL_616S is in alarm or Drain timer (TP001) is elapsed then go to step S0400                                                                                   |
| S0400  | Drain Wait    | A0010  | Closing DEV1                                                                                                                                                      |
|        |               | A0020  | Closing DEV2                                                                                                                                                      |
|        |               | A0030  | Closing DEV3                                                                                                                                                      |
|        |               | A0040  | Closing DEV4                                                                                                                                                      |
|        |               | A0050  | Closing and Disabling Conditional Logic for DEV5                                                                                                                  |
|        |               | A0060  | Closing DEV6                                                                                                                                                      |
|        |               | A0070  | Closing DEV7                                                                                                                                                      |
|        |               | A0080  | Opening DEV8                                                                                                                                                      |
| T0100B | Fill          |        | If BLEACH5_STEP is in step <Production> or <Pre Drain> or <Final Recirculation> or <Drain> or <Fill from Filter 616Ax> or <Empty> or <Blow> then go to step S0100 |
| T0500  | Close Devices |        | If VPRD2_OWNER is in step <Standby> and BLEACH5_OWNER in Step <Standby> then go to step S0500                                                                     |
| S0500  | Close Devices | A0010  | Closing DEV1                                                                                                                                                      |
|        |               | A0020  | Closing DEV2                                                                                                                                                      |
|        |               | A0030  | Closing DEV3                                                                                                                                                      |
|        |               | A0040  | Closing DEV4                                                                                                                                                      |
|        |               | A0050  | Closing DEV5                                                                                                                                                      |
|        |               | A0060  | Closing DEV6                                                                                                                                                      |
|        |               | A0070  | Closing DEV7                                                                                                                                                      |
|        |               | A0080  | Closing DEV8                                                                                                                                                      |
| T0600  | Disable HM    |        | If no pending actions in S0500 then go to step S0600                                                                                                              |
| S0600  | Disable HM    | A0010  | Disable HM 01 (Module Ownership Lost)                                                                                                                             |

| Step  | Description      | Action | Action Description                                   |
|-------|------------------|--------|------------------------------------------------------|
| T0700 | Release Devices  |        | If no pending actions in S0600 then go to step S0700 |
| S0700 | Release Devices  | A0010  | Release Private Devices                              |
|       |                  | A0020  | Release Private Devices                              |
|       |                  | A0030  | Release Private Devices                              |
|       |                  | A0040  | Release Private Devices                              |
|       |                  | A0050  | Release Private Devices                              |
|       |                  | A0060  | Release Private Devices                              |
|       |                  | A0070  | Release Private Devices                              |
|       |                  | A0080  | Release Private Devices                              |
| T9900 | Complete         |        | If no pending actions in S0700 then go to step S9900 |
| S9900 | Complete         | A0010  | Set Message: Filtration Command Complete             |
|       |                  | A0020  | Self release the EM                                  |
| END   | Command Complete |        | True                                                 |

Table 111: Filtration Sequence

## 14.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

| Parameter            | Description    | Type           | Units | Default Value/Reference |
|----------------------|----------------|----------------|-------|-------------------------|
| Operating Parameters |                |                |       |                         |
| N/A                  |                |                |       |                         |
| Tuning Parameters    |                |                |       |                         |
| TP001                | Drain time     | Floating Point | Sec   | 400                     |
| TP002                | DEV5 Open time | Floating Point | Sec   | 400                     |
| TP003                | DEV7 Open time | Floating Point | Sec   | 400                     |
| TP004                | PT_616S        | Floating Point | barg  | 3.5                     |

Table 112: Operating and Tuning Parameters

## 14.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

| Parameter Name | Parameter Type | Value/Reference |
|----------------|----------------|-----------------|
| MY_TAG         |                | RF1EQ_616S      |

| Parameter Name   | Parameter Type     | Value/Reference                    |
|------------------|--------------------|------------------------------------|
| OPR_CMD_MASK     |                    | 3                                  |
| HOLD             | External Reference | RF1EQ_STARTUP3/OP005_V<br>ALUE     |
| STEP_TIME        | Floating Point     | 0                                  |
| COMPLETED_TIME   | Floating Point     | 0                                  |
| RESET_COMP_TM    | Boolean            | False                              |
| VPRD2_OWNER      | External Reference | RF1EQ_682B/OWNER_ID                |
| BLEACH5_OWNER    | External Reference | RF1EQ_622_21/OWNER_ID              |
| LSH_616S         | External Reference | RF1_LSH1616S/DI1/DISC_AC<br>T      |
| LSL_616S         | External Reference | RF1_LSL1616S/DI1/DISC_ACT          |
| PT_616S          | External Reference | RF1_PT1616S/PV                     |
| BLEACH5_STEP     | External Reference | RF1EQ_622_21/STEP_NAME             |
| V1616S_FAIL_STAT | External Reference | RF1_V1616S1/EDC1/FAILURE<br>_STATE |

Table 113: Configurable Parameters

## 14.6 Named Set

Following Named set shall be used for configuration of EM.

| Name            | Description                                     |
|-----------------|-------------------------------------------------|
| NS_RF_FLT_VPRD1 | Refinery Area:Filtration & Vacuum Production EM |

Table 114: Named Set Details

Details of states for NS\_RF\_FLT\_VPRD1 named set are as follows,

| Named Set Value | Named Set Name |
|-----------------|----------------|
| 00              | Hold           |
| 01              | Filtration     |

Table 115: Name Set State Details

## 15 EQUIPMENT MODULE CLASS DESIGN – EM\_RF\_FLT\_VPRD2 (1-682B)

### 15.1 Functional Description

EM\_RF\_FLT\_VPRD2 is a class based equipment module that performs filling operation and drains the material to complete vacuum production. EM\_RF\_FLT\_VPRD2 is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. Operator shall be provided with Hold and Production commands. Hold Request shall be enabled for all acquired devices at the end of Hold command.

### 15.2 Instances

The following Table contains the module instances of the equipment module class.

| Device Name | Module Class    | Instance Name |
|-------------|-----------------|---------------|
| EM          | EM_RF_FLT_VPRD2 | RF1EQ_682B    |
| DEV1        | _MRSS11_130_SE  | RF1_PU1682B   |
| DEV2        | _V12_IP_130_SE  | RF1_V1692BD   |
| DEV3        | _V12_IP_130_SE  | RF1_V1682BBS  |
| DEV4        | _CSD_130_SE     | RF1_LT1622    |
| DEV5        | EM_RF_FLT_VPRD1 | RF1EQ_616S    |

Table 116: Instances

### 15.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

#### 15.3.1 Command 00 – Hold

| Step  | Description        | Action | Action Description          |
|-------|--------------------|--------|-----------------------------|
| H0010 | Command Hold Start | A0010  | Store Fail Message          |
|       |                    | A0020  | Reset Hold Request          |
|       |                    | A0030  | Save Previous Step Time     |
| T1000 | Actions Complete   |        | Go to step H1000            |
|       |                    |        |                             |
| H1000 | Hold Command 1     | A0010  | Stopping DEV1               |
|       |                    | A0020  | Closing DEV2                |
|       |                    | A0030  | Closing DEV3                |
|       |                    | A0040  | Set DEV4 Mode to Remote Out |
|       |                    | A0050  | Set DEV4 output to Closed   |

| Step   | Description                               | Action | Action Description      |
|--------|-------------------------------------------|--------|-------------------------|
| T9900A | Actions Complete or Time Out & Hold=True  |        | Go to step H9900A       |
| T9900B | Actions Complete or Time Out & Hold=False |        | Go to step H9900B       |
|        |                                           |        |                         |
| H9900A | Command Hold End                          | A0010  | Completed               |
| END    | Command Complete                          |        | True                    |
|        |                                           |        |                         |
| H9900B | Release Devices                           | A0010  | Release Private Devices |
|        |                                           | A0020  | Release Private Devices |
|        |                                           | A0030  | Release Private Devices |
|        |                                           | A0040  | Release Private Devices |
|        |                                           | A0050  | Reset Parameters        |
| END1   | Command Complete                          |        | True                    |

Table 117: Hold Sequence

## 15.3.2

## Command 01 – Production

| Step   | Description        | Action | Action Description                                   |
|--------|--------------------|--------|------------------------------------------------------|
| S0000  | Start              | A0010  | Set Message: Starting Production Command             |
| T0010  | First Pass = False |        | If First Pass = False then go to step S0010          |
| T0020A | First Pass = True  |        | If First Pass = True then go to step S0020           |
|        |                    |        |                                                      |
| S0010  | Initializations    | A0010  | First Pass Logic                                     |
|        |                    | A0020  | First Pass = True                                    |
| T0020B | Actions Complete   |        | If no pending actions in S0010 then go to step S0020 |
|        |                    |        |                                                      |
|        |                    |        |                                                      |
| S0020  | Acquire Devices    | A0010  | Acquire Devices Message                              |
|        |                    | A0020  | Acquire Private Devices                              |
|        |                    | A0030  | Acquire Shared Devices                               |
|        |                    | A0040  | Acquire Private Devices                              |
|        |                    | A0050  | Acquire Private Devices                              |
|        |                    | A0060  | Acquire Private Devices                              |
|        |                    | A0080  | Reset HOLD_REQ                                       |

| Step   | Description | Action | Action Description                                                                  |
|--------|-------------|--------|-------------------------------------------------------------------------------------|
| T0100  | Fill        |        | If Task Pointer = 1 then go to step S0100                                           |
| T0200A | Production  |        | If Task Pointer = 2 then go to step S0200                                           |
| T0300A | Drain       |        | If Task Pointer = 3 then go to step S0300                                           |
| T0400A | Drain wait  |        | If Task Pointer = 4 then go to step S0400                                           |
|        |             |        |                                                                                     |
| S0100  | Fill        | A0010  | Stopping DEV1                                                                       |
|        |             | A0020  | Opening DEV2                                                                        |
|        |             | A0030  | Closing DEV3                                                                        |
|        |             | A0040  | Set DEV4 Mode to Remote Out                                                         |
|        |             | A0050  | Set DEV4 output to Closed                                                           |
|        |             | A0055  | LSL_682B is not in alarm then S0200                                                 |
|        |             | A0060  | Set Delay Time                                                                      |
|        |             | A0070  | Start Delay Time                                                                    |
|        |             | A0080  | Delay Time Message                                                                  |
| T0200  | Production  |        | If LSL_682B is not in alarm and 1 minute (TP003) is completed then go to step S0200 |
|        |             |        |                                                                                     |
| S0200  | Production  | A0010  | Enabling Conditional Logic for DEV1                                                 |
|        |             | A0020  | Opening DEV2                                                                        |
|        |             | A0030  | Closing DEV3                                                                        |
|        |             | A0040  | Set DEV4 Mode to Remote Cascade                                                     |
|        |             | A0050  | Set DEV4 Setpoint to Remote Setpoint for LT_622                                     |
|        |             | A0060  | Checking Private Devices 5 Owner ID                                                 |
|        |             | A0070  | Acquire Private Devices DEV5                                                        |
|        |             | A0080  | Starting DEV5                                                                       |
|        |             | A0090  | Checking EM_RF_OF_BLEACH5 is in Fill from Filter-616Ax step                         |
| T0300  | Drain       |        | If RF_OF_BLEACH5 is in step <Fill From Filter 616Ax> then go to step S0300          |
|        |             |        |                                                                                     |
| S0300  | Drain       | A0010  | Enabling Conditional Logic for DEV1                                                 |
|        |             | A0020  | Opening DEV2                                                                        |
|        |             | A0030  | Closing DEV3                                                                        |
|        |             | A0040  | Set DEV4 Mode to Remote Out                                                         |
|        |             | A0050  | Set DEV4 output to Full open                                                        |
|        |             | A0060  | Delay Time                                                                          |
|        |             | A0070  | Delay Timer Message                                                                 |
|        |             | A0080  | Checking LSL_682B is in alarm                                                       |
| T0400  | Drain Wait  |        | If LSL_682B is in alarm and 10 secs is over then go to step S0400                   |
|        |             |        |                                                                                     |

| Step  | Description      | Action | Action Description                                                             |
|-------|------------------|--------|--------------------------------------------------------------------------------|
| S0400 | Drain Wait       | A0010  | Stopping and Disabling Conditional Logic for DEV1 (DEV1)                       |
|       |                  | A0020  | Closing DEV2                                                                   |
|       |                  | A0030  | Opening DEV3                                                                   |
|       |                  | A0040  | Set DEV4 Mode to Remote Out                                                    |
|       |                  | A0050  | Set DEV4 output to Full open                                                   |
|       |                  | A0060  | Wait for Blow Time                                                             |
|       |                  | A0070  | Set Blowing Time                                                               |
|       |                  | A0080  | Start Preset Blowing Time                                                      |
|       |                  | A0090  | Delay Timer Message                                                            |
| T0500 | Close Devices    |        | If DEV3 blowing timer elapsed and 5 secs (TP001) is over then go to step S0500 |
|       |                  |        |                                                                                |
| S0500 | Close Devices    | A0010  | Stopping and Disabling Conditional Logic for DEV1                              |
|       |                  | A0020  | Closing DEV2                                                                   |
|       |                  | A0030  | Closing DEV3                                                                   |
|       |                  | A0040  | Set DEV4 Mode to Remote Out                                                    |
|       |                  | A0050  | Set DEV4 output to Full close                                                  |
| T0600 | Disable HM       |        | If no pending action in S0500 then go to step S0600                            |
|       |                  |        |                                                                                |
| S0600 | Disable HM       | A0010  | Disable HM 01 (Module Ownership Lost)                                          |
| T0700 | Release Devices  |        | If no pending action in S0600 then go to step S0700                            |
|       |                  |        |                                                                                |
| S0700 | Release Devices  | A0010  | Release Private Devices                                                        |
|       |                  | A0020  | Release Private Devices                                                        |
|       |                  | A0030  | Release Private Devices                                                        |
|       |                  | A0040  | Release Private Devices                                                        |
| T9900 | Complete         |        | If no pending action in S0700 then go to step S9900                            |
|       |                  |        |                                                                                |
| S9900 | Complete         | A0010  | Set Message: Production Command Complete                                       |
|       |                  | A0020  | Self Release the EM                                                            |
| END1  | Command Complete |        | True                                                                           |

Table 118: Production Sequence

## 15.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

| Parameter            | Description    | Type           | Units | Default Value/Reference |
|----------------------|----------------|----------------|-------|-------------------------|
| Operating Parameters |                |                |       |                         |
| NA                   |                |                |       |                         |
| Tuning Parameters    |                |                |       |                         |
| TP001                | Delay Timer 1  | Floating Point | Sec   | 5                       |
| TP002                | Delay Timer 2  | Floating Point | Sec   | 10                      |
| TP003                | Delay Timer 3  | Floating Point | Sec   | 60                      |
| TP004                | Blow Time      | Floating Point | Sec   | 600                     |
| TP005                | RSP For LT_622 | Floating Point | %     | 10                      |

**Table 119: Operating and Tuning Parameters**

## 15.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

| Parameter Name | Parameter Type     | Value/Reference           |
|----------------|--------------------|---------------------------|
| MY_TAG         |                    | RF1EQ_682B                |
| OPR_CMD_MASK   |                    | 3                         |
| RF_FILTRTN1    | String             | RF1EQ_616A1               |
| RF_FILTRTN2    | String             | RF1EQ_616A2               |
| RF_FILTRTN3    | String             | RF1EQ_616A3               |
| RF_FILTRTN4    | String             | RF1EQ_616A4               |
| RF_FILTRTN5    | String             | RF1EQ_616A5               |
| RF_OF_BLEACH5  | String             | RF1EQ_622_21              |
| LSL_682B       | External Reference | RF1_LSL1682B/DI1/DISC_ACT |
| HOLD           | Boolean            | False                     |
| STEP_TIME      | Floating Point     | 0                         |
| COMPLETED_TIME | Floating Point     | 0                         |
| RESET_COMP_TM  | Boolean            | False                     |

**Table 120: Configurable Parameters**

## 15.6 Named Set

Following Named set shall be used for configuration of EM.

| Name            | Description                                        |
|-----------------|----------------------------------------------------|
| NS_RF_FLT_VPRD2 | Refinery Area: Filtration and Vacuum Production EM |

**Table 121: Named Set Details**

Details of states for NS\_RF\_FLT\_VPRD2 named set are as follows,

| Named Set Value | Named Set Name |
|-----------------|----------------|
| 00              | Hold           |
| 01              | Production     |

**Table 122: Name Set State Details**

## 16 EQUIPMENT MODULE CLASS DESIGN – EM\_RF\_FLT\_VPRD3 (1-682A)

### 16.1 Functional Description

EM\_RF\_FLT\_VPRD3 is a class based equipment module that performs filling operation and then drains the material on operator command. EM\_RF\_FLT\_VPRD3 is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. Initially Operator shall be provided with Hold and Fill Commands. Start drain and Stop drain commands enable during the execution of Fill command. Hold Request shall be enabled for all acquired devices at the end of HOLD command.

### 16.2 Instances

The following Table contains the module instances of the equipment module class.

| Device Name | Module Class    | Instance Name |
|-------------|-----------------|---------------|
| EM          | EM_RF_FLT_VPRD3 | RF1EQ_682A    |
| DEV1        | _V12_IP_130_SE  | RF1_V1682AD   |
| DEV2        | _V12_IP_130_SE  | RF1_V1682AB   |
| DEV3        | _V12_IP_130_SE  | RF1_V1682AD1  |
| DEV4        | _V12_IP_130_SE  | RF1_V1682AD2  |
| DEV5        | _MRSS11_130_SE  | RF1_PU1682A   |

**Table 123: Instances**

### 16.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

#### 16.3.1 Command 00 – Hold

| Step  | Description        | Action | Action Description                               |
|-------|--------------------|--------|--------------------------------------------------|
| H0010 | Command Hold Start | A0010  | Store Fail Message                               |
|       |                    | A0020  | Reset Hold Request                               |
|       |                    | A0030  | Save Previous Step Time                          |
| T1000 | Actions Complete   |        | Go to step H1000                                 |
| H1000 | Hold Command 1     | A0010  | Opening DEV1                                     |
|       |                    | A0020  | Closing and Disabling Conditional Logic for DEV2 |
|       |                    | A0030  | Closing and Disabling Conditional Logic for DEV3 |

| Step   | Description                               | Action | Action Description                                |
|--------|-------------------------------------------|--------|---------------------------------------------------|
|        |                                           | A0040  | Closing and Disabling Conditional Logic for DEV4  |
|        |                                           | A0050  | Stopping and Disabling Conditional Logic for DEV5 |
| T9900A | Actions Complete or Time Out & Hold=True  |        | Go to step H9900A                                 |
| T9900B | Actions Complete or Time Out & Hold=False |        | Go to step H9900B                                 |
|        |                                           |        |                                                   |
| H9900A | Command Hold End                          | A0010  | Completed                                         |
| END    | Command Complete                          |        | True                                              |
|        |                                           |        |                                                   |
| H9900B | Release Devices                           | A0010  | Release Devices Message                           |
|        |                                           | A0020  | Release Private Devices                           |
|        |                                           | A0030  | Release Private Devices                           |
|        |                                           | A0040  | Release Private Devices                           |
|        |                                           | A0050  | Release Private Devices                           |
|        |                                           | A0060  | Release Private Devices                           |
|        |                                           | A0070  | Reset Parameters                                  |
| END1   | Command Complete                          |        | True                                              |

Table 124: Hold Sequence

## 16.3.2

## Command 01 – Fill

| Step   | Description        | Action | Action Description                                   |
|--------|--------------------|--------|------------------------------------------------------|
| S0000  | Start              | A0010  | Set Message: Starting Fill Command                   |
| T0010  | First Pass = False |        | If First Pass = False then go to step S0010          |
| T0020A | First Pass = True  |        | If First Pass = True then go to step S0020           |
|        |                    |        |                                                      |
| S0010  | Initializations    | A0010  | First Pass Logic                                     |
|        |                    | A0020  | First Pass = True                                    |
| T0020B | Actions Complete   |        | If no pending actions in S0010 then go to step S0020 |
|        |                    |        |                                                      |
| S0020  | Acquire Devices    | A0010  | Acquire Devices Message                              |
|        |                    | A0020  | Acquire Private Devices                              |
|        |                    | A0030  | Acquire Private Devices                              |



| Step   | Description      | Action | Action Description                                                                                                           |
|--------|------------------|--------|------------------------------------------------------------------------------------------------------------------------------|
|        |                  | A0040  | Acquire Private Devices                                                                                                      |
|        |                  | A0050  | Acquire Private Devices                                                                                                      |
|        |                  | A0060  | Acquire Private Devices                                                                                                      |
|        |                  | A0070  | Reset HOLD_REQ                                                                                                               |
| T0100  | Start            |        | If Task Pointer = 1 then go to step S0100                                                                                    |
| T0200B | Fill             |        | If Task Pointer = 2 then go to step S0200                                                                                    |
| S0100  | Start            | A0010  | Closing DEV1                                                                                                                 |
|        |                  | A0020  | Closing and Disabling Conditional Logic for DEV2                                                                             |
|        |                  | A0030  | Closing and Disabling Conditional Logic for DEV3                                                                             |
|        |                  | A0040  | Closing and Disabling Conditional Logic for DEV4                                                                             |
|        |                  | A0050  | Stopping and Disabling Conditional Logic for DEV5                                                                            |
|        |                  | A0060  | Checking LSH_682A is not in alarm                                                                                            |
| T0200  | Fill             |        | If LSL_682A is not in alarm then go to step S0200                                                                            |
| S0200  | Fill             | A0010  | Closing DEV1                                                                                                                 |
|        |                  | A0020  | Closing and Disabling Conditional Logic for DEV2                                                                             |
|        |                  | A0030  | Closing and Disabling Conditional Logic for DEV3                                                                             |
|        |                  | A0040  | Closing and Disabling Conditional Logic for DEV4                                                                             |
|        |                  | A0050  | Stopping and Disabling Conditional Logic for DEV5                                                                            |
|        |                  | A0060  | Start Fill Time Counter                                                                                                      |
|        |                  | A0070  | Message                                                                                                                      |
|        |                  | A0080  | Set Preset Fill time                                                                                                         |
|        |                  | A0090  | Start Preset Fill Timer                                                                                                      |
|        |                  | A0100  | Preset Fill time Message                                                                                                     |
| T9900  | Complete         |        | If LSH_682A is in alarm or operator select <Drain> command (OP001) or Preset fill time elapsed (TP003) then go to step S9900 |
| S9900  | Complete         | A0010  | Set Message: Fill Command Complete                                                                                           |
|        |                  | A0020  | Setting Task pointer 3                                                                                                       |
|        |                  | A0030  | Disabling TP parameters                                                                                                      |
| END    | Command Complete |        | True                                                                                                                         |

Table 125: Fill Sequence

## 16.3.3 Command 02 – Start Drain

| Step  | Description      | Action | Action Description                                                                     |
|-------|------------------|--------|----------------------------------------------------------------------------------------|
| S0000 | Start            | A0010  | Set Message: Starting Drain Command                                                    |
|       |                  | A0020  | Reset HOLD_REQ                                                                         |
| T0100 | Drain            |        | If Task Pointer = 3 then go to step S0100                                              |
| S0100 | Drain            | A0010  | Opening DEV1                                                                           |
|       |                  | A0020  | Closing and Disabling Conditional Logic for DEV2                                       |
|       |                  | A0030  | Enabling Conditional Logic for DEV3                                                    |
|       |                  | A0040  | Enabling Conditional Logic for DEV4                                                    |
|       |                  | A0050  | Enabling Conditional Logic for DEV5                                                    |
|       |                  | A0060  | Checking LSL_682A is in alarm                                                          |
| T9900 | Complete         |        | If LSL_682A is in alarm or Operator command <Stop Drain> (OP004) then go to step S9900 |
| S9900 | Complete         | A0010  | Set Message: Start Drain Command Complete                                              |
|       |                  | A0020  | Disabling TP parameters                                                                |
| END   | Command Complete |        | True                                                                                   |

Table 126: Start Drain Sequence

## 16.3.4 Command 03 – Stop Drain

| Step  | Description | Action | Action Description                                          |
|-------|-------------|--------|-------------------------------------------------------------|
| S0000 | Start       | A0010  | Set Message: Executing Stop Drain Command                   |
| T0100 | Drain Wait  |        | If Task Pointer = 4 then go to step S0100                   |
| S0100 | Drain Wait  | A0010  | Closing DEV1                                                |
|       |             | A0020  | Enabling Conditional Logic for DEV2                         |
|       |             | A0030  | Enabling Conditional Logic for DEV3                         |
|       |             | A0040  | Enabling Conditional Logic for DEV4                         |
|       |             | A0050  | Stopping and Disabling Conditional Logic for DEV5           |
|       |             | A0060  | Message                                                     |
|       |             | A0070  | Set Blowing time                                            |
|       |             | A0080  | Start Blowing Timer                                         |
|       |             | A0090  | Preset Blowing time Message                                 |
| T0200 | Delay       |        | If DEV1 Blowing timer (TP002) elapsed then go to step S0200 |
| S0200 | Delay       | A0010  | Set Message: Waiting For Delay timer                        |

| Step  | Description      | Action | Action Description                                        |
|-------|------------------|--------|-----------------------------------------------------------|
| T0300 | Close Devices    |        | If 5 sec (TP001) delay time is over then go to step S0300 |
|       |                  |        |                                                           |
| S0300 | Close Devices    | A0010  | Closing DEV1                                              |
|       |                  | A0020  | Closing and Disabling Conditional Logic for DEV2          |
|       |                  | A0030  | Closing and Disabling Conditional Logic for DEV3          |
|       |                  | A0040  | Closing and Disabling Conditional Logic for DEV4          |
|       |                  | A0050  | Stopping and Disabling Conditional Logic for DEV5         |
|       |                  | A0060  | Reset Drain Selection Button (OP001)                      |
| T0400 | Disable HM       |        | If no pending actions in S0300 then go to step S0400      |
|       |                  |        |                                                           |
| S0400 | Disable HM       | A0010  | Disable HM 01 (Module Ownership Lost)                     |
| T0500 | Release Devices  |        | If no pending actions in S0400 then go to step S0500      |
|       |                  |        |                                                           |
| S0500 | Release Devices  | A0010  | Release Devices Message                                   |
|       |                  | A0020  | Release Private Devices                                   |
|       |                  | A0030  | Release Private Devices                                   |
|       |                  | A0040  | Release Private Devices                                   |
|       |                  | A0050  | Release Private Devices                                   |
|       |                  | A0060  | Release Private Devices                                   |
| T9900 | Complete         |        | Go to step S9900                                          |
|       |                  |        |                                                           |
| S9900 | Complete         | A0010  | Set Message: Stop Drain Command Complete                  |
|       |                  | A0020  | Self Release the EM                                       |
| END   | Command Complete |        | True                                                      |

Table 127: Stop Drain Sequence

## 16.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

| Parameter            | Description            | Type    | Units | Default Value/Reference |
|----------------------|------------------------|---------|-------|-------------------------|
| Operating Parameters |                        |         |       |                         |
| OP001                | Start Drain            | Boolean | NA    | False                   |
| OP002                | Bleacher 622 Selection | Boolean | NA    | False                   |

| Parameter         | Description             | Type           | Units | Default Value/Reference |
|-------------------|-------------------------|----------------|-------|-------------------------|
| OP003             | Slop oil tank selection | Boolean        | NA    | False                   |
| OP004             | Stop Drain              | Boolean        | NA    | False                   |
| Tuning Parameters |                         |                |       |                         |
| TP001             | Delay Timer             | Floating Point | Sec   | 5                       |
| TP002             | Blowing Preset time     | Floating Point | Sec   | 600                     |
| TP003             | Filling time            | Floating Point | Sec   | 1000                    |

Table 128: Operating and Tuning Parameters

## 16.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

| Parameter Name | Parameter Type     | Value/Reference            |
|----------------|--------------------|----------------------------|
| MY_TAG         |                    | RF1EQ_682A                 |
| OPR_CMD_MASK   |                    | 15                         |
| RF_OF_BLEACH5  | External Reference | RF1EQ_622_21/STEP_NAME     |
| LSH_682A       | External Reference | RF1_LSH1682A/DI1/DISC_ACT  |
| LSL_682A       | External Reference | RF1_LSL1682A/DI1/DISC_ACT  |
| LSH_622_21     | External Reference | RF1_LSH1622/DI1/DISC_ACT   |
| LT_622_21      | External Reference | RF1_LT1622/AI1/HI_HI_ACT   |
| DY_REF_1       | Dynamic Reference  |                            |
| HOLD           | External Reference | RF1EQ_STARTUP3/OP005_VALUE |
| STEP_TIME      | Floating Point     | 0                          |
| COMPLETED_TIME | Floating Point     | 0                          |
| RESET_COMP_TM  | Boolean            | False                      |

Table 129: Configurable Parameters

## 16.6 Named Set

Following Named set shall be used for configuration of EM.

| Name            | Description                                     |
|-----------------|-------------------------------------------------|
| NS_RF_FLT_VPRD3 | Refinery Area: Filtration and Vacuum Production |

Table 130: Named Set Details

Details of states for NS\_RF\_FLT\_VPRD3 named set are as follows,

| Named Set Value | Named Set Name |
|-----------------|----------------|
| 00              | Hold           |
| 01              | Fill           |
| 02              | Start Drain    |
| 03              | Stop Drain     |

**Table 131: Name Set State Details**

## 17 EQUIPMENT MODULE CLASS DESIGN – EM\_RF\_FLT\_VPRD4 (1-682A2)

### 17.1 Functional Description

EM\_RF\_FLT\_VPRD4 is a class based equipment module that performs filtration and vacuum production. This EM performs Filling Operation and drains the material based on the Operator command. EM\_RF\_FLT\_VPRD4 is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. Initially Operator shall be provided with Hold, Fill and Start Drain Command. All devices used in EM shall be acquired in Fill Command and shall be released on Completion of Stop Drain Command. Hold Request shall be enabled for all acquired devices at the end of Hold command.

### 17.2 Instances

The following Table contains the module instances of the equipment module class.

| Device Name | Module Class    | Instance Name |
|-------------|-----------------|---------------|
| EM          | EM_RF_FLT_VPRD4 | RF1EQ_682A2   |
| DEV1        | _V12_IP_130_SE  | RF1_V1682A2D  |
| DEV2        | _V12_IP_130_SE  | RF1_V1682A2B  |

**Table 132: Instances**

### 17.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

#### 17.3.1 Command 00 – Hold

| Step   | Description                              | Action | Action Description                               |
|--------|------------------------------------------|--------|--------------------------------------------------|
| H0010  | Command Hold Start                       | A0010  | Store Fail Message                               |
|        |                                          | A0020  | Reset Hold Request                               |
|        |                                          | A0030  | Save Previous Step Time                          |
| T1000  | Actions Complete                         |        | Go to step H1000                                 |
| H1000  | Hold Command 1                           | A0010  | Closing and Disabling Conditional Logic for DEV1 |
|        |                                          | A0020  | Closing DEV2                                     |
| T9900A | Actions Complete or Time Out & Hold=True |        | Go to step H9900A                                |
| T9900B | Actions Complete or                      |        | Go to step H9900B                                |

| Step   | Description           | Action | Action Description      |
|--------|-----------------------|--------|-------------------------|
|        | Time Out & Hold=False |        |                         |
|        |                       |        |                         |
| H9900A | Command Hold End      | A0010  | Completed               |
| END    | Command Complete      |        | True                    |
|        |                       |        |                         |
| H9900B | Release Devices       | A0010  | Release Devices Message |
|        |                       | A0020  | Release Private Devices |
|        |                       | A0030  | Release Private Devices |
|        |                       | A0040  | Reset Parametrs         |
| END1   | Command Complete      |        | True                    |

**Table 133: Hold Sequence**

## 17.3.2 Command 01 –Fill

| Step   | Description        | Action | Action Description                                   |
|--------|--------------------|--------|------------------------------------------------------|
| S0000  | Start              | A0010  | Set Message: Starting Start Fill Command             |
| T0010  | First Pass = False |        | If First Pass = False then go to step S0010          |
| T0020A | First Pass = True  |        | If First Pass = True then go to step S0020           |
|        |                    |        |                                                      |
| S0010  | Initializations    | A0010  | First Pass Logic                                     |
|        |                    | A0020  | First Pass = True                                    |
| T0020B | Actions Complete   |        | If no pending actions in S0010 then go to step S0020 |
|        |                    |        |                                                      |
| S0020  | Acquire Devices    | A0010  | Acquire Devices Message                              |
|        |                    | A0020  | Acquire Private Devices                              |
|        |                    | A0030  | Acquire Private Devices                              |
|        |                    | A0040  | Reset HOLD_REQ                                       |
|        |                    | A0050  | Checking LSL_682A2 is not in alarm                   |
| T0100  | Fill               |        | If LSL_682A2 is not in alarm then go to step S0100   |
|        |                    |        |                                                      |
| S0100  | Fill               | A0010  | Closing and Disabling Conditional Logic for DEV1     |
|        |                    | A0020  | Closing DEV2                                         |
|        |                    | A0030  | Set Preset Fill time                                 |
|        |                    | A0040  | Start Preset Fill Timer                              |
|        |                    | A0050  | Preset Fill time Message                             |
|        |                    | A0060  | Start Fill Time Counter                              |



| Step  | Description      | Action | Action Description                                                                                                            |
|-------|------------------|--------|-------------------------------------------------------------------------------------------------------------------------------|
| T9900 | Complete         |        | If LSM_682A2 is in alarm or operator select <Drain> command (OP001) or Preset Fill Time (TP001) elapsed then go to step S0100 |
|       |                  |        |                                                                                                                               |
| S9900 | Complete         | A0010  | Set Message: Start Fill Command Complete                                                                                      |
| END   | Command Complete |        | True                                                                                                                          |

**Table 134: Fill Sequence**

## 17.3.3 Command 02 – Start Drain

| Step  | Description      | Action | Action Description                                                                     |
|-------|------------------|--------|----------------------------------------------------------------------------------------|
| S0000 | Start            | A0010  | Set Message: Starting Start Drain Command                                              |
|       |                  | A0020  | Reset HOLD_REQ                                                                         |
| T0100 | Drain            |        | Go to step S0100                                                                       |
|       |                  |        |                                                                                        |
| S0100 | Drain            | A0010  | Enabling Conditional Logic for DEV1                                                    |
|       |                  | A0020  | Closing DEV2                                                                           |
|       |                  | A0030  | Checking LSL_682A2 is in alarm                                                         |
| T9900 | Complete         |        | If LSL_682A2 is in alarm or Operator select <Stop Drain> Command then go to step S9900 |
|       |                  |        |                                                                                        |
| S9900 | Complete         | A0010  | Set Message: Start Drain Command Complete                                              |
| END   | Command Complete |        | True                                                                                   |

**Table 135: Start Drain Sequence**

## 17.3.4 Command 03 – Stop Drain

| Step  | Description   | Action | Action Description                                         |
|-------|---------------|--------|------------------------------------------------------------|
| S0000 | Start         | A0010  | Set Message: Executing Stop Drain Command                  |
|       |               | A0020  | Reset HOLD_REQ                                             |
| T0100 | Blow          |        | Go to step S0100                                           |
|       |               |        |                                                            |
| S0100 | Blow          | A0010  | Closing and Disabling Conditional Logic for DEV1           |
|       |               | A0020  | Opening DEV2                                               |
|       |               | A0030  | Set Preset Blowing time                                    |
|       |               | A0040  | Start Preset Blowing Timer                                 |
|       |               | A0050  | Preset Blowing time Message                                |
| T0200 | Close Devices |        | If DEV2 blowing time elapsed (TP002) then go to step S0200 |

| Step  | Description      | Action | Action Description                                   |
|-------|------------------|--------|------------------------------------------------------|
| S0200 | Close Devices    | A0010  | Closing and Disabling Conditional Logic for DEV1     |
|       |                  | A0020  | Closing DEV2                                         |
|       |                  | A0030  | Reset Start/Stop Drain                               |
| T0300 | Disable HM       |        | If no pending actions in S0200 then go to step S0300 |
| S0300 | Disable HM       | A0010  | Disable HM 01 (Module Ownership Lost)                |
| T0400 | Release Devices  |        | If no pending actions in S0300 then go to step S0400 |
| S0400 | Release Devices  | A0010  | Release Devices Message                              |
|       |                  | A0020  | Release Private Devices                              |
|       |                  | A0030  | Release Private Devices                              |
| T9900 | Complete         |        | If no pending actions in S0400 then go to step S9900 |
| S9900 | Complete         | A0010  | Set Message: Stop Drain Command Complete             |
|       |                  | A0020  | Self Release the EM                                  |
| END   | Command Complete |        | True                                                 |

Table 136: Stop Drain Sequence

## 17.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

| Parameter            | Description      | Type    | Units | Default Value/Reference |
|----------------------|------------------|---------|-------|-------------------------|
| Operating Parameters |                  |         |       |                         |
| OP001                | Start Drain      | Boolean | NA    | False                   |
| OP002                | Stop Drain       | Boolean | NA    | False                   |
| Tuning Parameters    |                  |         |       |                         |
| TP001                | Preset Fill time | 15      | Sec   | Yes                     |
| TP002                | Blowing time     | 600     | Sec   | Yes                     |

Table 137: Operating and Tuning Parameters

## 17.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

| Parameter Name | Parameter Type     | Value/Reference                |
|----------------|--------------------|--------------------------------|
| MY_TAG         |                    | RF1EQ_682A2                    |
| OPR_CMD_MASK   |                    | 3                              |
| RF_OF_BLEACH5  | External Reference | RF1EQ_622_21/STEP_NAME         |
| LSL_682A2      | External Reference | RF1_LSL1682F/DI1/DISC_ACT      |
| LSM_682A2      | External Reference | RF1_LSM1682A/DI1/DISC_ACT      |
| LSH_622_21     | External Reference | RF1_LSH1622/DI1/DISC_ACT       |
| LT_622_21      | External Reference | RF1_LT1622/AI1/HI_HI_LIM       |
| VACCUM_ALM     | Boolean            | False                          |
| DY_REF_1       | Dynamic Reference  |                                |
| STEP_TIME      | Floating Point     | 0                              |
| COMPLETED_TIME | Floating Point     | 0                              |
| RESET_COMP_TM  | Boolean            | False                          |
| HOLD           | External Reference | RF1EQ_STARTUP3/OP005_VAL<br>UE |

Table 138: Configurable Parameters

## 17.6 Named Set

Following Named set shall be used for configuration of EM.

| Name            | Description                                      |
|-----------------|--------------------------------------------------|
| NS_RF_FLT_VPRD4 | Refinery Area: Filtration & Vacuum Production EM |

Table 139: Named Set Details

Details of states for NS\_RF\_FLT\_VPRD4 named set are as follows,

| Named Set Value | Named Set Name |
|-----------------|----------------|
| 00              | Hold           |
| 01              | Fill           |
| 02              | Start Drain    |
| 03              | Stop Drain     |

Table 140: Name Set State Details

## 18 EQUIPMENT MODULE CLASS DESIGN – EM\_RF\_FLT\_VPRD5 (1-606C1/C2)

### 18.1 Functional Description

EM\_RF\_FLT\_VPRD5 is a class based equipment module that performs filling and discharge for a pre-defined time. EM\_RF\_FLT\_VPRD5 module class drives two instances RF1EQ-1/606C1 and RF1EQ-1/606C2. This EM Class is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. Operator shall be provided only with Hold and Fill Commands. Discharge operation shall be executed as part of Fill State. Hold Request shall be enabled for all acquired devices at the end of Hold command.

### 18.2 Instances

The following Table contains the module instances of the equipment module class.

| Device Name | Module Class    | Instance Name |              |
|-------------|-----------------|---------------|--------------|
| EM          | EM_RF_FLT_VPRD5 | RF1EQ_606C1   | RF1EQ_606C2  |
| DEV1        | _V12_IP_130_SE  | RF1_V1606C1F  | RF1_V1606C2F |
| DEV2        | _V12_IP_130_SE  | RF1_V1606C1D  | RF1_V1606C2D |
| DEV3        | _V12_IP_130_SE  | RF1_V1630C    |              |

**Table 141: Instances**

### 18.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

#### 18.3.1 Command 00 – Hold

| Step  | Description                  | Action | Action Description                               |
|-------|------------------------------|--------|--------------------------------------------------|
| H0010 | Command Hold Start           | A0010  | Store Fail Message                               |
|       |                              | A0020  | Reset Hold Request                               |
|       |                              | A0030  | Save Previous Step Time                          |
| T1000 | Actions Complete or Time Out |        | Go to step H1000                                 |
|       |                              |        |                                                  |
| H1000 | Hold Command 1               | A0010  | Closing and Disabling Conditional Logic for DEV1 |
|       |                              | A0020  | Closing and Disabling Conditional Logic for DEV2 |

| Step   | Description                               | Action | Action Description                               |
|--------|-------------------------------------------|--------|--------------------------------------------------|
|        |                                           | A0030  | Closing and Disabling Conditional Logic for DEV3 |
| T9900A | Actions Complete or Time Out & Hold=True  |        | Go to step H9900A                                |
| T9900B | Actions Complete or Time Out & Hold=False |        | Go to step H9900B                                |
|        |                                           |        |                                                  |
| H9900A | Command Hold End                          | A0010  | Completed                                        |
| END    | Command Complete                          |        | True                                             |
|        |                                           |        |                                                  |
| H9900B | Release Devices                           | A0010  | Disable HM 01 (Module Ownership Lost)            |
|        |                                           | A0020  | Release Devices Message                          |
|        |                                           | A0030  | Release Private Devices                          |
|        |                                           | A0040  | Release Private Devices                          |
|        |                                           | A0050  | Release Private Devices                          |
|        |                                           | A0060  | Reset Parameters                                 |
| END1   | Command Complete                          |        | True                                             |

Table 142: Hold Sequence

## 18.3.2 Command 01 – Fill

| Step   | Description        | Action | Action Description                                   |
|--------|--------------------|--------|------------------------------------------------------|
| S0000  | Start              | A0010  | Set Message: Starting Fill Command                   |
| T0010  | First Pass = False |        | If First Pass = False then go to step S0010          |
| T0020A | First Pass = True  |        | If First Pass = True then go to step S0020           |
|        |                    |        |                                                      |
| S0010  | Initiaizations     | A0010  | First Pass Logicc                                    |
|        |                    | A0020  | First Pass = True                                    |
| T0020B | Actions Complete   |        | If no pending actions in S0010 then go to step S0020 |
|        |                    |        |                                                      |
| S0020  | Acquire Devices    | A0010  | Acquire Devices Message                              |
|        |                    | A0020  | Acquire Private Devices                              |
|        |                    | A0030  | Acquire Private Devices                              |
|        |                    | A0040  | Acquire Shared Devices                               |
|        |                    | A0050  | Reset HOLD_REQ                                       |

| Step   | Description | Action | Action Description                                                                                                                                                                                            |
|--------|-------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        |             | A0060  | Set Message: Waiting for 604C to be in Filter Aid Dosing                                                                                                                                                      |
|        |             | A0070  | Set Message: Checking Dosing Filter Aid is Enable                                                                                                                                                             |
| T0100  | Standby     |        | If RF_FLT_VPRD6 is in <Filter Aid Dosing> steps and Demand for the Dosing Filter Aid > 0 (DEMAND_DOSING ) and Dosing Filter Aid enabled for dosing (FILTER_ENABLE) and Task Pointer = 1 then go to step S0100 |
| T0200A | Fill        |        | If Task Pointer = 2 then go to step S0200                                                                                                                                                                     |
| T0300A | Wait        |        | If Task Pointer = 3 then go to step S0300                                                                                                                                                                     |
| T0400A | Discharge   |        | If Task Pointer = 4 then go to step S0400                                                                                                                                                                     |
| S0100  | Evacuate    | A0010  | Closing and Disabling Conditional Logic for DEV1                                                                                                                                                              |
|        |             | A0020  | Enabling Conditional Logic for DEV2                                                                                                                                                                           |
|        |             | A0030  | Closing and Disabling Conditional Logic for DEV3                                                                                                                                                              |
|        |             | A0040  | Set Preset Evacuate Time                                                                                                                                                                                      |
|        |             | A0050  | Start Preset Evacuate Time                                                                                                                                                                                    |
|        |             | A0060  | Preset Evacuate Time Message                                                                                                                                                                                  |
| T0200  | Fill        |        | If Preset Evacuate time (TP004) is over then go to step S0200                                                                                                                                                 |
| S0200  | Fill        | A0010  | Closing and Disabling Conditional Logic for DEV2                                                                                                                                                              |
|        |             | A0020  | Enabling Conditional Logic for DEV1                                                                                                                                                                           |
|        |             | A0030  | Enabling Conditional Logic for DEV3                                                                                                                                                                           |
|        |             | A0040  | Set Preset Fill Time                                                                                                                                                                                          |
|        |             | A0050  | Start Preset Fill Time                                                                                                                                                                                        |
|        |             | A0060  | Preset Fill Time Message                                                                                                                                                                                      |
| T0300  | Wait        |        | If Preset Fill time (TP003) is over then go to step S0300                                                                                                                                                     |
| S0300  | Wait        | A0010  | Closing and Disabling Conditional Logic for DEV1                                                                                                                                                              |
|        |             | A0020  | Closing and Disabling Conditional Logic for DEV2                                                                                                                                                              |
|        |             | A0030  | Closing and Disabling Conditional Logic for DEV3                                                                                                                                                              |
|        |             | A0040  | Set Preset Wait Time                                                                                                                                                                                          |
|        |             | A0050  | Start Preset Wait Time                                                                                                                                                                                        |
|        |             | A0060  | Preset Wait Time Message                                                                                                                                                                                      |
|        |             | A0070  | Integrate Quantity Dosed                                                                                                                                                                                      |
| T0400  | Discharge   |        | If Preset Wait time (TP005 or TP006) is over then go to step S0400                                                                                                                                            |

| Step  | Description      | Action | Action Description                                             |
|-------|------------------|--------|----------------------------------------------------------------|
| S0400 | Discharge        | A0010  | Closing and Disabling Conditional Logic for DEV1               |
|       |                  | A0020  | Enabling Conditional Logic for DEV2                            |
|       |                  | A0030  | Closing and Disabling Conditional Logic for DEV3               |
|       |                  | A0040  | Set Preset Discharge Time                                      |
|       |                  | A0050  | Start Preset Discharge Time                                    |
|       |                  | A0060  | Preset Discharge Time Message                                  |
|       |                  | A0070  | Waiting for Discharge Time                                     |
| T0500 | Close Devices    |        | If Preset Discharge time (TP007) is over then go to step S0500 |
| S0500 | Close Devices    | A0010  | Closing DEV2                                                   |
| T0600 | Disable HM       |        | If no pending actions in S0500 then go to step S0600           |
| S0600 | Disable HM       | A0010  | Disable HM 01 (Module Ownership Lost)                          |
| T0700 | Release Devices  |        | If no pending actions in S0600 then go to step S0700           |
| S0700 | Release Devices  | A0010  | Disable HM 01 (Module Ownership Lost)                          |
|       |                  | A0020  | Release Devices Message                                        |
|       |                  | A0030  | Release Private Devices                                        |
|       |                  | A0040  | Release Private Devices                                        |
|       |                  | A0050  | Release Private Devices                                        |
| T9900 | Complete         |        | If no pending actions in S0700 then go to step S9900           |
| S9900 | Complete         | A0010  | Set Message: Fill Command Complete                             |
|       |                  | A0020  | Self Release the EM                                            |
| END   | Command Complete |        | True                                                           |

Table 143: Fill Sequence

## 18.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

| Parameter            | Description | Type   | Units | Default Value/Reference |
|----------------------|-------------|--------|-------|-------------------------|
| Operating Parameters |             |        |       |                         |
| OP001                | Dosing Arm  | String | N/A   | 606C                    |

| Parameter         | Description              | Type           | Units | Default Value/Reference |
|-------------------|--------------------------|----------------|-------|-------------------------|
| OP002             | Reset dosed quantity     | Boolean        | N/A   | Yes                     |
| OP003             | Actual Step              | String         | N/A   | Fill                    |
| Tuning Parameters |                          |                |       |                         |
| TP001             | Delay Timer              | Floating Point | Sec   | 5                       |
| TP002             | Dosing Amount Set Point  | Floating Point | Kg    | 50                      |
| TP003             | Preset Fill time         | Floating Point | Sec   | 10                      |
| TP004             | Preset Evacuate time     | Floating Point | Sec   | 5                       |
| TP005             | Preset Waiting time Slow | Floating Point | Sec   | 5                       |
| TP006             | Preset Waiting time Fast | Floating Point | Sec   | 2                       |
| TP007             | Discharge time           | Floating Point | Sec   | 10                      |
| TP008             | Volume 606C              | Floating Point | Kg    | 4.0                     |
| TP009             | Call demand              | Floating Point | Kg    | 0.2                     |
| TP010             | Remain Time              | Floating Point | Sec   | 10                      |
| TP011             | Dosed Quantity           | Floating Point | Kg    | 100                     |
| TP012             | Initial Volume           | Floating Point | Kg    | 0                       |
| TP013             | Final Volume             | Floating Point | Kg    | 0                       |
| TP014             | Demand Limit             | Floating Point | Kg    | 0                       |

Table 144: Operating and Tuning Parameters

## 18.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

| Parameter Name | Parameter Type     | Value/Reference            |             |
|----------------|--------------------|----------------------------|-------------|
| MY_TAG         |                    | RF1EQ_606C1                | RF1EQ_606C2 |
| OPR_CMD_MASK   |                    | 3                          |             |
| RF_FLT_VPRD6   | String             | RF1EQ_604C                 |             |
| FILTER_ENABLE  | Boolean            | False                      |             |
| WT_630C        | External Reference | RF1_WE1630C/PV             |             |
| DEMAND_DOSING  | Floating Point     | 0                          |             |
| COMPLETED_TIME | Floating Point     | 0                          |             |
| RESET_COMP_TM  | Boolean            | False                      |             |
| STEP_TIME      | Floating Point     | 0                          |             |
| HOLD           | External Reference | RF1EQ_STARTUP3/OP005_VALUE |             |

| Parameter Name | Parameter Type    | Value/Reference |
|----------------|-------------------|-----------------|
| DY_REF_1       | Dynamic Reference |                 |

**Table 145: Configurable Parameters**

## 18.6 Named Set

Following Named set shall be used for configuration of EM.

| Name            | Description                                     |
|-----------------|-------------------------------------------------|
| NS_RF_FLT_VPRD5 | Refinery Area: Filtration and Vacuum Production |

**Table 146: Named Set Details**

Details of states NS\_RF\_FLT\_VPRD5 named set are as follows,

| Named Set Value | Named Set Name |
|-----------------|----------------|
| 00              | Hold           |
| 01              | Fill           |

**Table 147: Name Set State Details**

## 19 EQUIPMENT MODULE CLASS DESIGN – EM\_RF\_FLT\_VPRD6 (1-604C)

### 19.1 Functional Description

EM\_RF\_FLT\_VPRD6 is a class based equipment module that performs filling operation followed by filter aid dosing and then drains the material. EM\_RF\_FLT\_VPRD6 is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. All devices used in EM shall be acquired in Fill Command and shall be released on completion of Drain Operation. Hold Request shall be enabled for all acquired devices at the end of Hold command.

### 19.2 Instances

The following Table contains the module instances of the equipment module class.

| Device Name | Module Class    | Instance Name |
|-------------|-----------------|---------------|
| EM          | EM_RF_FLT_VPRD6 | RF1EQ_604C    |
| DEV1        | _V12_IP_130_SE  | RF1_V1692BD   |
| DEV2        | _V12_IP_130_SE  | RF1_V1604CV   |
| DEV3        | _V12_IP_130_SE  | Ignored       |
| DEV4        | _V12_IP_130_SE  | RF1_V1604CS   |
| DEV5        | _MRSS11_130_SE  | RF1_PU1604C   |
| DEV6        | _MRSS11_130_SE  | RF1_AG1604C   |
| DEV7        | EM_RF_FLT_VPRD5 | RF1EQ_606C1   |
| DEV8        | EM_RF_FLT_VPRD5 | RF1EQ_606C2   |

**Table 148: Instances**

### 19.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

#### 19.3.1 Command 00 – Hold

| Step  | Description                  | Action | Action Description                               |
|-------|------------------------------|--------|--------------------------------------------------|
| H0010 | Command Hold Start           | A0010  | Store Fail Message                               |
|       |                              | A0020  | Reset Hold Request                               |
| T1000 | Actions Complete & Hold=True |        | Go to step H1000                                 |
|       |                              |        |                                                  |
| H1000 | Hold Command 1               | A0010  | Closing and Disabling Conditional Logic for DEV1 |

| Step   | Description                               | Action | Action Description                                |
|--------|-------------------------------------------|--------|---------------------------------------------------|
|        |                                           | A0020  | Closing and Disabling Conditional Logic for DEV2  |
|        |                                           | A0030  | Closing and Disabling Conditional Logic for DEV3  |
|        |                                           | A0040  | Closing and Disabling Conditional Logic for DEV4  |
|        |                                           | A0050  | Stopping and Disabling Conditional logic for DEV5 |
|        |                                           | A0060  | Stopping and Disabling Conditional logic for DEV6 |
| T9900A | Actions Complete or Time Out & Hold=True  |        | Go to step H9900A                                 |
| T9900B | Actions Complete or Time Out & Hold=False |        | Go to step H9900B                                 |
|        |                                           |        |                                                   |
| H9900A | Command Hold End                          | A0010  | Completed                                         |
| END    | Command Complete                          |        | True                                              |
|        |                                           |        |                                                   |
| H9900B | Release Devices                           | A0010  | Release Devices Message                           |
|        |                                           | A0020  | Release Private Devices                           |
|        |                                           | A0030  | Release Private Devices                           |
|        |                                           | A0040  | Release Private Devices                           |
|        |                                           | A0050  | Release Private Devices                           |
|        |                                           | A0060  | Release Private Devices                           |
|        |                                           | A0070  | Release Private Devices                           |
|        |                                           | A0080  | Reset Parameters                                  |
| END1   | Command Complete                          |        | True                                              |

Table 149: Hold Sequence

## 19.3.2 Command 01 –Fill

| Step   | Description        | Action | Action Description                          |
|--------|--------------------|--------|---------------------------------------------|
| S0000  | Start              | A0010  | Set Message: Starting Fill Command          |
| T0010  | First Pass = False |        | If First Pass = False then go to step S0010 |
| T0020A | First Pass = True  |        | If First Pass = True then go to step S0020  |
|        |                    |        |                                             |
| S0010  | Initializations    | A0010  | First Pass Logic                            |

| Step   | Description       | Action | Action Description                                                              |
|--------|-------------------|--------|---------------------------------------------------------------------------------|
|        |                   | A0020  | First Pass = True                                                               |
| T0020B | Actions Complete  |        | If no pending actions in S0010 then go to step S0020                            |
|        |                   |        |                                                                                 |
| S0020  | Acquire Devices   | A0010  | Acquire Devices Message                                                         |
|        |                   | A0020  | Acquire Shared Devices                                                          |
|        |                   | A0030  | Acquire Private Devices                                                         |
|        |                   | A0040  | Acquire Private Devices                                                         |
|        |                   | A0050  | Acquire Private Devices                                                         |
|        |                   | A0060  | Acquire Shared Devices                                                          |
|        |                   | A0070  | Acquire Private Devices                                                         |
|        |                   | A0080  | Acquire Private Devices                                                         |
|        |                   | A0090  | Reset HOLD_REQ                                                                  |
| T0100  | Fill              |        | If Filling Button (OP001) is pressed and Task Pointer = 1 then go to step S0100 |
| T0200A | Filter Aid Dosing |        | If Task Pointer = 2 then go to step S0200                                       |
| T0300A | Ready             |        | If Task Pointer = 3 then go to step S0300                                       |
|        |                   |        |                                                                                 |
| S0100  | Fill              | A0010  | Enabling Conditional Logic for DEV1                                             |
|        |                   | A0020  | Enabling Conditional Logic for DEV2                                             |
|        |                   | A0030  | Closing and Disabling Conditional Logic for DEV3                                |
|        |                   | A0040  | Enabling Conditional Logic for DEV4                                             |
|        |                   | A0050  | Stopping and Disabling Conditional logic for DEV5                               |
|        |                   | A0060  | Enabling Conditional Logic for DEV6                                             |
|        |                   | A0070  | Msg: Waiting for LSM_604C or LSH_604C to be in alarm                            |
| T0200  | Filter Aid Dosing |        | If LSM_604C or LSH_604C is in alarm then go to step S0200                       |
|        |                   |        |                                                                                 |
| S0200  | Filter Aid Dosing | A0010  | Closing and Disabling Conditional Logic for DEV1                                |
|        |                   | A0020  | Closing and Disabling Conditional Logic for DEV2                                |
|        |                   | A0030  | Closing and Disabling Conditional Logic for DEV3                                |
|        |                   | A0040  | Enabling Conditional Logic for DEV4                                             |
|        |                   | A0050  | Stopping and Disabling Conditional logic for DEV5                               |
|        |                   | A0060  | Enabling Conditional Logic for DEV6                                             |
|        |                   | A0070  | Provide Fill Command to DEV7 and DEV8                                           |
|        |                   | A0080  | Waiting for Filter Aid Demand < 0 Kg                                            |

| Step  | Description      | Action | Action Description                                                                                                 |
|-------|------------------|--------|--------------------------------------------------------------------------------------------------------------------|
| T0300 | Ready            |        | If Filter Aid Demand (FILTER_AID_DEM) < 0 kg and Temperature (TT_604C) > = Set point (TP001) then go to step S0300 |
| S0300 | Ready            | A0010  | Closing and Disabling Conditional Logic for DEV1                                                                   |
|       |                  | A0020  | Enabling Conditional Logic for DEV2                                                                                |
|       |                  | A0030  | Closing and Disabling Conditional Logic for DEV3                                                                   |
|       |                  | A0040  | Enabling Conditional Logic for DEV4                                                                                |
|       |                  | A0050  | Stopping and Disabling Conditional logic for DEV5                                                                  |
|       |                  | A0060  | Enabling Conditional Logic for DEV6                                                                                |
| T9900 | Complete         |        | If no pending actions in S0300 then go to step S9900                                                               |
| S9900 | Complete         | A0010  | Set Message: Start Fill Command Complete                                                                           |
| END   | Command Complete |        | True                                                                                                               |

**Table 150: Fill Sequence**

## 19.3.3

## Command 02 – Drain

| Step  | Description   | Action | Action Description                               |
|-------|---------------|--------|--------------------------------------------------|
| S0000 | Start         | A0010  | Set Message: Starting Stop Fill Command          |
| T0100 | Drain         |        | Go to step S0100                                 |
| S0100 | Drain         | A0010  | Closing and Disabling Conditional Logic for DEV1 |
|       |               | A0020  | Closing and Disabling Conditional Logic for DEV2 |
|       |               | A0030  | Opening DEV3                                     |
|       |               | A0040  | Enabling Conditional Logic for DEV4              |
|       |               | A0050  | Enabling Conditional Logic for DEV5              |
|       |               | A0060  | Enabling Conditional Logic for DEV6              |
|       |               | A0070  | Waiting for LSL_604C to be in alarm              |
| T0200 | Close Devices |        | If LSL_604C is in alarm then go to step S0200    |
| S0200 | Close Devices | A0010  | Closing and Disabling Conditional Logic for DEV1 |
|       |               | A0020  | Closing and Disabling Conditional Logic for DEV2 |
|       |               | A0030  | Closing and Disabling Conditional Logic for DEV3 |

| Step  | Description      | Action | Action Description                                   |
|-------|------------------|--------|------------------------------------------------------|
|       |                  | A0040  | Closing and Disabling Conditional Logic for DEV4     |
|       |                  | A0050  | Stopping and Disabling Conditional logic for DEV5    |
|       |                  | A0060  | Stopping and Disabling Conditional logic for DEV6    |
|       |                  | A0070  | Reset Start & Stop Fill                              |
| T0300 | Disable HM       |        | If no pending actions in S0200 then go to step S0300 |
|       |                  |        |                                                      |
| S0300 | Disable HM       | A0010  | Disable HM 01 (Module Ownership Lost)                |
| T0400 | Release Devices  |        | If no pending actions in S0300 then go to step S0400 |
|       |                  |        |                                                      |
| S0400 | Release Devices  | A0010  | Release Devices Message                              |
|       |                  | A0020  | Release Private Devices                              |
|       |                  | A0030  | Release Private Devices                              |
|       |                  | A0040  | Release Private Devices                              |
|       |                  | A0050  | Release Private Devices                              |
|       |                  | A0060  | Release Private Devices                              |
|       |                  | A0070  | Release Private Devices                              |
| T9900 | Complete         |        | If no pending actions in S0400 then go to step S9900 |
|       |                  |        |                                                      |
| S9900 | Complete         | A0010  | Set Message: Stop Fill Command Complete              |
|       |                  | A0020  | Self Release the EM                                  |
| END   | Command Complete |        | True                                                 |

Table 151: Drain Sequence

## 19.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

| Parameter            | Description          | Type           | Units | Default Value/Reference |
|----------------------|----------------------|----------------|-------|-------------------------|
| Operating Parameters |                      |                |       |                         |
| OP001                | Start Fill           | Boolean        | NA    | False                   |
| OP002                | Stop Fill            | Boolean        | NA    | False                   |
| Tuning Parameters    |                      |                |       |                         |
| TP001                | Temperature Setpoint | Floating Point | °C    | 70                      |
| TP002                | Hysteresis           | Floating Point | °C    | 5                       |

**Table 152: Operating and Tuning Parameters****19.5 Configurable Parameters**

The following table provides the information of configurable parameters for the equipment module instances:

| Parameter Name | Parameter Type     | Value/Reference            |
|----------------|--------------------|----------------------------|
| MY_TAG         |                    | RF1EQ_604C                 |
| OPR_CMD_MASK   |                    | 7                          |
| DY_REF_1       | Dynamic Reference  |                            |
| LSL_604C       | External Reference | RF1_LSL1604C/DI1/DISC_ACT  |
| LSM_604C       | External Reference | RF1_LSM1604C/DI1/DISC_ACT  |
| LSH_604C       | External Reference | RF1_LSH1604C/DI1/DISC_ACT  |
| RF_FILTRTN     | External Reference | #IGNORE                    |
| TT_604C        | External Reference | RF1_TT1604C/PV             |
| FILTER_AID_DEM | External Reference | #IGNORE                    |
| VACUUM         | External Reference | #IGNORE                    |
| XV_616A1_1A    | External Reference | RF1_V1616A1A/PV_D          |
| P682B_RUN      | External Reference | RF1_PU1682B/PV_D           |
| COMPLETED_TIME | Floating Point     | 0                          |
| RESET_COMP_TM  | Boolean            | False                      |
| STEP_TIME      | Floating Point     | 0                          |
| HOLD           | External Reference | RF1EQ_STARTUP3/OP005_VALUE |

**Table 153: Configurable Parameters****19.6 Named Set**

Following Named set shall be used for configuration of EM.

| Name            | Description                                     |
|-----------------|-------------------------------------------------|
| NS_RF_FLT_VPRD6 | Refinery Area: Filtration and Vacuum Production |

**Table 154: Named Set Details**

Details of states for NS\_RF\_FLT\_VPRD6 named set are as follows,

| Named Set Value | Named Set Name |
|-----------------|----------------|
| 00              | Hold           |
| 01              | Fill           |
| 02              | Drain          |

**Table 155: Name Set State Details**

## 20 EQUIPMENT MODULE CLASS DESIGN – EM\_RF\_OF\_BLEACH6 (1-635)

### 20.1 Functional Description

EM\_RF\_OF\_BLEACH6 is a class based equipment module. EM\_RF\_OF\_BLEACH6 is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. All devices used in EM shall be acquired in Production Command and shall be released on completion of Production Operation. Hold Request shall be enabled for all acquired devices at the end of Hold command.

### 20.2 Instances

The following Table contains the module instances of the equipment module class.

| Device Name | Module Class     | Instance Name |
|-------------|------------------|---------------|
| EM          | EM_RF_OF_BLEACH6 | RF1EQ_635     |
| DEV1        | _V12_IP_130_SE   | RF1_V1635S    |
| DEV2        | _V12_IP_130_SE   | RF1_V1635     |
| DEV3        | _V12_IP_130_SE   | RF1_V1635B    |
| DEV4        | _MRSS11_130_SE   | RF1_PU1635    |
| DEV5        | _CSD_130_SE      | RF1_LT1635    |

Table 156: Instances

### 20.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

#### 20.3.1 Command 00 – Hold

| Step  | Description                  | Action | Action Description                                                   |
|-------|------------------------------|--------|----------------------------------------------------------------------|
| H0010 | Command Hold Start           | A0010  | Store Fail Message                                                   |
|       |                              | A0020  | Reset Hold Request                                                   |
|       |                              | A0030  | Save Previous Step Time                                              |
| T1000 | Actions Complete or Time Out |        | Go to step H1000                                                     |
| H1000 | Hold Command 1               | A0010  | Closing and Disabling Conditional Logic for DEV1and Set Task Pointer |
|       |                              | A0020  | Closing DEV2                                                         |
|       |                              | A0030  | Opening DEV3                                                         |

| Step   | Description                               | Action | Action Description                               |
|--------|-------------------------------------------|--------|--------------------------------------------------|
|        |                                           | A0040  | Closing and Disabling Conditional Logic for DEV4 |
|        |                                           | A0050  | Set DEV5 Mode to Remote Out                      |
|        |                                           | A0060  | Set DEV5 output to Closed                        |
| T9900A | Actions Complete or Time Out & Hold=True  |        | Go to step H9900A                                |
| T9900B | Actions Complete or Time Out & Hold=False |        | Go to step H9900B                                |
|        |                                           |        |                                                  |
| H9900A | Command Hold End                          | A0010  | Completed                                        |
| END    | Command Complete                          |        | True                                             |
|        |                                           |        |                                                  |
| H9900B | Release Devices                           | A0010  | Release Devices Message                          |
|        |                                           | A0020  | Release Private Device                           |
|        |                                           | A0030  | Release Private Device                           |
|        |                                           | A0040  | Release Private Device                           |
|        |                                           | A0050  | Release Private Device                           |
|        |                                           | A0060  | Release Private Device                           |
|        |                                           | A0170  | Reset Parameters                                 |
| END1   | Command Complete                          |        | True                                             |

Table 157: Hold Sequence

## 20.3.2

## Command 01 –Production

| Step   | Description        | Action | Action Description                                   |
|--------|--------------------|--------|------------------------------------------------------|
| S0000  | Start              | A0010  | Set Message: Starting Production Command             |
| T0010  | First Pass = False |        | If First Pass = False then go to step S0010          |
| T0020A | First Pass = True  |        | If First Pass = True then go to step S0020           |
|        |                    |        |                                                      |
| S0010  | Initializations    | A0010  | First Pass Logic                                     |
|        |                    | A0020  | First Pass = True                                    |
|        |                    | A0030  | Set Task Pointer                                     |
| T0020B | Actions Complete   |        | If no pending actions in S0010 then go to step S0020 |
|        |                    |        |                                                      |
| S0020  | Acquire Devices    | A0010  | Acquire Devices Message                              |
|        |                    | A0020  | Acquire Private Device                               |

| Step   | Description | Action | Action Description                                                 |
|--------|-------------|--------|--------------------------------------------------------------------|
|        |             | A0030  | Acquire Private Device                                             |
|        |             | A0040  | Acquire Private Device                                             |
|        |             | A0050  | Acquire Private Device                                             |
|        |             | A0060  | Acquire Private Device                                             |
|        |             | A0120  | Set Hold Request                                                   |
| T0100  | Standby     |        | If Task Pointer = 1 then go to step S0100                          |
| T0200B | Fill        |        | If Task Pointer = 2 then go to step S0200                          |
| T0300B | Production  |        | If Task Pointer = 3 then go to step S0300                          |
| T0400B | Drain       |        | If Task Pointer = 4 then go to step S0400                          |
| T0500B | Drain Wait  |        | If Task Pointer = 5 then go to step S0500                          |
|        |             |        |                                                                    |
| S0100  | Standby     | A0010  | Closing DEV1                                                       |
|        |             | A0020  | Closing DEV2                                                       |
|        |             | A0030  | Closing DEV3                                                       |
|        |             | A0040  | Closing DEV4                                                       |
|        |             | A0050  | Set DEV5 Mode to Remote Out                                        |
|        |             | A0060  | Set DEV5 output to Closed                                          |
|        |             | A0070  | Set Message                                                        |
| T0200A | Fill        |        | If EM_BLEACH3 is in step <Production> then go to step S0200        |
|        |             |        |                                                                    |
| S0200  | Fill        | A0010  | Enabling Conditional Logic for DEV1 and Set Task Pointer           |
|        |             | A0020  | Opening DEV2                                                       |
|        |             | A0030  | Closing DEV3                                                       |
|        |             | A0040  | Closing DEV4                                                       |
|        |             | A0050  | Set DEV5 Mode to Remote Out                                        |
|        |             | A0060  | Set DEV5 output to Closed                                          |
|        |             | A0070  | Set Message                                                        |
| T0300A | Production  |        | If DEV5 >= LT_635 reach the setpoint (TP001) then go to step S0300 |
|        |             |        |                                                                    |
| S0300  | Production  | A0010  | Enabling Conditional Logic for DEV1 and Set Task Pointer           |
|        |             | A0020  | Opening DEV2                                                       |
|        |             | A0030  | Closing DEV3                                                       |
|        |             | A0040  | Enabling Conditional Logic for DEV4                                |
|        |             | A0050  | Set DEV5 Mode to Remote Cascade                                    |
|        |             | A0060  | Setting DEV5 setpoint to LT_635 RSP                                |
| T0400A | Drain       |        | If EM_BLEACH3 is in step <Drain Wait> then go to step S0400        |
|        |             |        |                                                                    |

| Step   | Description     | Action | Action Description                                                                  |
|--------|-----------------|--------|-------------------------------------------------------------------------------------|
| S0400  | Drain           | A0010  | Enabling Conditional Logic for DEV1 and Set Task Pointer                            |
|        |                 | A0020  | Opening DEV2                                                                        |
|        |                 | A0030  | Closing DEV3                                                                        |
|        |                 | A0040  | Enabling Conditional Logic for DEV4                                                 |
|        |                 | A0050  | Set DEV5 Mode to Remote Out                                                         |
|        |                 | A0060  | Set DEV5 output to Open                                                             |
|        |                 | A0070  | Set Message: Checking LSL_635 is in alarm                                           |
|        |                 | A0080  | Set Scratch Time                                                                    |
| T0500A | Drain Wait      |        | If LSL_635 is in alarm and 30 secs (DELAY_TIMER) time is over then go to step S0500 |
| S0500  | Drain Wait      | A0010  | Closing and Disabling Conditional Logic for DEV1and Set Task Pointer                |
|        |                 | A0020  | Closing DEV2                                                                        |
|        |                 | A0030  | Enabling Conditional Logic for DEV3                                                 |
|        |                 | A0040  | Closing and Disabling Conditional Logic for DEV4                                    |
|        |                 | A0050  | Set DEV5 Mode to Remote Out                                                         |
|        |                 | A0060  | Set DEV5 output to Open                                                             |
|        |                 | A0070  | Set Blow Time                                                                       |
|        |                 | A0080  | Start Blow Time                                                                     |
|        |                 | A0090  | Blow Time Message                                                                   |
| T0600  | Close Devices   |        | If Blowing time XV 1-635-B (TP004) is elapsed then go to step S0600                 |
| S0600  | Close Devices   | A0010  | Closing and Disabling Conditional Logic for DEV1and Set Task Pointer                |
|        |                 | A0020  | Closing DEV2                                                                        |
|        |                 | A0030  | Closing and Disabling Conditional Logic for DEV3                                    |
|        |                 | A0040  | Closing and Disabling Conditional Logic for DEV4                                    |
|        |                 | A0050  | Set DEV5 Mode to Remote Out                                                         |
|        |                 | A0060  | Set DEV5 output to Closed                                                           |
| T0700  | Disable HM      |        | If no pending actions in S0600 then go to step S0700                                |
| S0700  | Disable HM      | A0010  | Disable HM 01 (Module Ownership Lost)                                               |
| T0800  | Release Devices |        | If no pending actions in S0700 then go to step S0800                                |
| S0800  | Release Devices | A0010  | Release Devices Message                                                             |
|        |                 | A0020  | Release Private Device                                                              |

| Step  | Description      | Action | Action Description                                   |
|-------|------------------|--------|------------------------------------------------------|
|       |                  | A0030  | Release Private Device                               |
|       |                  | A0040  | Release Private Device                               |
|       |                  | A0050  | Release Private Device                               |
|       |                  | A0060  | Release Private Device                               |
| T9900 | Complete         |        | If no pending actions in S0800 then go to step S9900 |
| S9900 | Complete         | A0010  | Set Message: Production Command Complete             |
|       |                  | A0020  | Self Release the EM                                  |
| END   | Command Complete |        | True                                                 |

Table 158: Production Sequence

## 20.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

| Parameter            | Description | Type           | Units | Default Value/Reference |
|----------------------|-------------|----------------|-------|-------------------------|
| Operating Parameters |             |                |       |                         |
| NA                   |             |                |       |                         |
| Tuning Parameters    |             |                |       |                         |
| TP001                | LT_635 RSP  | Floating Point | %     | 5                       |
| TP002                | TT_635 RSP  | Floating Point | °C    | 70                      |
| TP003                | Hysteresis  | Floating Point | NA    | 5                       |
| TP004                | Blow Time   | Floating Point | sec   | 0                       |

Table 159: Operating and Tuning Parameters

## 20.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

| Parameter Name | Parameter Type     | Value/Reference            |
|----------------|--------------------|----------------------------|
| MY_TAG         |                    | RF1EQ_635                  |
| OPR_CMD_MASK   |                    | 3                          |
| EM_BLEACH3     | String             | RF1EQ_T503                 |
| DY_REF_1       | Dynamic Reference  |                            |
| HOLD           | External Reference | RF1EQ_STARTUP3/OP005_VALUE |
| DELAY_TIMER    | Floating Point     | 30                         |

| Parameter Name | Parameter Type     | Value/Reference          |
|----------------|--------------------|--------------------------|
| LSL_635        | External Reference | RF1_LSL1635/DI1/DISC_ACT |
| STEP_TIME      | Floating Point     | 0                        |
| COMPLETED_TIME | Floating Point     | 0                        |
| RESET_COMP_TM  | Boolean            | False                    |

**Table 160: Configurable Parameters**

## 20.6 Named Set

Following Named set shall be used for configuration of EM.

| Name             | Description                                     |
|------------------|-------------------------------------------------|
| NS_RF_OF_BLEACH6 | Refinery Area: Filtration and Vacuum Production |

**Table 161: Named Set Details**

Details of states for NS\_RF\_OF\_BLEACH6 named set are as follows,

| Named Set Value | Named Set Name |
|-----------------|----------------|
| 00              | Hold           |
| 01              | Production     |

**Table 162: Name Set State Details**

## 21 EQUIPMENT MODULE CLASS DESIGN – EM\_RF\_STARTUP2 (STARTUP)

### 21.1 Functional Description

EM\_RF\_STARTUP2 is a class based equipment module that drives two other sequences to complete the washing process for different products as per the selected jobs. EM\_RF\_STARTUP2 is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. Initially Operator shall be provided only with Hold and Aux On. Aux Off and Production Command shall be enabled during the execution of Aux On State. All devices used in EM shall be acquired in Aux On Command and shall be released in Aux Off Command. Hold Request shall be enabled for all acquired devices at the end of Hold command.

### 21.2 Instances

The following Table contains the module instances of the equipment module class.

| Device Name | Module Class     | Instance Name |
|-------------|------------------|---------------|
| EM          | EM_RF_STARTUP2   | RF1EQ_500     |
| DEV1        | _MRSS11_130_SE   | RF1_PU1078HW  |
| DEV2        | _MVS11_IL_130_SO | Ignored       |
| DEV3        | _MRSS11_130_SE   | RF1_PU1032C   |
| DEV4        | _V12_IP_130_SE   | RF1_V1034CAF  |
| DEV5        | _V12_IP_130_SE   | RF1_V1034NAF  |
| DEV6        | EM_RF1_CPO_WASH1 | RF1EQ_501     |
| DEV7        | EM_RF1_CPO_WASH2 | RF1EQ_582W    |

**Table 163: Instances**

### 21.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

#### 21.3.1 Command 00 – Hold

| Step  | Description                              | Action | Action Description |
|-------|------------------------------------------|--------|--------------------|
| H0010 | Command Hold Start                       | A0010  | Store Fail Message |
|       |                                          | A0020  | Reset Hold Request |
| T1000 | Actions Complete or Time Out & Hold=True |        | Go to step H1000   |

| Step   | Description                               | Action | Action Description                                |
|--------|-------------------------------------------|--------|---------------------------------------------------|
| T2000  | Actions Complete or Time Out & Hold=False |        | Go to step H2000                                  |
|        |                                           |        |                                                   |
| H1000  | Hold Command 1                            | A0010  | Stopping and Disabling Conditional Logic for DEV3 |
| T9900A | Actions Complete or Time Out              |        | Go to step H9900A                                 |
|        |                                           |        |                                                   |
| H2000  | General reset                             | A0010  | Stopping DEV1                                     |
|        |                                           | A0020  | Stopping and Disabling Conditional Logic for DEV2 |
|        |                                           | A0030  | Stopping and Disabling Conditional Logic for DEV3 |
|        |                                           | A0040  | Closing and Disabling Conditional Logic for DEV4  |
|        |                                           | A0050  | Closing and Disabling Conditional Logic for DEV5  |
| T9900B | Actions Complete or Time Out              |        | Go to step H9900B                                 |
|        |                                           |        |                                                   |
| H9900A | Common Hold End                           | A0010  | Completed                                         |
| END    | Command Complete                          |        | True                                              |
|        |                                           |        |                                                   |
| H9900B | Release Devices                           | A0010  | Release Devices Message                           |
|        |                                           | A0020  | Release Private Devices                           |
|        |                                           | A0030  | Release Private Devices                           |
|        |                                           | A0040  | Release Private Devices                           |
|        |                                           | A0050  | Release Private Devices                           |
|        |                                           | A0060  | Reset Parameters                                  |
| END1   | Command Complete                          |        | True                                              |

**Table 164: Hold Sequence**

## 21.3.2

## Command 01 – Aux On

| Step   | Description        | Action | Action Description                          |
|--------|--------------------|--------|---------------------------------------------|
| S0000  | Start              | A0010  | Set Message: Starting Aux On Command        |
| T0010  | First Pass = False |        | If First Pass = False then go to step S0010 |
| T0020A | First Pass = True  |        | If First Pass = True then go to step S0020  |

| Step   | Description     | Action | Action Description                                          |
|--------|-----------------|--------|-------------------------------------------------------------|
|        |                 |        |                                                             |
| S0010  | Initializations | A0010  | First Pass Logic                                            |
|        |                 | A0020  | First Pass = True                                           |
|        |                 | A0030  | Set Task Pointer                                            |
| T0020B | Acquire Devices |        | If no pending actions in S0010 then go to step S0020        |
|        |                 |        |                                                             |
| S0020  | Acquire Devices | A0010  | Acquire Devices Message                                     |
|        |                 | A0020  | Acquire Private Devices                                     |
|        |                 | A0030  | Acquire Private Devices                                     |
|        |                 | A0040  | Acquire Private Devices                                     |
|        |                 | A0050  | Acquire Private Devices                                     |
|        |                 | A0060  | Reset Hold Request                                          |
| T0100  | Standby         |        | If Task Pointer = 1 then go to step S0100                   |
| T0200B | Start Aux       |        | If Task Pointer = 2 then go to step S0200                   |
| T0300B | Ready           |        | If Task Pointer = 3 then go to step S0300                   |
|        |                 |        |                                                             |
| S0100  | Standby         | A0010  | Stopping DEV1                                               |
|        |                 | A0020  | Stopping and Disabling Conditional Logic for DEV2           |
|        |                 | A0030  | Stopping and Disabling Conditional Logic for DEV3           |
|        |                 | A0040  | Closing and Disabling Conditional Logic for DEV4            |
|        |                 | A0050  | Closing and Disabling Conditional Logic for DEV5            |
| T0200  | Start Aux       |        | Go to the step S0200                                        |
|        |                 |        |                                                             |
| S0200  | Start Aux       | A0010  | Starting DEV1                                               |
|        |                 | A0020  | Enabling Conditional Logic for DEV2                         |
|        |                 | A0030  | Enabling Conditional Logic for DEV3                         |
|        |                 | A0040  | Enabling Conditional Logic for DEV4                         |
|        |                 | A0050  | Enabling Conditional Logic for DEV5                         |
| T0300  | Ready           |        | If DEV1 running feedback is available then go to step S0300 |
|        |                 |        |                                                             |
| S0300  | Ready           | A0010  | Starting DEV1                                               |
|        |                 | A0020  | Enabling Conditional Logic for DEV2                         |
|        |                 | A0030  | Enabling Conditional Logic for DEV3                         |
|        |                 | A0040  | Enabling Conditional Logic for DEV4                         |
|        |                 | A0050  | Enabling Conditional Logic for DEV5                         |
| T9900  | Complete        |        | Go to the step S9900                                        |
|        |                 |        |                                                             |

| Step  | Description      | Action | Action Description                   |
|-------|------------------|--------|--------------------------------------|
| S9900 | Complete         | A0010  | Set Message: Aux On Command Complete |
| END   | Command Complete |        | True                                 |

**Table 165: Aux On Sequence**

## 21.3.3 Command 02 – Aux Off

| Step  | Description     | Action | Action Description                                          |
|-------|-----------------|--------|-------------------------------------------------------------|
| S0000 | Start           | A0010  | Set Message: Starting Aux Off Command                       |
| T0100 | Aux Off         |        | Go to step S0100                                            |
| S0100 | Aux Off         | A0010  | Starting DEV1                                               |
|       |                 | A0020  | Stopping and Disabling Conditional Logic for DEV2           |
|       |                 | A0030  | Stopping and Disabling Conditional Logic for DEV3           |
|       |                 | A0040  | Closing and Disabling Conditional Logic for DEV4            |
|       |                 | A0050  | Closing and Disabling Conditional Logic for DEV5            |
|       |                 | A0060  | Reset Drain Command                                         |
| T0200 | Standby         |        | If Aux Off button (OP001) is selected then go to step S0200 |
| S0200 | Standby         | A0010  | Stopping DEV1                                               |
|       |                 | A0020  | Stopping and Disabling Conditional Logic for DEV2           |
|       |                 | A0030  | Stopping and Disabling Conditional Logic for DEV3           |
|       |                 | A0040  | Closing and Disabling Conditional Logic for DEV4            |
|       |                 | A0050  | Closing and Disabling Conditional Logic for DEV5            |
| T0300 | Disable HM      |        | If no actions pending in S0200 then go to step S0300        |
| S0300 | Disable HM      | A0010  | Disable HM 01 (Module Ownership Lost)                       |
| T0400 | Release Devices |        | Go to step S0400                                            |
| S0400 | Release Devices | A0010  | Release Devices Message                                     |
|       |                 | A0020  | Release Private Devices                                     |
|       |                 | A0030  | Release Private Devices                                     |
|       |                 | A0040  | Release Private Devices                                     |
|       |                 | A0050  | Release Private Devices                                     |
| T9900 | Complete        |        | Go to step S9900                                            |



| Step  | Description      | Action | Action Description                    |
|-------|------------------|--------|---------------------------------------|
| S9900 | Complete         | A0010  | Set Message: Aux Off Command Complete |
| END   | Command Complete |        | True                                  |

**Table 166: Aux Off Sequence**

## 21.3.4 Command 03 – Production

| Step  | Description      | Action | Action Description                                                                                           |
|-------|------------------|--------|--------------------------------------------------------------------------------------------------------------|
| S0000 | Start            | A0010  | Set Message: Starting Production Command                                                                     |
| T0100 | Production       |        | If Task Pointer = 3 then go to step S0100                                                                    |
| S0100 | Production       | A0010  | Starting DEV1                                                                                                |
|       |                  | A0020  | Enabling Conditional Logic for DEV2                                                                          |
|       |                  | A0030  | Enabling Conditional Logic for DEV3                                                                          |
|       |                  | A0040  | Enabling Conditional Logic for DEV4                                                                          |
|       |                  | A0050  | Enabling Conditional Logic for DEV5                                                                          |
|       |                  | A0060  | Acquire DEV6                                                                                                 |
|       |                  | A0070  | Acquire DEV7                                                                                                 |
|       |                  | A0080  | Start DEV6                                                                                                   |
|       |                  | A0090  | Start DEV7                                                                                                   |
| T9900 | Complete         |        | If <Drain> command (OP003) selected by operator and Hold button (OP004) is not pressed then go to step S9900 |
| S9900 | Complete         | A0010  | Set Message: Production Command Complete                                                                     |
|       |                  | A0020  | Set Message: Press Drain Button to start Drain (OP001)                                                       |
| END   | Command Complete |        | True                                                                                                         |

**Table 167: Production Sequence**

## 21.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

| Parameter            | Description | Type    | Units | Default Value/Reference |
|----------------------|-------------|---------|-------|-------------------------|
| Operating Parameters |             |         |       |                         |
| OP001                | AUX         | Boolean | NA    | OFF                     |
| OP002                | RUN         | Boolean | NA    | OFF                     |
| OP003                | DRAIN       | Boolean | NA    | OFF                     |

| Parameter         | Description | Type    | Units | Default Value/Reference |
|-------------------|-------------|---------|-------|-------------------------|
| OP004             | HOLD        | Boolean | NA    | OFF                     |
| Tuning Parameters |             |         |       |                         |
| NA                |             |         |       |                         |

**Table 168: Operating and Tuning Parameters**

## 21.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

| Parameter Name | Parameter Type | Value Reference |
|----------------|----------------|-----------------|
| MY_TAG         |                | RF1EQ_500       |
| OPR_CMD_MASK   |                | 15              |

**Table 169: Configurable Parameters**

## 21.6 Named Set

Following Named set shall be used for configuration of EM.

| Name           | Description                               |
|----------------|-------------------------------------------|
| NS_RF_STARTUP2 | Refinery Area: Startup EM for CPO Washing |

**Table 170: Named Set Details**

Details of states for NS\_RF\_STARTUP2 named set are as follows,

| Named Set Value | Named Set Name |
|-----------------|----------------|
| 00              | Hold           |
| 01              | Aux On         |
| 02              | Aux Off        |
| 03              | Production     |

**Table 171: Name Set State Details**

## 22 EQUIPMENT MODULE CLASS DESIGN – EM\_RF\_CPO\_WASH1 (1-501)

### 22.1 Functional Description

EM\_RF\_CPO\_WASH1 is a class based equipment module that performs production operation and later blows out the material to complete the draining process. EM\_RF\_CPO\_WASH1 is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. All devices used in EM shall be acquired in Production Command and shall be released on completion of Drain Command. Hold Request shall be enabled for all acquired devices at the end of Hold command.

### 22.2 Instances

The following Table contains the module instances of the equipment module class.

| Device Name | Module Class     | Instance Name |
|-------------|------------------|---------------|
| EM          | EM_RF_CPO_WASH1  | RF1EQ_501     |
| DEV1        | _MVS11_IL_130_SO | RF1_PU1001    |
| DEV2        | _MVS11_IL_130_SO | RF1_PU1034CA  |
| DEV3        | _MVS11_IL_130_SO | RF1_PU1034NA  |
| DEV4        | _MRSS11_130_CUR  | RF1_MX1004A   |
| DEV5        | _MVS11_IL_130_SO | Ignored       |
| DEV6        | _V12_IP_130_SE   | RF1_V1001B    |
| DEV7        | _V12_IP_130_SE   | RF2_V2024F    |
| DEV8        | _V12_IP_130_SE   | RF1_V1024F    |
| DEV9        | _V12_IP_130_SE   | RF1_V1034CA   |
| DEV10       | _V12_IP_130_SE   | RF1_V1034NA   |
| DEV11       | _V12_IP_130_SE   | RF1_V1001D    |
| DEV12       | _MRSS11_130_SE   | RF1_AG1001    |
| DEV13       | _MVS11_IL_130_SO | RF1_PU1034NA  |
| DEV14       | _MRSS11_130_SE   | RF1_AG1034CA  |
| DEV15       | _CSD_130_SE      | RF1_TT1021D   |
| DEV16       | _CSD_130_SE      | RF1_FT1978    |
| DEV17       | _CMT_130_SE      | RF1_FT1034NA  |

Table 172: Instances

### 22.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

## 22.3.1 Command 00 – Hold

| Step   | Description                               | Action | Action Description                                 |
|--------|-------------------------------------------|--------|----------------------------------------------------|
| H0010  | Command Hold Start                        | A0010  | Store Fail Message                                 |
|        |                                           | A0020  | Reset Hold Request                                 |
|        |                                           | A0030  | Save Previous Step Time                            |
| T1000  | Actions Complete or Time Out & Hold=True  |        | Go to step H1000                                   |
| T2000  | Actions Complete or Time Out & Hold=False |        | Go to step H2000                                   |
|        |                                           |        |                                                    |
| H1000  | Hold Command 1                            | A0010  | Stopping and Disabling Conditional logic for DEV1  |
|        |                                           | A0020  | Stopping and Disabling Conditional logic for DEV2  |
|        |                                           | A0030  | Stopping and Disabling Conditional logic for DEV3  |
|        |                                           | A0040  | Stopping and Disabling Conditional logic for DEV4  |
| T9900A | Actions Complete or Time Out              |        | Go to step H9900A                                  |
|        |                                           |        |                                                    |
| H2000  | General Reset                             | A0010  | Stopping and Disabling Conditional logic for DEV1  |
|        |                                           | A0020  | Stopping and Disabling Conditional logic for DEV2  |
|        |                                           | A0030  | Stopping and Disabling Conditional logic for DEV3  |
|        |                                           | A0040  | Stopping and Disabling Conditional logic for DEV4  |
|        |                                           | A0050  | Stopping and Disabling Conditional logic for DEV5  |
|        |                                           | A0060  | Closing and Disabling Conditional Logic for DEV6   |
|        |                                           | A0070  | Closing and Disabling Conditional Logic for DEV7   |
|        |                                           | A0080  | Closing and Disabling Conditional Logic for DEV8   |
|        |                                           | A0090  | Closing and Disabling Conditional Logic for DEV9   |
|        |                                           | A0100  | Closing and Disabling Conditional Logic for DEV10  |
|        |                                           | A0110  | Closing DEV11                                      |
|        |                                           | A0120  | Stopping and Disabling Conditional logic for DEV12 |

| Step   | Description                  | Action | Action Description                                       |
|--------|------------------------------|--------|----------------------------------------------------------|
|        |                              | A0130  | Stopping and Disabling Conditional logic for DEV13       |
|        |                              | A0140  | Stopping and Disabling Conditional logic for DEV14       |
|        |                              | A0150  | Set DEV15 Mode to Remote Out                             |
|        |                              | A0160  | Set DEV15 output to Closed and Disable Conditional Logic |
|        |                              | A0170  | Set DEV16 Mode to Remote Out                             |
|        |                              | A0180  | Set DEV16 output to Closed and Disable Conditional Logic |
|        |                              | A0190  | Set DEV17 Mode to Remote Out                             |
|        |                              | A0200  | Set DEV17 output to Closed and Disable Conditional Logic |
|        |                              | A0210  | Set DEV18 Mode to Remote Out                             |
|        |                              | A0220  | Set DEV18 output to Closed and Disable Conditional Logic |
|        |                              | A0230  | Set DEV19 Mode to Remote Out                             |
|        |                              | A0240  | Set DEV19 output to Closed and Disable Conditional Logic |
|        |                              | A0250  | Stopping and Disabling Conditional logic for DEV23       |
| T9900B | Actions Complete or Time Out |        | Go to step H9900B                                        |
|        |                              |        |                                                          |
| H9900A | Command Hold End             | A0010  | Completed                                                |
| END    | Command Complete             |        | True                                                     |
|        |                              |        |                                                          |
| H9900B | Release Devices              | A0010  | Release Devices Message                                  |
|        |                              | A0020  | Release Private Devices                                  |
|        |                              | A0030  | Release Private Devices                                  |
|        |                              | A0040  | Release Private Devices                                  |
|        |                              | A0050  | Release Private Devices                                  |
|        |                              | A0060  | Release Private Devices                                  |
|        |                              | A0070  | Release Private Devices                                  |
|        |                              | A0080  | Release Shared Devices                                   |
|        |                              | A0090  | Release Shared Devices                                   |
|        |                              | A0100  | Release Private Devices                                  |
|        |                              | A0110  | Release Private Devices                                  |
|        |                              | A0120  | Release Private Devices                                  |
|        |                              | A0130  | Release Private Devices                                  |
|        |                              | A0140  | Release Private Devices                                  |
|        |                              | A0150  | Release Private Devices                                  |

| Step | Description      | Action | Action Description      |
|------|------------------|--------|-------------------------|
|      |                  | A0160  | Release Private Devices |
|      |                  | A0170  | Release Private Devices |
|      |                  | A0180  | Release Private Devices |
|      |                  | A0190  | Release Private Devices |
|      |                  | A0200  | Release Private Devices |
|      |                  | A0210  | Release Private Devices |
|      |                  | A0220  | Release Private Devices |
|      |                  | A0230  | Release Private Devices |
|      |                  | A0240  | Reset Parameters        |
| END1 | Command Complete |        | True                    |

**Table 173: Hold Sequence**

## 22.3.2 Command 01 – Production

| Step   | Description        | Action | Action Description                                   |
|--------|--------------------|--------|------------------------------------------------------|
| S0000  | Start              | A0010  | Set Message: Starting Production Command             |
| T0010  | First Pass = False |        | If First Pass = False then go to step S0010          |
| T0020A | First Pass = True  |        | If First Pass = True then go to step S0020           |
| S0010  | Initializations    | A0010  | First Pass Logic                                     |
|        |                    | A0020  | First Pass = True                                    |
|        |                    | A0030  | Set Task Pointer                                     |
| T0020B | Actions Complete   |        | If no pending actions in S0010 then go to step S0020 |
| S0020  | Acquire Devices    | A0010  | Acquire Devices Message                              |
|        |                    | A0020  | Acquire Private Devices                              |
|        |                    | A0030  | Acquire Private Devices                              |
|        |                    | A0040  | Acquire Private Devices                              |
|        |                    | A0050  | Acquire Private Devices                              |
|        |                    | A0060  | Acquire Private Devices                              |
|        |                    | A0070  | Acquire Private Devices                              |
|        |                    | A0080  | Acquire Shared Devices                               |
|        |                    | A0090  | Acquire Shared Devices                               |
|        |                    | A0100  | Acquire Private Devices                              |
|        |                    | A0110  | Acquire Private Devices                              |
|        |                    | A0120  | Acquire Private Devices                              |
|        |                    | A0130  | Acquire Private Devices                              |
|        |                    | A0140  | Acquire Private Devices                              |
|        |                    | A0150  | Acquire Private Devices                              |

| Step   | Description | Action | Action Description                                       |
|--------|-------------|--------|----------------------------------------------------------|
|        |             | A0160  | Acquire Private Devices                                  |
|        |             | A0170  | Acquire Private Devices                                  |
|        |             | A0180  | Acquire Private Devices                                  |
|        |             | A0190  | Acquire Private Devices                                  |
|        |             | A0200  | Acquire Private Devices                                  |
|        |             | A0210  | Acquire Private Devices                                  |
|        |             | A0220  | Acquire Private Devices                                  |
|        |             | A0230  | Acquire Private Devices                                  |
|        |             | A0240  | Reset Hold Request                                       |
| T0100  | Standby     |        | If Task Pointer = 3 then go to step S0100                |
| T0200B | Fill        |        | If Task Pointer = 4 then go to step S0200                |
| T0300B | Production  |        | If Task Pointer = 5 then go to step S0300                |
| S0100  | Standby     | A0010  | Stopping and Disabling Conditional logic for DEV1        |
|        |             | A0020  | Stopping and Disabling Conditional logic for DEV2        |
|        |             | A0030  | Stopping and Disabling Conditional logic for DEV3        |
|        |             | A0040  | Stopping and Disabling Conditional logic for DEV4        |
|        |             | A0050  | Stopping and Disabling Conditional logic for DEV5        |
|        |             | A0060  | Closing and Disabling Conditional Logic for DEV6         |
|        |             | A0070  | Closing and Disabling Conditional Logic for DEV7         |
|        |             | A0080  | Closing and Disabling Conditional Logic for DEV8         |
|        |             | A0090  | Closing and Disabling Conditional Logic for DEV9         |
|        |             | A0100  | Closing and Disabling Conditional Logic for DEV10        |
|        |             | A0110  | Closing DEV11                                            |
|        |             | A0120  | Stopping and Disabling Conditional logic for DEV12       |
|        |             | A0130  | Stopping and Disabling Conditional logic for DEV13       |
|        |             | A0140  | Stopping and Disabling Conditional logic for DEV14       |
|        |             | A0150  | Set DEV15 Mode to Remote Out                             |
|        |             | A0160  | Set DEV15 output to Closed and Disable Conditional Logic |
|        |             | A0170  | Set DEV16 Mode to Remote Out                             |
|        |             | A0180  | Set DEV16 output to Closed and Disable Conditional Logic |

| Step  | Description | Action | Action Description                                       |
|-------|-------------|--------|----------------------------------------------------------|
|       |             | A0190  | Set DEV17 Mode to Remote Out                             |
|       |             | A0200  | Set DEV17 output to Closed and Disable Conditional Logic |
|       |             | A0210  | Set DEV18 Mode to Remote Out                             |
|       |             | A0220  | Set DEV18 output to Closed and Disable Conditional Logic |
|       |             | A0230  | Set DEV19 Mode to Remote Out                             |
|       |             | A0240  | Set DEV19 output to Closed and Disable Conditional Logic |
|       |             | A0250  | Stopping and Disabling Conditional logic for DEV23       |
| T0200 | Fill        |        | If no pending actions in S0100 then go to step S0200     |
| S0200 | Fill        | A0010  | Stopping and Disabling Conditional logic for DEV1        |
|       |             | A0020  | Enabling Conditional logic for DEV2                      |
|       |             | A0030  | Stopping and Disabling Conditional logic for DEV3        |
|       |             | A0040  | Enabling Conditional logic for DEV4                      |
|       |             | A0050  | Stopping and Disabling Conditional logic for DEV5        |
|       |             | A0060  | Closing and Disabling Conditional Logic for DEV6         |
|       |             | A0070  | Enabling Conditional Logic for DEV7                      |
|       |             | A0080  | Enabling Conditional Logic for DEV8                      |
|       |             | A0090  | Enabling Conditional Logic for DEV9                      |
|       |             | A0100  | Closing and Disabling Conditional Logic for DEV10        |
|       |             | A0110  | Opening DEV11                                            |
|       |             | A0120  | Stopping and Disabling Conditional logic for DEV12       |
|       |             | A0130  | Enabling Conditional logic for DEV13                     |
|       |             | A0140  | Stopping and Disabling Conditional logic for DEV14       |
|       |             | A0150  | Set DEV15 Mode to Remote Out                             |
|       |             | A0160  | Set DEV15 output to Closed and Disable Conditional Logic |
|       |             | A0170  | Set DEV16 Mode to CAS                                    |
|       |             | A0180  | Enabling Conditional Logic for DEV16                     |
|       |             | A0190  | Set DEV17 Mode to Remote Out                             |
|       |             | A0200  | Set DEV17 output to Closed and Disable Conditional Logic |
|       |             | A0210  | Set DEV18 Mode to Remote Out                             |
|       |             | A0220  | Set DEV18 output to Closed and Disable Conditional Logic |

| Step  | Description | Action | Action Description                                                                                                                                                          |
|-------|-------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |             | A0230  | Set DEV19 Mode to Cascade                                                                                                                                                   |
|       |             | A0240  | Enabling Conditional Logic For DEV19                                                                                                                                        |
|       |             | A0250  | Set DEV20 Mode to Cascade                                                                                                                                                   |
|       |             | A0260  | Set DEV21 Mode to Cascade                                                                                                                                                   |
|       |             | A0270  | Set DEV22 Mode to Manual                                                                                                                                                    |
|       |             | A0280  | Enabling Conditional logic for DEV23                                                                                                                                        |
|       |             | A0290  | Waiting for Level                                                                                                                                                           |
|       |             | A0300  | Set Scratch Time                                                                                                                                                            |
| T0300 | Production  |        | If (LSL_001 is not in alarm & LT_001_VLL is not in VLL alarm and 2 minutes (TP007) completed) or ( LSH_001 is in alarm Or LT_001_VHL is in VHL alarm) then go to step S0300 |
|       |             |        |                                                                                                                                                                             |
| S0300 | Production  | A0010  | Enabling Conditional logic for DEV1                                                                                                                                         |
|       |             | A0020  | Enabling Conditional logic for DEV2                                                                                                                                         |
|       |             | A0030  | Enabling Conditional logic for DEV3                                                                                                                                         |
|       |             | A0040  | Enabling Conditional logic for DEV4                                                                                                                                         |
|       |             | A0050  | Enabling Conditional logic for DEV5                                                                                                                                         |
|       |             | A0060  | Closing and Disabling Conditional Logic for DEV6                                                                                                                            |
|       |             | A0070  | Enabling Conditional Logic for DEV7                                                                                                                                         |
|       |             | A0080  | Enabling Conditional Logic for DEV8                                                                                                                                         |
|       |             | A0090  | Enabling Conditional Logic for DEV9                                                                                                                                         |
|       |             | A0100  | Enabling Conditional Logic for DEV10                                                                                                                                        |
|       |             | A0110  | Opening DEV11                                                                                                                                                               |
|       |             | A0120  | Enabling Conditional logic for DEV12                                                                                                                                        |
|       |             | A0130  | Enabling Conditional logic for DEV13                                                                                                                                        |
|       |             | A0140  | Enabling Conditional logic for DEV14                                                                                                                                        |
|       |             | A0150  | Set DEV15 Mode to Remote Cascade                                                                                                                                            |
|       |             | A0160  | Setting DEV15 setpoint (TT_021D Setpoint) and Enabling Conditional Logic                                                                                                    |
|       |             | A0170  | Set DEV16 Mode to Cascade                                                                                                                                                   |
|       |             | A0180  | Enabling Conditional Logic for DEV16                                                                                                                                        |
|       |             | A0190  | Set DEV17 Mode to Remote Cascade                                                                                                                                            |
|       |             | A0200  | Enabling Conditional Logic for DEV17                                                                                                                                        |
|       |             | A0210  | Set DEV18 Mode to Remote Cascade                                                                                                                                            |
|       |             | A0220  | Enabling Conditional Logic for DEV18                                                                                                                                        |
|       |             | A0230  | Set DEV19 Mode to Cascade                                                                                                                                                   |
|       |             | A0240  | Enabling Conditional Logic for DEV19                                                                                                                                        |
|       |             | A0250  | Set DEV20 Mode to Cascade                                                                                                                                                   |
|       |             | A0260  | Set DEV21 Mode to Cascade                                                                                                                                                   |
|       |             | A0270  | Set DEV22 Mode to Cascade                                                                                                                                                   |

| Step  | Description      | Action | Action Description                       |
|-------|------------------|--------|------------------------------------------|
|       |                  | A0280  | Enabling Conditional logic for DEV23     |
|       |                  | A0290  | Set Scratch Time                         |
| T9900 | Complete         |        | Go to step S9900                         |
|       |                  |        |                                          |
| S9900 | Complete         | A0010  | Set Message: Production Command complete |
| END   | Command Complete |        | True                                     |

**Table 174: Production Sequence**

## 22.3.3 Command 02 – Drain

| Step   | Description  | Action | Action Description                                                       |
|--------|--------------|--------|--------------------------------------------------------------------------|
| S0000  | Start        | A0010  | Set Message: Starting Drain Command                                      |
| T0100  | Drain        |        | If Task Pointer = 3 then go to step S0100                                |
| T0200B | Blow         |        | If Task Pointer = 4 then go to step S0200                                |
| T0300B | Close Device |        | If Task Pointer = 5 then go to step S0300                                |
|        |              |        |                                                                          |
| S0100  | Drain        | A0010  | Enabling Conditional logic for DEV1                                      |
|        |              | A0020  | Stopping and Disabling Conditional logic for DEV2                        |
|        |              | A0030  | Enabling Conditional logic for DEV3                                      |
|        |              | A0040  | Enabling Conditional logic for DEV4                                      |
|        |              | A0050  | Enabling Conditional logic for DEV5                                      |
|        |              | A0060  | Closing and Disabling Conditional Logic for DEV6                         |
|        |              | A0070  | Closing and Disabling Conditional Logic for DEV7                         |
|        |              | A0080  | Closing and Disabling Conditional Logic for DEV8                         |
|        |              | A0090  | Closing and Disabling Conditional Logic for DEV9                         |
|        |              | A0100  | Enabling Conditional Logic for DEV10                                     |
|        |              | A0110  | Opening DEV11                                                            |
|        |              | A0120  | Enabling Conditional logic for DEV12                                     |
|        |              | A0130  | Enabling Conditional logic for DEV13                                     |
|        |              | A0140  | Stopping DEV14 and Disabling Conditional logic                           |
|        |              | A0150  | Set DEV15 Mode to Remote Cascade                                         |
|        |              | A0160  | Setting DEV15 setpoint (TT_021D Setpoint) and Enabling Conditional Logic |
|        |              | A0170  | Set DEV16 Mode to Remote Out                                             |
|        |              | A0180  | Set DEV16 output to Closed and Disable Conditional Logic                 |
|        |              | A0190  | Set DEV17 Mode to Cascade                                                |

| Step  | Description | Action | Action Description                                                                                      |
|-------|-------------|--------|---------------------------------------------------------------------------------------------------------|
|       |             | A0200  | Enabling Conditional Logic for DEV17                                                                    |
|       |             | A0210  | Set DEV18 Mode to Cascade                                                                               |
|       |             | A0220  | Enabling Conditional Logic for DEV18                                                                    |
|       |             | A0230  | Set DEV19 Mode to Remote Out                                                                            |
|       |             | A0240  | Set DEV19 output to Closed and Disable Conditional Logic                                                |
|       |             | A0250  | Enabling Conditional logic for DEV23                                                                    |
|       |             | A0260  | Waiting for LSL_001, LT_001_VLL and FT_001_VLL to be in alarm                                           |
| T0200 | Blow        |        | If LSL_001 is in alarm & LT_001_VLL is in VLL alarm and FT_001_LL is in VLL alarm then go to step S0200 |
| S0200 | Blow        | A0010  | Stopping and Disabling Conditional logic for DEV1                                                       |
|       |             | A0020  | Stopping and Disabling Conditional logic for DEV2                                                       |
|       |             | A0030  | Stopping and Disabling Conditional logic for DEV3                                                       |
|       |             | A0040  | Stopping and Disabling Conditional logic for DEV4                                                       |
|       |             | A0050  | Stopping and Disabling Conditional logic for DEV5                                                       |
|       |             | A0060  | Enabling Conditional Logic for DEV6                                                                     |
|       |             | A0070  | Closing and Disabling Conditional Logic for DEV7                                                        |
|       |             | A0080  | Closing and Disabling Conditional Logic for DEV8                                                        |
|       |             | A0090  | Closing and Disabling Conditional Logic for DEV9                                                        |
|       |             | A0100  | Closing and Disabling Conditional Logic for DEV10                                                       |
|       |             | A0110  | Closing DEV11                                                                                           |
|       |             | A0120  | Stopping and Disabling Conditional logic for DEV12                                                      |
|       |             | A0130  | Stopping and Disabling Conditional logic for DEV13                                                      |
|       |             | A0140  | Stopping and Disabling Conditional logic for DEV14                                                      |
|       |             | A0150  | Set DEV15 Mode to Remote Out                                                                            |
|       |             | A0160  | Set DEV15 output to Closed and Disable Conditional Logic                                                |
|       |             | A0170  | Set DEV16 Mode to Remote Out                                                                            |
|       |             | A0180  | Set DEV16 output to Closed and Disable Conditional Logic                                                |
|       |             | A0190  | Set DEV17 Mode to Remote Out                                                                            |
|       |             | A0200  | Set DEV17 output to Closed and Disable Conditional Logic                                                |

| Step  | Description   | Action | Action Description                                         |
|-------|---------------|--------|------------------------------------------------------------|
|       |               | A0210  | Set DEV18 Mode to Remote Out                               |
|       |               | A0220  | Set DEV18 output to Closed and Disable Conditional Logic   |
|       |               | A0230  | Set DEV19 Mode to Remote Out                               |
|       |               | A0240  | Set DEV19 output to Closed and Disable Conditional Logic   |
|       |               | A0250  | Stopping and Disabling Conditional logic for DEV23         |
|       |               | A0260  | Set Blow time and Set Dynamic References                   |
|       |               | A0270  | Start Blow Timer                                           |
|       |               | A0280  | Blow time Message                                          |
|       |               | A0290  | Waiting for CPO wash to complete Production Command        |
|       |               | A0300  | Set Dynamic References                                     |
|       |               | A0310  | Starting CPO wash 2                                        |
| T0300 | Close Devices |        | If DEV6 Blowing Time (TP001) Elapsed then go to step S0300 |
| S0300 | Close Devices | A0010  | Stopping and Disabling Conditional logic for DEV1          |
|       |               | A0020  | Stopping and Disabling Conditional logic for DEV2          |
|       |               | A0030  | Stopping and Disabling Conditional logic for DEV3          |
|       |               | A0040  | Stopping and Disabling Conditional logic for DEV4          |
|       |               | A0050  | Stopping and Disabling Conditional logic for DEV5          |
|       |               | A0060  | Closing and Disabling Conditional Logic for DEV6           |
|       |               | A0070  | Closing and Disabling Conditional Logic for DEV7           |
|       |               | A0080  | Closing and Disabling Conditional Logic for DEV8           |
|       |               | A0090  | Closing and Disabling Conditional Logic for DEV9           |
|       |               | A0100  | Closing and Disabling Conditional Logic for DEV10          |
|       |               | A0110  | Closing DEV11                                              |
|       |               | A0120  | Stopping and Disabling Conditional logic for DEV12         |
|       |               | A0130  | Stopping and Disabling Conditional logic for DEV13         |
|       |               | A0140  | Stopping and Disabling Conditional logic for DEV14         |
|       |               | A0150  | Set DEV15 Mode to Remote Out                               |
|       |               | A0160  | Set DEV15 output to Closed and Disable Conditional Logic   |

| Step  | Description     | Action | Action Description                                       |
|-------|-----------------|--------|----------------------------------------------------------|
|       |                 | A0170  | Set DEV16 Mode to Remote Out                             |
|       |                 | A0180  | Set DEV16 output to Closed and Disable Conditional Logic |
|       |                 | A0190  | Set DEV17 Mode to Remote Out                             |
|       |                 | A0200  | Set DEV17 output to Closed and Disable Conditional Logic |
|       |                 | A0210  | Set DEV18 Mode to Remote Out                             |
|       |                 | A0220  | Set DEV18 output to Closed and Disable Conditional Logic |
|       |                 | A0230  | Set DEV19 Mode to Remote Out                             |
|       |                 | A0240  | Set DEV19 output to Closed and Disable Conditional Logic |
|       |                 | A0250  | Stopping and Disabling Conditional logic for DEV23       |
| S0400 | Disable HM      | A0010  | Disable HM 01 (Module Ownership Lost)                    |
| T0500 | Release Devices |        | If no pending actions in S0400 then go to step S0500     |
| S0500 | Release Devices | A0010  | Release Devices Message                                  |
|       |                 | A0020  | Release Private Devices                                  |
|       |                 | A0030  | Release Private Devices                                  |
|       |                 | A0040  | Release Private Devices                                  |
|       |                 | A0050  | Release Private Devices                                  |
|       |                 | A0060  | Release Private Devices                                  |
|       |                 | A0070  | Release Private Devices                                  |
|       |                 | A0080  | Release Shared Devices                                   |
|       |                 | A0090  | Release Shared Devices                                   |
|       |                 | A0100  | Release Private Devices                                  |
|       |                 | A0110  | Release Private Devices                                  |
|       |                 | A0120  | Release Private Devices                                  |
|       |                 | A0130  | Release Private Devices                                  |
|       |                 | A0140  | Release Private Devices                                  |
|       |                 | A0150  | Release Private Devices                                  |
|       |                 | A0160  | Release Private Devices                                  |
|       |                 | A0170  | Release Private Devices                                  |
|       |                 | A0180  | Release Private Devices                                  |
|       |                 | A0190  | Release Private Devices                                  |
|       |                 | A0200  | Release Private Devices                                  |
|       |                 | A0210  | Release Private Devices                                  |
|       |                 | A0220  | Release Private Devices                                  |
|       |                 | A0230  | Release Private Devices                                  |
| T9900 | Complete        |        | Go to step S9900                                         |
|       |                 |        |                                                          |

| Step  | Description      | Action | Action Description                  |
|-------|------------------|--------|-------------------------------------|
| S9900 | Complete         | A0010  | Set Message: Drain Command Complete |
|       |                  | A0020  | Self Release the EM                 |
| END   | Command Complete |        | True                                |

**Table 175: Drain Sequence**

## 22.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

| Parameter            | Description                | Type           | Units  | Default Value/Reference |
|----------------------|----------------------------|----------------|--------|-------------------------|
| Operating Parameters |                            |                |        |                         |
| OP001                | Plant                      | Name Set       | NA     | MNA1                    |
| OP002                | Pure Acid Dosed            | Floating Point | Kg/Ton | 1                       |
| OP003                | Available Concentration    | Floating Point | %      | 30                      |
| OP004                | Density Solution           | Floating Point | Kg/L   | 1.1118                  |
| OP005                | Degumming Water            | Floating Point | Kg/Ton | 10                      |
| OP006                | Pure Caustic Soda Dosed    | Floating Point | Kg/Ton | 1                       |
| OP007                | Available Concentration    | Floating Point | %      | 30                      |
| OP008                | Density Solution           | Floating Point | Kg/L   | 1.1118                  |
| OP009                | Acid Dosing Enable         | Name Set       | NA     | NO                      |
| OP010                | Caustic Soda Dosing Enable | Name Set       | NA     | NO                      |
| Tuning Parameters    |                            |                |        |                         |
| TP001                | Delay Timer                | Floating Point | Sec    | 5                       |
| TP002                | Blowing Time               | Floating Point | Sec    | 10                      |
| TP003                | RSP for TT_021D            | Floating Point | C      | 30                      |
| TP004                | RSP for FT_978             | Floating Point | Kg/hr  | 40                      |
| TP005                | RSP for FT_011             | Floating Point | Kg/hr  | 5000                    |
| TP006                | Delay Timer                | Floating Point | Sec    | 20                      |
| TP007                | Delay Timer                | Floating Point | Sec    | 120                     |
| TP008                | RSP for FT_001             | Floating Point | Kg/hr  | 0                       |

**Table 176: Operating and Tuning Parameters**

## 22.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

| Parameter Name | Parameter Type     | Value\Reference           |
|----------------|--------------------|---------------------------|
| MY_TAG         |                    | RF1EQ_501                 |
| OPR_CMD_MASK   |                    | 3                         |
| LSL_001        | External Reference | RF1_LSL1001/DI1/DISC_ACT  |
| LSH_001        | External Reference | RF1_LSH1001/DI1/DISC_ACT  |
| LT_001_VHL     | External Reference | RF1_LT1001/AI1/HI_HI_ACT  |
| LT_001_VLL     | External Reference | RF1_LT1001/AI1/LO_LO_ACT  |
| CPO_WASH2      | String             | RF1EQ_582W                |
| DY_REF_1       | Dynamic Reference  |                           |
| STEP_TIME      | Floating Point     | 0                         |
| COMPLETED_TIME | Floating Point     | 0                         |
| RESET_COMP_TM  | Boolean            | False                     |
| FT_001_LL      | External Reference | RF1_FT1001/PID1/LO_LO_ACT |
| HOLD           | External Reference | RF1EQ_500/OP004_VALUE     |

**Table 177: Configurable Parameters**

## 22.6 Named Set

Following Named set shall be used for configuration of EM.

| Name            | Description               |
|-----------------|---------------------------|
| NS_RF_CPO_WASH1 | Refinery Area: CPO Wash 1 |

**Table 178: Named Set Details**

Details of states for NS\_RF\_CPO\_WASH1 named set are as follows,

| Named Set Value | Named Set Name |
|-----------------|----------------|
| 00              | Hold           |
| 01              | Production     |
| 02              | Drain          |

**Table 179: Name Set State Details**

## 23 EQUIPMENT MODULE CLASS DESIGN – EM\_RF\_CPO\_WASH2 (1-582W)

### 23.1 Functional Description

EM\_RF\_CPO\_WASH2 is a class based equipment module that is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. Initially Operator shall be provided only with Hold and Production Commands and the Drain Command will be activated after completion of Production command. All devices used in EM shall be acquired in Production Command and shall be released on completion of the Drain Command. Hold Request shall be enabled for all acquired devices at the end of Hold command. Commands which require operator intervention shall be configured as Operator Prompts wherever possible.

### 23.2 Instances

The following Table contains the module instances of the equipment module class.

| Device Name | Module Class    | Instance Name |
|-------------|-----------------|---------------|
| EM          | EM_RF_CPO_WASH2 | RF1EQ_582W    |
| DEV1        | _MRSS11_130_SE  | RF1_PU1082W   |
| DEV2        | _CSD_130_SE     | RF1_LT1082W   |

**Table 180: Instances**

### 23.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

#### 23.3.1 Command 00 – Hold

| Step  | Description                               | Action | Action Description                                |
|-------|-------------------------------------------|--------|---------------------------------------------------|
| H0010 | Command Hold Start                        | A0010  | Store Fail Message                                |
|       |                                           | A0020  | Reset Hold Request                                |
| T1000 | Actions Complete or Time Out & Hold=True  |        | Go to step H1000                                  |
| T2000 | Actions Complete or Time Out & Hold=False |        | Go to step H2000                                  |
|       |                                           |        |                                                   |
| H1000 | Hold Command 1                            | A0010  | Stopping and Disabling Conditional Logic for DEV1 |

| Step   | Description                  | Action | Action Description                                      |
|--------|------------------------------|--------|---------------------------------------------------------|
| T9900A | Actions Complete or Time Out |        | Go to step H9900A                                       |
|        |                              |        |                                                         |
| H2000  | General Reset                | A0010  | Stopping and Disabling Conditional Logic for DEV1       |
|        |                              | A0020  | Set DEV2 Mode to Remote Out                             |
|        |                              | A0030  | Set DEV2 output to Closed and Disable Conditional Logic |
| T9900B | Actions Complete or Time Out |        | Go to step H9900B                                       |
|        |                              |        |                                                         |
| H9900A | Command Hold End             | A0010  | Completed                                               |
| END    | Command Complete             |        | True                                                    |
|        |                              |        |                                                         |
| H9900B | Release Devices              | A0010  | Release Devices Message                                 |
|        |                              | A0020  | Release Private Devices                                 |
|        |                              | A0030  | Release Private Devices                                 |
|        |                              | A0040  | Reset Parameters                                        |
| END1   | Command Complete             |        | True                                                    |

**Table 181: Hold Sequence**

## 23.3.2

## Command 01 – Production

| Step   | Description        | Action | Action Description                                   |
|--------|--------------------|--------|------------------------------------------------------|
| S0000  | Start              | A0010  | Set Message: Starting Production Command             |
| T0010  | First Pass = False |        | If First Pass = False then go to step S0010          |
| T0020A | First Pass = True  |        | If First Pass = True then go to step S0020           |
|        |                    |        |                                                      |
| S0010  | Initializations    | A0010  | First Pass Logic                                     |
|        |                    | A0020  | First Pass = True                                    |
|        |                    | A0030  | Set Task Pointer                                     |
| T0020B | Actions Complete   |        | If no pending actions in S0010 then go to step S0020 |
|        |                    |        |                                                      |
| S0020  | Acquire Devices    | A0010  | Acquire Devices Message                              |
|        |                    | A0020  | Acquire Private Devices                              |
|        |                    | A0030  | Acquire Private Devices                              |
|        |                    | A0040  | Reset Hold Request                                   |

| Step   | Description      | Action | Action Description                                                                                              |
|--------|------------------|--------|-----------------------------------------------------------------------------------------------------------------|
| T0100  | Standby          |        | If Task Pointer = 1 then go to step S0100                                                                       |
| T0200B | Fill             |        | If Task Pointer = 2 then go to step S0200                                                                       |
| T0300B | Production       |        | If Task Pointer = 3 then go to step S0300                                                                       |
|        |                  |        |                                                                                                                 |
| S0100  | Standby          | A0010  | Stopping and Disabling Conditional Logic for DEV1                                                               |
|        |                  | A0020  | Set DEV2 Mode to Remote Out                                                                                     |
|        |                  | A0030  | Set DEV2 output to Closed and Disable Conditional Logic                                                         |
| T0200A | Fill             |        | If no pending actions in S0100 then go to step S0200                                                            |
|        |                  |        |                                                                                                                 |
| S0200  | Fill             | A0010  | Stopping and Disabling Conditional Logic for DEV1                                                               |
|        |                  | A0020  | Set DEV2 Mode to Remote Out                                                                                     |
|        |                  | A0030  | Set DEV2 output to Closed and Disable Conditional Logic                                                         |
|        |                  | A0040  | Waiting for Level                                                                                               |
|        |                  | A0050  | Set Scratch Time                                                                                                |
|        |                  | A0060  | Set Message: Starting Fill Command                                                                              |
| T0300A | Production       |        | If (LSL_082W is not in alarm and two minutes completed ) or (LT_082W_VHL is in VHL alarm) then go to step S0300 |
|        |                  |        |                                                                                                                 |
| S0300  | Production       | A0010  | Enabling Conditional Logic for DEV1                                                                             |
|        |                  | A0020  | Set DEV2 Mode to Remote Cascade                                                                                 |
|        |                  | A0030  | Set DEV2 setpoint ( LT_082W Setpoint ) and Enabling Conditional Logic                                           |
| T9900  | Complete         |        | If CPO_WASH1 is in <Blow> then go to step S9900                                                                 |
|        |                  |        |                                                                                                                 |
| S9900  | Complete         | A0010  | Set Message: Production Command Complete                                                                        |
| END    | Command Complete |        | True                                                                                                            |

**Table 182: Production Sequence**

### 23.3.3 Command 02 – Drain

| Step   | Description | Action | Action Description                        |
|--------|-------------|--------|-------------------------------------------|
| S0000  | Start       | A0010  | Set Message: Starting Drain Command       |
| T0100  | Drain       |        | If Task Pointer = 1 then go to step S0100 |
| T0200B | Blow        |        | If Task Pointer = 2 then go to step S0200 |
| T0300B | Aux Off     |        | If Task Pointer = 3 then go to step S0300 |
|        |             |        |                                           |

| Step  | Description      | Action | Action Description                                                            |
|-------|------------------|--------|-------------------------------------------------------------------------------|
| S0100 | Drain            | A0010  | Enabling Conditional Logic for DEV1                                           |
|       |                  | A0020  | Set DEV2 Mode to Remote Cascade                                               |
|       |                  | A0030  | Set DEV2 setpoint ( LT_082W Setpoint ) and Enabling Conditional Logic         |
| T0200 | Blow             |        | If LSL_082W is in alarm and LT_082W_VLL is in VLL alarm then go to step S0200 |
| S0200 | Blow             | A0010  | Stopping and Disabling Conditional Logic for DEV1                             |
|       |                  | A0020  | Set DEV2 Mode to Remote Out                                                   |
|       |                  | A0030  | Set DEV2 output to Open and Disable Conditional Logic                         |
|       |                  | A0040  | Prompt: Blow Complete                                                         |
| T0300 | Aux Off          |        | If Blow Complete confirmation button (OP001) is pressed then go to step S0300 |
| S0300 | Aux Off          | A0010  | Stopping and Disabling Conditional Logic for DEV1                             |
|       |                  | A0020  | Set DEV2 Mode to Remote Out                                                   |
|       |                  | A0030  | Set DEV2 output to Closed and Disable Conditional Logic                       |
| T0400 | Disable HM       |        | If no pending actions in S0300 then go to step S0400                          |
| S0400 | Disable HM       | A0010  | Disable HM 01 (Module Ownership Lost)                                         |
| T0500 | Release Devices  |        | If no pending actions in S0400 then go to step S0500                          |
| S0500 | Release Devices  | A0010  | Release Devices Message                                                       |
|       |                  | A0020  | Release Private Devices                                                       |
|       |                  | A0030  | Release Private Devices                                                       |
| T9900 | Complete         |        | Go to step S9900                                                              |
| S9900 | Complete         | A0010  | Set Message: Drain Command Complete                                           |
|       |                  | A0020  | Self Release the EM                                                           |
| END   | Command Complete |        | True                                                                          |

Table 183: Drain Sequence

## 23.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

| Parameter            | Description     | Type           | Units | Default Value/Reference |
|----------------------|-----------------|----------------|-------|-------------------------|
| Operating Parameters |                 |                |       |                         |
| OP001                | Blow Complete   | Name Set       | NA    | NO                      |
| Tuning Parameters    |                 |                |       |                         |
| TP001                | Delay Timer     | Floating Point | Sec   | 5                       |
| TP002                | Blowing Time    | Floating Point | Sec   | XX                      |
| TP003                | RSP for LT_082W | Floating Point | %     | 50                      |

**Table 184: Operating and Tuning Parameters**

## 23.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

| Parameter Name | Parameter Type     | Value/Reference                   |
|----------------|--------------------|-----------------------------------|
| MY_TAG         |                    | RF1EQ_582W                        |
| OPR_CMD_MASK   |                    | 3                                 |
| LSL_082W       | External Reference | RF1_LSL1082W/DI1/DISC_ACT         |
| LSH_032C       | External Reference | RF1_LSH1032C/DI1/DISC_ACT         |
| CPO_WASH1      | External Reference | RF1EQ_501/COMMAND_00002/ACT_STEPS |
| HOLD           | External Reference | RF1EQ_500/OP004_VALUE             |
| LT_082W_VHL    | External Reference | RF1_LT1082W/PID1/HI_HI_ACT        |
| LT_082W_VLL    | External Reference | RF1_LT1082W/PID1/LO_LO_ACT        |

**Table 185: Configurable Parameters**

## 23.6 Named Set

Following Named set shall be used for configuration of EM.

| Name            | Description               |
|-----------------|---------------------------|
| NS_RF_CPO_WASH2 | Refinery Area: CPO Wash 2 |

**Table 186: Named Set Details**

Details of states for NS\_RF\_CPO\_WASH2 named set are as follows,

| Named Set Value | Named Set Name |
|-----------------|----------------|
| 00              | Hold           |
| 01              | Production     |
| 02              | Drains         |

**Table 187: Named Set State Details**

## 24 MODULE INTERLOCKS

### 24.1 Functional Description

| Module       | Name        | Description | Expression                                                                                                                                                                                                                    | Force SP State | Delay On (s) |
|--------------|-------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------|
| RF1_PU1082W  | Condition 1 | Auto Logic  | ('/TP01.CV' = TRUE) AND <CR>('/RF1_LSL1082W/DI1/DISC_ACT' != TRUE OR <CR>'/RF1_LSH1032C/DI1/DISC_ACT' != TRUE)                                                                                                                | START          | NO           |
| RF1_PU1082W  | Condition 2 | Auto Logic  | ('/TP01.CV' = TRUE) AND <CR>('/RF1_LSL1082W/DI1/DISC_ACT' = TRUE AND <CR>'/RF1_LSH1032C/DI1/DISC_ACT' = TRUE)                                                                                                                 | STOP           | NO           |
| RF1_PU1034NA | Condition 1 | Auto Logic  | ('/TP01.CV' = TRUE) AND <CR>('/RF1EQ_501/OP010_VALUE' = TRUE AND '/RF1_LT1034NA/AI1/LO_LO_ACT.CV' != TRUE AND <CR>'/RF1_PU1001/PV_D.CV' = '_EDC_MTR2_PV:RUNNING' AND '/RF1_FT1001/PV.CV' > 5000)                              | START          | NO           |
| RF1_PU1034NA | Condition 2 | Auto Logic  | ('/TP01.CV' = TRUE) AND <CR>('/RF1EQ_501/OP010_VALUE' != TRUE OR '/RF1_LT1034NA/AI1/LO_LO_ACT.CV' = TRUE OR <CR>'/RF1_FT1001/PV.CV' <= 5000 OR '/RF1_PU1001/PV_D.CV' != '_EDC_MTR2_PV:RUNNING')                               | STOP           | NO           |
| RF1_PU1034CA | Condition 1 | Auto Logic  | ('/TP01.CV' = TRUE) AND <CR>('/RF1_LT1034CA/AI1/LO_LO_ACT.CV' != TRUE AND '/RF1EQ_501/OP009_VALUE.CV' = TRUE AND <CR>('/RF1_FT_011_SL/PV.CV' > 5000)))                                                                        | START          | NO           |
| RF1_PU1034CA | Condition 2 | Auto Logic  | ('/TP01.CV' = TRUE) AND <CR>('/RF1_LT1034CA/AI1/LO_LO_ACT.CV' = TRUE OR '/RF1EQ_501/OP009_VALUE.CV' = FALSE OR <CR>'/RF1_FT_011_SL/PV.CV' <= 5000)                                                                            | STOP           | NO           |
| RF1_PU1001   | Condition 1 | Auto Logic  | ('/TP01.CV' = TRUE) AND <CR>('/RF1_LSL1001/DI1/DISC_ACT.CV' != TRUE AND <CR>'/RF1_LT1001/AI1/LO_LO_ACT.CV' != TRUE AND <CR>'/RF1_V1001D/PV_D.CV' != '_EDC_VLV_PV:CLOSED' AND <CR>('/RF1_FT1978/PID1/DV_HI_ACT.CV' != TRUE AND | START          | NO           |

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| Module       | Name        | Description | Expression                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Force SP State | Delay On (s) |
|--------------|-------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------|
|              |             |             | '//RF1_FT1978/PID1/DV_LO_ACT.CV' != TRUE) AND<CR><br>( '//RF1_FT1034NA/PID1/DV_HI_ACT.CV' != TRUE AND<br>'//RF1_FT1034NA/PID1/DV_LO_ACT.CV' != TRUE) AND<CR><br>( '//RF1_FT1034CA/PID1/DV_HI_ACT.CV' != TRUE AND<br>'//RF1_FT1034CA/PID1/DV_LO_ACT.CV' !=<br>TRUE))AND<CR>('//RF1_LSH1006/DI1/DISC_ACT.CV' != TRUE OR<br>'//RF1_LT1006/PID1/HI_HI_ACT.CV' != TRUE))                                                                                                                                                                                                                                                                                                                          |                |              |
| RF1_PU1001   | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND <CR>('//RF1_LSL1001/DI1/DISC_ACT.CV'<br>= TRUE OR <CR>'//RF1_LT1001/AI1/LO_LO_ACT.CV' = TRUE<br>OR<CR>'//RF1_V1001D/PV_D.CV' = ' _EDC_VLV_PV:CLOSED'<br>OR<CR>(('//RF1_FT1978/PID1/DV_HI_ACT.CV' = TRUE OR<br>'//RF1_FT1978/PID1/DV_LO_ACT.CV' = TRUE) OR<CR><br>( '//RF1_FT1034NA/PID1/DV_HI_ACT.CV' = TRUE OR<br>'//RF1_FT1034NA/PID1/DV_LO_ACT.CV' = TRUE) OR<CR><br>( '//RF1_FT1034CA/PID1/DV_HI_ACT.CV' = TRUE OR<br>'//RF1_FT1034CA/PID1/DV_LO_ACT.CV' =<br>TRUE))OR<CR>('//RF1_LSH1006/DI1/DISC_ACT.CV' = TRUE AND<br>'//RF1_LT1006/PID1/HI_HI_ACT.CV' = TRUE))<br>OR<CR>('//RF1_LSH1082W/DI1/DISC_ACT' = TRUE OR<br>'//RF1_LT1082W/PID1/HI_HI_ACT.CV' = TRUE) | STOP           | NO           |
| RF1_MX1004A  | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND <CR>('//RF1_FT_011_SL/PV.CV' > 5000)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | START          | 20           |
| RF1_MX1004A  | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND <CR>('//RF1_FT_011_SL/PV.CV' <= 5000)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | STOP           | NO           |
| RF1_AG1034CA | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND<br><CR>('//RF1_LT1034CA/AI1/LO_LO_ACT.CV' != TRUE AND<br>'//RF1EQ_501/OP009_VALUE.CV' = TRUE AND<br><CR>('//RF1_FT_011_SL/PV.CV' > 5000)))                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | START          | NO           |
| RF1_AG1034CA | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND<br><CR>('//RF1_LT1034NA/AI1/LO_LO_ACT.CV' = TRUE OR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | STOP           | NO           |

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| Module      | Name        | Description | Expression                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Force SP State | Delay On (s) |
|-------------|-------------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------|
|             |             |             | '//RF1EQ_501/OP009_VALUE.CV'= FALSE OR<br><CR>('//RF1_FT_011_SL/PV.CV' <= 5000)))                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                |              |
| RF1_AG1001  | Condition 1 | Auto Logic  | ('//TP01.CV' = TRUE) AND <CR>('//RF1_LSL1001/DI1/DISC_ACT.CV' != TRUE AND <CR>'//RF1_LT1001/AI1/LO_LO_ACT.CV' != TRUE AND<CR>'//RF1_V1001D/PV_D.CV' != '_EDC_VLV_PV:CLOSED' AND<CR>('//RF1_FT1978/PID1/DV_HI_ACT.CV' != TRUE AND '//RF1_FT1978/PID1/DV_LO_ACT.CV' != TRUE) AND<CR>('//RF1_FT1034NA/PID1/DV_HI_ACT.CV' != TRUE AND '//RF1_FT1034NA/PID1/DV_LO_ACT.CV' != TRUE) AND<CR>('//RF1_FT1034CA/PID1/DV_HI_ACT.CV' != TRUE AND '//RF1_FT1034CA/PID1/DV_LO_ACT.CV' != TRUE))AND<CR>('//RF1_LSH1006/DI1/DISC_ACT.CV' != TRUE OR '//RF1_LT1006/PID1/HI_HI_ACT.CV' != TRUE))                                                                       | START          | NO           |
| RF1_AG1001  | Condition 2 | Auto Logic  | ('//TP01.CV' = TRUE) AND <CR>('//RF1_LSL1001/DI1/DISC_ACT.CV' = TRUE OR <CR>'//RF1_LT1001/AI1/LO_LO_ACT.CV' = TRUE OR<CR>'//RF1_V1001D/PV_D.CV' = '_EDC_VLV_PV:CLOSED' OR<CR>('//RF1_FT1978/PID1/DV_HI_ACT.CV' = TRUE OR '//RF1_FT1978/PID1/DV_LO_ACT.CV' = TRUE) OR<CR>('//RF1_FT1034NA/PID1/DV_HI_ACT.CV' = TRUE OR '//RF1_FT1034NA/PID1/DV_LO_ACT.CV' = TRUE) OR<CR>('//RF1_FT1034CA/PID1/DV_HI_ACT.CV' = TRUE OR '//RF1_FT1034CA/PID1/DV_LO_ACT.CV' = TRUE))OR<CR>('//RF1_LSH1006/DI1/DISC_ACT.CV' = TRUE AND '//RF1_LT1006/PID1/HI_HI_ACT.CV' = TRUE)) OR<CR>('//RF1_LSH1082W/DI1/DISC_ACT' = TRUE OR '//RF1_LT1082W/PID1/HI_HI_ACT.CV' = TRUE) | STOP           | NO           |
| RF1_PU1032C | Condition 1 | Auto Logic  | ('//TP01.CV' = TRUE) AND <CR>('//RF1_LSH1032C/DI1/DISC_ACT' = TRUE AND<CR>'//RF1_LSL1032C/DI1/DISC_ACT' = FALSE AND<CR>'//RF1_LSH1001/DI1/DISC_ACT' = FALSE AND <CR>'//RF1_LT1001/AI1/HI_HI_ACT.CV' = FALSE)                                                                                                                                                                                                                                                                                                                                                                                                                                         | START          | NO           |

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| Module       | Name        | Description | Expression                                                                                                                                                                                                                                                                           | Force SP State | Delay On (s) |
|--------------|-------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------|
| RF1_PU1032C  | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND <CR>('RF1_LSL1032C/DI1/DISC_ACT' = TRUE OR <CR>'RF1_LSH1001/DI1/DISC_ACT' = TRUE OR <CR>'RF1_LT1001/AI1/HI_HI_ACT.CV' = TRUE)                                                                                                                                 | STOP           | NO           |
| RF1_PU1781IB | Condition 1 | Auto Logic  | 'TP01.CV' = TRUE AND 'RF1_V1778/PV_D.CV' = ' EDC_VLV_SP:OPEN'                                                                                                                                                                                                                        | START          | NO           |
| RF1_PU1781IB | Condition 2 | Auto Logic  | 'TP01.CV' = TRUE AND 'RF1_V1778/PV_D.CV' != ' EDC_VLV_SP:OPEN'                                                                                                                                                                                                                       | STOP           | NO           |
| RF1_PU1781IB | Condition 3 | Auto Logic  | 'TP02.CV' = TRUE AND <CR>('RF1_V1719ICF/PV_D.CV' = ' EDC_VLV_SP:OPEN' OR 'RF1_V1719ICL/PV_D.CV' = ' EDC_VLV_SP:OPEN')                                                                                                                                                                | START          | NO           |
| RF1_PU1781IB | Condition 4 | Auto Logic  | 'TP02.CV' = TRUE AND <CR>('RF1_V1719ICF/PV_D.CV' != ' EDC_VLV_SP:OPEN' AND 'RF1_V1719ICL/PV_D.CV' != ' EDC_VLV_SP:OPEN')                                                                                                                                                             | STOP           | NO           |
| RF1_PU1750   | Condition 1 | Auto Logic  | 'TP01.CV' = TRUE AND <CR>('RF1_PU1714AG/PV_D.CV' = ' EDC_MTR2_PV:RUNNING')                                                                                                                                                                                                           | START          | NO           |
| RF1_PU1750   | Condition 2 | Auto Logic  | 'TP01.CV' = TRUE AND <CR>('RF1_PU1714AG/PV_D.CV' != ' EDC_MTR2_PV:RUNNING')                                                                                                                                                                                                          | STOP           | NO           |
| RF1_PU1722C2 | Condition 1 | Auto Logic  | ('TP02.CV' = TRUE) AND <CR>('RF1_LSL1722C/DI1/DISC_ACT' != TRUE AND <CR>'RF1_LT1722C/PID1/LO_LO_ACT.CV' != TRUE AND <CR>'RF1_LSH1780B/DI1/DISC_ACT' != TRUE AND <CR>'RF1EQ_822QE/OP001_VALUE.CV' = 'NS_DEOD1_PUMP:RF1_PU1722C2' AND <CR>'RF1_V1722CD1/PV_D.CV' = ' EDC_VLV_PV:OPEN') | START          | NO           |
| RF1_PU1722C2 | Condition 2 | Auto Logic  | ('TP02.CV' = TRUE) AND <CR>('RF1_LSL1722C/DI1/DISC_ACT' = TRUE OR <CR>'RF1_LT1722C/PID1/LO_LO_ACT.CV' = TRUE OR <CR>'RF1_LSH1780B/DI1/DISC_ACT' = TRUE OR <CR>'RF1EQ_822QE/OP001_VALUE.CV' != 'NS_DEOD1_PUMP:RF1_PU1722C2' OR <CR>'RF1_V1722CD1/PV_D.CV' != ' EDC_VLV_PV:OPEN')      | STOP           | NO           |

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| Module       | Name        | Description | Expression                                                                                                                                                                                                                                                                      | Force SP State | Delay On (s) |
|--------------|-------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------|
| RF1_PU1722C1 | Condition 1 | Auto Logic  | ('TP02.CV' = TRUE) AND <CR>('RF1_LSL1722C/DI1/DISC_ACT' != TRUE AND <CR>'RF1_LT1722C/PID1/LO_LO_ACT' != TRUE AND <CR>'RF1_LSH1780B/DI1/DISC_ACT' != TRUE AND<CR>'RF1EQ_822QE/OP001_VALUE.CV' = 'NS_DEOD1_PUMP:RF1_PU1722C1' AND<CR>'RF1_V1722CD1/PV_D.CV' = ' EDC VLV PV:OPEN') | START          | NO           |
| RF1_PU1722C1 | Condition 2 | Auto Logic  | ('TP02.CV' = TRUE) AND <CR>('RF1_LSL1722C/DI1/DISC_ACT' = TRUE OR<CR>'RF1_LT1722C/PID1/LO_LO_ACT.CV' = TRUE OR<CR>'RF1_LSH1780B/DI1/DISC_ACT' = TRUE OR<CR>'RF1EQ_822QE/OP001_VALUE.CV' != 'NS_DEOD1_PUMP:RF1_PU1722C1' OR<CR>'RF1_V1722CD1/PV_D.CV' != ' EDC VLV PV:OPEN')     | STOP           | NO           |
| RF1_PU17222  | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND <CR>('RF1_LSL1780B/DI1/DISC_ACT' != TRUE AND <CR>'RF1_LT1780B/PID1/LO_LO_ACT.CV' != TRUE AND<CR>'RF1EQ_822QE/OP002_VALUE.CV' = 'NS_DEOD2_PUMP:RF1_PU17222' AND<CR>'RF1_V1780BD/PV_D.CV' = ' EDC VLV PV:OPEN')                                            | START          | NO           |
| RF1_PU17222  | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND <CR>('RF1_LSL1780B/DI1/DISC_ACT' = TRUE OR <CR>'RF1_LT1780B/PID1/LO_LO_ACT.CV' = TRUE OR<CR>'RF1EQ_822QE/OP002_VALUE.CV' != 'NS_DEOD2_PUMP:RF1_PU17221' OR<CR>'RF1_V1780BD/PV_D.CV' != ' EDC VLV PV:OPEN')                                               | STOP           | NO           |
| RF1_PU17221  | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND <CR>('RF1_LSL1780B/DI1/DISC_ACT' != TRUE AND <CR>'RF1_LT1780B/PID1/LO_LO_ACT.CV' != TRUE AND<CR>'RF1EQ_822QE/OP002_VALUE.CV' = 'NS_DEOD2_PUMP:RF1_PU17221' AND<CR>'RF1_V1780BD/PV_D.CV' = ' EDC VLV PV:OPEN')                                            | START          | NO           |
| RF1_PU17221  | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND <CR>('RF1_LSL1780B/DI1/DISC_ACT' = TRUE OR <CR>'RF1_LT1780B/PID1/LO_LO_ACT.CV' = TRUE OR<CR>'RF1EQ_822QE/OP002_VALUE.CV' != 'NS_DEOD2_PUMP:RF1_PU17221' OR<CR>'RF1_V1780BD/PV_D.CV' != ' EDC VLV PV:OPEN')                                               | STOP           | NO           |

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| Module     | Name        | Description | Expression                                                                                                                                                                                                                                                                                                                                                      | Force SP State | Delay On (s) |
|------------|-------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------|
|            |             |             | 'NS_DEOD2_PUMP:RF1_PU17221'<br>OR<CR>'//RF1_V1780BD/PV_D.CV' != ' _EDC_VLV_PV:OPEN')                                                                                                                                                                                                                                                                            |                |              |
| RF1_PU1702 | Condition 1 | Auto Logic  | ('//RF1EQ_802/COMMAND_00001/S0300/ACTIVE.CV' = TRUE) AND<br><CR>'//RF1_LSL1702/DI1/DISC_ACT.CV' != TRUE) AND<br><CR>'//TP01.CV' = TRUE) AND <CR>'//RF1_V1712D/PV_D.CV' =<br>' _EDC_VLV_PV:OPEN') AND<br><CR>'//RF1_LSH1722C/DI1/DISC_ACT.CV' != TRUE ) AND<br><CR>'//RF1_LT1722C/PID1/HI_HI_ACT.CV' != TRUE)<br>AND<CR>'//RF1_LT1702/AI1/LO_LO_ACT.CV' != TRUE) | START          | NO           |
| RF1_PU1702 | Condition 2 | Auto Logic  | ('//RF1EQ_802/COMMAND_00001/S0300/ACTIVE.CV' != TRUE OR<br><CR>'//RF1_LSL1702/DI1/DISC_ACT.CV' = TRUE OR<br><CR>'//RF1_LT1702/AI1/LO_LO_ACT.CV' = TRUE OR<br><CR>'//RF1_V1712D/PV_D.CV' != ' _EDC_VLV_PV:OPEN' OR<br><CR>'//RF1_LSH1722C/DI1/DISC_ACT.CV' = TRUE OR<br><CR>'//RF1_LT1722C/PID1/HI_HI_ACT.CV' = TRUE) AND<br><CR>'//TP01.CV' = TRUE)             | STOP           | NO           |
| RF1_PU1702 | Condition 3 | Auto Logic  | ('//RF1EQ_802/COMMAND_00001/S0300/ACTIVE.CV' = TRUE AND<br><CR>'//RF1_LSH1722C/DI1/DISC_ACT.CV' != TRUE AND<br><CR>'//RF1_LSL1702/DI1/DISC_ACT.CV' != TRUE<br>AND<CR>'//RF1_V1712D/PV_D.CV' = ' _EDC_VLV_PV:OPEN')<br>AND<CR>'//TP02.CV' = TRUE) AND<br><CR>'//RF1_LT1722C/PID1/HI_HI_ACT.CV' != TRUE) AND<br><CR>'//RF1_LT1702/AI1/LO_LO_ACT.CV' != TRUE)      | START          | NO           |
| RF1_PU1702 | Condition 4 | Auto Logic  | ('//RF1EQ_802/COMMAND_00001/S0300/ACTIVE.CV' != TRUE OR<br><CR>'//RF1_LSH1722C/DI1/DISC_ACT.CV' = TRUE OR<br><CR>'//RF1_LT1702/AI1/LO_LO_ACT.CV' = TRUE OR<br><CR>'//RF1_LSL1702/DI1/DISC_ACT.CV' = TRUE<br>OR<CR>'//RF1_V1712D/PV_D.CV' != ' _EDC_VLV_PV:OPEN' OR<br><CR>'//RF1_LT1722C/PID1/HI_HI_ACT.CV' = TRUE) AND<br><CR>'//TP02.CV' = TRUE)              | STOP           | NO           |

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| Module       | Name        | Description | Expression                                                                                                                                                                          | Force SP State | Delay On (s) |
|--------------|-------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------|
| RF1_PU1702   | Condition 5 | Auto Logic  | ('//RF1EQ_802/COMMAND_00001/S0400/ACTIVE.CV' = TRUE AND<CR>'//RF1_LSH1722C/DI1/DISC_ACT.CV' != TRUE AND<CR>'//RF1_V1712D/PV_D.CV' = '_EDC_VLV_PV:OPEN') AND <CR>'//TP03.CV' = TRUE) | START          | NO           |
| RF1_PU1702   | Condition 6 | Auto Logic  | ('//RF1EQ_802/COMMAND_00001/S0400/ACTIVE.CV' != TRUE OR<CR>'//RF1_LSH1722C/DI1/DISC_ACT.CV' = TRUE OR<CR>'//RF1_V1712D/PV_D.CV' != '_EDC_VLV_PV:OPEN') AND <CR>'//TP03.CV' = TRUE)  | STOP           | NO           |
| RF1_PU1741X2 | Condition 1 | Auto Logic  | '//TP01.CV' = TRUE AND<CR>'//RF1EQ_STARTUP1/OP002_VALUE.CV'='NS_VAC_PMP_SEL:PU1741X2' AND<CR>'//RF1EQ_STARTUP1/OP001_VALUE.CV'='NS_VACCUM_SELECTION:PU1741X'                        | START          | NO           |
| RF1_PU1741X2 | Condition 2 | Auto Logic  | '//TP01.CV' = TRUE AND<CR>'//RF1EQ_STARTUP1/OP002_VALUE.CV' != 'NS_VAC_PMP_SEL:PU1741X2' OR<CR>'//RF1EQ_STARTUP1/OP001_VALUE.CV' != 'NS_VACCUM_SELECTION:PU1741X')                  | STOP           | NO           |
| RF1_PU1741X1 | Condition 1 | Auto Logic  | '//TP01.CV' = TRUE AND<CR>'//RF1EQ_STARTUP1/OP002_VALUE.CV'='NS_VAC_PMP_SEL:PU1741X1' AND<CR>'//RF1EQ_STARTUP1/OP001_VALUE.CV'='NS_VACCUM_SELECTION:PU1741X'                        | START          | NO           |
| RF1_PU1741X1 | Condition 2 | Auto Logic  | '//TP01.CV' = TRUE AND<CR>'//RF1EQ_STARTUP1/OP002_VALUE.CV' != 'NS_VAC_PMP_SEL:PU1741X1' OR<CR>'//RF1EQ_STARTUP1/OP001_VALUE.CV' != 'NS_VACCUM_SELECTION:PU1741X')                  | STOP           | NO           |
| RF1_PU1734CA | Condition 1 | Auto Logic  | ('//TP01.CV' = TRUE) AND ('//RF1_LT1734CA/AI1/LO_LO_ACT.CV' != TRUE);                                                                                                               | START          | NO           |

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| Module       | Name        | Description | Expression                                                                                                                                                                                                                            | Force SP State | Delay On (s) |
|--------------|-------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------|
| RF1_PU1734CA | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND ('/RF1_LT1734CA/AI1/LO_LO_ACT.CV' = TRUE);                                                                                                                                                                     | STOP           | NO           |
| RF1_PU1732B  | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND ('/RF1_LSL1732A/DI1/DISC_ACT' != TRUE)                                                                                                                                                                         | START          | NO           |
| RF1_PU1732B  | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND ('/RF1_LSL1732A/DI1/DISC_ACT' = TRUE)                                                                                                                                                                          | STOP           | NO           |
| RF1_PU1734NA | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND ('/RF1_LT1734CA/AI1/LO_LO_ACT.CV' != TRUE);                                                                                                                                                                    | START          | 60           |
| RF1_PU1734NA | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND ('/RF1_LT1734CA/AI1/LO_LO_ACT.CV' = TRUE);                                                                                                                                                                     | STOP           | NO           |
| RF1_PU1635   | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND ('/RF1_LSL1635/DI1/DISC_ACT' != TRUE AND '/RF1_LSH1622/DI1/DISC_ACT' != TRUE)                                                                                                                                  | START          | NO           |
| RF1_PU1635   | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND ('/RF1_LSL1635/DI1/DISC_ACT' = TRUE OR '/RF1_LSH1622/DI1/DISC_ACT' = TRUE)                                                                                                                                     | STOP           | NO           |
| RF1_AG1003   | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND <CR>('/RF1_MX1004AC/EDC1/PV_D.CV' = '_EDC_MTR2_PV:RUNNING')                                                                                                                                                    | START          | NO           |
| RF1_AG1003   | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND <CR>('/RF1_MX1004AC/EDC1/PV_D.CV' = '_EDC_MTR2_PV:STOPPED')                                                                                                                                                    | STOP           | NO           |
| RF1_AG1003   | Condition 3 | Auto Logic  | ('TP02.CV' = TRUE) AND <CR>('/RF1_LSL1003/DI1/DISC_ACT.CV' != TRUE)                                                                                                                                                                   | START          | NO           |
| RF1_AG1003   | Condition 4 | Auto Logic  | ('TP02.CV' = TRUE) AND <CR>('/RF1_LSL1003/DI1/DISC_ACT.CV' = TRUE)                                                                                                                                                                    | STOP           | NO           |
| RF1_PU1044CA | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE AND<CR>'/RF1_LT1006/PID1/OUT' > 30 AND<CR>'/RF1_PU1006/PV_D.CV' = '_EDC_MTR2_PV:RUNNING' AND<CR>'/RF1_LT1044CA/AI1/LO_LO_ACT.CV' != TRUE) AND <CR>('/RF1EQ_T506/OP001_VALUE.CV' = 'NS_RF_ACID_SELECTN:Citric Acid') | START          | NO           |
| RF1_PU1044CA | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND <CR>('/RF1_LT1006/PID1/OUT' <= 30 OR<CR>'/RF1_PU1006/PV_D.CV' != '_EDC_MTR2_PV:RUNNING' OR <CR>'/RF1_LT1044CA/AI1/LO_LO_ACT.CV' = TRUE) OR                                                                     | STOP           | NO           |

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| Module       | Name        | Description | Expression                                                                                                                                                                                                                                       | Force SP State | Delay On (s) |
|--------------|-------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------|
|              |             |             | <CR>('//RF1EQ_T506/OP001_VALUE.CV' != 'NS_RF_ACID_SELECTN:Citric Acid')                                                                                                                                                                          |                |              |
| RF1_PU1034PA | Condition 1 | Auto Logic  | ('//TP01.CV' = TRUE AND<CR>'//RF1_LT1006/PID1/OUT' > 30 AND<CR>'//RF1_PU1006/PV_D.CV' = 'EDC_MTR2_PV:RUNNING' AND <CR>'//RF1_LT1034PA/AI1/LO_LO_ACT.CV' != TRUE) AND <CR>('//RF1EQ_T506/OP001_VALUE.CV' = 'NS_RF_ACID_SELECTN:Phosphoric Acid')  | START          | NO           |
| RF1_PU1034PA | Condition 2 | Auto Logic  | ('//TP01.CV' = TRUE) AND <CR>('//RF1_LT1006/PID1/OUT' <= 30 OR<CR>'//RF1_PU1006/PV_D.CV' != 'EDC_MTR2_PV:RUNNING'OR <CR>'//RF1_LT1034PA/AI1/LO_LO_ACT.CV' = TRUE) OR <CR>('//RF1EQ_T506/OP001_VALUE.CV' != 'NS_RF_ACID_SELECTN:Phosphoric Acid') | STOP           | NO           |
| RF1_PU1006   | Condition 1 | Auto Logic  | ('//TP01.CV' = TRUE) AND <CR>('//RF1_LSL1006/DI1/DISC_ACT' != TRUE AND <CR>'//RF1_LSH1622/DI1/DISC_ACT' != TRUE AND <CR>'//RF1_V1006D/PV_D.CV' != 'EDC_VLV_PV:CLOSED')                                                                           | START          | NO           |
| RF1_PU1006   | Condition 2 | Auto Logic  | ('//TP01.CV' = TRUE) AND <CR>('//RF1_LSL1006/DI1/DISC_ACT' = TRUE OR <CR>'//RF1_LSH1622/DI1/DISC_ACT' = TRUE OR <CR>'//RF1_V1006D/PV_D.CV' = 'EDC_VLV_PV:CLOSED')                                                                                | STOP           | NO           |
| RF1_PU1006   | Condition 3 | Auto Logic  | ('//TP02.CV' = TRUE) AND <CR>'//RF1_LSL1006/DI1/DISC_ACT' != TRUE                                                                                                                                                                                | START          | NO           |
| RF1_PU1006   | Condition 4 | Auto Logic  | ('//TP02.CV' = TRUE) AND <CR>'//RF1_LSL1006/DI1/DISC_ACT' = TRUE                                                                                                                                                                                 | STOP           | NO           |
| RF1_MX1004AC | Condition 1 | Auto Logic  | ('//TP01.CV' = TRUE AND<CR>'//RF1_LT1006/PID1/OUT' > 30 AND<CR>'//RF1_PU1006/PV_D.CV' = 'EDC_MTR2_PV:RUNNING')                                                                                                                                   | START          | NO           |
| RF1_MX1004AC | Condition 2 | Auto Logic  | ('//TP01.CV' = TRUE) AND <CR>('//RF1_LT1006/PID1/OUT' <= 30 OR<CR>'//RF1_PU1006/PV_D.CV' = 'EDC_MTR2_PV:STOPPED')                                                                                                                                | STOP           | NO           |
| RF1_PU1682B  | Condition 1 | Auto Logic  | ('//TP01.CV' = TRUE) AND <CR>('//RF1_LSL1682B/DI1/DISC_ACT.CV' != TRUE) AND <CR>'//RF1_V1682BD/PV_D' = 'EDC_VLV_PV:OPEN'                                                                                                                         | START          | NO           |

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| Module      | Name        | Description | Expression                                                                                                                                                                   | Force SP State | Delay On (s) |
|-------------|-------------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------|
| RF1_PU1682B | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND<br><CR>('RF1_LSL1682B/DI1/DISC_ACT.CV' = TRUE) OR<br>( 'RF1_V1682BD/PV_D' = ' EDC_VLV_PV:CLOSED'))                                                    | STOP           | NO           |
| RF1_PU1682A | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE)AND<CR>('RF1_LSH1622/DI1/DISC_ACT.CV'<br>!= TRUE AND<CR>'RF1_LT1622/PID1/HI_HI_ACT.CV' != TRUE<br>AND <CR>'RF1_PT1622/AI1/HI_HI_ACT.CV' != TRUE)           | START          | NO           |
| RF1_PU1682A | Condition 2 | Auto Logic  | ('TP01.CV' =<br>TRUE)AND<CR><CR>('RF1_LSH1622/DI1/DISC_ACT.CV' = TRUE<br>OR<CR><CR>'RF1_LT1622/PID1/HI_HI_ACT.CV' = TRUE OR<br><CR><CR>'RF1_PT1622/AI1/HI_HI_ACT.CV' = TRUE) | STOP           | NO           |
| RF1_PU16225 | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND<br><CR>('RF1_LT1622/PID1/LO_LO_ACT.CV' != TRUE AND<br><CR>'RF1_LSL1622/DI1/DISC_ACT' != TRUE<br>AND<CR>'RF1_V1622C/EDC1/PV_D' = ' EDC_VLV_PV:OPEN')   | START          | NO           |
| RF1_PU16225 | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND<br><CR>('RF1_LT1622/PID1/LO_LO_ACT.CV' = TRUE OR<br><CR>'RF1_LSL1622/DI1/DISC_ACT' = TRUE OR<br><CR>'RF1_V1622C/EDC1/PV_D' = ' EDC_VLV_PV:CLOSED')    | STOP           | NO           |
| RF1_PU16224 | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND<br><CR>('RF1_LT1622/PID1/LO_LO_ACT.CV' != TRUE AND<br><CR>'RF1_LSL1622/DI1/DISC_ACT' != TRUE<br>AND<CR>'RF1_V1622C/EDC1/PV_D' = ' EDC_VLV_PV:OPEN')   | START          | NO           |
| RF1_PU16224 | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND<br><CR>('RF1_LT1622/PID1/LO_LO_ACT.CV' = TRUE OR<br><CR>'RF1_LSL1622/DI1/DISC_ACT' = TRUE OR<br><CR>'RF1_V1622C/EDC1/PV_D' = ' EDC_VLV_PV:CLOSED')    | STOP           | NO           |
| RF1_PU16222 | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND<br><CR>('RF1_LT1622/PID1/LO_LO_ACT.CV' != TRUE AND<br><CR>'RF1_LSL1622/DI1/DISC_ACT' != TRUE<br>AND<CR>'RF1_V1622C/EDC1/PV_D' = ' EDC_VLV_PV:OPEN')   | START          | NO           |

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| Module      | Name        | Description | Expression                                                                                                                                                                                                     | Force SP State | Delay On (s) |
|-------------|-------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------|
| RF1_PU16222 | Condition 2 | Auto Logic  | (/TP01.CV' = TRUE) AND<br><CR>('//RF1_LT1622/PID1/LO_LO_ACT.CV' = TRUE OR<br><CR>'//RF1_LSL1622/DI1/DISC_ACT' = TRUE OR<br><CR>'//RF1_V1622C/EDC1/PV_D' = ' EDC_VLV_PV:CLOSED')                                | STOP           | NO           |
| RF1_PU16221 | Condition 1 | Auto Logic  | (/TP01.CV' = TRUE) AND<br><CR>('//RF1_LT1622/PID1/LO_LO_ACT.CV' != TRUE AND<br><CR>'//RF1_LSL1622/DI1/DISC_ACT' != TRUE<br>AND<CR>'//RF1_V1622C/EDC1/PV_D' = ' EDC_VLV_PV:OPEN')                               | START          | NO           |
| RF1_PU16221 | Condition 2 | Auto Logic  | (/TP01.CV' = TRUE) AND<br><CR>('//RF1_LT1622/PID1/LO_LO_ACT.CV' = TRUE OR<br><CR>'//RF1_LSL1622/DI1/DISC_ACT' = TRUE OR<br><CR>'//RF1_V1622C/EDC1/PV_D' = ' EDC_VLV_PV:CLOSED')                                | STOP           | NO           |
| RF1_AG1604C | Condition 1 | Auto Logic  | (/TP01.CV' = TRUE) AND<br><CR>('//RF1_LSL1604C/DI1/DISC_ACT.CV' != TRUE)                                                                                                                                       | START          | NO           |
| RF1_AG1604C | Condition 2 | Auto Logic  | (/TP01.CV' = TRUE) AND<br><CR>('//RF1_LSL1604C/DI1/DISC_ACT.CV' = TRUE)                                                                                                                                        | STOP           | NO           |
| RF1_V1734NA | Condition 1 | Auto Logic  | (/TP01.CV' = TRUE AND '//RF1_LT1734NA/AI1/LO_ACT.CV' = TRUE)                                                                                                                                                   | OPEN           | NO           |
| RF1_V1734NA | Condition 2 | Auto Logic  | (/TP01.CV' = TRUE AND '//RF1_LT1734NA/AI1/HI_HI_ACT.CV' = TRUE)                                                                                                                                                | CLOSE          | NO           |
| RF1_V1034NA | Condition 1 | Auto Logic  | (/TP01.CV' = TRUE) AND <CR>('//RF1EQ_501/OP010_VALUE.CV' = TRUE AND <CR>'//RF1_LT1034NA/AI1/LO_LO_ACT.CV' != TRUE AND <CR>'//RF1_FT1001/PV.CV' > 5000 AND<CR>'//RF1_PU1001/PV_D.CV' = ' EDC_MTR2_PV:RUNNING' ) | OPEN           | NO           |
| RF1_V1034NA | Condition 2 | Auto Logic  | (/TP01.CV' = TRUE) AND <CR>('//RF1EQ_501/OP010_VALUE.CV' != TRUE OR <CR>'//RF1_LT1034NA/AI1/LO_LO_ACT.CV' = TRUE OR <CR>'//RF1_FT1001/PV.CV' <= 5000 OR<CR>'//RF1_PU1001/PV_D.CV' != ' EDC_MTR2_PV:RUNNING' )  | CLOSE          | NO           |
| RF1_V1034CA | Condition 1 | Auto Logic  | (/TP01.CV' = TRUE) AND<br><CR>('//RF1_LT1034CA/AI1/LO_LO_ACT.CV' != TRUE AND                                                                                                                                   | OPEN           | NO           |

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| Module       | Name        | Description | Expression                                                                                                                                                                                   | Force SP State | Delay On (s) |
|--------------|-------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------|
|              |             |             | '//RF1EQ_501/OP009_VALUE.CV' = TRUE AND<br><CR>('//RF1_FT_011_SL/PV.CV' > 5000)))                                                                                                            |                |              |
| RF1_V1034CA  | Condition 2 | Auto Logic  | ('//TP01.CV' = TRUE) AND<br><CR>('//RF1_LT1034CA/AI1/LO_LO_ACT.CV' = TRUE OR<br>'//RF1EQ_501/OP009_VALUE.CV' = FALSE OR<br><CR>('//RF1_FT_011_SL/PV.CV' <= 5000)))                           | CLOSE          | NO           |
| RF1_V1001B   | Condition 1 | Auto Logic  | ('//TP01.CV' = TRUE) AND ('//PV_D.CV' = '_EDC_VLV_PV:CLOSED')<br>AND ('//RF1_V1001D/PV_D.CV' = '_EDC_VLV_PV:CLOSED')                                                                         | OPEN           | 10           |
| RF1_V1001B   | Condition 2 | Auto Logic  | ('//TP01.CV' = TRUE) AND ('//RF1_V1001D/PV_D.CV' !=<br>'_EDC_VLV_PV:CLOSED')                                                                                                                 | CLOSE          | 0            |
| RF1_V1001B   | Condition 3 | Auto Logic  | ('//TP01.CV' = TRUE) AND ('//PV_D.CV' = '_EDC_VLV_PV:OPEN')                                                                                                                                  | CLOSE          | 10           |
| RF1_V1034NAF | Condition 1 | Auto Logic  | ('//TP01.CV' = TRUE) AND <CR>('//RF1_LT1034NA/AI1/LO_LO_ACT'<br>= TRUE)                                                                                                                      | OPEN           | NO           |
| RF1_V1034NAF | Condition 2 | Auto Logic  | ('//TP01.CV' = TRUE) AND<br><CR>('//RF1_LT1034NA/AI1/HI_HI_ACT.CV' = TRUE)                                                                                                                   | CLOSE          | NO           |
| RF1_V1034CAF | Condition 1 | Auto Logic  | ('//TP01.CV' = TRUE) AND<br><CR>('//RF1_LT1034CA/AI1/LO_LO_ACT.CV' = TRUE)                                                                                                                   | OPEN           | NO           |
| RF1_V1034CAF | Condition 2 | Auto Logic  | ('//TP01.CV' = TRUE) AND<br><CR>('//RF1_LT1034CA/AI1/HI_HI_ACT.CV' = TRUE)                                                                                                                   | CLOSE          | NO           |
| RF1_V1719ICN | Condition 1 | Auto Logic  | ('//TP01.CV' = TRUE) AND <CR>('//RF1_V1719ICL/PV_D.CV' =<br>'_EDC_VLV_PV:CLOSED' AND<CR>'//RF1_V1719ICK/PV_D.CV' =<br>'_EDC_VLV_PV:CLOSED' AND<br><CR>'//RF1_PSE1719B/DI1/DISC_ACT' != TRUE) | OPEN           | NO           |
| RF1_V1719ICN | Condition 2 | Auto Logic  | ('//TP01.CV' = TRUE) AND <CR>('//RF1_V1719ICL/PV_D.CV' =<br>'_EDC_VLV_PV:OPEN' OR<CR>'//RF1_V1719ICK/PV_D.CV' =<br>'_EDC_VLV_PV:OPEN' OR <CR>'//RF1_PSE1719B/DI1/DISC_ACT' =<br>TRUE)        | CLOSE          | NO           |
| RF1_V1719ICM | Condition 1 | Auto Logic  | ('//TP01.CV' = TRUE) AND <CR>('//RF1_V1719ICL/PV_D.CV' =<br>'_EDC_VLV_PV:CLOSED' AND '//RF1_V1719ICK/PV_D.CV' =                                                                              | OPEN           | NO           |

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| Module       | Name        | Description | Expression                                                                                                                                                                                                                                                                                                                                                              | Force SP State | Delay On (s) |
|--------------|-------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------|
|              |             |             | '_EDC_VLV_PV:CLOSED') AND<br><CR>('//RF1_PSE1719B/DI1/DISC_ACT' != TRUE)                                                                                                                                                                                                                                                                                                |                |              |
| RF1_V1719ICM | Condition 2 | Auto Logic  | ('//TP01.CV' = TRUE) AND <CR>('//RF1_V1719ICL/PV_D.CV' =<br>'_EDC_VLV_PV:OPEN' OR<CR>'//RF1_V1719ICK/PV_D.CV' =<br>'_EDC_VLV_PV:OPEN' OR <CR>'//RF1_PSE1719B/DI1/DISC_ACT' =<br>TRUE)                                                                                                                                                                                   | CLOSE          | NO           |
| RF1_V1719ICL | Condition 1 | Auto Logic  | ('//TP01.CV' = TRUE) AND <CR>('//RF1_V1719ICK/PV_D.CV' =<br>'_EDC_VLV_PV:OPEN' AND<CR>'//RF1_V1719ICM/PV_D.CV' =<br>'_EDC_VLV_PV:CLOSED' AND<CR>'//RF1_V1719ICN/PV_D.CV' =<br>'_EDC_VLV_PV:CLOSED' AND<CR>'//RF1_V1719ICJ/PV_D.CV' =<br>'_EDC_VLV_PV:CLOSED' AND<CR>'//RF1_V1719ICI/PV_D.CV' =<br>'_EDC_VLV_PV:CLOSED'<br>AND<CR>'//RF1EQ_819IC2/READY_MELT.CV' = TRUE) | OPEN           | NO           |
| RF1_V1719ICL | Condition 2 | Auto Logic  | ('//TP01.CV' = TRUE) AND <CR>('//RF1_V1719ICK/PV_D.CV' =<br>'_EDC_VLV_PV:CLOSED' OR<CR>'//RF1_V1719ICM/PV_D.CV' =<br>'_EDC_VLV_PV:OPEN' OR<CR>'//RF1_V1719ICN/PV_D.CV' =<br>'_EDC_VLV_PV:OPEN' OR<CR>'//RF1_V1719ICJ/PV_D.CV' =<br>'_EDC_VLV_PV:OPEN' OR<CR>'//RF1_V1719ICI/PV_D.CV' =<br>'_EDC_VLV_PV:OPEN'<br>OR<CR>'//RF1EQ_819IC2/READY_MELT.CV' != TRUE)           | CLOSE          | NO           |
| RF1_V1719ICK | Condition 1 | Auto Logic  | ('//TP01.CV' = TRUE) AND <CR>('//RF1_V1719ICM/PV_D.CV' =<br>'_EDC_VLV_PV:CLOSED' AND<CR>'//RF1_V1719ICN/PV_D.CV' =<br>'_EDC_VLV_PV:CLOSED' AND<CR>'//RF1_V1719ICJ/PV_D.CV' =<br>'_EDC_VLV_PV:CLOSED' AND<CR>'//RF1_V1719ICI/PV_D.CV' =<br>'_EDC_VLV_PV:CLOSED'<br>AND<CR>'//RF1EQ_878_32IC/COMMAND_00002/S0200/ACTIVE.CV'<br>= TRUE)                                    | OPEN           | NO           |
| RF1_V1719ICK | Condition 2 | Auto Logic  | ('//TP01.CV' = TRUE) AND <CR>('//RF1_V1719ICM/PV_D.CV' =<br>'_EDC_VLV_PV:OPEN' OR<CR>'//RF1_V1719ICN/PV_D.CV' =<br>'_EDC_VLV_PV:OPEN' OR<CR>'//RF1_V1719ICJ/PV_D.CV' =                                                                                                                                                                                                  | CLOSE          | NO           |

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| Module       | Name        | Description | Expression                                                                                                                                                                                                                                            | Force SP State | Delay On (s) |
|--------------|-------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------|
|              |             |             | '_EDC_VLV_PV:OPEN' OR<CR>'//RF1_V1719ICI/PV_D.CV' = '_EDC_VLV_PV:OPEN' OR<CR>'//RF1EQ_878_32IC/COMMAND_00002/S0200/ACTIVE.CV' != TRUE)                                                                                                                |                |              |
| RF1_V1719ICJ | Condition 1 | Auto Logic  | ('/TP01.CV' = TRUE) AND <CR>'//RF1_V1719ICL/PV_D.CV' = '_EDC_VLV_PV:CLOSED' AND<CR>'//RF1_V1719ICK/PV_D.CV' = '_EDC_VLV_PV:CLOSED')                                                                                                                   | OPEN           | NO           |
| RF1_V1719ICJ | Condition 2 | Auto Logic  | ('/TP01.CV' = TRUE) AND <CR>'//RF1_V1719ICL/PV_D.CV' = '_EDC_VLV_PV:OPEN' OR <CR>'//RF1_V1719ICK/PV_D.CV' = '_EDC_VLV_PV:OPEN')                                                                                                                       | CLOSE          | NO           |
| RF1_V1719ICI | Condition 1 | Auto Logic  | ('/TP01.CV' = TRUE) AND <CR>'//RF1_V1719ICL/PV_D.CV' = '_EDC_VLV_PV:CLOSED' AND<CR>'//RF1_V1719ICK/PV_D.CV' = '_EDC_VLV_PV:CLOSED')                                                                                                                   | OPEN           | NO           |
| RF1_V1719ICI | Condition 2 | Auto Logic  | ('/TP01.CV' = TRUE) AND <CR>'//RF1_V1719ICL/PV_D.CV' = '_EDC_VLV_PV:OPEN' OR <CR>'//RF1_V1719ICK/PV_D.CV' = '_EDC_VLV_PV:OPEN')                                                                                                                       | CLOSE          | NO           |
| RF1_V1719ICH | Condition 1 | Auto Logic  | ('/TP01.CV' = TRUE) AND <CR>'//RF1_V1719ICF/PV_D.CV' = '_EDC_VLV_PV:CLOSED' AND '//'RF1_V1719ICE/PV_D.CV' = '_EDC_VLV_PV:CLOSED') AND <CR>'//RF1_PT1722AB/AI1/HI_HI_ACT.CV' != TRUE)AND <CR>'//RF1EQ_819IC1/OP003_VALUE.CV' = 'NS_RF_PLANT_SEL:MNA2') | OPEN           | NO           |
| RF1_V1719ICH | Condition 2 | Auto Logic  | ('/TP01.CV' = TRUE) AND <CR>'//RF1_V1719ICF/PV_D.CV' = '_EDC_VLV_PV:OPEN' OR<CR>'//RF1_V1719ICE/PV_D.CV' = '_EDC_VLV_PV:OPEN' OR <CR>'//RF1_PT1722AB/AI1/HI_HI_ACT.CV'= TRUE OR <CR>'//RF1EQ_819IC1/OP003_VALUE.CV' != 'NS_RF_PLANT_SEL:MNA2')        | CLOSE          | NO           |
| RF1_V1719ICG | Condition 1 | Auto Logic  | ('/TP01.CV' = TRUE) AND <CR>'//RF1_V1719ICF/PV_D.CV' = '_EDC_VLV_PV:CLOSED' AND '//'RF1_V1719ICE/PV_D.CV' =                                                                                                                                           | OPEN           | NO           |

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| Module       | Name        | Description | Expression                                                                                                                                                                                                                                                                                                                                                              | Force SP State | Delay On (s) |
|--------------|-------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------|
|              |             |             | '_EDC_VLV_PV:CLOSED') AND<br><CR>('//RF1_PT1722AB/AI1/HI_HI_ACT.CV' != TRUE) AND<br><CR>('//RF1EQ_819IC1/OP003_VALUE.CV' =<br>'NS_RF_PLANT_SEL:MNA1')                                                                                                                                                                                                                   |                |              |
| RF1_V1719ICG | Condition 2 | Auto Logic  | ('//TP01.CV' = TRUE) AND <CR>('//RF1_V1719ICF/PV_D.CV' =<br>'_EDC_VLV_PV:OPEN' OR '//RF1_V1719ICE/PV_D.CV' =<br>'_EDC_VLV_PV:OPEN'<br>OR<CR>'//RF1_PT1722AB/AI1/HI_HI_ACT.CV' = TRUE OR<br><CR>('//RF1EQ_819IC1/OP003_VALUE.CV' !=<br>'NS_RF_PLANT_SEL:MNA1'))                                                                                                          | CLOSE          | NO           |
| RF1_V1719ICF | Condition 1 | Auto Logic  | ('//TP01.CV' = TRUE) AND <CR>('//RF1_V1719ICE/PV_D.CV' =<br>'_EDC_VLV_PV:OPEN' AND<CR>'//RF1_V1719ICG/PV_D.CV' =<br>'_EDC_VLV_PV:CLOSED' AND<CR>'//RF1_V1719ICH/PV_D.CV' =<br>'_EDC_VLV_PV:CLOSED' AND<CR>'//RF1_V1719ICC/PV_D.CV' =<br>'_EDC_VLV_PV:CLOSED' AND<CR>'//RF1_V1719ICD/PV_D.CV' =<br>'_EDC_VLV_PV:CLOSED'<br>AND<CR>'//RF1EQ_819IC1/READY_MELT.CV' = TRUE) | OPEN           | NO           |
| RF1_V1719ICF | Condition 2 | Auto Logic  | ('//TP01.CV' = TRUE) AND <CR>('//RF1_V1719ICE/PV_D.CV' =<br>'_EDC_VLV_PV:CLOSED' OR<CR>'//RF1_V1719ICG/PV_D.CV' =<br>'_EDC_VLV_PV:OPEN' OR<CR>'//RF1_V1719ICH/PV_D.CV' =<br>'_EDC_VLV_PV:OPEN' OR<CR>'//RF1_V1719ICC/PV_D.CV' =<br>'_EDC_VLV_PV:OPEN' OR<CR>'//RF1_V1719ICD/PV_D.CV' =<br>'_EDC_VLV_PV:OPEN'<br>OR<CR>'//RF1EQ_819IC1/READY_MELT.CV' != TRUE)           | CLOSE          | NO           |
| RF1_V1719ICE | Condition 1 | Auto Logic  | ('//TP01.CV' = TRUE) AND <CR>('//RF1_V1719ICG/PV_D.CV' =<br>'_EDC_VLV_PV:CLOSED' AND<CR>'//RF1_V1719ICH/PV_D.CV' =<br>'_EDC_VLV_PV:CLOSED' AND<CR>'//RF1_V1719ICC/PV_D.CV' =<br>'_EDC_VLV_PV:CLOSED' AND<CR>'//RF1_V1719ICD/PV_D.CV' =<br>'_EDC_VLV_PV:CLOSED'                                                                                                          | OPEN           | NO           |

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| Module       | Name        | Description | Expression                                                                                                                                                                                                                                                                                              | Force SP State | Delay On (s) |
|--------------|-------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------|
|              |             |             | AND<CR>'//RF1EQ_878_32IC/COMMAND_00002/S0200/ACTIVE.CV' = TRUE)                                                                                                                                                                                                                                         |                |              |
| RF1_V1719ICE | Condition 2 | Auto Logic  | ('//TP01.CV' = TRUE) AND <CR>('//RF1_V1719ICG/PV_D.CV' = '_EDC_VLV_PV:OPEN' OR<CR>'//RF1_V1719ICH/PV_D.CV' = '_EDC_VLV_PV:OPEN' OR<CR>'//RF1_V1719ICC/PV_D.CV' = '_EDC_VLV_PV:OPEN' OR<CR>'//RF1_V1719ICD/PV_D.CV' = '_EDC_VLV_PV:OPEN' OR<CR>'//RF1EQ_878_32IC/COMMAND_00002/S0200/ACTIVE.CV' != TRUE) | CLOSE          | NO           |
| RF1_V1719ICD | Condition 1 | Auto Logic  | ('//TP01.CV' = TRUE) AND <CR>('//RF1_V1719ICF/PV_D.CV' = '_EDC_VLV_PV:CLOSED' AND '//RF1_V1719ICE/PV_D.CV' = '_EDC_VLV_PV:CLOSED')AND <CR>('//RF1EQ_819IC1/OP003_VALUE.CV' = 'NS_RF_PLANT_SEL:MNA2')                                                                                                    | OPEN           | NO           |
| RF1_V1719ICD | Condition 2 | Auto Logic  | ('//TP01.CV' = TRUE) AND <CR>('//RF1_V1719ICF/PV_D.CV' = '_EDC_VLV_PV:OPEN' OR<CR>'//RF1_V1719ICE/PV_D.CV' = '_EDC_VLV_PV:OPEN' OR <CR>('//RF1EQ_819IC1/OP003_VALUE.CV' != 'NS_RF_PLANT_SEL:MNA2'))                                                                                                     | CLOSE          | NO           |
| RF1_V1719ICC | Condition 1 | Auto Logic  | ('//TP01.CV' = TRUE) AND <CR>('//RF1_V1719ICF/PV_D.CV' = '_EDC_VLV_PV:CLOSED' AND '//RF1_V1719ICE/PV_D.CV' = '_EDC_VLV_PV:CLOSED')AND <CR>('//RF1EQ_819IC1/OP003_VALUE.CV' = 'NS_RF_PLANT_SEL:MNA1')                                                                                                    | OPEN           | NO           |
| RF1_V1719ICC | Condition 2 | Auto Logic  | ('//TP01.CV' = TRUE) AND <CR>('//RF1_V1719ICF/PV_D.CV' = '_EDC_VLV_PV:OPEN' OR<CR>'//RF1_V1719ICE/PV_D.CV' = '_EDC_VLV_PV:OPEN' OR <CR>('//RF1EQ_819IC1/OP003_VALUE.CV' != 'NS_RF_PLANT_SEL:MNA1'))                                                                                                     | CLOSE          | NO           |

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| Module       | Name        | Description | Expression                                                                                                          | Force SP State | Delay On (s) |
|--------------|-------------|-------------|---------------------------------------------------------------------------------------------------------------------|----------------|--------------|
| RF1_V1714AGB | Condition 1 | Auto Logic  | '/TP01.CV' = TRUE AND '/PV_D.CV' = '_EDC_VLV_PV:CLOSED'                                                             | OPEN           | 5            |
| RF1_V1714AGB | Condition 2 | Auto Logic  | '/TP01.CV' = TRUE AND '/PV_D.CV' = '_EDC_VLV_PV:OPEN'                                                               | CLOSE          | 30           |
| RF1_V1781AB  | Condition 1 | Auto Logic  | '/TP01.CV' = TRUE AND '/PV_D.CV' = '_EDC_VLV_PV:CLOSED'                                                             | OPEN           | 5            |
| RF1_V1781AB  | Condition 2 | Auto Logic  | '/TP01.CV' = TRUE AND '/PV_D.CV' = '_EDC_VLV_PV:OPEN'                                                               | CLOSE          | 30           |
| RF1_V1780BB  | Condition 1 | Auto Logic  | '/TP01.CV' = TRUE AND '/PV_D.CV' = '_EDC_VLV_PV:CLOSED'                                                             | OPEN           | 5            |
| RF1_V1780BB  | Condition 2 | Auto Logic  | '/TP01.CV' = TRUE AND '/PV_D.CV' = '_EDC_VLV_PV:OPEN'                                                               | CLOSE          | 30           |
| RF1_V1780A2B | Condition 1 | Auto Logic  | ('/TP01.CV' = TRUE) AND <CR>'//RF1_LSH1780B/DI1/DISC_ACT' != TRUE OR <CR>'//RF1_LT1780B/PID1/HI_HI_ACT.CV' != TRUE) | OPEN           | NO           |
| RF1_V1780A2B | Condition 2 | Auto Logic  | ('/TP01.CV' = TRUE) AND <CR>'//RF1_LSH1780B/DI1/DISC_ACT' = TRUE AND <CR>'//RF1_LT1780B/PID1/HI_HI_ACT.CV' = TRUE)  | CLOSE          | NO           |
| RF1_V1780A2A | Condition 1 | Auto Logic  | ('/TP01.CV' = TRUE) AND <CR>'//RF1_LSH1780B/DI1/DISC_ACT' != TRUE OR <CR>'//RF1_LT1780B/PID1/HI_HI_ACT.CV' != TRUE) | OPEN           | NO           |
| RF1_V1780A2A | Condition 2 | Auto Logic  | ('/TP01.CV' = TRUE) AND <CR>'//RF1_LSH1780B/DI1/DISC_ACT' = TRUE AND <CR>'//RF1_LT1780B/PID1/HI_HI_ACT.CV' = TRUE)  | CLOSE          | NO           |
| RF1_V1780A1B | Condition 1 | Auto Logic  | ('/TP01.CV' = TRUE) AND <CR>'//RF1_LSH1780B/DI1/DISC_ACT' != TRUE OR <CR>'//RF1_LT1780B/PID1/HI_HI_ACT.CV' != TRUE) | OPEN           | NO           |
| RF1_V1780A1B | Condition 2 | Auto Logic  | ('/TP01.CV' = TRUE) AND <CR>'//RF1_LSH1780B/DI1/DISC_ACT' = TRUE AND <CR>'//RF1_LT1780B/PID1/HI_HI_ACT.CV' = TRUE)  | CLOSE          | NO           |
| RF1_V1780A1A | Condition 1 | Auto Logic  | ('/TP01.CV' = TRUE) AND <CR>'//RF1_LSH1780B/DI1/DISC_ACT' != TRUE OR <CR>'//RF1_LT1780B/PID1/HI_HI_ACT.CV' != TRUE) | OPEN           | NO           |
| RF1_V1780A1A | Condition 2 | Auto Logic  | ('/TP01.CV' = TRUE) AND <CR>'//RF1_LSH1780B/DI1/DISC_ACT' = TRUE AND <CR>'//RF1_LT1780B/PID1/HI_HI_ACT.CV' = TRUE)  | CLOSE          | NO           |
| RF1_V1722QED | Condition 1 | Auto Logic  | '//RF1EQ_822QE/TP017_VALUE.CV' = 1 AND '/TP01.CV' = TRUE                                                            | OPEN           | NO           |
| RF1_V1722QED | Condition 2 | Auto Logic  | '//RF1EQ_822QE/TP017_VALUE.CV' != 1 AND '/TP01.CV' = TRUE                                                           | CLOSE          | NO           |

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| Module       | Name        | Description | Expression                                                                                                         | Force SP State | Delay On (s) |
|--------------|-------------|-------------|--------------------------------------------------------------------------------------------------------------------|----------------|--------------|
| RF1_V1722QEC | Condition 1 | Auto Logic  | '/TP01.CV' = TRUE AND '/RF1_V1722QED/PV_D.CV' = ' EDC_VLV_PV:CLOSED'                                               | OPEN           | NO           |
| RF1_V1722QEC | Condition 2 | Auto Logic  | '/TP01.CV' = TRUE AND '/RF1_V1722QED/PV_D.CV' = ' EDC_VLV_PV:OPEN'                                                 | CLOSE          | NO           |
| RF1_V1722QEA | Condition 1 | Auto Logic  | '/TP01.CV' = TRUE AND '/PV_D.CV' = ' _EDC_VLV_PV:CLOSED'                                                           | OPEN           | 5            |
| RF1_V1722QEA | Condition 2 | Auto Logic  | '/TP01.CV' = TRUE AND '/PV_D.CV' = ' _EDC_VLV_PV:OPEN'                                                             | CLOSE          | 30           |
| RF1_V1722CB  | Condition 1 | Auto Logic  | '/TP01.CV' = TRUE AND '/PV_D.CV' = ' _EDC_VLV_PV:CLOSED'                                                           | OPEN           | 5            |
| RF1_V1722CB  | Condition 2 | Auto Logic  | '/TP01.CV' = TRUE AND '/PV_D.CV' = ' _EDC_VLV_PV:OPEN'                                                             | CLOSE          | 30           |
| RF1_V1722A6D | Condition 1 | Auto Logic  | ('/TP01.CV' = TRUE) AND <CR>('/RF1_LSH1780B/DI1/DISC_ACT' != TRUE OR <CR>'/RF1_LT1780B/PID1/HI_HI_ACT.CV' != TRUE) | OPEN           | NO           |
| RF1_V1722A6D | Condition 2 | Auto Logic  | ('/TP01.CV' = TRUE) AND <CR>('/RF1_LSH1780B/DI1/DISC_ACT' = TRUE AND <CR>'/RF1_LT1780B/PID1/HI_HI_ACT.CV' = TRUE)  | CLOSE          | NO           |
| RF1_V1722A5D | Condition 1 | Auto Logic  | ('/TP01.CV' = TRUE) AND <CR>('/RF1_LSH1780B/DI1/DISC_ACT' != TRUE OR <CR>'/RF1_LT1780B/PID1/HI_HI_ACT.CV' != TRUE) | OPEN           | NO           |
| RF1_V1722A5D | Condition 2 | Auto Logic  | ('/TP01.CV' = TRUE) AND <CR>('/RF1_LSH1780B/DI1/DISC_ACT' = TRUE AND <CR>'/RF1_LT1780B/PID1/HI_HI_ACT.CV' = TRUE)  | CLOSE          | NO           |
| RF1_V1722A4D | Condition 1 | Auto Logic  | ('/TP01.CV' = TRUE) AND <CR>('/RF1_LSH1780B/DI1/DISC_ACT' != TRUE OR <CR>'/RF1_LT1780B/PID1/HI_HI_ACT.CV' != TRUE) | OPEN           | NO           |
| RF1_V1722A4D | Condition 2 | Auto Logic  | ('/TP01.CV' = TRUE) AND <CR>('/RF1_LSH1780B/DI1/DISC_ACT' = TRUE AND <CR>'/RF1_LT1780B/PID1/HI_HI_ACT.CV' = TRUE)  | CLOSE          | NO           |
| RF1_V1722A3D | Condition 1 | Auto Logic  | ('/TP01.CV' = TRUE) AND <CR>('/RF1_LSH1780B/DI1/DISC_ACT' != TRUE OR <CR>'/RF1_LT1780B/PID1/HI_HI_ACT.CV' != TRUE) | OPEN           | NO           |
| RF1_V1722A3D | Condition 2 | Auto Logic  | ('/TP01.CV' = TRUE) AND <CR>('/RF1_LSH1780B/DI1/DISC_ACT' = TRUE AND <CR>'/RF1_LT1780B/PID1/HI_HI_ACT.CV' = TRUE)  | CLOSE          | NO           |
| RF1_V1722A2D | Condition 1 | Auto Logic  | ('/TP01.CV' = TRUE) AND <CR>('/RF1_LSH1780B/DI1/DISC_ACT' != TRUE OR <CR>'/RF1_LT1780B/PID1/HI_HI_ACT.CV' != TRUE) | OPEN           | NO           |

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| Module       | Name        | Description | Expression                                                                                                         | Force SP State | Delay On (s) |
|--------------|-------------|-------------|--------------------------------------------------------------------------------------------------------------------|----------------|--------------|
| RF1_V1722A2D | Condition 2 | Auto Logic  | ('/TP01.CV' = TRUE) AND <CR>('/RF1_LSH1780B/DI1/DISC_ACT' = TRUE AND <CR>'/RF1_LT1780B/PID1/HI_HI_ACT.CV' = TRUE)  | CLOSE          | NO           |
| RF1_V1722A1D | Condition 1 | Auto Logic  | ('/TP01.CV' = TRUE) AND <CR>('/RF1_LSH1780B/DI1/DISC_ACT' != TRUE OR <CR>'/RF1_LT1780B/PID1/HI_HI_ACT.CV' != TRUE) | OPEN           | NO           |
| RF1_V1722A1D | Condition 2 | Auto Logic  | ('/TP01.CV' = TRUE) AND <CR>('/RF1_LSH1780B/DI1/DISC_ACT' = TRUE AND <CR>'/RF1_LT1780B/PID1/HI_HI_ACT.CV' = TRUE)  | CLOSE          | NO           |
| RF1_V1721A2D | Condition 1 | Auto Logic  | ('/TP01.CV' = TRUE) AND <CR>('/RF1_LSH1722C/DI1/DISC_ACT' != TRUE OR <CR>'/RF1_LT1722C/PID1/HI_HI_ACT.CV' != TRUE) | OPEN           | NO           |
| RF1_V1721A2D | Condition 2 | Auto Logic  | ('/TP01.CV' = TRUE) AND <CR>('/RF1_LSH1722C/DI1/DISC_ACT' = TRUE AND <CR>'/RF1_LT1722C/PID1/HI_HI_ACT.CV' = TRUE)  | CLOSE          | NO           |
| RF1_V1721A1D | Condition 1 | Auto Logic  | ('/TP01.CV' = TRUE) AND <CR>('/RF1_LSH1722C/DI1/DISC_ACT' != TRUE OR <CR>'/RF1_LT1722C/PID1/HI_HI_ACT.CV' != TRUE) | OPEN           | NO           |
| RF1_V1721A1D | Condition 2 | Auto Logic  | ('/TP01.CV' = TRUE) AND <CR>('/RF1_LSH1722C/DI1/DISC_ACT' = TRUE AND <CR>'/RF1_LT1722C/PID1/HI_HI_ACT.CV' = TRUE)  | CLOSE          | NO           |
| RF1_V1702B   | Condition 1 | Auto Logic  | '/TP01.CV' = TRUE AND '/PV_D.CV' = '_EDC_VLV_PV:CLOSED'                                                            | OPEN           | 5            |
| RF1_V1702B   | Condition 2 | Auto Logic  | '/TP01.CV' = TRUE AND '/PV_D.CV' = '_EDC_VLV_PV:OPEN'                                                              | CLOSE          | 30           |
| RF1_V1721XS  | Condition 1 | Auto Logic  | '/TP01.CV' = TRUE AND <CR>'/RF1EQ_802/OP001_VALUE.CV' = TRUE                                                       | OPEN           | NO           |
| RF1_V1721XS  | Condition 2 | Auto Logic  | '/TP01.CV' = TRUE AND <CR>'/RF1EQ_802/OP001_VALUE.CV' = FALSE                                                      | CLOSE          | NO           |
| RF1_V1702F   | Condition 1 | Auto Logic  | ((('/RF1_LT1702/AI1/HI_HI_ACT.CV' != TRUE AND '/RF1_LSH1702/DI1/DISC_ACT' != TRUE) AND <CR>'/TP01.CV' = TRUE)      | OPEN           | NO           |
| RF1_V1702F   | Condition 2 | Auto Logic  | ((('/RF1_LT1702/AI1/HI_HI_ACT.CV' = TRUE OR '/RF1_LSH1702/DI1/DISC_ACT' = TRUE) AND <CR>'/TP01.CV' = TRUE)         | CLOSE          | NO           |
| RF1_V1702F   | Condition 3 | Auto Logic  | '/RF1_LSH1702/DI1/DISC_ACT' != TRUE AND <CR>'/TP02.CV' = TRUE AND                                                  | OPEN           | NO           |

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| Module       | Name        | Description | Expression                                                                                                                                                           | Force SP State | Delay On (s) |
|--------------|-------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------|
|              |             |             | <CR>'//RF1EQ_822QE/COMMAND_00002/S0500/ACTIVE.CV' != TRUE                                                                                                            |                |              |
| RF1_V1702F   | Condition 4 | Auto Logic  | ('//RF1_LSH1702/DI1/DISC_ACT' = TRUE OR <CR>'//RF1EQ_822QE/COMMAND_00002/S0500/ACTIVE.CV' = TRUE)<CR>AND <CR>'//TP02.CV' = TRUE                                      | CLOSE          | NO           |
| RF1_V1714ICB | Condition 1 | Auto Logic  | '//TP01.CV' = TRUE AND<CR>'//RF1EQ_STARTUP1/OP001_VALUE.CV'='NS_VACCUM_SELECTION:EJECTOR'                                                                            | OPEN           | NO           |
| RF1_V1714ICB | Condition 2 | Auto Logic  | '//TP01.CV' = TRUE AND<CR>'//RF1EQ_STARTUP1/OP001_VALUE.CV' != 'NS_VACCUM_SELECTION:EJECTOR'                                                                         | CLOSE          | NO           |
| RF1_V1635S   | Condition 1 | Auto Logic  | ('//TP01.CV' = TRUE) AND ('//RF1_TT1635/PV' < ('//RF1EQ_635/TP002_VALUE.CV'+ '//RF1EQ_635/TP003_VALUE.CV' ))                                                         | OPEN           | NO           |
| RF1_V1635S   | Condition 2 | Auto Logic  | ('//TP01.CV' = TRUE) AND ('//RF1_TT1635/PV' >= ('//RF1EQ_635/TP002_VALUE.CV' + '//RF1EQ_635/TP003_VALUE.CV' ))                                                       | CLOSE          | NO           |
| RF1_V1635B   | Condition 1 | Auto Logic  | ('//TP01.CV' = TRUE) AND ('//RF1_PT1622/AI1/HI_HI_ACT.CV' != TRUE)                                                                                                   | OPEN           | NO           |
| RF1_V1635B   | Condition 2 | Auto Logic  | ('//TP01.CV' = TRUE) AND ('//RF1_PT1622/AI1/HI_HI_ACT.CV' = TRUE)                                                                                                    | CLOSE          | NO           |
| RF1_V1616S1  | Condition 1 | Auto Logic  | '//TP01.CV' = TRUE AND '//RF1_V1616S1/PV_D' = ' EDC VLV PV:CLOSED'                                                                                                   | OPEN           | 5            |
| RF1_V1616S1  | Condition 2 | Auto Logic  | '//TP01.CV' = TRUE AND '//RF1_V1616S1/PV_D' = ' EDC VLV PV:OPEN'                                                                                                     | CLOSE          | 5            |
| RF1_V1682A2D | Condition 1 | Auto Logic  | ('//TP01.CV' = TRUE)AND<CR>('//RF1_LSH1622/DI1/DISC_ACT.CV' != TRUE AND<CR>'//RF1_LT1622/PID1/HI_HI_ACT.CV' != TRUE AND <CR>'//RF1_PT1622/AI1/HI_HI_ACT.CV' != TRUE) | OPEN           | NO           |

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| Module       | Name        | Description | Expression                                                                                                                                                    | Force SP State | Delay On (s) |
|--------------|-------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------|
| RF1_V1682A2D | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND<CR>('RF1_LSH1622/DI1/DISC_ACT.CV' = TRUE OR <CR>'RF1_LT1622/PID1/HI_HI_ACT.CV' = TRUE OR <CR>'RF1_PT1622/AI1/HI_HI_ACT.CV' = TRUE)     | CLOSE          | NO           |
| RF1_V1682AD2 | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND 'RF1EQ_682A/OP003_VALUE' = 1                                                                                                           | OPEN           | NO           |
| RF1_V1682AD2 | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND 'RF1EQ_682A/OP003_VALUE' = 0                                                                                                           | CLOSE          | NO           |
| RF1_V1682AD1 | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND 'RF1EQ_682A/OP002_VALUE' = 1                                                                                                           | OPEN           | NO           |
| RF1_V1682AD1 | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND 'RF1EQ_682A/OP002_VALUE' = 0                                                                                                           | CLOSE          | NO           |
| RF1_V1682AB  | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND<CR>('RF1_LSH1622/DI1/DISC_ACT.CV' != TRUE AND<CR>'RF1_LT1622/PID1/HI_HI_ACT.CV' != TRUE AND <CR>'RF1_PT1622/AI1/HI_HI_ACT.CV' != TRUE) | OPEN           | NO           |
| RF1_V1682AB  | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND<CR>('RF1_LSH1622/DI1/DISC_ACT.CV' = TRUE OR <CR>'RF1_LT1622/PID1/HI_HI_ACT.CV' = TRUE OR <CR>'RF1_PT1622/AI1/HI_HI_ACT.CV' = TRUE)     | CLOSE          | NO           |
| RF1_V1616A5N | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_PU16225/PV_D' = 'EDC_MTR2_PV:STOPPED')                                                                                           | OPEN           | NO           |
| RF1_V1616A5N | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_PU16225/PV_D' != 'EDC_MTR2_PV:STOPPED')                                                                                          | CLOSE          | NO           |
| RF1_V1616A5M | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND<CR>('RF1_V1616A5J/PV_D.CV' = 'EDC_VLV_PV:OPEN') AND <CR>('PV_D.CV' = 'EDC_VLV_PV:CLOSED')                                              | OPEN           | 10           |
| RF1_V1616A5M | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND <CR>('RF1_V1616A5J/PV_D.CV' = 'EDC_VLV_PV:OPEN') AND <CR>('PV_D.CV' = 'EDC_VLV_PV:OPEN')                                               | CLOSE          | 10           |
| RF1_V1616A5M | Condition 3 | Auto Logic  | ('TP01.CV' = TRUE) AND <CR>('RF1_V1616A5J/PV_D.CV' = 'EDC_VLV_PV:CLOSED')                                                                                     | CLOSE          | NO           |

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| Module       | Name        | Description | Expression                                                                                                                 | Force SP State | Delay On (s) |
|--------------|-------------|-------------|----------------------------------------------------------------------------------------------------------------------------|----------------|--------------|
| RF1_V1616A5L | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND<CR>('RF1_V1616A5K/PV_D.CV' = 'EDC_VLV_PV:OPEN') AND <CR>('PV_D.CV' = 'EDC_VLV_PV:CLOSED')           | OPEN           | 10           |
| RF1_V1616A5L | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND <CR>('RF1_V1616A5K/PV_D.CV' = 'EDC_VLV_PV:OPEN') AND <CR>('PV_D.CV' = 'EDC_VLV_PV:OPEN')            | CLOSE          | 10           |
| RF1_V1616A5L | Condition 3 | Auto Logic  | ('TP01.CV' = TRUE) AND <CR>('RF1_V1616A5K/PV_D.CV' = 'EDC_VLV_PV:CLOSED')                                                  | CLOSE          | NO           |
| RF1_V1616A5K | Condition 1 | Auto Logic  | 'TP01.CV' = TRUE AND 'RF1EQ_616A5/OP003_VALUE.CV' = 'NS_YES_NO:YES'                                                        | OPEN           | NO           |
| RF1_V1616A5K | Condition 2 | Auto Logic  | 'TP01.CV' = TRUE AND 'RF1EQ_616A1/OP003_VALUE.CV' = 'NS_YES_NO:NO'                                                         | CLOSE          | NO           |
| RF1_V1616A5I | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_LSH1622/DI1/DISC_ACT' != TRUE)                                                                | OPEN           | NO           |
| RF1_V1616A5I | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_LSH1622/DI1/DISC_ACT' = TRUE)                                                                 | CLOSE          | NO           |
| RF1_V1616A5H | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND<CR>('RF1_V1616A5G/PV_D.CV' = 'EDC_VLV_PV:OPEN' AND<CR>'RF1_V1616A5I/PV_D.CV' = 'EDC_VLV_PV:OPEN')   | OPEN           | 3            |
| RF1_V1616A5H | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND<CR>('RF1_V1616A5G/PV_D.CV' != 'EDC_VLV_PV:OPEN' OR <CR>'RF1_V1616A5I/PV_D.CV' != 'EDC_VLV_PV:OPEN') | CLOSE          | NO           |
| RF1_V1616A5H | Condition 1 | Auto Logic  | ('TP02.CV' = TRUE) AND<CR>('RF1_V1616A1J/PV_D.CV' = 'EDC_VLV_PV:OPEN')                                                     | OPEN           | 3            |
| RF1_V1616A5H | Condition 2 | Auto Logic  | ('TP02.CV' = TRUE) AND<CR>('RF1_V1616A1J/PV_D.CV' != 'EDC_VLV_PV:OPEN')                                                    | CLOSE          | NO           |
| RF1_V1616A5G | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND<CR>('RF1_V1616A5I/PV_D.CV' = 'EDC_VLV_PV:OPEN')                                                     | OPEN           | NO           |
| RF1_V1616A5G | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND<CR>('RF1_V1616A5I/PV_D.CV' != 'EDC_VLV_PV:OPEN')                                                    | CLOSE          | NO           |

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| Module       | Name        | Description | Expression                                                                                                               | Force SP State | Delay On (s) |
|--------------|-------------|-------------|--------------------------------------------------------------------------------------------------------------------------|----------------|--------------|
| RF1_V1616A5F | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A5N/PV_D' = ' EDC_VLV_PV:CLOSED')                                                      | OPEN           | NO           |
| RF1_V1616A5F | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A5N/PV_D' != ' EDC_VLV_PV:CLOSED')                                                     | CLOSE          | NO           |
| RF1_V1616A5E | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A5N/PV_D' = ' EDC_VLV_PV:CLOSED')                                                      | OPEN           | NO           |
| RF1_V1616A5E | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A5N/PV_D' = ' EDC_VLV_PV:CLOSED')                                                      | CLOSE          | NO           |
| RF1_V1616A5D | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A5N/PV_D' = ' EDC_VLV_PV:CLOSED')                                                      | CLOSE          | NO           |
| RF1_V1616A5D | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A5N/PV_D' != ' EDC_VLV_PV:CLOSED')                                                     | OPEN           | NO           |
| RF1_V1616A5B | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A5N/PV_D' = ' EDC_VLV_PV:CLOSED')                                                      | OPEN           | NO           |
| RF1_V1616A5B | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A5N/PV_D' != ' EDC_VLV_PV:CLOSED')                                                     | CLOSE          | NO           |
| RF1_V1616A5A | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A5N/PV_D' = ' EDC_VLV_PV:CLOSED')                                                      | OPEN           | NO           |
| RF1_V1616A5A | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A5N/PV_D' != ' EDC_VLV_PV:CLOSED')                                                     | CLOSE          | NO           |
| RF1_V1616A4M | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_PU16224/PV_D.CV' = ' EDC_MTR2_PV:STOPPED')                                                  | OPEN           | NO           |
| RF1_V1616A4M | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_PU16224/PV_D.CV' != ' EDC_MTR2_PV:STOPPED')                                                 | CLOSE          | NO           |
| RF1_V1616A4L | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A4J/PV_D.CV' = ' EDC_VLV_PV:OPEN') AND ('RF1_V1616A4J/PV_D.CV' = ' EDC_VLV_PV:CLOSED') | OPEN           | 10           |
| RF1_V1616A4L | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A4J/PV_D.CV' = ' EDC_VLV_PV:OPEN') AND ('RF1_V1616A4J/PV_D.CV' = ' EDC_VLV_PV:OPEN')   | CLOSE          | 10           |

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| Module       | Name        | Description | Expression                                                                                                              | Force SP State | Delay On (s) |
|--------------|-------------|-------------|-------------------------------------------------------------------------------------------------------------------------|----------------|--------------|
| RF1_V1616A4L | Condition 3 | Auto Logic  | ('TP01.CV' = TRUE) AND <CR>('RF1_V1616A4J/PV_D.CV' = 'EDC_VLV_PV:CLOSED')                                               | CLOSE          | NO           |
| RF1_V1616A4K | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND<CR>('RF1_V1616A4J/PV_D.CV' = 'EDC_VLV_PV:OPEN') AND <CR>('PV_D.CV' = 'EDC_VLV_PV:CLOSED')        | OPEN           | 10           |
| RF1_V1616A4K | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND <CR>('RF1_V1616A4J/PV_D.CV' = 'EDC_VLV_PV:OPEN') AND <CR>('PV_D.CV' = 'EDC_VLV_PV:OPEN')         | CLOSE          | 10           |
| RF1_V1616A4K | Condition 3 | Auto Logic  | ('TP01.CV' = TRUE) AND <CR>('RF1_V1616A4J/PV_D.CV' = 'EDC_VLV_PV:CLOSED')                                               | CLOSE          | NO           |
| RF1_V1616A4J | Condition 1 | Auto Logic  | 'TP01.CV' = TRUE AND 'RF1EQ_616A4/OP003_VALUE.CV' = 'NS_YES_NO:YES'                                                     | OPEN           | NO           |
| RF1_V1616A4J | Condition 2 | Auto Logic  | 'TP01.CV' = TRUE AND 'RF1EQ_616A4/OP003_VALUE.CV' = 'NS_YES_NO:NO'                                                      | CLOSE          | NO           |
| RF1_V1616A4H | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND <CR>('RF1_LSH1622/DI1/DISC_ACT' != TRUE) AND <CR>('RF1_V1616A3H/PV_D.CV' = 'EDC_VLV_PV:CLOSED')  | OPEN           | NO           |
| RF1_V1616A4H | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND <CR>('RF1_LSH1622/DI1/DISC_ACT' = TRUE OR<CR>'RF1_V1616A3H/PV_D.CV' != 'EDC_VLV_PV:CLOSED')      | CLOSE          | NO           |
| RF1_V1616A4G | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND<CR>('RF1_V1616A4H/PV_D.CV' = 'EDC_VLV_PV:OPEN' AND<CR>'RF1_V1616A3N/PV_D' = 'EDC_VLV_PV:OPEN')   | OPEN           | NO           |
| RF1_V1616A4G | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND<CR>('RF1_V1616A4H/PV_D.CV' != 'EDC_VLV_PV:OPEN' OR <CR>'RF1_V1616A3N/PV_D' != 'EDC_VLV_PV:OPEN') | CLOSE          | NO           |
| RF1_V1616A4G | Condition 3 | Auto Logic  | ('TP02.CV' = TRUE) AND<CR>('RF1_V1616A4I/PV_D.CV' = 'EDC_VLV_PV:OPEN')                                                  | OPEN           | NO           |
| RF1_V1616A4G | Condition 4 | Auto Logic  | ('TP02.CV' = TRUE) AND<CR>('RF1_V1616A4I/PV_D.CV' != 'EDC_VLV_PV:OPEN')                                                 | CLOSE          | NO           |

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| Module       | Name        | Description | Expression                                                                                                 | Force SP State | Delay On (s) |
|--------------|-------------|-------------|------------------------------------------------------------------------------------------------------------|----------------|--------------|
| RF1_V1616A4F | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A4M/PV_D.CV' = 'EDC_VLV_PV:CLOSED')                                      | OPEN           | NO           |
| RF1_V1616A4F | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A4M/PV_D.CV' != 'EDC_VLV_PV:CLOSED')                                     | CLOSE          | NO           |
| RF1_V1616A4E | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A4M/PV_D.CV' = 'EDC_VLV_PV:CLOSED')                                      | OPEN           | NO           |
| RF1_V1616A4E | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A4M/PV_D.CV' != 'EDC_VLV_PV:CLOSED')                                     | CLOSE          | NO           |
| RF1_V1616A4D | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A4M/PV_D.CV' = 'EDC_VLV_PV:CLOSED')                                      | OPEN           | NO           |
| RF1_V1616A4D | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A4M/PV_D.CV' != 'EDC_VLV_PV:CLOSED')                                     | CLOSE          | NO           |
| RF1_V1616A4B | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A4M/PV_D.CV' = 'EDC_VLV_PV:CLOSED')                                      | OPEN           | NO           |
| RF1_V1616A4B | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A4M/PV_D.CV' != 'EDC_VLV_PV:CLOSED')                                     | CLOSE          | NO           |
| RF1_V1616A4A | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A4M/PV_D.CV' = 'EDC_VLV_PV:CLOSED')                                      | OPEN           | NO           |
| RF1_V1616A4A | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A4M/PV_D.CV' != 'EDC_VLV_PV:CLOSED')                                     | CLOSE          | NO           |
| RF1_V1616A3M | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A3J/PV_D.CV' = 'EDC_VLV_PV:OPEN') AND ('PV_D.CV' = 'EDC_VLV_PV:CLOSED')  | OPEN           | 10           |
| RF1_V1616A3M | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A3J/PV_D.CV' != 'EDC_VLV_PV:CLOSED') AND ('PV_D.CV' = 'EDC_VLV_PV:OPEN') | CLOSE          | 10           |
| RF1_V1616A3M | Condition 3 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A3J/PV_D.CV' = 'EDC_VLV_PV:CLOSED')                                      | CLOSE          | NO           |

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| Module       | Name        | Description | Expression                                                                                                             | Force SP State | Delay On (s) |
|--------------|-------------|-------------|------------------------------------------------------------------------------------------------------------------------|----------------|--------------|
| RF1_V1616A3L | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND<CR>('RF1_V1616A3J/PV_D.CV' = 'EDC_VLV_PV:OPEN') AND <CR>('PV_D.CV' = 'EDC_VLV_PV:CLOSED')       | OPEN           | 10           |
| RF1_V1616A3L | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND <CR>('RF1_V1616A3J/PV_D.CV' != 'EDC_VLV_PV:CLOSED') AND <CR>('PV_D.CV' = 'EDC_VLV_PV:OPEN')     | CLOSE          | 10           |
| RF1_V1616A3L | Condition 3 | Auto Logic  | ('TP01.CV' = TRUE) AND <CR>('RF1_V1616A3J/PV_D.CV' = 'EDC_VLV_PV:CLOSED')                                              | CLOSE          | NO           |
| RF1_V1616A3K | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_PU16223/PV_D' = 'EDC_MTR2_PV:STOPPED')                                                    | OPEN           | NO           |
| RF1_V1616A3K | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_PU16223/PV_D' != 'EDC_MTR2_PV:STOPPED')                                                   | CLOSE          | NO           |
| RF1_V1616A3J | Condition 1 | Auto Logic  | 'TP01.CV' = TRUE AND 'RF1EQ_616A3/OP003_VALUE.CV' = 'NS_YES_NO:YES'                                                    | OPEN           | NO           |
| RF1_V1616A3J | Condition 2 | Auto Logic  | 'TP01.CV' = TRUE AND 'RF1EQ_616A3/OP003_VALUE.CV' = 'NS_YES_NO:NO'                                                     | CLOSE          | NO           |
| RF1_V1616A3H | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_LSH1622/DI1/DISC_ACT' != TRUE) AND <CR>('RF1_V1616A4H/PV_D.CV' = 'EDC_VLV_PV:CLOSED')     | OPEN           | NO           |
| RF1_V1616A3H | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND <CR>('RF1_LSH1622/DI1/DISC_ACT' = TRUE OR <CR>'RF1_V1616A4H/PV_D.CV' = 'EDC_VLV_PV:CLOSED')     | CLOSE          | NO           |
| RF1_V1616A3G | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND<CR>('RF1_V1616A3H/PV_D.CV' = 'EDC_VLV_PV:OPEN' AND <CR>'RF1_V1616A3N/PV_D' = 'EDC_VLV_PV:OPEN') | OPEN           | NO           |
| RF1_V1616A3G | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND<CR>('RF1_V1616A3H/PV_D' != 'EDC_VLV_PV:OPEN' OR <CR>'RF1_V1616A3N/PV_D' != 'EDC_VLV_PV:OPEN')   | CLOSE          | NO           |
| RF1_V1616A3G | Condition 3 | Auto Logic  | ('TP02.CV' = TRUE) AND<CR>('RF1_V1616A3I/PV_D.CV' = 'EDC_VLV_PV:OPEN')                                                 | OPEN           | NO           |

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| Module       | Name        | Description | Expression                                                                                                                                                                 | Force SP State | Delay On (s) |
|--------------|-------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------|
| RF1_V1616A3G | Condition 4 | Auto Logic  | ('TP02.CV' = TRUE) AND<CR>('RF1_V1616A3I/PV_D.CV' != ' EDC_VLV_PV:OPEN')                                                                                                   | CLOSE          | NO           |
| RF1_V1616A3F | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A3K/PV_D' = ' EDC_VLV_PV:CLOSED')                                                                                                        | OPEN           | NO           |
| RF1_V1616A3F | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A3K/PV_D' != ' EDC_VLV_PV:CLOSED')                                                                                                       | CLOSE          | NO           |
| RF1_V1616A3E | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A3K/PV_D' = ' EDC_VLV_PV:CLOSED')                                                                                                        | OPEN           | NO           |
| RF1_V1616A3E | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A3K/PV_D' != ' EDC_VLV_PV:CLOSED')                                                                                                       | CLOSE          | NO           |
| RF1_V1616A3D | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A3K/PV_D' = ' EDC_VLV_PV:CLOSED')                                                                                                        | OPEN           | NO           |
| RF1_V1616A3D | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A3K/PV_D' != ' EDC_VLV_PV:CLOSED')                                                                                                       | CLOSE          | NO           |
| RF1_V1616A3B | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A3K/PV_D' = ' EDC_VLV_PV:CLOSED')                                                                                                        | OPEN           | NO           |
| RF1_V1616A3B | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A3K/PV_D' != ' EDC_VLV_PV:CLOSED')                                                                                                       | CLOSE          | NO           |
| RF1_V1616A3A | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A3K/PV_D' = ' EDC_VLV_PV:CLOSED')                                                                                                        | OPEN           | NO           |
| RF1_V1616A3A | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A3K/PV_D' != ' EDC_VLV_PV:CLOSED')                                                                                                       | CLOSE          | NO           |
| RF1_PU16223  | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND<br><CR>('RF1_LT1622/PID1/LO_LO_ACT.CV' != TRUE AND<br><CR>'RF1_LSL1622/DI1/DISC_ACT' != TRUE<br>AND<CR>'RF1_V1622C/EDC1/PV_D' = ' EDC_VLV_PV:OPEN') | OPEN           | NO           |
| RF1_PU16223  | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND<br><CR>('RF1_LT1622/PID1/LO_LO_ACT.CV' = TRUE OR<br><CR>'RF1_LSL1622/DI1/DISC_ACT' = TRUE OR<br><CR>'RF1_V1622C/EDC1/PV_D' = ' EDC_VLV_PV:CLOSED')  | CLOSE          | NO           |

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| Module       | Name        | Description | Expression                                                                                                           | Force SP State | Delay On (s) |
|--------------|-------------|-------------|----------------------------------------------------------------------------------------------------------------------|----------------|--------------|
| RF1_V1616A2M | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND<CR>('RF1_V1616A2J/PV_D.CV' = 'EDC_VLV_PV:OPEN') AND <CR>('PV_D.CV' = 'EDC_VLV_PV:CLOSED')     | OPEN           | 10           |
| RF1_V1616A2M | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND <CR>('RF1_V1616A2J/PV_D.CV' = 'EDC_VLV_PV:OPEN') AND <CR>('PV_D.CV' = 'EDC_VLV_PV:OPEN')      | CLOSE          | 10           |
| RF1_V1616A2M | Condition 3 | Auto Logic  | ('TP01.CV' = TRUE) AND <CR>('RF1_V1616A2J/PV_D.CV' = 'EDC_VLV_PV:CLOSED')                                            | CLOSE          | NO           |
| RF1_V1616A2L | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND<CR>('RF1_V1616A2J/PV_D.CV' = 'EDC_VLV_PV:OPEN') AND <CR>('PV_D.CV' = 'EDC_VLV_PV:CLOSED')     | OPEN           | 10           |
| RF1_V1616A2L | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND <CR>('RF1_V1616A2J/PV_D.CV' = 'EDC_VLV_PV:OPEN') AND <CR>('PV_D.CV' = 'EDC_VLV_PV:OPEN')      | CLOSE          | 10           |
| RF1_V1616A2L | Condition 3 | Auto Logic  | ('TP01.CV' = TRUE) AND <CR>('RF1_V1616A2J/PV_D.CV' = 'EDC_VLV_SP:CLOSE')                                             | CLOSE          | NO           |
| RF1_V1616A2K | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_PU16222/PV_D' = 'EDC_MTR2_PV:STOPPED')                                                  | OPEN           | NO           |
| RF1_V1616A2K | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_PU16222/PV_D' != 'EDC_MTR2_PV:STOPPED')                                                 | CLOSE          | NO           |
| RF1_V1616A2J | Condition 1 | Auto Logic  | 'TP01.CV' = TRUE AND 'RF1EQ_616A2/OP003_VALUE.CV' = 'NS_YES_NO:YES'                                                  | OPEN           | NO           |
| RF1_V1616A2J | Condition 2 | Auto Logic  | 'TP01.CV' = TRUE AND 'RF1EQ_616A2/OP003_VALUE.CV' = 'NS_YES_NO:NO'                                                   | CLOSE          | NO           |
| RF1_V1616A2H | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_LSH1622/DI1/DISC_ACT' != TRUE) AND <CR>('RF1_V1616A1H/PV_D.CV' = 'EDC_VLV_PV:CLOSED')   | OPEN           | NO           |
| RF1_V1616A2H | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND <CR>('RF1_LSH1622/DI1/DISC_ACT' = TRUE OR<CR>('RF1_V1616A1H/PV_D.CV' != 'EDC_VLV_PV:CLOSED')) | CLOSE          | NO           |

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| Module       | Name        | Description | Expression                                                                                                                | Force SP State | Delay On (s) |
|--------------|-------------|-------------|---------------------------------------------------------------------------------------------------------------------------|----------------|--------------|
| RF1_V1616A2G | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND<CR>('RF1_V1616A2H/PV_D.CV' = 'EDC_VLV_PV:OPEN' AND <CR>'RF1_V1616A1N/PV_D.CV' = 'EDC_VLV_PV:OPEN') | OPEN           | NO           |
| RF1_V1616A2G | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND<CR>('RF1_V1616A2H/PV_D.CV' != 'EDC_VLV_PV:OPEN' OR <CR>'RF1_V1616A1N/PV_D.CV' = 'EDC_VLV_PV:OPEN') | CLOSE          | NO           |
| RF1_V1616A2G | Condition 3 | Auto Logic  | ('TP02.CV' = TRUE) AND<CR>('RF1_V1616A2I/PV_D.CV' = 'EDC_VLV_PV:OPEN')                                                    | OPEN           | NO           |
| RF1_V1616A2G | Condition 4 | Auto Logic  | ('TP02.CV' = TRUE) AND<CR>('RF1_V1616A2I/PV_D.CV' != 'EDC_VLV_PV:OPEN')                                                   | CLOSE          | 3            |
| RF1_V1616A2F | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A2K/PV_D.CV' = 'EDC_VLV_PV:CLOSED')                                                     | OPEN           | NO           |
| RF1_V1616A2F | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A2K/PV_D.CV' != 'EDC_VLV_PV:CLOSED')                                                    | CLOSE          | NO           |
| RF1_V1616A2E | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A2K/PV_D.CV' = 'EDC_VLV_PV:CLOSED')                                                     | OPEN           | NO           |
| RF1_V1616A2E | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A2K/PV_D.CV' != 'EDC_VLV_PV:CLOSED')                                                    | CLOSE          | NO           |
| RF1_V1616A2D | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A2K/PV_D.CV' = 'EDC_VLV_PV:CLOSED')                                                     | OPEN           | NO           |
| RF1_V1616A2D | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A2K/PV_D.CV' != 'EDC_VLV_PV:CLOSED')                                                    | CLOSE          | NO           |
| RF1_V1616A2B | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A2K/PV_D.CV' = 'EDC_VLV_PV:CLOSED')                                                     | OPEN           | NO           |
| RF1_V1616A2B | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A2K/PV_D.CV' != 'EDC_VLV_PV:CLOSED')                                                    | CLOSE          | NO           |
| RF1_V1616A2A | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A2K/PV_D.CV' = 'EDC_VLV_PV:CLOSED')                                                     | OPEN           | NO           |
| RF1_V1616A2A | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A2K/PV_D.CV' != 'EDC_VLV_PV:CLOSED')                                                    | CLOSE          | NO           |

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| Module       | Name        | Description | Expression                                                                                                         | Force SP State | Delay On (s) |
|--------------|-------------|-------------|--------------------------------------------------------------------------------------------------------------------|----------------|--------------|
| RF1_V1616A1M | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_PU16221/PV_D' = 'EDC_MTR2_PV:STOPPED')                                                | OPEN           | NO           |
| RF1_V1616A1M | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_PU16221/PV_D' != 'EDC_MTR2_PV:STOPPED')                                               | CLOSE          | NO           |
| RF1_V1616A1L | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND<CR>('RF1_V1616A1J/PV_D.CV' = 'EDC_VLV_PV:OPEN') AND <CR>('PV_D.CV' = 'EDC_VLV_PV:CLOSED')   | OPEN           | NO           |
| RF1_V1616A1L | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND<CR>('RF1_V1616A1J/PV_D.CV' = 'EDC_VLV_PV:OPEN') AND <CR>('PV_D.CV' = 'EDC_VLV_PV:OPEN')     | CLOSE          | NO           |
| RF1_V1616A1L | Condition 3 | Auto Logic  | ('TP01.CV' = TRUE) AND <CR>('RF1_V1616A1J/PV_D.CV' = 'EDC_VLV_PV:CLOSED')                                          | CLOSE          | NO           |
| RF1_V1616A1K | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND<CR>('RF1_V1616A1J/PV_D.CV' = 'EDC_VLV_PV:OPEN') AND <CR>('PV_D.CV' = 'EDC_VLV_PV:CLOSED')   | OPEN           | NO           |
| RF1_V1616A1K | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND <CR>('RF1_V1616A1J/PV_D.CV' = 'EDC_VLV_PV:OPEN') AND <CR>('PV_D.CV' = 'EDC_VLV_PV:OPEN')    | CLOSE          | NO           |
| RF1_V1616A1K | Condition 3 | Auto Logic  | ('TP01.CV' = TRUE) AND <CR>('RF1_V1616A1J/PV_D.CV' = 'EDC_VLV_PV:CLOSED')                                          | CLOSE          | NO           |
| RF1_V1616A1J | Condition 1 | Auto Logic  | 'TP01.CV' = TRUE AND 'RF1EQ_616A1/OP003_VALUE.CV' = 'NS_YES_NO:YES'                                                | OPEN           | NO           |
| RF1_V1616A1J | Condition 2 | Auto Logic  | 'TP01.CV' = TRUE AND 'RF1EQ_616A1/OP003_VALUE.CV' = 'NS_YES_NO:NO'                                                 | CLOSE          | NO           |
| RF1_V1616A1H | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_LSH1622/DI1/DISC_ACT' != TRUE) AND <CR>('RF1_V1616A2H/PV_D.CV' = 'EDC_VLV_PV:CLOSED') | OPEN           | NO           |
| RF1_V1616A1H | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND<CR>('RF1_LSH1622/DI1/DISC_ACT' = TRUE OR <CR>'RF1_V1616A2H/PV_D.CV' != 'EDC_VLV_PV:CLOSED') | CLOSE          | NO           |

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| Module       | Name        | Description | Expression                                                                                                                 | Force SP State | Delay On (s) |
|--------------|-------------|-------------|----------------------------------------------------------------------------------------------------------------------------|----------------|--------------|
| RF1_V1616A1G | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND<CR>('RF1_V1616A1N/PV_D.CV' = 'EDC_VLV_PV:OPEN' AND<CR>'RF1_V1616A1H/PV_D.CV' = 'EDC_VLV_PV:OPEN')   | OPEN           | 3            |
| RF1_V1616A1G | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND<CR>('RF1_V1616A1N/PV_D.CV' != 'EDC_VLV_PV:OPEN' OR <CR>'RF1_V1616A1H/PV_D.CV' != 'EDC_VLV_PV:OPEN') | CLOSE          | NO           |
| RF1_V1616A1G | Condition 3 | Auto Logic  | ('TP02.CV' = TRUE) AND<CR>('RF1_V1616A1I/PV_D.CV' != 'EDC_VLV_PV:OPEN')                                                    | OPEN           | 3            |
| RF1_V1616A1G | Condition 4 | Auto Logic  | ('TP02.CV' = TRUE) AND<CR>('RF1_V1616A1I/PV_D.CV' = 'EDC_VLV_PV:OPEN')                                                     | CLOSE          | NO           |
| RF1_V1616A1F | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A1M/PV_D.CV' = 'EDC_VLV_PV:CLOSED')                                                      | OPEN           | NO           |
| RF1_V1616A1F | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A1M/PV_D.CV' != 'EDC_VLV_PV:CLOSED')                                                     | CLOSE          | NO           |
| RF1_V1616A1E | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A1M/PV_D.CV' = 'EDC_VLV_PV:CLOSED')                                                      | OPEN           | NO           |
| RF1_V1616A1E | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A1M/PV_D.CV' != 'EDC_VLV_PV:CLOSED')                                                     | CLOSE          | NO           |
| RF1_V1616A1D | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A1M/PV_D.CV' = 'EDC_VLV_PV:CLOSED')                                                      | OPEN           | NO           |
| RF1_V1616A1D | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A1M/PV_D.CV' != 'EDC_VLV_PV:CLOSED')                                                     | CLOSE          | NO           |
| RF1_V1616A1B | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A1M/PV_D.CV' = 'EDC_VLV_PV:CLOSED')                                                      | OPEN           | NO           |
| RF1_V1616A1B | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A1M/PV_D.CV' != 'EDC_VLV_PV:CLOSED')                                                     | CLOSE          | NO           |
| RF1_V1616A1A | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A1M/PV_D.CV' = 'EDC_VLV_PV:CLOSED')                                                      | OPEN           | NO           |
| RF1_V1616A1A | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_V1616A1M/PV_D.CV' != 'EDC_VLV_PV:CLOSED')                                                     | CLOSE          | NO           |

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| Module       | Name        | Description | Expression                                                                                           | Force SP State | Delay On (s) |
|--------------|-------------|-------------|------------------------------------------------------------------------------------------------------|----------------|--------------|
| RF1_V1622B   | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND<br>('RF1_LSH1622/DI1/DISC_ACT' != TRUE) AND<br>'PV_D.CV' = ' VLVNC_PV:CLOSED' | OPEN           | 5            |
| RF1_V1622B   | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE OR 'TP02.CV' = TRUE) AND<br>('RF1_LSH1622/DI1/DISC_ACT' = TRUE)                    | CLOSE          | NO           |
| RF1_V1622B   | Condition 3 | Auto Logic  | ('TP01.CV' = TRUE) AND<br>('RF1_LSH1622/DI1/DISC_ACT' != TRUE) AND<br>'PV_D.CV' = ' VLVNC_PV:OPEN'   | OPEN           | 3            |
| RF1_V1622B   | Condition 4 | Auto Logic  | ('TP02.CV' = TRUE) AND<br>('RF1_LSH1622/DI1/DISC_ACT' != TRUE)                                       | CLOSE          | NO           |
| RF1_V1606B2F | Condition 1 | Auto Logic  | 'TP01.CV' = TRUE AND <CR>'RF1_V1606B2D/PV_D' =<br>' EDC_VLV_PV:CLOSED'                               | OPEN           | NO           |
| RF1_V1606B2F | Condition 2 | Auto Logic  | 'TP01.CV' = TRUE AND <CR>'RF1_V1606B2D/PV_D' !=<br>' EDC_VLV_PV:CLOSED'                              | CLOSE          | NO           |
| RF1_V1606B1F | Condition 1 | Auto Logic  | 'TP01.CV' = TRUE AND <CR>'RF1_V1606B1D/PV_D' =<br>' EDC_VLV_PV:CLOSED'                               | OPEN           | NO           |
| RF1_V1606B1F | Condition 2 | Auto Logic  | 'TP01.CV' = TRUE AND <CR>'RF1_V1606B1D/PV_D' !=<br>' EDC_VLV_PV:CLOSED'                              | CLOSE          | NO           |
| RF1_V1606B1D | Condition 1 | Auto Logic  | 'TP01.CV' = TRUE AND <CR>'RF1_V1606B1F/PV_D' =<br>' EDC_VLV_PV:CLOSED'                               | OPEN           | NO           |
| RF1_V1606B1D | Condition 2 | Auto Logic  | 'TP01.CV' = TRUE AND <CR>'RF1_V1606B1F/PV_D' !=<br>' EDC_VLV_PV:CLOSED'                              | CLOSE          | NO           |
| RF1_V1606A6F | Condition 1 | Auto Logic  | 'TP01.CV' = TRUE AND <CR>'RF1_V1606A6D/PV_D.CV' =<br>' EDC_VLV_PV:CLOSED'                            | OPEN           | NO           |
| RF1_V1606A6F | Condition 2 | Auto Logic  | 'TP01.CV' = TRUE AND <CR>'RF1_V1606A6D/PV_D.CV' =<br>' EDC_VLV_PV:OPEN'                              | CLOSE          | NO           |
| RF1_V1606A6D | Condition 1 | Auto Logic  | 'TP01.CV' = TRUE AND <CR>'RF1_V1606A6F/PV_D.CV' =<br>' EDC_VLV_PV:CLOSED'                            | OPEN           | NO           |
| RF1_V1606A6D | Condition 2 | Auto Logic  | 'TP01.CV' = TRUE AND <CR>'RF1_V1606A6F/PV_D.CV' =<br>' EDC_VLV_PV:OPEN'                              | CLOSE          | NO           |

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| Module       | Name        | Description | Expression                                                                | Force SP State | Delay On (s) |
|--------------|-------------|-------------|---------------------------------------------------------------------------|----------------|--------------|
| RF1_V1606A5F | Condition 1 | Auto Logic  | '/TP01.CV' = TRUE AND <CR>'//RF1_V1606A5D/PV_D.CV' = ' EDC_VLV_PV:CLOSED' | OPEN           | NO           |
| RF1_V1606A5F | Condition 2 | Auto Logic  | '/TP01.CV' = TRUE AND <CR>'//RF1_V1606A5D/PV_D.CV' = ' EDC_VLV_PV:OPEN'   | CLOSE          | NO           |
| RF1_V1606A5D | Condition 1 | Auto Logic  | '/TP01.CV' = TRUE AND <CR>'//RF1_V1606A5F/PV_D.CV' = ' EDC_VLV_PV:CLOSED' | OPEN           | NO           |
| RF1_V1606A5D | Condition 2 | Auto Logic  | '/TP01.CV' = TRUE AND <CR>'//RF1_V1606A5F/PV_D.CV' = ' EDC_VLV_PV:OPEN'   | CLOSE          | NO           |
| RF1_V1606A4F | Condition 1 | Auto Logic  | '/TP01.CV' = TRUE AND <CR>'//RF1_V1606A4D/PV_D.CV' = ' EDC_VLV_PV:CLOSED' | OPEN           | NO           |
| RF1_V1606A4F | Condition 2 | Auto Logic  | '/TP01.CV' = TRUE AND <CR>'//RF1_V1606A4D/PV_D.CV' = ' EDC_VLV_PV:OPEN'   | CLOSE          | NO           |
| RF1_V1606A4D | Condition 1 | Auto Logic  | '/TP01.CV' = TRUE AND <CR>'//RF1_V1606A4F/PV_D.CV' = ' EDC_VLV_PV:CLOSED' | OPEN           | NO           |
| RF1_V1606A4D | Condition 2 | Auto Logic  | '/TP01.CV' = TRUE AND <CR>'//RF1_V1606A4F/PV_D.CV' = ' EDC_VLV_PV:OPEN'   | CLOSE          | NO           |
| RF1_V1606A3F | Condition 1 | Auto Logic  | '/TP01.CV' = TRUE AND <CR>'//RF1_V1606A3D/PV_D.CV' = ' EDC_VLV_PV:CLOSED' | OPEN           | NO           |
| RF1_V1606A3F | Condition 2 | Auto Logic  | '/TP01.CV' = TRUE AND <CR>'//RF1_V1606A3D/PV_D.CV' = ' EDC_VLV_PV:OPEN'   | CLOSE          | NO           |
| RF1_V1606A3D | Condition 1 | Auto Logic  | '/TP01.CV' = TRUE AND <CR>'//RF1_V1606A3F/PV_D.CV' = ' EDC_VLV_PV:CLOSED' | OPEN           | NO           |
| RF1_V1606A3D | Condition 2 | Auto Logic  | '/TP01.CV' = TRUE AND <CR>'//RF1_V1606A3F/PV_D.CV' = ' EDC_VLV_PV:OPEN'   | CLOSE          | NO           |
| RF1_V1606A2F | Condition 1 | Auto Logic  | '/TP01.CV' = TRUE AND <CR>'//RF1_V1606A2D/PV_D.CV' = ' EDC_VLV_PV:CLOSED' | OPEN           | NO           |
| RF1_V1606A2F | Condition 2 | Auto Logic  | '/TP01.CV' = TRUE AND <CR>'//RF1_V1606A2D/PV_D.CV' = ' EDC_VLV_PV:OPEN'   | CLOSE          | NO           |
| RF1_V1606A2D | Condition 1 | Auto Logic  | '/TP01.CV' = TRUE AND <CR>'//RF1_V1606A2F/PV_D.CV' = ' EDC_VLV_PV:CLOSED' | OPEN           | NO           |

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| Module       | Name        | Description | Expression                                                                                                                                                                                         | Force SP State | Delay On (s) |
|--------------|-------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------|
| RF1_V1606A2D | Condition 2 | Auto Logic  | '/TP01.CV' = TRUE AND <CR>'//RF1_V1606A2F/PV_D.CV' = ' _EDC_VLV_PV:OPEN'                                                                                                                           | CLOSE          | NO           |
| RF1_V1606A1F | Condition 1 | Auto Logic  | '/TP01.CV' = TRUE AND <CR>'//RF1_V1606A1D/PV_D.CV' = ' _EDC_VLV_PV:CLOSED'                                                                                                                         | OPEN           | NO           |
| RF1_V1606A1F | Condition 2 | Auto Logic  | '/TP01.CV' = TRUE AND <CR>'//RF1_V1606A1D/PV_D.CV' = ' _EDC_VLV_PV:OPEN'                                                                                                                           | CLOSE          | NO           |
| RF1_V1606A1D | Condition 1 | Auto Logic  | '/TP01.CV' = TRUE AND <CR>'//RF1_V1606A1F/PV_D.CV' = ' _EDC_VLV_PV:CLOSED'                                                                                                                         | OPEN           | NO           |
| RF1_V1606A1D | Condition 2 | Auto Logic  | '/TP01.CV' = TRUE AND <CR>'//RF1_V1606A1F/PV_D.CV' = ' _EDC_VLV_PV:OPEN'                                                                                                                           | CLOSE          | NO           |
| RF1_V1606C2D | Condition 1 | Auto Logic  | '^/TP01'=1 AND '//RF1_V1606C2F/PV_D' = ' _EDC_VLV_PV:CLOSED'                                                                                                                                       | OPEN           | NO           |
| RF1_V1606C2D | Condition 2 | Auto Logic  | '^/TP01'=1 AND '//RF1_V1606C2F/PV_D' = ' _EDC_VLV_PV:OPEN'                                                                                                                                         | CLOSE          | NO           |
| RF1_V1606C1F | Condition 1 | Auto Logic  | '^/TP01'=1 AND '//RF1_V1606C1D/PV_D' = ' _EDC_VLV_PV:CLOSED'                                                                                                                                       | OPEN           | NO           |
| RF1_V1606C1F | Condition 2 | Auto Logic  | '^/TP01'=1 AND '//RF1_V1606C1D/PV_D' = ' _EDC_VLV_PV:OPEN'                                                                                                                                         | CLOSE          | NO           |
| RF1_V1606C1D | Condition 1 | Auto Logic  | '^/TP01'=1 AND '//RF1_V1606C1F/PV_D' = ' _EDC_VLV_PV:CLOSED'                                                                                                                                       | OPEN           | NO           |
| RF1_V1606C1D | Condition 2 | Auto Logic  | '^/TP01'=1 AND '//RF1_V1606C1F/PV_D' = ' _EDC_VLV_PV:OPEN'                                                                                                                                         | CLOSE          | NO           |
| RF1_V1692BD  | Condition 1 | Auto Logic  | ('/TP01.CV' = TRUE) AND<br><CR>('//RF1_LSM1604C/DI1/DISC_ACT.CV' != TRUE OR<br><CR>'//RF1_LSH1604C/DI1/DISC_ACT.CV' != TRUE) AND<br><CR>'//RF1_PU1682B/EDC1/PV_D.CV' =<br>' _EDC_MTR2_PV:RUNNING') | OPEN           | NO           |
| RF1_V1692BD  | Condition 2 | Auto Logic  | ('/TP01.CV' = TRUE) AND<br><CR>('//RF1_LSM1604C/DI1/DISC_ACT.CV' = TRUE AND<br><CR>'//RF1_LSH1604C/DI1/DISC_ACT.CV' = TRUE) OR<br><CR>'//RF1_PU1682B/EDC1/PV_D.CV' =<br>' _EDC_MTR2_PV:STOPPED')   | CLOSE          | NO           |

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| Module       | Name        | Description | Expression                                                                                                                                                                                                                                                                                                                   | Force SP State | Delay On (s) |
|--------------|-------------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------|
| RF1_V1604CV  | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_PT1622/AI1/HI_HI_ACT.CV' != TRUE)                                                                                                                                                                                                                                                               | OPEN           | NO           |
| RF1_V1604CV  | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND ('RF1_PT1622/AI1/HI_HI_ACT.CV' = TRUE)                                                                                                                                                                                                                                                                | CLOSE          | NO           |
| RF1_V1604CS  | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND <CR>('RF1_TT1604C/PV' < ('RF1EQ_604C/TP001_VALUE.CV' + 'RF1EQ_604C/TP002_VALUE.CV'))                                                                                                                                                                                                                  | OPEN           | NO           |
| RF1_V1604CS  | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND <CR>('RF1_TT1604C/PV' >= ('RF1EQ_604C/TP001_VALUE.CV' + 'RF1EQ_604C/TP002_VALUE.CV'))                                                                                                                                                                                                                 | CLOSE          | NO           |
| RF1_PU1604C  | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND <CR>('RF1_LSL1604C/DI1/DISC_ACT.CV' != TRUE AND <CR>('RF1_V1616A1B/PV_D' = 'EDC_VLV_PV:OPEN' OR <CR> 'RF1_V1616A2B/PV_D' = 'EDC_VLV_PV:OPEN' OR <CR> 'RF1_V1616A3B/PV_D' = 'EDC_VLV_PV:OPEN' OR <CR> 'RF1_V1616A4B/PV_D' = 'EDC_VLV_PV:OPEN' OR <CR> 'RF1_V1616A5B/PV_D' = 'EDC_VLV_PV:OPEN'))        | OPEN           | NO           |
| RF1_PU1604C  | Condition 2 | Auto Logic  | ('TP01.CV' = TRUE) AND <CR>('RF1_LSL1604C/DI1/DISC_ACT.CV' = TRUE OR <CR>('RF1_V1616A1B/PV_D' != 'EDC_VLV_PV:OPEN' AND <CR> 'RF1_V1616A2B/PV_D' != 'EDC_VLV_PV:OPEN' AND <CR> 'RF1_V1616A3B/PV_D' != 'EDC_VLV_PV:OPEN' AND <CR> 'RF1_V1616A4B/PV_D' != 'EDC_VLV_PV:OPEN' AND <CR> 'RF1_V1616A5B/PV_D' != 'EDC_VLV_PV:OPEN')) | CLOSE          | NO           |
| RF1_V1616A3N | Condition 1 | Auto Logic  | 'MODE.ACTUAL' = CAS AND <CR>('RF1_V1616A3H/PV_D.CV' = 'VLVNC_PV:OPEN' OR <CR>'RF1_V1616A4H/PV_D.CV' = 'VLVNC_PV:OPEN')                                                                                                                                                                                                       | OPEN           | NO           |
| RF1_V1616A3N | Condition 2 | Auto Logic  | 'MODE.ACTUAL' = CAS AND <CR>('RF1_V1616A3H/PV_D.CV' != 'VLVNC_PV:OPEN' AND <CR>'RF1_V1616A4H/PV_D.CV' != 'VLVNC_PV:OPEN')                                                                                                                                                                                                    | CLOSE          | NO           |

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| Module       | Name        | Description | Expression                                                                                                                                                                                                                                                                                                                                                    | Force SP State | Delay On (s) |
|--------------|-------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------|
| RF1_V1616A1N | Condition 1 | Auto Logic  | '/MODE.ACTUAL' = CAS AND <CR>('/RF1_V1616A1H/PV_D.CV' = ' VLVNC_PV:OPEN' OR <CR>'/RF1_V1616A2H/PV_D.CV' = ' VLVNC_PV:OPEN')                                                                                                                                                                                                                                   | OPEN           | NO           |
| RF1_V1616A1N | Condition 2 | Auto Logic  | '/MODE.ACTUAL' = CAS AND <CR>('/RF1_V1616A1H/PV_D.CV' != ' VLVNC_PV:OPEN' AND <CR>'/RF1_V1616A2H/PV_D.CV' != ' VLVNC_PV:OPEN')                                                                                                                                                                                                                                | CLOSE          | NO           |
| RF1_V1630C   | Condition 1 | Auto Logic  | '^/TP01'=1 AND ('/RF1_V1606C1D/PV_D' = ' EDC_VLV_PV:CLOSED' OR '/RF1_V1606C2D/PV_D' = ' EDC_VLV_PV:CLOSED')                                                                                                                                                                                                                                                   | OPEN           | NO           |
| RF1_V1630C   | Condition 2 | Auto Logic  | '^/TP01'=1 AND ('/RF1_V1606C1D/PV_D' = ' EDC_VLV_PV:OPEN' OR '/RF1_V1606C2D/PV_D' = ' EDC_VLV_PV:OPEN')                                                                                                                                                                                                                                                       | CLOSE          | NO           |
| RF1_V1630B   | Condition 1 | Auto Logic  | ('/TP01.CV' = TRUE AND <CR>'/RF1_V1606B1D/PV_D' = ' EDC_VLV_PV:CLOSED' AND <CR>'/RF1_V1606B2D/PV_D' = ' EDC_VLV_PV:CLOSED')                                                                                                                                                                                                                                   | OPEN           | NO           |
| RF1_V1630B   | Condition 2 | Auto Logic  | '/TP01.CV' = TRUE AND <CR>('/RF1_V1606B1D/PV_D' != ' EDC_VLV_PV:CLOSED' OR <CR>'/RF1_V1606B2D/PV_D' != ' EDC_VLV_PV:CLOSED')                                                                                                                                                                                                                                  | CLOSE          | NO           |
| RF1_V1630A34 | Condition 1 | Auto Logic  | '/TP01.CV' = TRUE AND <CR>('/RF1_V1606A1D/PV_D.CV' = ' EDC_VLV_PV:CLOSED' AND <CR>'/RF1_V1606A2D/PV_D.CV' = ' EDC_VLV_PV:CLOSED' AND <CR>'/RF1_V1606A3D/PV_D.CV' = ' EDC_VLV_PV:CLOSED' AND <CR>'/RF1_V1606A4D/PV_D.CV' = ' EDC_VLV_PV:CLOSED' AND <CR>'/RF1_V1606A5D/PV_D.CV' = ' EDC_VLV_PV:CLOSED' AND <CR>'/RF1_V1606A6D/PV_D.CV' = ' EDC_VLV_PV:CLOSED') | OPEN           | NO           |
| RF1_V1630A34 | Condition 2 | Auto Logic  | '/TP01.CV' = TRUE AND <CR>('/RF1_V1606A1D/PV_D.CV' = ' EDC_VLV_PV:OPEN' OR <CR>'/RF1_V1606A2D/PV_D.CV' = ' EDC_VLV_PV:OPEN' OR <CR>'/RF1_V1606A3D/PV_D.CV' = ' EDC_VLV_PV:OPEN' OR <CR>'/RF1_V1606A4D/PV_D.CV' = ' EDC_VLV_PV:OPEN' OR <CR>'/RF1_V1606A5D/PV_D.CV' = ' EDC_VLV_PV:OPEN' OR <CR>'/RF1_V1606A6D/PV_D.CV' = ' EDC_VLV_PV:OPEN')                  | CLOSE          | NO           |

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| Module       | Name        | Description | Expression                                                                                                                                                                                                                                                                                                                                                        | Force SP State | Delay On (s) |
|--------------|-------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------|
|              |             |             | '_EDC_VLV_PV:OPEN' OR <CR>'//RF1_V1606A6D/PV_D.CV' = '_EDC_VLV_PV:OPEN')                                                                                                                                                                                                                                                                                          |                |              |
| RF1_V1630A12 | Condition 1 | Auto Logic  | '/TP01.CV' = TRUE AND <CR>'//RF1_V1606A1D/PV_D.CV' = '_EDC_VLV_PV:CLOSED' AND <CR>'//RF1_V1606A2D/PV_D.CV' = '_EDC_VLV_PV:CLOSED' AND <CR>'//RF1_V1606A3D/PV_D.CV' = '_EDC_VLV_PV:CLOSED' AND <CR>'//RF1_V1606A4D/PV_D.CV' = '_EDC_VLV_PV:CLOSED' AND <CR>'//RF1_V1606A5D/PV_D.CV' = '_EDC_VLV_PV:CLOSED' AND <CR>'//RF1_V1606A6D/PV_D.CV' = '_EDC_VLV_PV:CLOSED' | OPEN           | NO           |
| RF1_V1630A12 | Condition 2 | Auto Logic  | '/TP01.CV' = TRUE AND <CR>'//RF1_V1606A1D/PV_D.CV' = '_EDC_VLV_PV:OPEN' OR <CR>'//RF1_V1606A2D/PV_D.CV' = '_EDC_VLV_PV:OPEN' OR <CR>'//RF1_V1606A3D/PV_D.CV' = '_EDC_VLV_PV:OPEN' OR <CR>'//RF1_V1606A4D/PV_D.CV' = '_EDC_VLV_PV:OPEN' OR <CR>'//RF1_V1606A5D/PV_D.CV' = '_EDC_VLV_PV:OPEN' OR <CR>'//RF1_V1606A6D/PV_D.CV' = '_EDC_VLV_PV:OPEN')                 | CLOSE          | NO           |
| RF1_V1024F   | Condition 1 | Auto Logic  | ('/TP01.CV' = TRUE AND '/RF1_LSH1006/DI1/DISC_ACT' != TRUE);                                                                                                                                                                                                                                                                                                      | OPEN           | NO           |
| RF1_V1024F   | Condition 2 | Auto Logic  | ('/TP01.CV' = TRUE AND '/RF1_LSH1006/DI1/DISC_ACT' = TRUE);                                                                                                                                                                                                                                                                                                       | CLOSE          | NO           |
| RF1_V1024F   | Condition 3 | Auto Logic  | ('/TP02.CV' = TRUE) AND ('//RF1EQ_501/OP001_VALUE.CV' = 'NS_MNA_SELECTION:MNA1' AND ('//RF1_LSH1001/DI1/DISC_ACT' != TRUE OR '/RF1_LT1001/AI1/HI_HI_ACT.CV' != TRUE))                                                                                                                                                                                             | OPEN           | NO           |
| RF1_V1024F   | Condition 4 | Auto Logic  | ('/TP02.CV' = TRUE) AND ('//RF1EQ_501/OP001_VALUE.CV' = 'NS_MNA_SELECTION:MNA2' OR ('//RF1_LSH1001/DI1/DISC_ACT' = TRUE AND '/RF1_LT1001/AI1/HI_HI_ACT.CV' = TRUE))                                                                                                                                                                                               | CLOSE          | NO           |
| RF1_LT1082W  | Condition 1 | Auto Logic  | ('/TP01.CV' = TRUE) AND <CR>'//RF1_LSL1082W/DI1/DISC_ACT' = TRUE AND <CR>'//RF1_LSH1032C/DI1/DISC_ACT' = TRUE) AND <CR>'//MODELOCK_OVR.CV' = FALSE                                                                                                                                                                                                                | 0              | NO           |

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| Module       | Name        | Description | Expression                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Force SP State | Delay On (s) |
|--------------|-------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------|
| RF1_TT1021D  | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND<CR>('RF1_PU1001/PV_D.CV' != 'EDC_MTR2_PV:RUNNING')                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0              | NO           |
| RF1_FT1978   | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND <CR>('RF1_PU1078HW/PV_D.CV' != 'EDC_MTR2_PV:RUNNING' OR 'RF1EQ_501/OP011_VALUE' != TRUE OR ('RF1_FT_011_SL/PV.CV' <= 5000)) AND <CR>'MODELOCK_OVR.CV' = FALSE                                                                                                                                                                                                                                                                                                                                                                                  | 0              | NO           |
| RF1_FT1034NA | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND <CR>('RF1EQ_501/OP010_VALUE' != TRUE OR 'RF1_LT1034NA/AI1/LO_LO_ACT.CV' = TRUE OR 'RF1_PU1001/PV_D.CV' = 'EDC_MTR2_PV:STOPPED' OR <CR>'RF1_FT1001/PV.CV' <= 5000) AND 'MODELOCK_OVR.CV' = FALSE                                                                                                                                                                                                                                                                                                                                                | 0              | NO           |
| RF1_FT1034CA | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND <CR>('RF1_LT1034CA/AI1/LO_LO_ACT.CV' = TRUE OR 'RF1EQ_501/OP009_VALUE.CV' != TRUE OR <CR>'RF1_FT_011_SL/PV.CV' <= 5000) AND <CR>'MODELOCK_OVR.CV' = FALSE                                                                                                                                                                                                                                                                                                                                                                                      | 0              | NO           |
| RF1_FT1001   | Condition 1 | Auto Logic  | ('TP01.CV' = TRUE) AND <CR>('RF1_LSL1001/DI1/DISC_ACT.CV' = TRUE OR <CR>'RF1_LT1001/AI1/LO_LO_ACT.CV' = TRUE OR <CR>'RF1_V1001D/PV_D.CV' = 'EDC_VLV_PV:CLOSED' OR<CR>('RF1_FT1978/PID1/DV_HI_ACT.CV' = TRUE OR 'RF1_FT1978/PID1/DV_LO_ACT.CV' = TRUE) OR<CR>('RF1_FT1034NA/PID1/DV_HI_ACT.CV' = TRUE OR 'RF1_FT1034NA/PID1/DV_LO_ACT.CV' = TRUE) OR<CR>('RF1_FT1034CA/PID1/DV_HI_ACT.CV' = TRUE OR 'RF1_FT1034CA/PID1/DV_LO_ACT.CV' = TRUE) OR<CR>('RF1_LSH1006/DI1/DISC_ACT.CV' = TRUE AND 'RF1_LT1006/PID1/HI_HI_ACT.CV' = TRUE)) AND <CR>'MODELOCK_OVR.CV' = FALSE | 0              | NO           |
| RF1_TT1721IC | Condition 1 | Auto Logic  | 'RF1_PU1781IB/PV_D.CV' = 'EDC_MTR2_PV:STOPPED' AND<CR>'TP01.CV' = TRUE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 0              | NO           |

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| Module       | Name        | Description | Expression                                                                                                                                                                                       | Force SP State | Delay On (s) |
|--------------|-------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------|
| RF1_TT1781AA | Condition 1 | Auto Logic  | '/TP01.CV' = TRUE AND<CR>('/RF1_PU1714AG/PV_D.CV' != ' _EDC_MTR2_PV:RUNNING') AND<CR>'/MODELOCK_OVR.CV' = FALSE                                                                                  | 0              | NO           |
| RF1_LT1714   | Condition 1 | Auto Logic  | '/TP01.CV' = TRUE AND<CR>'/RF1_LSL1714/DI1/DISC_ACT' = TRUE AND<CR>'/MODELOCK_OVR.CV' = FALSE                                                                                                    | 0              | NO           |
| RF1_TT1781B  | Condition 1 | Auto Logic  | (<CR>'/RF1_PU17221/PV_D.CV' != ' _EDC_MTR2_PV:RUNNING' AND <CR>'/RF1_PU17222/PV_D.CV' != ' _EDC_MTR2_PV:RUNNING') AND<CR>'/TP01.CV' = TRUE AND <CR>'/MODELOCK_OVR.CV' = FALSE                    | 0              | NO           |
| RF1_TT1722C  | Condition 1 | Auto Logic  | '/TP01.CV' = TRUE AND <CR>'/PV.CV' >= '/RF1EQ_822QE/TP012_VALUE.CV' AND <CR>'/RF1EQ_822QE/OP009_VALUE.CV' = TRUE AND <CR>'/RF1EQ_822QE/OP010_VALUE.CV' = TRUE AND<CR>'/MODELOCK_OVR.CV' = FALSE; | 0              | NO           |
| RF1_TT1722C  | Condition 2 | Auto Logic  | '/TP01.CV' = TRUE AND '/RF1EQ_822QE/OP009_VALUE.CV' != TRUE AND <CR>('/MODELOCK_OVR.CV' = FALSE);                                                                                                | 0              | NO           |
| RF1_LT1780B  | Condition 1 | Auto Logic  | (<CR>'/RF1_PU17221/PV_D.CV' != ' _EDC_MTR2_PV:RUNNING' AND <CR>'/RF1_PU17222/PV_D.CV' != ' _EDC_MTR2_PV:RUNNING') AND<CR>'/TP01.CV' = TRUE AND <CR>'/MODELOCK_OVR.CV' = FALSE                    | 0              | NO           |
| RF1_LT1722C  | Condition 1 | Auto Logic  | (<CR>'/RF1_PU1722C1/PV_D.CV' != ' _EDC_MTR2_PV:RUNNING' AND <CR>'/RF1_PU1722C2/PV_D.CV' != ' _EDC_MTR2_PV:RUNNING') AND<CR>'/TP01.CV' = TRUE AND <CR>'/MODELOCK_OVR.CV' = FALSE                  | 0              | NO           |
| RF1_FT1702   | Condition 1 | Auto Logic  | '/RF1_PU1702/PV_D.CV' != ' _EDC_MTR2_PV:RUNNING' AND<CR>'/TP01.CV' = TRUE AND <CR>'/MODELOCK_OVR.CV' = FALSE                                                                                     | 0              | NO           |
| RF1_FT1734CA | Condition 1 | Auto Logic  | (<CR>'/TP01.CV' = TRUE) AND (<CR>'/RF1_LT1734CA/AI1/LO_LO_ACT.CV' = TRUE) AND <CR>('/MODELOCK_OVR.CV' = FALSE);                                                                                  | 0              | NO           |

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|-------------|-------------|-------------|-----------------------------------------------------------------------------------------------------------|----------------|--------------|
| RF1_TT1021B | Condition 1 | Auto Logic  | '/TP01.CV' = TRUE AND '/RF1_FT1011/PID1/PV.CV' <= 1000 AND <CR>' /MODELOCK_OVR.CV' = FALSE;               | 0              | NO           |
| RF1_LT1006  | Condition 1 | Auto Logic  | ((('/TP01.CV' = TRUE) AND '/RF1_PU1006/PV_D.CV' = ' EDC_MTR2_PV:STOPPED' AND '/MODELOCK_OVR.CV' = FALSE ) | 0              | NO           |
| RF1_FT1011  | Condition 1 | Auto Logic  | ('/TP01.CV' = TRUE AND '/RF1_LSH1006/DI1/DISC_ACT' = TRUE AND <CR>' /MODELOCK_OVR.CV' = FALSE);           | 0              | NO           |

**Table 188: Module Force Setpoints**